52. How many mL conc. HNO_3 and how many mL of water are required to prepare 500 mL of 0.1 M HNO_3 from a conc. 13M HNO_3?

A. 1 mL HNO_3 and 496.15 mL of H_2O
B. 3 mL HNO_3 and 500 mL H_2O
C. 3.85 mL HNO_3 and 500 mL H_2O
D. 3.85 mL HNO_3 and 496.15 mL H_2O

Solution:

V_1 =? V_2 = 500 mL
M_1 = 13 M M_2 = 0.1M

From law of dilution, we know that

$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{0.1M \times 500mL}{13M} = 3.85mL$$

V_{H_2O} added $= V_2 - V_1$

V_{H_2O} added = 500 mL – 3.85 mL = 496.15 mL **Answer: D**

53. Which of the following reactions is expected to have negative value of entropy change (Δs)?

A. $C_6H_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$
B. $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
C. $N_2O_4(g) + Cl_2(g) \longrightarrow 2NOCl(g) + O_2(g)$
D. $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Explanation:

Entropy (s) is the thermodynamic property of a system that is related to its degree of randomness of disorder. The entropy of gas >> liquid > solid. If the number of moles of gaseous product is greater than that of gaseous reactant, the entropy change is expected to be positive on the other hand, if the number of moles of gaseous reactant is greater than the number of moles of gaseous product, the entropy change will be expected to be negative. **Answer: D**

54. Gaseous petrol in a combustion system has done 375 kJ of work during its expansion in the piston. Simultaneous, it absorbed 586 kJ of heat from the engine. What is the internal energy change during the process?

A. +211 kJ C. -211 Kj
B. +961 kJ D. -961 kJ

Solution:

Work done by the system has a negative sign while work done by the surroundings has a positive sign. Heat absorbed by a system is positive while heat released by the system is negative.

$\Delta E = W + q$

ΔE = -375 kJ + 586 kJ = +211 kJ **Answer: A**

55. Consider the following gaseous reaction at 25°C:

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$

ΔH = -802 kJ

Which energy change would occur if 3.2 moles of CH_4 is completely combusted?

A. 2.57×10^2 kJ will be released
B. 2.57×10^2 kJ will be absorbed
C. 6.43×10^3 kJ will be released
D. 6.43×10^3 kJ will be absorbed

Solution:

Combustion reaction is exothermic reaction. The energy released when 1 mole of CH_4 is combusted is 802 kJ. Therefore, when 3.2 moles of CH_4 is combusted the energy released 3.2 × 802 kJ = 2566.4 kJ = 2.57×10^2 kJ. **Answer: A**

56. What is the quantity of heat required to raise the temperature of 80 g of ethanol from 25°C to 75°C?
(Specific heat of ethanol = $2.46 Jg^{-1}k^{-1}$)

A. 2.46 kJ C. 9.84 kJ
B. 4.0 kJ D. 18.68 kJ

Solution:

$\Delta T = T_f - T_i$ = 75°C - 25°C = 50°C = 50 K
m = 80 g
c = 2.46 $Jg^{-1}k^{-1}$
$q = mc\Delta T$
q = 80 g × 2.46 J $g^{-1}k^{-1}$ × 50 k = 9840 J
q = 9.84 kJ **Answer: C**

57. How many kilo joules of heat are absorbed when 20 g of NaCl (s) is decomposed into Na (s) and Cl_2 (g) at constant pressure according to the following reaction?

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$, ΔH -821.8 kJ

A. -281.0 C. +140.5
B. -140.5 D. +281.0

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Solution:
When we reverse the given reaction, the sign of ΔH will be changed.

$\Rightarrow 2NaCl\,(s) \longrightarrow 2Na\,(s) + Cl_2\,(g),\ \Delta H = +821.8\ kJ$

$2 \times 58.5\ g = 821.8\ kJ$
NaCl

20 g of =?
NaCl

$$? = \frac{20g \times 821.8kJ}{117g} = 140.5kJ$$

⇒ The amount of heat absorbed to decompose 20 g of NaCl in to Na and Cl_2 is 140.5 kJ. **Answer: C**

58. Enthalpy is defined as the heat content of the system at constant:
A. Heat B. Moles C. pressure D. Volume

Explanation:
Enthalpy is defined as the sum of the internal energy and the pressure – volume product of a system.

$H = E = PV$

$\Delta H = \Delta E = P\Delta V$ (at constant pressure)

but $\Delta E = q - P\Delta V$

$\Rightarrow \Delta H = q - P\Delta V + P\Delta V;\ \Delta H = q_P$

Enthalpy change is heat transferred at constant pressure.

Answer: C

59. Which of the following metals is NOT obtained by commercial electrolytic process?
A. Ag B. Al C. Cu D. Na

Explanation:
Most silver (Ag) produced now a days is as a by – product of the production of other metals such as zinc, copper and lead. Cyanidation process is also used to manufacture silver.

$$Zn\,(s) + 2Ag(CN)_2^- + 4OH^- \longrightarrow 2Ag + 4CN^- + Zn(OH)_4^{2-}$$

Answer: A

60. What is the equivalent weight of HNO_3, as an oxidizing agent, in the following balanced reaction?

$$3Fe^{2+} + 4H^+ + NO_3^- \longrightarrow 3Fe^{3+} + NO + 2H_2O$$

A. 10.50 B. 15.75 C. 21.00 D. 31.50

Solution:
The oxidation state of N in NO_3^- is +5 while in NO is +2. The total electrons gained will be 3.

$$E_{HNO_3} = \frac{M_{HNO_3}}{Z}$$

E – equivalent mass
Z – electrons gained or lost
M – molar mass

$$E_{HNO_3} = \frac{63\ g/mol}{3\ eq/mol} = 21.00\ g/eq$$

Answer: C

61. In the electroplating of nickel from a solution containing Ni^{2+} ion, what will be the weight of the metal deposited on the cathode by a current of 8A flowing for 500 minutes?
A. 73 g B. 103 g C. 117.4 g D. 145 g

Solution:
From Faraday's first law of electrolysis:

$$m = \frac{MIt}{ZF},\ m - mass$$

M – molar mass
I – current
t – time
z – electrons gained or lost
F – Faraday's constant

$Ni^{2+} + 2e^- \longrightarrow Ni \Rightarrow Z = 2$

$t = 500\ min = 500 \times 60\ sec. = 30{,}000\ sec.$

$$m = \frac{5g \times 8A \times 30{,}000sec}{2 \times 96500A.C} = 73.37\ g\ of\ Ni$$

Answer: A

62. A 1 M solution of $Cu(NO_3)_2$ is placed in a beaker with a strip of Cu metal. A 1M of $SnSO_4$ is placed in a second beaker with strip of Sn metal then connected by a salt bridge and the two metal electrodes are connected by wires to a voltmeter. Which of the following electrode serve as the anode and which electrode gains mass?
Given that $E°\ Cu^{2+}/Cu = 0.34\ V$ and $E°\ Sn^{+2}/Sn = -0.14V$
A. Anode, Sn, Sn electrode gains mass
B. Anode, Sn, Cu electrode gains mass

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C. Anode, Cu, Sn electrode gains mass
D. Anode, Cu, Cu electrode gains mass

Explanation:
The more reactive metal, with more negative reduction potential serves as anode. The less reactive metal with more positive reduction potential used as cathode.
⇒ Sn is anode and Cu is cathode. Mass of cathode increases since copper is deposited on the surface of cathode.

63. Consider the following unbalanced, redox reacton in acidic **Answer: B**
Solution:

$MnO_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$

What is the change in oxidation state for both the substances oxidized and reduced and the coefficients of Fe^{2+} and Mn^{2+}, respectively after balancing?

A. 2 and 7, 2 and 5
B. 3 and 2, and 5 and 1D.
C. 1 and 5, and 5 and 1
1 and 5, and 5 and 2

Solution: $(8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^{2+} + 4H_2O) \times 1$

$Fe^{2+} \longrightarrow Fe^{3+} + 3 \times 5$

$8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^+ + 4H_2O$

$5Fe^{2+} \longrightarrow 5Fe^{3+} + 5e^-$

$8H^+ + 5Fe^{2+} + MnO_4^- \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

64. What reactions occur at the anode and cathode when an aqueous solution of Na_2SO_4 is electrolyzed? **Answer: C**

I. $S_2O_8^{2-} + 2e^- \longrightarrow 2SO_4^{2-}$ E°red
II. $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$ 2.01V
III. $2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$ 1.23 V
IV. $Na^+ + e^- \longrightarrow Na$

A. H_2 at cathode and O_2 at anode
B. Na at cathode and $S_2O_8^{2-}$ at anode
C. H_2 at cathode and $S_2O_8^{2-}$ at anode
D. Na at cathode and O_2 at anode

Explanation:
Anode reaction: $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$
Cathode reaction: $4H_2O + 4e^- \longrightarrow 2H_2 + 4OH^-$
Cell reaction: $2H_2O \longrightarrow 2H_2 + O_2$

Answer: A

65. For the following hypothetical equation, in aqueous solution, what is the correct representation of the cell notation?

$A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$

A. $A(s) \mid A^{2+}(aq) \parallel B^{2+}(aq) \mid B(s)$
B. $B^{2+}(aq) \mid B(s) \parallel A^{2+}(aq) \mid A(s)$
C. $A^{2+}(aq) \mid A(s) \parallel B(s) \mid B^{2+}(aq)$
D. $B(s) \mid B^{2+}(aq) \parallel A^{2+}(aq) \mid A(s)$

Explanation:
Cell notation is short hand representation of a galvanic cell without drawing the diagram. It is represent by the following general formula.
Anode | Anode electrolyte (M) || Cathode electrolyte (M)| cathode
The anode is the above hypothetical reaction is A since it is oxidized and cathode is B since it is reduced.
⇒ $A(s) \mid A^{2+}(aq) \parallel B^{2+}(aq) \mid B(s)$ **Answer: A**

66. Which of the following is the most important ingredients used for production of DAP fertilizers?

A. Ammonia and phosphoric acid.
B. Nitric acid, urea and phospheric acid
C. Phosperic acid, urea and ammonia
D. Sulphuric acid, ammonia and urea

Explanation:
Diammonium mono hydrogen phosphate (DAP) a white crystalline compound that is completely soluble in water. It is produced when ammonia reacts with phosphoric acid.

$3NH_3 + 2H_3PO_4 \longrightarrow NH_4H_2PO_4 + (NH_4)_2HPO_4$

$NH_4H_2PO_4 + NH_3 \longrightarrow (NH_4)HPO_4$
DAP

DAP is used as fertilizer and as a yeast nutrient in wine making and beer brewing. **Answer: A**

67. Which of the following metal alloys does NOT contain tin?

A. Brass
B. Bronze
C. Pewter
D. Plumber's solder

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Explanation:
Brass is an alloy of copper and zinc that does not contain tin. Bronze is an alloy of copper and tin pewter is an alloy that contains 85 – 90% tin with the remainder consisting of copper, antimony, bismuth. Plumber's solder is an alloy of tin and lead.

Answer: A

68. Which of the following plant nutrient will be produced as a result of nitrogen fixation?
A. Carbohydrates C. Mineral
B. Cellulose D. Protein

Explanation:
Nitrogen is used to make proteins.

Answer: D

69. Which of the following elements is the second most abundant element in the earth's crust?
A. Aluminum C. Oxygen
B. Iron D. Silicon

Explanation:
The first four abundant elements in the earth crust in decreasing order are oxygen, silicon, aluminum and iron respectively.

Answer: D

70. Which of the following metals has the highest electrical and thermal conductivities?
A. Ag B. Co C. Cu D. Ni

Solution:
Silver (Ag) is the best conductor of heat and electricity so far known. However, we do not use silver metal for this purpose because it is expensive.

Answer: A

71. Which of the following gases is manufactured using the Haber process?
A. Ammonia C. Nitrogen
B. Nitric oxide D. Nitrogen dioxide

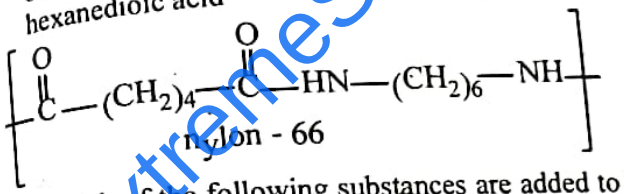
Explanation:
Ammonia is industrially manufactured from hydrogen and nitrogen gases in the presence of iron catalyst at high pressure and moderate temperature by Haber process.

$$3H_2(g) + N_2(g) \underset{}{\overset{Fe}{\rightleftharpoons}} 2NH_3(g)$$

Answer: A

72. What are the raw materials required to synthesize nylon – 66, a specific kind of nylon?
A. Diacids C. Diacids and Diamines
B. Diamines D. Polyethylene

Solution:
Nylons are polyamides. One of the most polyamides is nylon – 66. It is prepared from 1, 6 – amino hexane and 1, 6 – hexanedioic acid



Answer: C

73. Which of the following substances are added to natural rubber to toughen it?
A. Calcium C. Nitrogen
B. Carbon D. Sulfur

Explanation:
The substance used to vulcanize rubber is sulphur: Vulcanized rubber has the following properties:
- It is hard and non – stickey.
- It has higher elasticity.
- It has high tensil strength.
- It is insoluble in solvents.
- It is not oxidized readily.

Answer: D

74. What structural feature is usually needed to present in order for an addition polymer to be produced?
A. A carbon – carbon sigma bond.
B. A carbon – oxygen pi bond.
C. A carbon – oxygen sigma bond.
D. A carbon – carbon pi bond.

Explanation:
Addition polymerization requires a large number of monomers which can be polymerized by addition reactions. The monomers usually contain carbon – carbon multiple bonds. Pi bonds occur in double and triple bonds. All single bonds are sigma bonds.

Answer: D

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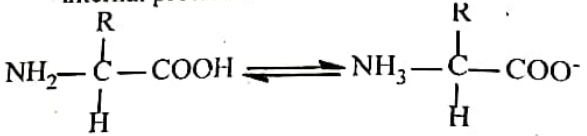
75. Which of the following is molecules forms a Zwitter ion at higher or lower pH?
A. Cellulose C. Protein
B. Glucose D. Starch

Solution:
Amino acids are monomers of proteins that form zwitter ion by internal protonation.

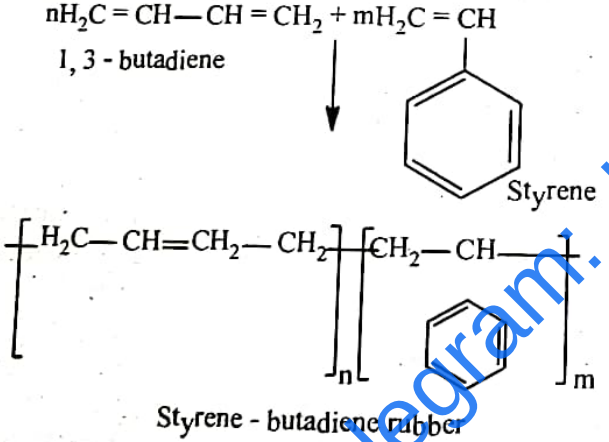


Zwitter ion

Answer: C

76. Which of the following form of synthetic rubbers can be vulcanized to greatly enhance its mechanical strength?
A. Neoprene C. Butyl rubber
B. Isoprene D. Styrene – butadiene rubber

Explanation:
Styrene – butadiene rubber is a copolymer of styrene and butadiene.



Styrene - butadiene rubber

It is vulcanized in a similar manner as natural rubber. This rubber is superior to natural rubber with regard to mechanical strength.

Answer: D

77. What is the range in the number of carbon atoms of the mono saccharides that are found in nature?
A. 3 to 6 C. 4 to 12
B. 4 to 10 D. 5 to 12

Solution:
Mono saccharides are simple sugars that cannot be hydrolyzed in to smaller compounds. They usually contain 3 to 6 carbon atoms.

Answer: A

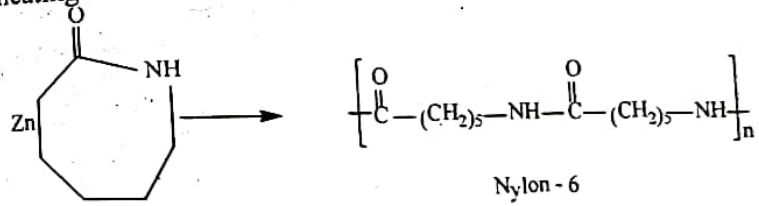
78. What are the two principal poly saccharides forms of starch?
A. Alde hexose and keto pentose
B. Amylose and amylopectin
C. Maltose and cellobiose
D. Sucrose and lactose

Explanation:
Starch is used for energy storage in plants. It is separated into two principal poly saccharides, amylose and amylopectin.

Answer: B

79. Which synthetic polymer is produced from caprolactam?
A. Nylon – 6 C. Teflon
B. Nylon 6, 10 D. Terylene

Solution:
Nylon – 6 is made from caprolactam, which polymenzes on heating in water.



Caprolacton

Nylon - 6

The use and properties of nylon 6 is similar with that of nylon 66.

Answer: A

80. How many types of cubic unit cells are known?
A. 2 B. 3 C. 4 D. 5

Explanation:
There are three types of cubic unit cells. These are:
i. A simple cubic unit cell or a primitive cubic unit cell – has lattice points only at the corners.
ii. A body centered cubic unit cell – has lattice points at the corners plus a lattice point in the center of the cell.
iii. A face centered cubic unit cell – has lattice points at the eight corners and one in the center of each of its six faces.

Answer: B

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1. Which of the following is correct?
   A. $1L = 1dm^3$ C. $10L = 1dm^3$
   B. $1L = 10dm^3$ D. $1L = 1m^3$
   Solution: $1m^3 = 1000dm^3 = 1000L = 10^6cm^3 = 10^6mL$
   $1cm^3 = 1mL$
   Answer: A

2. Which of the following is NOT a conjugate acid – base pair?
   A. $HNO_3|NO^-_3$ C. $NH_3|NH^-_2$
   B. $H_2SO_4|HSO^-_4$ D. $H_3O^+|OH^-$
   Explanation: The conjugate acid – base pairs differ by a single hydrogen. The acid has one more hydrogen than its conjugate base. ⇒The conjugate base of H_3O^+ is H_2O but not OH^-
   Answer: D

3. Commercially, liquid vegetable oils are converted to solid fats such as margarine by:
   A. Hydrogenation C. saponification
   B. Hydration D. oxidation
   Explanation: Vegetable oils are converted to fats by adding hydrogen to them. This reaction is termed as hydrogenation reaction.
   Answer: A

4. What is the chemical name for Aspirin?
   A. Acetyl salicylic acid C. methyl salicylate
   B. Salicylic acid D. sodium salicylate
   Explanation: Aspirin is formed when salicylic acid reacts with acetic acid. Its chemical name is acetyl salicylic acid.
   Answer: A

5. Which of the following statements (s) is |are| true of an ideal liquid – liquid solution?
   I. It obeys PV = nRT
   II. It obeys Raoult's law
   III. Solute – solute, solvent – solvent and solute solvent interactions are very similar
   IV. Solute – solute, solvent – solvent and solute – solvent interactions are quite different

   A. I, II and III C. II and III
   B. I, II and IV D. II and IV
   Explanation: Ideal solutions have the following properties
   - There is no expansion or contraction on mixing
   - Solute – solute, solvent – solvent and solute – solvent interactions are similar
   - Obey Raoult's law
   - The components have similarity in their structure
   - The vapor pressure above the solution is equal to the sum of the individual particle vapor pressures.

   Answer: C

6. The unit cell in a certain lattice consists of a cube formed by an anion at each corner, an anion in the center and a cation at the center of each face. How many cations and how many anions does the unit cell have?
   A. 5 anions and 6 cations C. 2 anions and 3 cations
   B. 5 anion and 3 canions D. 3 anions and 4 cations
   Answer: A

7. In a reaction, A + B ⟶ product, the rate is doubled when the concentration of B is doubled and the rate increases by a factor of 8 when the concentration of both the reactants A and B are doubled, the rate law for the reaction can be written as:
   A. Rate = k[A] [B] C. Rate = $K[A]^2[B]$
   B. Rate = $K[A] [B]^2$ C. Rate = $K[A]^2 [B]^2$
   Solution: Let the rate law be:rate = $K[A]^x [B]^y$ when the concentration of B is doubled, the rate is doubled. This is true for first order reaction. Thus, the value of Y is 1.
   $rate_1 = K[A]^x_1 [B]_1$
   when $[A]_2 = 1[A]_1$ and $[B]_2 = 2[B]_2$
   $rate_2 = 8\ rate_1$

$$\frac{rate_1}{8rate_1} = \frac{k[A]_1^x[B]_1}{k(2[A]_1)^x 2[B]_1}$$

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$$\frac{1}{8}=\frac{1}{2^x.2}$$

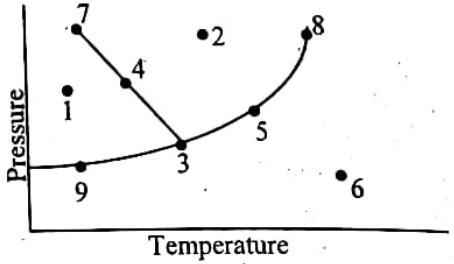
$$\left(\frac{1}{2}\right)^3=\left(\frac{1}{2}\right)^{x+1}$$

$$x+1=3\Rightarrow x=2$$

Therefore, the rate law is: rate = $k[A]^2[B]$.

Answer: C

(8.) Answer the following question using the phase diagram below.



At which point can only the solid and liquid phase coexist?

A. 1 B. 2 C. 3 D. 4

Explanation: Solid and liquid coexist at point 4, solid and vapor coexist at point 9 liquid and vapor coexist at point 5 and the three states coexists at equilibrium at point 3.

Answer: D

(9.) Which electron transition in a hydrogen atom releases the largest energy?

A. $n=2\longrightarrow n=1$ C. $n=6\longrightarrow n=3$

B. $n=4\longrightarrow n=2$ D. $n=7\longrightarrow n=6$

Solution:

A. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right)$

$$\Delta E=2.18\times10^{-18}J\left(\frac{1}{2^2}-\frac{1}{12}\right)=-1.625\times10^{-18}J$$

B. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{4^2}-\frac{1}{2^2}\right)=-4.09\times10^{-18}Js$

C. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{6^2}-\frac{1}{3^2}\right)=-1.82\times10^{-19}J$

D. $\Delta E=2.18\times10^{-18}J\left(\frac{1}{7^2}-\frac{1}{6^2}\right)=-1.66\times10^{-20}J$

Answer: A

(10.) The simplest formula for a compound containing Mn^{4+} and O^{2-} is

A. MnO B. MnO_2 C. Mn_2O_4 D. Mn_4O_2

Explanation: $Mn^{4}O^{2}=Mn_2O_4$

The simplest formula is MnO_2

Answer: B

(11.) What is the ratio of the energy of a photon of 300nm wavelength radiation to that of 600nm radiation?

A. 1:2 B. 1:1 C. 2:1 D. 3:1

Solution:

$E=h\nu$ but $\nu=\frac{C}{\lambda}$

$$E=\frac{hC}{\lambda}$$

$$E_1=\frac{hC}{\lambda_1}\text{ and }E_2=\frac{hC}{\lambda_2}\Rightarrow\frac{E_1}{E_2}=\frac{hC/\lambda_1}{hC/\lambda_2}$$

$$\frac{E_1}{E_2}=\frac{\lambda_1}{\lambda_2}=\frac{600nm}{300nm}=\frac{2}{1}$$

$E_1:E_2=2:1$

Answer: C

(12.) Which one of the following chemicals is used to disinfect water?

A. Fluorine C. oxygen

B. Nitrogen D. chlorine

Explanation:

Chlorine is added to water to kill bacteria and other microorganisms.

Answer: D

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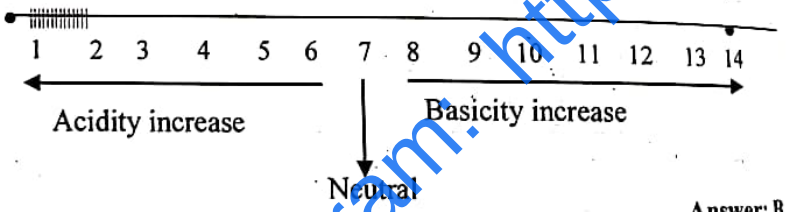
13. During the titration of a known volume of a strong acid with a strong base, there is
A. A steady increase in pH
B. A sharp increase in pH around the end point.
C. A steady decrease in pH
D. A sharp decrease in pH around the end point.

Explanation: Features of the titration of a strong acid with a strong base:
- The pH is low at the beginning of the titration
- The pH change slowly until just before the equivalent point.
- Just before the equivalence point, the pH rises sharply.
- At the equivalence point, the pH is 7.
- Just past the equivalence point, the pH continues its sharp rise.

Answer: B

14. A solution with a pH of 7.5 would be described as:
A. Very basic
B. Slightly basic
C. slightly acidic
D. very acidic

Explanation:



Answer: B

15. The electrolysis of molten NaCl is an industrial process. What does the electrolysis produce?
A. Na and Cl_2
B. H_2 and O_2
C. Na^+ and Cl^-
D. NaOH and Cl_2

Explanation: $NaCl \xrightarrow{\text{Melting}} Na^+ + Cl^-$

Anode reaction: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode reaction: $2Na^+ + 2e^- \longrightarrow 2Na$

Cell reaction: $2Na^+ + 2Cl^- \longrightarrow 2Na + Cl_2$

Answer: A

16. Butane burns in oxygen according to the equation below.
$2C_4H_{10}(g) + 13O_2(g) \longrightarrow 8CO_2(g) + 10H2O(1)$
If 11.6g of butane is burned in 11.6g of oxygen, which is the limiting reactant?
A. Butane
B. Oxygen
C. Neither
D. Both oxygen and butane

Solution: Limiting reactant is a reactant that determines the yield and has smaller total number of moles.

$$\text{Total number of mole } (n_T) = \frac{\textit{given mass}}{\textit{coefficient} \times \textit{Molar mass}}$$

$$n_T(C_4H_{10}) = \frac{11.6g}{2 \times 58g/mol} = 0.10mol$$

$$n_T(O_2) = \frac{11.6g}{13 \times 32g/mol} = 0.03mol$$

Thus, the limiting reactant is oxygen (O_2). **Answer: B**

17. A beaker filled to the 100mL mark with salt (the salt has a mass of 100g) and another beaker to the 100mL mark with water (the water has a mass of 100g) are mixed together in a bigger beaker until the salt is completely dissolved. What will be the mass of the solution?
A. It will be much more than 200g
B. It will be much smaller than 200g
C. It will be exactly 200g
D. It will be slightly more than 200g

Explanation:
mass of solution = mass of solute + mass of solvent mass of solution = 100g + 100g = 200g. But, volume is not always additive.

Answer: C

18. Which one of the following is NOT an intensive property?
A. Mass
B. Temperature
C. Colour
D. Density

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Explanation:

Extensive properties	Intensive properties
Depend on size of sample of matter.	Independent on size
Example	**Example**
• volume	• density
• mass	• melting point
• area	• conductivity
• thickness	• color
	• temperature

Answer: A

19. Which of the following is natural polymer?
A. Nylon B. PVC C. Cotton D. Dacron

Explanation: Natural polymers include:
- Polysaccharides such as cellulose, starch
- Proteins
- Natural rubber

Cotton is a cellulose **Answer: C**

20. Which of the following metals forms a volatile compound during extraction process?
A. Fe B. Co C. Ni D. Cu

Explanation: Volatile nickel carbonyl, $Ni(CO)_4$, a poisonous liquid of boiling point 43^0C is formed during the production of nickel. **Answer: C**

21. Which one of the following atoms in its ground state has the greatest number of unpaired electrons?
A. $_{13}Al$ B. $_{14}Si$ C. $_{15}P$ D. $_{16}S$

Explanation: $_{13}Al$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^0{}_y\ 3P^0{}_z$
Number of unpaired electron = 1

$_{14}Si$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^1{}_y\ 3P^0{}_z$
Number of unpaired electrons = 2

$_{15}P$: $[Ne]\ 3S^2\ 3P^1{}_x\ 3P^1{}_y\ 3P^1{}_z$
Number of unpaired electrons = 3

$_{16}S$: $[Ne]\ 3S^2\ 3P^2{}_x\ 3P^1{}_y\ 3P^1{}_z$
Number of unpaired electrons = 2

Answer: C

22. Which statement is true about chemical reactions at equilibrium?
A. The forward and back ward reactions proceed at equal rates
B. The forward and backward reactions have stopped
C. The concentrations of the reactants and products are equal.
D. The forward reaction is exothermic **Answer: A**

23. Which compound contains both covalent and ionic bonds?
A. Sodium carbonate, Na_2CO_3
B. Magnesium bromide, $MgBr_2$
C. Dichloromethane, CH_2Cl_2
D. Ethanoic acid, CH_3COOH

Explanation: Compounds containing metal ion and polyatomic ions have both ionic and covalent bonds. **Answer: A**

24. Which set of quantum numbers (n, ℓ, mℓ, m_s) is not possible?
A. $1, 0, 0, \frac{1}{2}$ C. $1, 0, 0 \frac{-1}{2}$
B. $1, 1, 0, \frac{1}{2}$ D. $2, 1, -1, \frac{1}{2}$

Explanation: n = 1, 2, 3, 4,....
$\ell = 0$ to $n-1$, its maximum value is $n-1$
me = $-\ell$ to $+\ell$
$ms = \pm\frac{1}{2}$ **Answer: B**

25. Which of the following particles contains more electrons than neutrons?
I. $^{1}_{1}H$ II. $^{35}_{17}Cl^-$ III. $^{39}_{19}K^+$
A. I only C. I and II only
B. II only D. II and III only

Solution: $^{1}_{1}H$: e = 1 , n = A − p = 1 − 1 = 0, thus e > n
$^{35}_{17}Cl^-$: e = 18 , n = 35 − 17 = 18, thus, e = n
$^{39}_{19}K^+$: e = 18, n = 39 − 19 = 20, thus, e < n **Answer: A**

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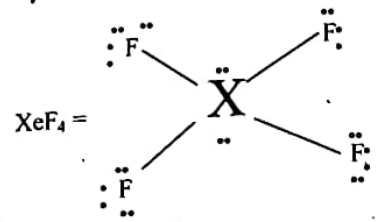
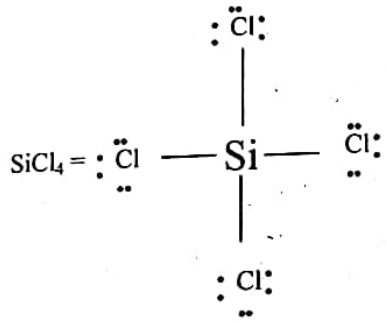
(26) Which molecule or ion does NOT have a tetrahedral shape?

A. XeF_4 B. $SiCl_4$ C. BF^-_4 D. NH^+_4

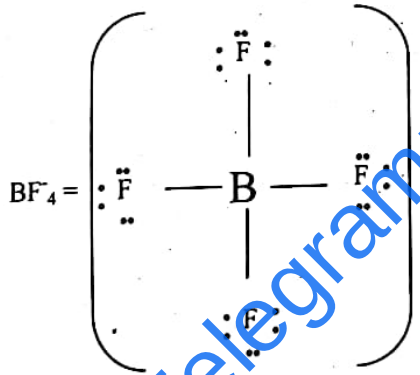
Explanation:

XeF_4 = 

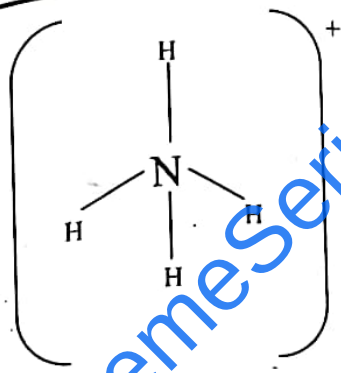
, square planar

$SiCl_4$ = 

, tetrahedral

BF^-_4 = 

, tetrahedral

NH^+_4 = 

, tetrahedral

Answer: A

(27) Why are metals soft and malleable?

A. Because they are very shiny

B. Because they experience electrostatic repulsion

C. Because of the presence of mobile electrons

D. Because the metal cations can slip over each other fairly easily.

Answer: D

(28) In which region of the periodic table would the element with the electronic structure below be location?

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^6 5s^2$

A. Group 6 C. s – block

B. Noble gases D. d – block

Explanation: write its correct electron configuration based on building up principle $\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6 4s^2 3d^{10}\ 4p^6 5s^2 4d^6$

Group = 2 + 6 = VIIIB

Period = 5

Block = d – block

Answer: D

(29) Which factors will influence the rate of the reaction shown below? $NO_2(g) + CO(g) \rightleftarrows NO(g) + CO_2(g)$

I. The number of collisions per second

II. The energy of collisions

III. The geometry with which the molecules collide

A. I only C. I and II only

B. II only D. I, II and III

Answer: A

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How many π bonds are present in CO_2?

A. One B. Two C. Three D. Four

Explanation: Draw the Lewis structure for $CO_2 \Rightarrow O = C = O$

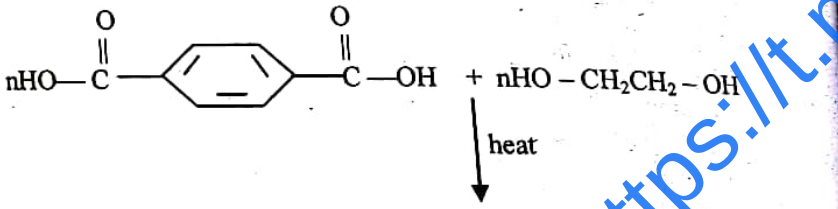
One double contains one π and one σbonds. Since there are two double bonds in CO_2, 2 π and 2σ bonds are present.

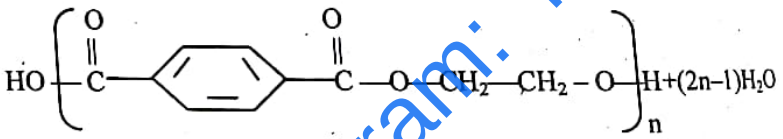
Answer: B

31. Which pair of monomers forms polyesters?

A. Difunctional alcohol, difunctional organic acid
B. Difunctional alcohol, difunctional amino acid
C. Ethylene, ethylene
D. Amino acid, amino acid

Explanation: Polyester (polyethylene tarephthalate or Dacron) is a copolymer formed when 1, 4 – benzene dicarboxylic acid and 1,2 – ethnodiol react





Polyester

Answer: A

32. Which changes will increase the amount of $SO_3(g)$ at equilibrium? $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -197KJ$

I. Increasing the temperature
II. Decreasing volume
III. Adding a catalyst

A. I only C. I and II only
B. II only D. I, II and III

Explanation: The amount of SO_3 is increased by:

- decreasing temperature
- increasing pressure
- increasing amount of SO_2 or O_2 or both
- decreasing volume

Answer: B

33. Which compound is a carboxylic acid?

A. CH_3COOH C. $(CH_3)_2CHOOCH_3$
B. $(CH_3CO)_2O$ D. $(CH_3)_2O$

Explanation: The general formula for carboxylic acid is RCOOH

Answer: A

34. The mechanism of a reaction is shown below.

$HCOOH + I^- \longrightarrow HOI + OH^-$ (slow)

$HOI + I^- \longrightarrow I_2 + OH^-$ (fast)

$2OH^- + 2H_3O^+ \longrightarrow 4H_2O$ (fast)

What is the rate law based on this mechanism?

A. Rate = K[HCOOH] $[I^-]$
B. Rate = K$[HCOOH]^2$ $[I^-]$
C. Rate = K[HCOOH] $[I^-]^2$
D. Rate = K[HCOOH]

Explanation:The slowest step is the rate determing step.

⇒ Rate law, rate = K[HCOOH] $[I^-]$ **Answer: A**

35. What is the equilibrium constant expression for the following reaction?

$2Hg(g) + O_2(g) \rightleftharpoons 2HgO(s)$

A. $K = \dfrac{1}{[Hg]^2[O_2]}$ C. $K = \dfrac{[HgO]^2}{[Hg]^2[O_2]}$

B. $K = [Hg]^2[O_2]$ D. $K = \dfrac{[2HgO]}{[2Hg][O_2]}$

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Explanation: Pure solids and liquids (solvent) do not appear in the equilibrium constant expression.

$$\Rightarrow K = \frac{1}{[Hg]^2[O_2]}$$

Answer: A

36. A triacylglycerol that is solid at room temperature is called
A. Lecithin B. Fat C. Wax D. Oil

Answer: B

37. Which compound is an ester?
A. CH_3COOH C. C_2H_5CHO
B. $CH_3O\,C_2H_5$ D. $HCOOCH_3$

Explanation: The general formula of an ester is $RCOOR^1$

Answer: D

38. If a solute dissolves in an endothermic process,
A. H bonds must exist between solvent and solute
B. Strong ion – dipole forces must exist in solution
C. The entropy of the solution must be greater than that of its pure components
D. The solute must be a gas.

Explanation: If a solute dissolves in an endothermic process the entropy of the solution must be greater than that of pure components.

Answer: C

39. A solution is made by dissolving 250g of potassium chromate crystals (K_2CrO_4 , molar mass, 194.2g) in 1kg of water. What will be the freezing point of the solution? (K_f for water is $1.86^0c\ m^{-1}$)
A. -8.87^0c b. -7.18^0c c. -5.73^0c d. -1.86^0c

Solution: $\Delta T_f = Tf^0 - Tf = imk_f$

For electrolytes, i ≈ number of moles of ions.

$$\Rightarrow K_2CrO_4 \longrightarrow 2K^+ + CrO^{2-}_4 \Rightarrow i \approx 3$$

$$T_f = O°C - 3 \times \left(\frac{350g}{194.2\,g/mol \times 1kg}\right) \times 1.86^0c\ m^{-1} = -7.18°C$$

Answer: B

40. How many moles of sodium hydroxide are present in 2.5L of 0.5M aqueous solution?
A. 0.2 B. 0.5 C. 1.25 D. 12.5

Solution: $M = \frac{n}{v}$

$$n = m \times v = 0.5\,\frac{mol}{L} \times 2.5L = 1.25mol$$

Answer: C

41. Which species cannot act as a Lewis acid?
A. NH_3 B. BF_3 C. Fe^{2+} D. $AlCl_3$

Explanation: Lewis acids are electron pair acceptors. Cations, molecules below and above octet are Lewis acids.

Answer: A

42. Which of the following statements is true?
A. A universal indicator is a mixtures of indicators that will give a different color for a different pH.
B. Phenolphthalein is a universal indicator
C. A universal indicator can only be used in either strongly acidic or basic solution
D. The color of a universal indicator is red in a weak acid.

Explanation:
A universal indicator gives different color for different pH.For example, in strong acid red, in medium orange and in weak acid yellow.

Answer: A

43. Three acids, HA, HB, HC have the following ka values ka(HA) = 1×10^{-5}, Ka (HB) = 1×10^{-4} (H C) = 1×10^{-6} what is the correct order of increasing acid strength (weakest first)?
A. HA, HB. HC C. HC, HA, HB
B. HC, HB, HA D. HB, HA, HC

Explanation:
As Ka increases, the strength of acid also increases.
$\Rightarrow HC < HA < HB$

Answer: C

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44. If the solute – solvent interaction are greater than the solute – solute and solvent – solvent interactions, what will be the total vapor pressure of the solution?

A. Greater than that calculated from Raoult's law
B. Less than that calculated from Raoult's law
C. The same as calculated from Raoult's law
D. Raoult's law cannot be applied for such calculations.

Explanation: If the solute – solvent interactions is greater than the solute – solute and solvent – solvent interactions, the total vapor pressure is less than the sum of the vapor pressures as predicted by Raoult's law.

Answer: B

45. Which of the following procedures will produce a buffered solution?

I. Equal volumes of 0.5M NaOH and 1M HCl solutions are mixed
II. Equal volumes of 0.5MNaOH and 1M CH_3COOH solutions are mixed
III. Equal volumes of 1M $NaCH_3CO_2$ and 1M CH_3COOH solutions are mixed
IV. Equal volumes of 1.0MNaOH and 1M HCl solutions are mixed.

A. I B. III C. I and II D. II and III

Explanation:

Buffered solution is formed from:

i. Weak acid and its conjugate base
ii. Weak base and its conjuoate acid
iii. Excess weak acid and limited amount of strong base
Example: 1M CH_3COOH and 0.5MNaOH
iv. Excess weak base and limited amount of strong acid
Example: 1MNH_3 and 0.5MHCl

Answer: D

46. For which conversion is an oxidizing agent required?

A. $2H^+(aq) \longrightarrow H_2(g)$ C. $SO_3(g) \longrightarrow SO^{2-}{}_4$
B. $2Br^-(aq) \longrightarrow Br_2(aq)$ D. $MnO_2(s) \longrightarrow Mn^{2+}(aq)$

Explanation: Oxidizing agent is a species being reduced but cause oxidation

$2H^+ + 2e^- \longrightarrow H_2$ is reduction and needs reducing agent

$2Br^- \longrightarrow Br_2 + 2e$ is oxidation half – reaction and requires oxidizing agent.

$SO_3 \longrightarrow SO^{2-}{}_4$ is neither oxidation nor reduction. It does not need reducing and oxidizing agent.

$MnO_2 + 2e \longrightarrow Mn^{2+}$ is reduction half reaction and needs reducing agent

Answer: B

47. What volume of 0.500M NaOH is required to neutralize 25.0mL of 1.2M H_2SO_4? (Assume complete ionization of the acid)

A. 60mL B. 90mL C. 100mL D. 120mL

Solution: From the balanced chemical equation find the number of moles of the base (n_b).

$n_a = M_a \times V_a$

$n_a = 1.2m \times 25mL = 30mmol$

30mmol nb

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 2

$$\frac{30mmol}{1} = \frac{n_b}{2}$$

$n_b = 60mmol$

$$Mb = \frac{n_b}{V_b}$$

$$Vb = \frac{n_b}{V_b} = \frac{60mmol}{0.5M} = 120mL$$

Answer: D

48. A gas is confined to a cylinder under a constant atmospheric pressure. When the gas undergoes a particular chemical reaction, it releases 135kJ of heat to its surrounding and does 63kJ of P – V work on its surroundings. What are the values of ΔH and ΔE for the process

A. $\Delta H = 125KJ, \Delta E = 63KJ$
B. $\Delta H = 135KJ, \Delta E = -63KJ$
C. $\Delta H = 135KJ, \Delta E = 198KJ$
D. $\Delta H = 135KJ, \Delta E = -198KJ$

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Explanation: ΔH is heat transferred at constant pressure.

$\Rightarrow \Delta H = qp = -135kJ - 63kJ = -198kJ$

$\Delta H = q + w = -135kJ - 63kJ = -198kJ$ **Answer: D**

49. The oxidation number of nitrogen in NH_3, HNO_3 and NO_2 are _____ respectively.

A. $-3, -5, +4$ C. $-3, +5, -4$

B. $+3, +5, +4$ D. $-3, +5, +4$

Solution:

$\overset{x}{N}\overset{+1}{H_3}$ $\quad$ $\overset{+1}{H}\overset{x}{N}\overset{-2}{O_3}$ $\quad$ $\overset{x}{N}\overset{-2}{O_2}$

$x + 3(+) = 0$ $\quad$ $1 + x + 3(-2) = 0$ $\quad$ $x + 2(-2) = 0$

$x + 3 = 0$ $\quad$ $1 + x - 6 = 0$ $\quad$ $x - 4 = 0$

$x = -3$ $\quad$ $x = +5$ $\quad$ $x = +4$

Thus, the oxidation states of N in NH_3, HNO_3 and NO_2 are -3, $+5$ and $+4$ respectively. **Answer: D**

50. Which statement about the following reaction is correct?

$2Fe(s) + 3CO_2(g) \longrightarrow Fe_2O_3(s) + 3CO(g)$,

$\Delta H^O = +26.6kJ$

A. 26.6KJ of energy are released for every mole of Fe reacted

B. 26.6KJ of energy are absorbed for every mole of iron reacted.

C. 53.2kJ of energy are released for every mole of Fe reacted

D. 13.3KJ of energy are absorbed for every mole of Fe reacted.

Solution: The reaction is endothermic because of ΔH^0 is positive. 26.6kJ of heat is absorbed for 2 moles of iron reacted. As a result, 13.3kJ of heat is absorbed for every mole of iron reacted.

Answer: D

51. Which of the following metals is extracted by thermal reduction process?

A. Cu B. Fe C. Al D. Mg

Answer: A

52. The two standard electrode potentials involved in the nickel – cadmium rechargeable cell are given below, calculate the ΔG^0 in kJ of the cell.

$NiO_2(s) + 2H_2O(l) + 2e \longrightarrow Ni(OH)_2(s) + 2OH^-$

$E^0 = +0.49V$

$Cd(OH)_2 + 2e \longrightarrow Cd(s) + 2OH^-(aq)$

$E^0 = -0.76V$

A. -184 B. -153 C. -241 D. -206

Solution:

E^0 cell $= E^\circ$ cathode $- E^0$ anode

E^0 cell $= 0.49V - (-0.76V) = 1.25V$

$\Delta G^\circ = -nFE^0$ cell

$\Delta G^\circ = -2 \times 96500 \times 1.25 = -241250J$

$\Delta G^\circ = -241.250kJ$ **Answer: C**

53. Which of the following is a synthetic rubber produced from caprolatam (CPL)?

A. Nylon 6, 10 C. Teflon

B. Teryene D. Nylon 6 **Answer: D**

54. Which one of the following reactions is NOT a redox × reaction?

A. $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

B. $2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$

C. $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$

D. $Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2-}(aq)$

Explanation: Redox reaction involves transfer of electrons. The presence of free atom or molecule indicates that the reaction is redox **Answer: A**

55. The half – reaction for formation of magnesium metal upon electrolysis of molten $MgCl_2$ is

$Mg^{2+} + 2e^- \longrightarrow Mg$

What is the mass of magnesium formed upon a passage of a current of 60A for a period of 2.00×10^3S?

A. 5g B. 10g C. 15.1g D. 30.2g

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Solution: $m = \frac{MIt}{2F}$

$$m = \frac{24g/mol \times 60A \times 2 \times 10^3 S}{2 \times 96500 \frac{A.S}{mol}} = 15g$$

Answer: C

56. An aqueous solution is 70% nitric acid (HNO_3) by mass. What is the concentration of HNO_3 expressed in molality?

A. 0.55gm B. 8.62m C. 11.1m D. 37m

Solution:

70% by mass HNO_3 implies 70g HNO_3 in 30g of H_2O

$$\Rightarrow m = \frac{n_{HNO_3}}{kgH_2O} = \frac{70g/63g/mol}{0.030kg}$$

m – 37.04m

Answer: D

57. Which of the following is natural polymer?

A. Polyethylene C. Nylon

B. Polysaccharides D. Terylene

Explanation: Natural polymers include the following

- Proteins
- Polysaccharides such as cellulose, starch and glycogen
- Natural rubber

Answer: B

58. What is the ionization energy of all ion atom if it requires a radiation of 276nm to completely remove its outermost electron in the gaseous state? (plank's constant, h = 6.626 ×10J.S, speed of light, $C=3\times10^8 ms^{-1}$)

A. 7.21×10^{-19}J C. 7.21×10^{19}J

B. 7.21×10^{-19}kJ D. 7.21×10^{19}kJ

Solution: $E = hu$ but $v = \frac{c}{\lambda}$

$$\Rightarrow E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34} J.s \times 3\times10^8 m/s}{276\times10^{-9} m} = 7.2 \times 10^{-19}J$$

Answer: A

59. Which of the following configurations describes the ground state electron configuration of Cl^-?

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ C. $1s^2\ 2s^2\ 2p^6\ 3p^1$

B. $1s^2\ 2s^2\ 2p^6\ 3s^1$ D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2 3p_4^1$

Explanation: Total number of electrons = 17 + 1 = 18

Electron configuration = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Answer: A

60. When 0.68 is divided by 14.364, the actual answer is 0.0473405. what will be the correct answer:

A. 0.05 C. 0.0473

B. 0.047 D. 0.04734

Explanation: During multiplication and division of a measured quantity, the last answer is reported with the least number of significant figures. Thus, the answer is written with two significant figures.

Answer: B

61. What is the correct molecular electronic configuration for the molecular ion, B^+_2?

A. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\sigma2p^2$

B. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_y^2$

C. $\sigma1s^2\sigma^*1s^2\sigma2s^2\delta2s^{*2}\pi2p_x^1$

D. $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p_x^1\pi2^1p_y$

Explanation: Total electrons = 2 × 5 – 1 = 9

$\Rightarrow \sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\pi2p^1$

Answer: C

62. Which of the following molecular ions will exhibit delocalized bonding?

NO^-_2, NH^+_4, N^-_3

A. NO^-_2 and N^-_3 C. NO^-_2

B. NH^+_4 and N^-_3 D. NO^-_2 and NH^+_4

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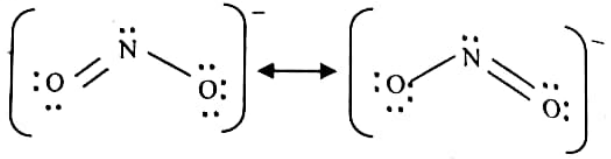
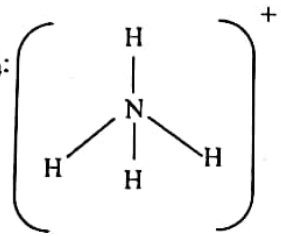
Explanation:

NO^-_2 :



NH^+_4:



Answer: A

(63.) Based on molecular orbital theory, the bond orders of H_2, H_2^+, and H_2^-, are _____ respectively.

A. 1,0 and 0 C. 1,0, and ½

B. 1,1/2 and 0 D. 1,1/2 and ½

Solution:

H_2: Total electrons = 2XI = $2\sigma ls^2$

B0= ½ (nb-na)= ½ (2-0) =1

H_2+: total electrons =2X1-1=1,σls^1

Bo=1/2 (nb-na) = ½(1-0)=1/2

H_2-: Total electrons =2X1+1=3,σls^2,σs^1

B0=1/2 (nb-na) =1/2(2-1) =1/2

Answer: D

(64.) How many 3d electrons are present in the ground state of chromium atom?

A. 4 B.5 C.6 D.10

Explanation

24Cr: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

Number of electrons in 3d is 5

Answer: B

(63) When the following substances are arranged in the order of increasing melting point (lowest melting point first),the correct order is :

A. $CH_3CH_2\ CH_3, CH_3COCH_3, CH_3\ CH_2\ CH_2OH$

B. $CH_3CH_2CH_3, CH_3CH_2OH, CH_3COCH_3$

C. $CH_3COCH_3, CH_3CH_2CH_2OH, CH_3CH_2CH_3$

D. $CH_3CH_2CH_2OH, CH_3CH_2CH_3, CH_3COCH_3$

Explanation: Boiling Point of :

Alkane < Ketone < Alcohol <Carboxylicacid **Answer: A**

(66) The half-life for the first order decomposition of nitromethane, CH_3NH_2at 500K is 650 seconds. If the initial concentration of CH_3NO_2 is 0.5 M, What will its concentration be in (M) after 1300 seconds have elapsed ?

A. 0.125 B. 0.140 C. 0.250 D. 0.425

Solution: $t_{1/2} = \dfrac{0.693}{k}$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{650} s^{-1}$$

$$\log \frac{[CH_3NO_2]_o}{(CH_3NO_2)_t} = \frac{kt}{2.303}$$

$$\frac{[CH_3NO_2]_o}{(CH_3NO_2)_t} = 10^{kt/2.303}$$

$$[CH_3NO_2]_t = \frac{0.50M}{10^{\frac{0.693}{650}\times 1300/2.303}} = \frac{0.50M}{10^{0.6}} = 0.25M$$

Answer: C

(67) In a zero-order reaction for every 10^0c rise of temperature, the rate is double. If the temperature is increased from 10^0C to 100^0C, the rate of the reaction will become.

A. 164 times C. 256 times

B. 128 times D. 512 times

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Solution: Determine the number of 10^0c

10^0C to 100^0C.

^0C	rate	^0c	rate
1.	10-20=2	6.	61-70=2
2.	21-30=2	7.	71-80=2
3.	31-40=2	8.	81-90=2
4.	41-50=2	9.	81-90=2
5.	51-60=2	10.	91-100=2

Rate = 2^9 =512 times

Answer: D

(68) Which of the following mathematical relationships between K, K_1 and K_2 is correct?

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$ K

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)$ K_1

$Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g)$ K_2

A. $K = K_1 + K_2$ C. $K = K1/K_2$

B. $K = K_1 X K_2$ D. $K = K_2/K_1$

Solution:

To obtain the first reverse the third and add it with the second.

$$Fe(s) + CO_2 \overrightarrow{\leftarrow} FeO(s) + CO(g) k_1$$

$$FeO(s) + H_2(g) \overrightarrow{\leftarrow} Fe(s) + H_2O(g) \frac{1}{k_2}$$

$$CO_2(g) + H_2(g) \overrightarrow{\leftarrow} CO(g) + H_2O(g)$$

$$K = k_1 \times \frac{1}{k_2} = \frac{k_1}{k_2}$$

Answer: C

(69) Which of the following gives the correct order of decreasing acidity of carboxylic acids?

A. $Cl_3CCOOH, Cl_2CHCOOH, FCH_2COOH, CH_3COOH$

B. $FCH_2COOH, CH_3COOH, Cl_2CHCOOH, Cl_3CCOOH$

C. $CH_3COOH, FCH_2COOH, Cl_2CHCOOH, Cl_3CCOOH$

D. $Cl_2CHCOOH, CH_3COOH, FCH_2COOH, Cl_3CCOOH$

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Explanation

If halogens replace hydrogen from the carboxylic acids, their strength increased

Example

$CH_3COOH < CH_2C1COOH < CHCl_2COOH < CC1_3COOH$

Answer: C

(70) The value of k_c for the following reaction is 4.0 at a temperature of 373k. $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

What mass of ethyl ester ($CH_3COOC_2H_5$) would be present in the equilibrium was established at this temperature

A. 5.2 B. 10.1 C. 12.6 D. 14.1

Solution: $k_c = \frac{[CH_3COOC_2H_5]}{[CH_3COOH][C_2H_5]}$

$$k_c = \frac{n_{CH_3COOC_2H_5}}{n_{CH_3COOH} \cdot n_{C_2H_5OH}}$$

$$4 = \frac{n_{CH_3COOC_2H_5}}{n_{CH_3COOH} \cdot n_{C_2H_5OH}}$$

$n_{CH_3COOC_2H_5} = 0.25mol$

$m_{CH_3COOC_2H_5} = 0.25mol \times 88g/mol = 22g$

Answer: 22g

71. Which of the following is true about of an open system?

A. A system that exchanges both energy and matter with its surroundings.

B. A system that cannot exchange both matter and energy with its surrounds.

C. A system that only exchanges matter with its surroundings

D. A system that only exchanges energy with its surroundings

Explanation:

- Open system exchanges both matter and energy
- Close system exchanges energy only
- Isolated system cannot exchanges both energy and matter.

Answer: A

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72. Which of the following statements is true about the percent ionization of a weak acid?

A. It decreases with increasing concentration
B. It increases with decreasing concentration
C. It increases with increasing concentration
D. It decreases with decreasing concentration

Explanation:

$$\text{Percent of ionization of acid} = \frac{\text{concentration of } H^+}{\text{Initial concentration of acid}} \times 100$$

Answer: A

73. Which is the value of ΔH for the reaction: $S(s) \longrightarrow S(g)$

$S(s) + O_2(g) \longrightarrow SO_2(g)$ $\quad \Delta H = -395KJ$

$S(g) + O_2(g) \longrightarrow SO_2(g)$ $\quad \Delta H = -618KJ$

A. $-1013KJ$ C. $+223KJ$
B. $-223KJ$ D. $+1013KJ$

Solution:

Reverse the second and add it with the first

$$S(s) + O_2(g) \rightarrow SO_2(g) \quad \Delta H_1 = -395KJ$$
$$SO_2(g) \rightarrow S(g) + O_2(g) \quad \Delta H^1 = +618KJ$$
$$S(s) \rightarrow S(g), \Delta H = \Delta H_1 + \Delta H^1$$

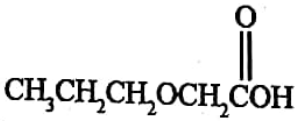
$\Delta H = -395KJ + 618KJ$

$\Delta H = +223KJ$

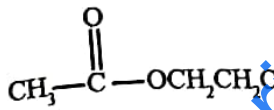
Answer: C

74. Which of these compounds is the ester formed from the reaction of acetic and 1 – Propanol?

A. $CH_3CH_2OOCH_2CH_3$
B. $CH_3CH_2CH_2CH_2COH$
C.

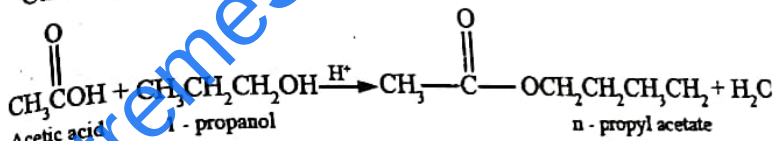


D.



Explanation:

Carboxylic acid + Alcohol $\xrightarrow{H^+}$ $\xrightarrow{\text{Ester}}$ + water



Answer: D

75. A lab instructor is preparing 5.0liters of a 0.10MPb $(NO_3)_2$ (molecular mass 331) solution. What is the mass required?

A. 165.5g of $Pb(NO_3)_2$ and add 5.0kg of H_2O
B. 165.5 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters
C. 33.1g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters
D. 33.1g of $Pb(NO_3)_2$ and add 5.0 liters of H_2O

Solution:

$m = M \times M_{wt} \times V$

$$m = 0.10 \frac{mol}{L} \times 331 \frac{g}{mol} \times 5.0L = 165.5g$$

Thus, 165.5g of Pb $(NO_3)_2$ and add water until the solution has a volume of 5.0 liters.

Answer: B

76. Which of the following is NOT a characteristic of theelectrolytic cell containing aqueous solution of NaCl used in the manufacture of sodium hydroxide?

A. The sodium hydroxide solution is produced in the electrolytic cell.
B. The electrolyte must be a dilute solution of NaCl
C. Hydrogen is produced at the cathode
D. The production of chlorine gas occurs at the anode.

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Explanation:

- The electrolyte must be concentrated NaCl (brine)
- At anode, Cl_2 gas is librated.
- At cathode, H_2 gas is given off
- NaOH is formed as residue

Answer: D

77. The kinetic data below are for the reaction: $A + B \longrightarrow C$
From these data what are the orders of the reaction with respect to A and B?

[A]	[B]	Initial rate ($moldm^{-3}$ sec^{-1})
0.1	0.1	1×10^{-5}
0.1	0.2	4×10^{-5}
0.2	0.1	1×10^{-5}

A. Order of A = 1, order of B = O
B. Order of A = O, order of B = 2
C. Order of A = O, order of B = 4
D. Order of A = 1, order of B = 2

Solution: Late the rate law be: $r = K[A]^x [B]^y$
Find the value of x by using the first and third data and Y using the first and second data.

$$\frac{rate\,1}{rate\,3} = \frac{K[A]_1^x [B]_1^y}{K[A]_3^x [B]_3^y}$$

$$\frac{1\times10^{-5}}{1\times10^{-5}} = \frac{(0.1)^x (0.1)^y}{K(0.2)^x (0.1)^y} \Rightarrow \left(\frac{1}{2}\right)^x = 1 = \left(\frac{1}{2}\right)^0$$

X = O

$$\frac{rate\,1}{rate\,2} = \frac{[B]_1^y}{[B]_2^y} \Rightarrow \left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^2 = \frac{1\times10^{-5}}{4\times10^{-5}} = \left(\frac{0.1}{0.2}\right)^y$$

y = 2

Thus, rate = K $[A]^o[B]^2 = K[B]^2$ **Answer: B**

78. Which of the following metals has the highest electrical and thermal conductivities of any metal?

A. Ag B. Cu C. Ni D. Co

Explanation: Silver is known to be the best conductors of heat and electricity. **Answer: A**

79. Which of the following is a linear polymer?

A. High density polythene (HDPt)
B. Low density polythene (LDPt)
C. Bakelite
D. Vulcanized rubber

Explanation: Polymers can be linear or branched i.e. cross – linked.

- Linear polymers are held together by London dispersion forces and are flexible, low strength and low melting point.
- Branched or cross – linked polymers are hard and non – ductile.Bakelite, vulcanized rubber and high density polythene are cross – linked polymers. **Answer: B**

80. Which of the following is the most important source for the extraction of iron

A. Hematite C. chalcopyrite
B. Bauxite D. Sphalerite

Answer: A

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1. According to valence bond theory, which orbitals on bromine atoms overlap in the formation of the bond in Br_2?

A. 3s B. 3p C. 4s D. 4P

Explanation: According to VBT, covalent bond is formed when half-filled orbitals overlap. The half-filled orbital is found in 4p orbital. $Br(z=35):[Ar]4s^2 3d^{10} 4p_x^2 4p_y^2 4p_z^1$

Answer: D

2. The type of compound that is most likely to contain a covalent bond is one that is

A. A solid metal
B. Composed of only non-metals
C. Composed of a metal from the far left and a non-metal from far right of the periodic table
D. Held together by the electrostatic forces between oppositely charged ions.

Explanation: Covalent bond is most likely formed between non-metals.

Answer: B

3. The first step of the scientific method involves

A. forming a hypothesis
B. making observations
C. performing an experiment
D. predicting the result of an experiment

Explanation: Steps in scientific method

Observation → hypothesis → experiment → theory → law

Answer: B

4. The process of solute particles being surrounded by solvent particles is known as

A. saturation C. solvation
B. agglomeration D. dehydration

Answer: C

5. Which of the following represents an acceptable possible set of quantum numbers (in the order n, ℓ, m_ℓ, m_s) for an electron an atom?

A. 2, 1, 0, 0 C. 2, 1, -1, $-\frac{1}{2}$
B. 2, 0, 2, $+\frac{1}{2}$ D. 2, 0, 1, $-\frac{1}{2}$

Explanation: n = 1, 2, 3, …

$\ell = 0$ to n − 1

$m_\ell = -\ell$ to $+\ell$

$m_s = \pm \frac{1}{2}$

Answer: C

6. Of the three types of radioactivity characterized by Rutherford, which of the following are particles?

A. γ – rays
B. β - rays
C. α - rays and β - rays
D. α - rays and β - rays and γ– rays

Explanation:

- α - rays and β - rays are particles.
- α - rays have mass and positive charge
- β - rays have negative charge but no mass
- γ– rays are radiation without mass and charge

Answer: C

7. What kind of energy is converted in a galvanic cell?

A. Chemical energy is converted into electrical energy
B. Chemical energy is converted to heat.
C. Chemical energy is obtained from heat
D. Electrical energy is converted into chemical energy.

Explanation:

Electro chemical cell

Electrolytic cell		Galvanic cell	
Electrical energy	→ Chemical energy	Chemical energy	→ Electrical energy

Answer: A

8. The conversion of nitrogen gas to nitrates by bacteria is called

A. nitrification C. execration
B. nitrogen fixation D. denitrification

Answer: B

9. According to the Arrhenius concept, an acid is a substance that?

A. is capable of donating one or more H+ to any solvent
B. causes an increase in concentration of H^+ in aqueous solutions

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C. can accept a pair of electrons to form a coordinate covalent bond

D. reacts with the solvent to form the cation formed by autoionization of that solvent.

Explanation: Arrhenius definition

- Acid is a substance that releases a hydrogen ion in water
- It causes an increase in the concentration of hydrogen ion in aqueous solutions.

Answer: B

10. A system which can exchange both matter and energy with its surroundings is said to be a/an

A. Isolated system C. ideal system

B. open system D. closed system

Explanation:

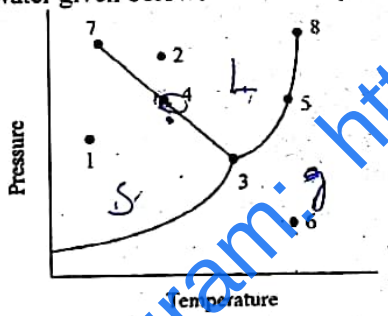
Open system – exchanged matter and energy

Isolated system – exchanges neither mass nor energy

Closed system – exchanges energy but not matter

Answer: B

11. At which point can only the solid and liquid phases coexist in the phase diagram of water given below?



A. 3 B. 4 C. 5 D. 8

Explanation:

→ Solid and liquid phases coexist at equilibrium on fusion curve (point 4)

→ Liquid and vapor phases coexist at equilibrium on vaporization curve (point 5)

→ Solid and vapor phases coexist at equilibrium on sublimation curve.

Answer: B

12. What type of solute-solvent interaction should be the most important in a solution of iodine in carbon tetrachloride?

A. London forces C. Ion-dipole forces

B. Ionic bonding D. Dipole-dipole forces.

Explanation: Iodine is non-polar solute and carbon tetrachloride is non-polar solvent. The solute – solvent interactions responsible for the solubility of non-polar solute in non-polar solvent is London force.

Answer: A

13. The distance between two carbon atoms in a diamond is 154pm. What is the distance between the carbon atoms in millimeters?

A. 7.7×10^{-5} C. 1.54×10^{-7}

B. 7.7×10^{-7} D. 1.54×10^{-9}

Solution:

$$\frac{1\text{pm}}{1\text{mm}}=\frac{10^{-12}\text{m}}{10^{-3}\text{m}}$$

$$\frac{1\text{pm}}{1\text{mm}}=10^{-9}$$

$$1\text{pm}=10^{-9}\text{mm}$$

$$\Rightarrow 154\text{pm}=154\times10^{-9}\text{mm}=1.54\times10^{-7}\text{mm}$$

Answer: C

14. For the acid-base equilibrium,

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

The Bronsted-Lowry acids are;

A. H_2O and OH^-

B. HCO_3^- and OH^-

C. H_2O and H_2CO_3

D. HCO_3^- and H_2CO_3

Explanation:

- Bronsted-lowry acids are proton (H^+) donors. For the forward reaction, H_2O is proton donor but for the reverse reaction H_2CO_3 is proton donor.

Answer: C

15. How many sigma and pi bonds are present in the following molecule?

$$H_3C-CH=CH-CH_3$$

A. 8σ bonds and 1π bonds

B. 8σ bonds and 2π bonds

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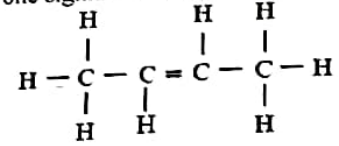
C. 10σ bonds and 2π bonds
D. 11σ bonds and 1π bonds

Solution: Draw detail structural formula to observe all the single bonds. All single bonds are sigma bonds. One double bond contains one sigma and one pi bond.



There are 11σ and 1π bonds. **Answer: D**

16. How many orbitals are there in an atom with $n = 4$?
A. 2 B. 8 C. 16 D. 25

Solution: Number of orbitals in main shell $= n^2 = 4^2 = 16$

Answer: C

17. The most abundant metal on the earth's crust is:
A. Fe B. Al C. Ca D. Na

Explanation:
Al > Fe > Ca > Na

Answer: B

18. Which of the following is correct?
A. $1L = 1dm^3$ C. $10L = 1dm^3$
B. $1L = 10dm^3$ D. $1L = 0.1dm^3$

Solution:

$$1m^3 = 10^3 L = 10^3 dm^3 = 10^6 mL = 10^6 cm^3$$
$$\Rightarrow 1L = 1dm^3,\ 1mL = 1cm^3$$

Answer: A

19. The enthalpy of combustion of solid carbon to form carbondioxide is -383.5KJ/mol carbon, and the enthalpy of combustion of carbon monoxide to form carbondioxide is -283.3KJ/mol CO. what will be the enthalpy change, ΔH for the reaction below?

$$2C(s) + O_2(g) \rightarrow 2CO(g)$$

A. -110.4KJ C. +172.9KJ
B. -220.8KJ D. +1354.0KJ

Solution: $C(s) + O_2(g) \rightarrow CO_2(g), \Delta H_1 = -393.5KJ$

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g), \Delta H_2 = -283.3KJ$$

To obtained the required reaction:

Multiply the first by 2

Reverse the second and multiply by 2

$$\Rightarrow 2C(s) + 2O_2(g) \longrightarrow 2CO_2(g),\ \Delta H_1^1 = 2\Delta H_2$$
$$\Delta H_1^1 = -787KJ$$
$$2CO_2(g) \longrightarrow 2CO(g) + O_2(g)\ \Delta H_2^1 = -2\Delta H_2$$
$$\Delta H_2^1 = -2x(-283.3KJ)$$
$$= 566.6\ KJ$$
$$2C(s) + O_2(g) \longrightarrow 2CO(g),\ \Delta H = \Delta H_1^1 + \Delta H_2^1$$
$$\Delta H = -787kg + 566.6KJ$$
$$\Delta H = -220.4KJ$$

Answer: B

20. The decreasing order of electrochemical characteristics of some metals is given as:
Mg > Al > Zn > Cu > Ag. What will happen if a copper spoon is used to stir a solution of aluminum nitrate ($Al(NO_3)_3$)?
A. There is no reaction
B. The solution becomes blue
C. The spoon will get coated with aluminum
D. An alloy of copper and aluminum is formed.

Explanation:
- There is no reaction because more active metal displaces less active metal but not the reverse.

$Cu + Al(NO_3)_3 \longrightarrow$ no reaction because Cu is less reactive that Al.

$2Al + 3CuSO_4 \longrightarrow Al_2(SO_4)_3 + 3Cu$

Answer: A

21. What hybridization change does the carbon atom undergo in the combustion of methane?

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$

A. $sp \longrightarrow sp^2$ C. $sp^3 \longrightarrow sp$
B. $sp^2 \longrightarrow sp^3$ D. $sp^2 \longrightarrow sp$

Explanation: $CH_4 \longrightarrow CO_2$

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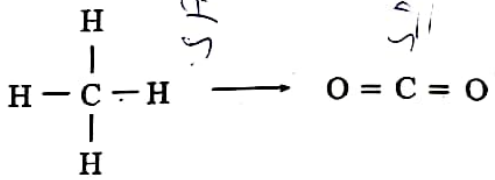


- Four sets of electoms hybridization: sp3
- Two sets of electrons
- Hybridization: sp

Answer: C

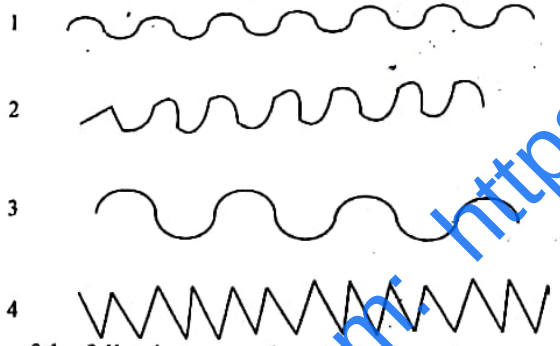
22. Which of the following ionic compounds has the greatest lattice energy?

A. LiF
B. LiCl
C. LiBr
D. LiI

Explanation: Lattice energy is affected by size and magnitude of charge
As size increases, lattice energy decreases.
As magnitude of change increases as, lattice energy also increase.

Answer: A

23. Consider the four electromagnetic waves shown below



Which of the following waves has the highest frequency?

A. 1 B. 2 C. 3 D. 4

Explanation:
As the number of waves increase, frequency also increases.

Answer: D

24. Considering the reaction below in which of the following will the effect of concentration and temperature simultaneous cause an increase in the rate at which products are formed?

$$CaCO_3(s) + 2HCl(aq) \rightarrow CO_2(g) + CaCl_2(aq) + H_2O(l) + heat$$

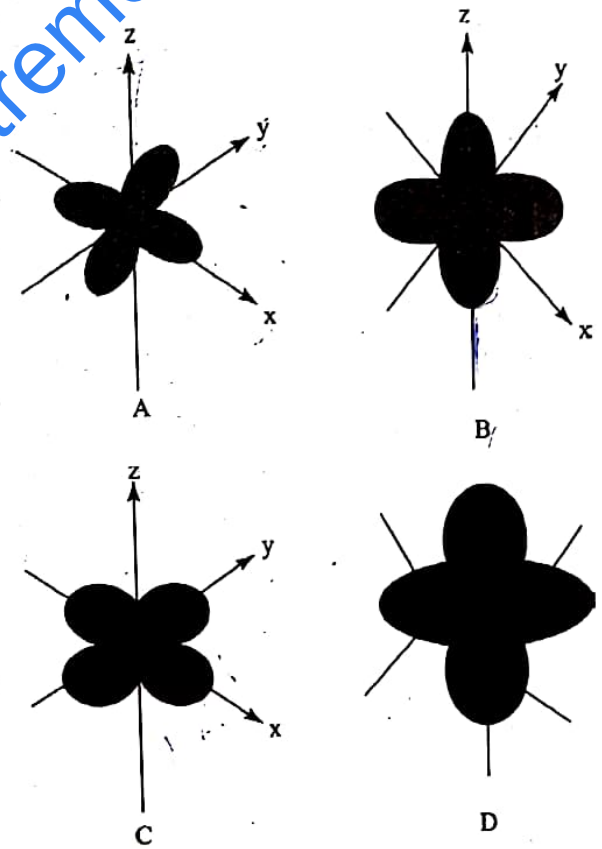
A. Decrease [HCl] and decrease temperature
B. Increase [HCl] and Increase temperature

C. Increase [HCl] and decrease temperature
D. Grind up the $CaCO_3$ and decrease temperature

Explanation:
For exothermic reaction, increasing temperature decrease the rate of a reaction.
Increasing concentration of aqueous solution and gases increase the rate of a reaction.

Answer: C

25. Which of the following diagrams describes the electron density in the dxy orbital?



Solution:

A. dxy
B. dz^2
C. $dx^2 - y^2$
D. dz^2

Answer: A

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(26) The wave number of an electromagnetic radiation is $1 \times 10^5 cm^{-1}$. The frequency of the –radiation would be

A. $3 \times 10^8 s^{-1}$ C. $3 \times 10^{10} s^{-1}$
B. $3 \times 10^6 s^{-1}$ D. $3 \times 10^{15} s^{-1}$

Solution: Wave number $= \frac{1}{\lambda}$

$$c = \lambda v = v = \frac{c}{\lambda} = C \times \text{wave number}$$

$$v = 3\times10^{10}\frac{cm}{s}\times1\times10^5 cm^{-1} = 3\times10^{15} s^{-1}$$

Answer: D

(27) For the gas phase reaction

$$N_2 + O_2 \rightleftharpoons 2NO,\ \Delta H = +180 KJ/mol$$

The value of K changes with the

A. change in pressure
B. introduction of NO
C. change in concentration of N_2
D. change in temperature

Explanation:

The value of equilibrium constant is affected by:

- temperature
- reversing the reaction
- change in coefficients of reactants and products

Answer: D

(28) In the reaction ($2SO_2 + O_2 \rightleftharpoons 2SO_3$, $K_{eq} = 100$). What will be the concentration of O_2, if the concentration of SO_2 is the same as that of SO_3?

A. $[O_2] = [SO_2]$ C. $[O_2] = 100M$
B. $[O_2] = 0.01M$ D. $[O_2] = 0.1M$

Solution: Given $[SO_2] = [SO_3]$

$$K_{eq} = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

$$K_{eq} = \frac{1}{[O_2]}$$

$$[O_2] = \frac{1}{K_{eq}} = \frac{1}{100} = 0.01\ M$$

Answer: B

(29) The decomposition of nitrosyl chloride was studied as:

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

The following data were obtained where

$$Rate = \frac{-\Delta[NOCl]}{\Delta t}$$

$[NOCl]_o$ (molecules/cm^3)	Initial Rate (molecules/cm^3.s)
3.0×10^{16}	5.98×10^4
2.0×10^{16}	2.66×10^4
1.0×10^{16}	6.64×10^3
4.0×10^{16}	1.06×10^5

What is the rate law in the above decomposition

A. $r = K[NOCl]^2$ C. $r = K[NOCl]\ [NO]$
B. $r = K[NOCl]$ D. $r = K[NOCl]\ [Cl]$

Solution:

Select any two simple data to find the order of the reaction with respect to NOCl.

$$Rate = K[NOCl]^x$$

For example use data 2 and 4

$$\frac{r_2}{r_4} = \frac{K[NOCl]_2^x}{K[NOCl]_4^x}$$

$$\frac{2.66\times10^4}{10.6\times10^4} = \frac{(2.0\times10^{16})^x}{(4\times10^4)^x}$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^x \Rightarrow \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^x$$

$x = 2$

Thus, rate $= K[NOCl]^2$

Answer: A

(30) Considering the mechanism for a reaction below, which of the following statement is correct?

Step1: $HBr + O_2 \rightarrow HOOBr$
Step 2: $HBr + HOOBr \rightarrow 2HOBr$
Step 3: $2HOOBr + 2HBr \rightarrow 2Br_2 + 2H_2O$

A. Br_2 is a reactant
B. HBr is a product

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C. HOBr is a catalyst
D. HOOBr is a reaction intermediate

Explanation:
The over all reaction is
$4HBr + O_2 \rightarrow 2Br_2 + 2H_2O$
$\Rightarrow$ HBr is reactant
Br_2 is product
HOBr and HOOBr are intermediates.

Answer: D

31. The acid-base indicator bromocresol green is a weak acid. The yellow acid and blue base forms of the indicator are present in equal concentrations in a solution when the PH is 4.68. What is the PKa of bromocresol green?
A. 4.48 B. 4.68 C. 5.68 D. 6.68

Solution: Let the formula of the indicator be HIn
$HIn \rightleftharpoons H^+ + In^-$
Given, $[HIn] = [In^-]$

$$P^H = PKa + 10g\frac{[In^-]}{[HIn]}$$

$4.68 = PKa + 10^1$
$\Rightarrow PKa = 4.68$

Answer: B

32. If $NaNO_2$ is added to a solution of HNO_2, which of the following statement is true?
A. The pH of the solution will increase
B. The pH of the solution will decrease
C. The pH will remain the same
D. The equilibrium will not be affected

Explanation: $NaNO_2$ is basic salt, because it is obtained from strong base. NaOH and weak acid, HNO_2. When a basic salt is added to a given solution, the pH of the solution will increase. **Answer: A**

33. A lipid is any substance of biochemical origin that is
A. soluble in both water and non-polar solvents
B. insoluble in both water and non-polar solvents
C. soluble in both water but insoluble in non-polar solvents.
D. soluble in non-polar solvents and insoluble in water

Explanation: Fats and soils are collectively known as lipids. They are organic compounds that dissolve in non-polar solvents but not in polar solvents such as water.

Answer: D

34. A solution in an electrolytic cell contains Cu^{2+} $(\epsilon^\circ = 0.34V)$. Ag^+ $(\epsilon^\circ = 0.80V)$ and Zn^{2+} $(\epsilon^\circ = -0.76V)$. If the voltage is initially very low and slowly increased, in which order will the metals be plated out onto the cathode?
A. $Zn^{2+} > Cu^{2+} > Ag^+$
B. $Cu^{2+} > Zn^{2+} > Ag$
C. $Ag^+ > Zn^{2+} > Cu^{2+}$
D. $Ag^+ > Cu^{2+} > Zn^{2+}$

Explanation: The more positive the reduction potential, the easier is to be reduced at cathode.
$\Rightarrow Ag^+ > Cu^{2+} > Zn^{2+}$

Answer: D

35. What is the concentration of sodium chloride in water needed in order to produce an aqueous solution that has an identical Osmotic pressure (isotonic) with blood ($\pi = 7.70$ atm at 25°C)?
A. $0.003molL^{-1}$ C. $0.1575molL^{-1}$
B. $0.006molL^{-1}$ D. $0.315molL^{-1}$

Solution: $\pi = i$ MRT
- The solution is electrolytic: $NaCl \rightarrow Na^+ + Cl^-$
- The value of Van't Hoff factor, i is almost equal to the number of ions produced.

$i = 2$, $T = 25°C = 298K$

$$R = 0.082\ \frac{atm.L}{mol.K}$$

$$M = \frac{\pi}{iRT} = \frac{7.70atm}{2 \times 0.827\ \frac{atm.L}{mol.K} \times 298K}$$

$M = 0.1575mol.L^{-1}$

Answer: C

36. What is the molarity of a 5g hydrogen peroxide (H_2O_2) in 100mL solution that is used for hair bleaching?
A. 0.015M C. 1.5M
B. 0.15M D. 3M

Solution: $M = \frac{m}{M_{wt} \times V}$

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$$M = \frac{5g}{34\frac{g}{mol} \times 0.1L} = 1.47M$$

Answer: C

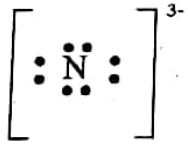
37. The maximum number of electrons in p-orbital with $n = 6$, $m\ell = 0$ is
A. 2 B. 6 C. 16 D. 14

Explanation:
The maximum number of electrons in a p-orbital is 2. The given quantum numbers represent one of three 6p orbital.

Answer: A

38. How many unpaired electrons are there in the Lewis structures of a N^{3-} ion?
A. 0 B. 1 C. 2 D. 3

Explanation: structure for N^{3-} is:



$\Rightarrow$ No unpaired electron

Answer: A

39. Which one of the following metals is extracted by thermal reduction process:
A. Al B. Cu C. Fe D. Mg

Answer: C

40. Which of the following transitions will omit maximum energy in the hydrogen atom?
A. $n = 4 \rightarrow n = 3$ C. $n = 2 \rightarrow n = 1$
B. $n = 4 \rightarrow n = 2$ D. $n = 3 \rightarrow n = 2$

Solution: $\Delta \epsilon = 2.18 \times 10^{-10} J\left(\frac{1}{4^2} - \frac{1}{3^2}\right) = -1.06 \times 10^{-19} J$

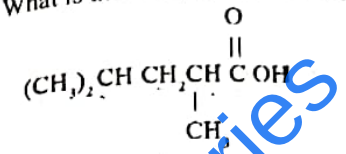
$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{4^2} - \frac{1}{2^2}\right) = -4.10 \times 10^{-19} J$

$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = -1.635 \times 10^{-18} J$

$\Delta \epsilon = 2.18 \times 10^{-18} J\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = -1.40 \times 10^{-19} J$

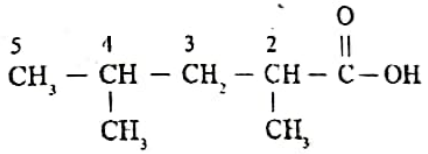
Answer: C

41. What is the IUPAC name for the compound



A. 2, 4 – dimethyl pentanoic acid
B. 1, 1, 3-trimethyl butanoic acid
C. 1-hydroxyl -2, 4 – dimethyl pentanone
D. 2-carboxy isohexone

Explanation:
The compound is carboxylic acid. Start form the carbonyl carbon to give IUPAC name.
The above structural formula is the same as the structure below



2, 4 – dimethyl pentanoic acid

Answer: A

42. The conventional equilibrium constant expression (K_C) for the system.

$2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$ is

A. $\frac{[I_2][Cl_2]}{[ICl]^2}$ C. $[Cl_2]$

B. $\frac{[I_2][Cl_2]}{2[ICl]}$ D. $\frac{[I_2]+[Cl_2]}{2[ICl]}$

Explanation: Pure solids and liquids /solvents/ are not included in the equilibrium constant expression.

$\Rightarrow K_C = [Cl_2]$

Answer: C

43. Which of the following compound does Not follow the octet rule?
A. CS_2 B. PBr_3 C. IBr D. BrF_5

Explanation: If the number of electrons on the centeral atom is not equal to eight, such type of compounds donot obey the octet rule.

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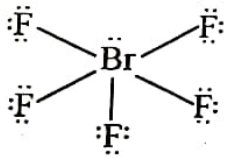
CS_2 : S = C = S :, obeys the octet rule

PBr_3 : Br — P — : Br : , obeys the octet rule
(with : Br : below P)

IBr : I — Br : , obeys the octet rule

BrF_5 

, does not obey the octet rule because the number of electrons on Br are 12

Answer: D

44. Chemically, fats and oils are
A. Acids
B. Alcohols
C. Esters
D. Alkenes

Answer: C

45. The reaction A + 3B → 2C + D is first order with respect to reactant A and second order with respect to reactant B. If the concentration of A is doubled and the concentration of B is halved, the rate of the reaction would _____ by a factor of _____
A. Increase, 2
B. decrease, 2
C. Increase, 4
D. decrease, 4

Solution: Given $[A]_2 = 2[A]$, $[B]_2 = \frac{1}{2}[B]_1$

$r_1 = K[A]_1 [B]_1^2$

$r_2 = K[A]_2 [B]_2^2$

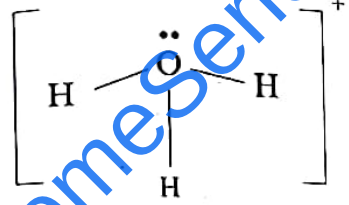
$r_2 = K2[A]_1\left(\frac{1}{2}[B]_1\right)^2 = K2[A]_1 \frac{1}{4}[B]_1^2$

$r_2 = \frac{1}{2} r_1$

Answer: B

46. The molecular geometry of the ion H_3O^+ is
A. linear
B. tetrahedral
C. bent
D. trigonal pyramidal

Explanation:

H_3O^+ 

, trigonal pyramidal

Answer: D

47. If a student wishes to prepare approximately 100mL of an aqueous solution of 6M HCl using 12M HCl. Which procedure is correct?
A. Adding 50mL of 12M HCl to 50mL of water while stirring the mixture steadily.
B. Adding 25mL of 12M HCl to 50mL of water while stirring the mixture steadily.
C. Adding 50mL of water to 50mLof 12M HCl while stirring the mixture steadily.
D. Adding 25mL of water to 50 mL of 12M HCl while stirring the mixture steadily.

Solution:

$M_1 = 12M$ $M_2 = 6M$
$V_1 = ?$ $V_2 = 100mL$
$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{6M \times 100mL}{12M} = 50mL$$

V_{H2O} added $= V_2 - V_1$
$= 100mL - 50mL = \mathbf{50mL}$

Answer: A

48. During dilution of concentrated acid, acid is added to water not water to acid.
What is the final concentration of Cl^- ion when 250mL of 0.20M $CaCl_2$ solution is mixed with 250mL of 0.40M KCl solution? (Assume additive volumes)
A. 1.60M
B. 0.40M
C. 0.20M
D. 0.60M

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Solution:

$$n_{CaCl_2} = M \times V = 0.20\frac{mol}{L} \times 250mL = 50mmol$$

$$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$$

$$\Rightarrow n_{Cl^-} = 2 \times 50m\ mol = 100m\ mol$$

$$n_{KCl} = M \times V = 0.40\frac{mol}{L} \times 250mL = 100m\ mol$$

$$KCl \longrightarrow K^+ + Cl^-$$

$V_T = (250 + 250)\ mL = 500mL$

$n_T = 100m\ mol + 100m\ mol = 200m\ mol$

$$\left[Cl^-\right] = \frac{n_T}{V_T} = \frac{200m\ mol}{500mL} = 0.40mol/L$$

Answer: B

49. Which of the following is used in the reaction called saponification?

A. Strong base C. Hydrogen
B. Strong acid D. Nickel

Explanation:
Long chain organic acids + NaOH/KOH → Soap strong bases NaOH and KOH are used to make hard and soft soaps respectively.

Answer: A

50. The concentration of nitrate ion in a solution that contains 0.900M aluminum nitrate is:

A. 0.90 B. 0.45 C. 0.30 D. 2.70

Solution: The formula of aluminum nitrate is $Al(NO_3)_3$

$$\left[NO_3^-\right] = 0.900M \times 3 = 2.70M$$

Answer: D

51. Electrolysis of dilute aqueous NaCl solution was carried out by passing 10milliampere current. The time required to liberate 0.01mol of H_2 gas at the cathode is

A. $9.65 \times 10^4 s$ C. $28.95 \times 10^4 s$
B. $19.3 \times 10^4 s$ D. $38.6 \times 10^4 s$

Solution: $2H^+ + 2e \longrightarrow H_2$

$2F = 1mol$

$? = 0.01mol$

2 Farad of electricity are required to produce 1mole of H_2

$2F = 2 \times 96500C = 193000C$

To produce 0.01mol of H_2, the quantity of electricity required, Q is

$$Q = \frac{0.01mol \times 193000C}{1mol}$$

$Q = 1930C$

$Q = It$

$I = 10mA = 10 \times 10^{-3}A = 1 \times 10^{-2}A$

$$t = \frac{Q}{I}$$

$$t = \frac{1930Q}{1 \times 10^{-2}A} = 1.93 \times 10^5 s$$

Answer: B

52. How many electrons will appear when the following half-reaction is balanced?

$$S_4O_6^{2-} \longrightarrow S_2^{2-}O_3$$

A. 3 B. 2 C. 4 D. 1

Explanation:
Balance the atoms first by inspection method.

$$S_4O_6^{2-} \longrightarrow 2S_2^{2-}O_3$$

Balance the charge by adding electrons to the deficient side

$$2e^- + S_4O_6^{2-} \longrightarrow 2S_2O_3^{2-}$$

Answer: B

53. PKa values of three acids X, Y and Z are 4.5, 3.5 and 6.5 respectively. Which of the following represents the correct order of acid strength?

A. X > Y > Z C. Y > X > Z
B. Z > X > Y D. Z > Y > X

Explanation: PKa and strength of acid are inversely related.

Answer: C

54. Standard electrode potential for Sn^{4+}/Sn^{2+} couple is +0.15V and that of the Cr^{3+}/Cr couple is -0.74V. These two couples in their standard state are connected to make a spontaneous electrochemical reaction. The cell potential will be

A. +1.83V C. +0.89V
B. +1.19V D. +0.18V

Solution: For spontaneous reaction, the more positive reduction potential used as cathode and the other as anode

$$E^\circ_{cell} = E^\circ_{(cathode)} - E^\circ_{(anode)}$$
$$= 0.15V - (-0.74V) = +0.89V$$

Answer: C

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55. The enthalpies of formation of gaseous N_2O and NO at 298K are 82 and 90KJ/mol, respectively. The enthalpy change for the reaction

$$N_2O(g) + \frac{1}{2}O_2(g) \longrightarrow 2NO(g)$$

A. −8KJ C. −74KJ
B. 98KJ D. 8KJ

Solution:

$$\Delta H° = \sum n\Delta H^o_f \text{ products} - \sum m\Delta H^o_f \text{ reactants}$$

$$\Delta H° = 2\times\Delta H^o_f NO - 1\times\Delta H^o_f N_2O$$

$$\Delta H° = 2\times 90 - 1\times 82$$

$$\Delta H° = 98KJ$$

Answer: B

56. The following data was collected at the end point of a titration performed to find the molarity of an HCl solution

Volume of acid (HCL) used = 14.4mL
Volume of base (NaOH) used = 22.4mL
Molarity of standard base (NaOH) = 0.20M

On the basis of the above data, what is the molarity of the acid solution?

A. 1.6M C. 0.31M
B. 0.64M D. 0.13M

Solution: $NaOH + HCl \longrightarrow NaCl + H_2O$

$$n_{NaOH} = M_{NaoH} \times V_{NaOH} = 22.4mL \times 0.20\frac{mol}{L}$$

$$n_{NaOH} = 4.48m\ mol$$

$$\overset{4.48m\ mol}{\underset{1}{NaOH}} + \overset{x}{\underset{1}{HCl}} \longrightarrow NaCl + H_2O$$

$$\frac{4.48m\ mol}{1} = \frac{x}{1}$$

$$\Rightarrow n_{HCl} = 4.48m\ mol$$

$$M_{HCl} = \frac{n_{HCl}}{V_{HCl}} = \frac{4.48m\ mol}{14.4mL} = 0.31M$$

Answer: C

57. The hybridization of the centeral atom in the XeF_4 molecule is

A. sp^2 B. sp^3 C. sp^3d D. sp^3d^2

Explanation: Draw the Lewis structure and the number of sets of electrons is equal to the number of hybrid orbitals

There are eight sets of electrons on Xe. Hybridization: sp^3d^2

Answer: D

58. For the reaction

$$N_2 + 3H_2 \longrightarrow 2NH_3$$

The rate of disappearance of H_2 is $0.01molL^{-1}.min^{-1}$. The rate of appearance of NH_3 would be:

A. $0.01\ molL^{-1}\ min^{-1}$
B. $0.02\ molL^{-1}\ min^{-1}$
C. $0.007\ molL^{-1}\ min^{-1}$
D. $0.02\ molL^{-1}\ min^{-1}$

Solution: $r_{NH_3} = \frac{coeff.NH_3}{coeff.H_2} \times r_{H_2}$

$$r_{NH_3} = \frac{2}{3} \times 0.01molL^{-1}min^{-1}$$

$$r_{NH_3} = 0.0067molL^{-1}min^{-1}$$

Answer: C

59. Bakelite is obtained from phenol by reacting with

A. HCHO C. CH_3CHO
B. $(CH_2OH)_2$ D. CH_3COCH_3

Explanation:
Bakelite is formed from phenol and formaldehyde. **Answer: A**

60. In which one of the following numbers are all the zeros significant?

A. 100.090090 C. 0.1000
B. 0.143290 D. 00.0030020

Explanation: Zeros before non-zero digits are not significant, however, after and between non-zero digits they are significant

Answer: A

61. An aqueous solution contains 0.100M NaOH at 25°C. the P^H of the solution is:

A. 0.100 C. 7.00
B. 1.00 D. 13.0

Solution: $[OH^-] = [NaOH] = 0.10M = 10^{-1}M$

$$P^H = 14 + 10g[OH^-]$$

$$P^H = 14 + 10g10^{-1} = 13$$

Answer: D

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62. The solubility of oxygen gas in water at 25°C and 1.0atm pressure of oxygen is 0.04g/L. the solubility of oxygen in water at 3.0atm and 25°C is _____ g/L

A. 0.014 B. 0.31 C. 0.041 D. 0.123

Solution: $C_{O2} = K_H \cdot P_{O2}$ KH - Henry's constant

$$K_H = \frac{C_{O2}}{P_{O2}} = \frac{0.041 g/L}{1 atm} = 0.041 g/L$$

When $P_{O2} = 3atm$, $C_{O2} = ?$

$$C_{O2} = K_{H(O2)} \cdot P_{O2} = 0.041 \frac{g}{L.atm} \times 3atm = 0.123\ g/L$$

Answer: D

63. The decomposition of carbondisulfide, CS_2, to carbon monosulfide, CS, and sulphur is first order with $K = 2.8 \times 10^{-7} S^{-1}$ at 1000°C. What is the half-life of the reaction below at 1000°C?

A. $5.0 \times 10^7 s$ C. $3.8 \times 10^5 s$

B. $4.7 \times 10^{-6} s$ D. $2.5 \times 10^6 s$

Solution: $t_{1/2} = \frac{0.693}{K}$

$$t_{1/2} = \frac{0.693}{2.8 \times 10^{-7} S^{-1}} = 2.475 \times 10^6 s$$

Answer: D

64. The PKa of a weak monoprotic acid is 4.8. What should be the ratio of [Acid]/[salt] of a buffer, if PH = 5.8 is required.

A. 0.1 B. 1.0 C. 2.0 D. 10

Solution: $PH = PKa + \log \frac{[salt]}{[Acid]}$

$$\log \frac{[salt]}{[Acid]} = PH - PKa = 5.8 - 4.8 = 1$$

$$\frac{[salt]}{[Acid]} = 10$$

$$\frac{[Acid]}{[Salt]} = \frac{1}{10} = 0.1$$

Answer: A

65. The dissolution of water in octane (C_8H_{18}) is prevented by

A. dipole – dipole attraction between octane molecules

B. hydrogen bonding between water molecules

C. London dispersion forces between octane molecules.

D. Repulsion between like-charged water and octane molecules

Explanation: water is polar and octane is non-polar. As a result, the two do not mix due to the like dissolves like principle. The weak London force in octane prevents the mixing of water in it.

Answer: C

66. At 445°C, K_c for the following reaction is 0.020.

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

A mixture of H_2, I_2 and HI in a vessels at 445°C has the following concentrations: $[HI] = 2.0M$; $[H_2] = 0.50M$ and $[I_2] = 0.10M$. Which of the following statements concerning the reaction quotient, Q_c, is true for the above system?

A. Q_c, is less than K_c; more HI will be produced

B. Q_c, is greater than K_c; more HI will be produced

C. Q_c, is less than K_c; more H_2 and I_2 will be produced

D. Q_c, is greater than K_c; more H_2 and I_2 will be produced

Solution: $Q_c = \frac{[H_2][I_2]}{[HI]^2}$

$$Q_c = \frac{(0.50)(0.10)}{(2.0)^2} = 0.0125$$

$\Rightarrow Q_c < K_c$, formation of products is favored

Answer: C

67. Natural rubber is a polymer of

A. butadiene C. neoprene

B. isoprene D. styrene

Explanation: Natural rubber, obtained from the milky sap or latex of a rubber tree, is a polymer of 2-methyl -1, 3-butadiene which has a common name **isoprene.**

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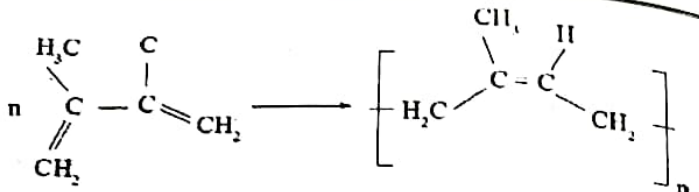


Isoprene natural rubber

Answer: B

68. Which of the following statements is true?

A. All forms of electromagnetic radiation are visible.
B. Radio waves have shorter wavelengths than visible light.
C. Ultraviolet light has longer wavelengths than visible light
D. The frequency of radiation increases as the wavelengths decreases

Explanation: Natural rubber, obtained from the milky sap or latex of a rubber tree, is a polymer of 2-methyl -1, 3-butadiene which has a common name isoprene.

Frav, x-ray, UV, Visible, IR, Microwaves, Radio waves

- Wave length increase
- Frequency and energy decreases

Answer: D

69. If the enthalpy change for a certain reaction A → B is -2kJ at 300K. What would be the entropy change in the surroundings?

A. -40J/K C. -3.6×10^6J/K
B. 40J/K D. 3.6×10^6J/K

Solution: $\Delta G = \Delta H - T\Delta S$

Dividing by T gives all values pertaining to the system.

$$\frac{\Delta G}{T} = \frac{\Delta H}{T} - \Delta S$$

It can be shown that

$$\boxed{\Delta S_{sur} = -\frac{\Delta H_{sys}}{T}} \text{ (at constant P and T)}$$

$$\frac{\Delta G_{sys}}{T} = \Delta S_{sur} - \Delta S_{sys}$$

$$-\frac{\Delta G_{sys}}{T} = \Delta S_{sur} + \Delta S_{sys}$$

$\Delta S_{unit} = -\frac{\Delta G_{sys}}{T}$ because

$\Delta S_{unit} = \Delta S_{sur} + \Delta S_{sys}$

ΔS_{unit} - change in entropy of the universe

ΔS_{sur} - change in entropy of the surrounding

ΔS_{sys} - change in entropy of the system

ΔG - Gibb's free energy change

T- Kelvin temperature

$$\Delta S_{sur} = -\frac{\Delta H_{sys}}{T} = -\frac{(-2\times10^3 J)}{300K} = 1.7\frac{J}{K}$$

70. During the electrolysis of an aqueous solution of copper sulphate using platinum electrodes, the reaction takes place at the anode is?

A. $Cu^{2+} + 2e^- \longrightarrow Cu$
B. $Cu \longrightarrow Cu^{2+} + 2e^-$
C. $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$
D. $4H^+ + O_2 + 4e^- \longrightarrow H_2O$

Explanation:

→ Using platinum electrode:

Anode reaction: $2H_2O \longrightarrow 4H^+ + O_2 + 4e^-$

Cathode reaction: $Cu^{2-} + 2e^- \longrightarrow Cu$

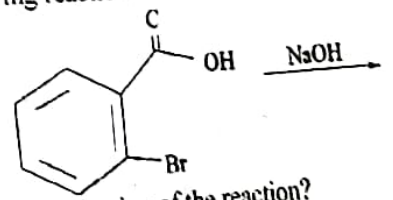
→ Using copper electrode

Anode reaction: $Cu \longrightarrow Cu^{2-} + 2e^-$

Cathode reaction: $Cu^{2+} + 2e^- \longrightarrow Cu$

Answer: C

71. Given the following reaction



What is the major production of the reaction?

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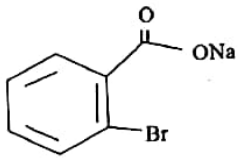
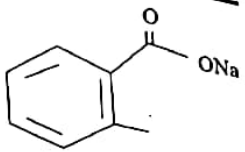
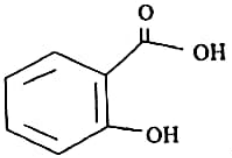
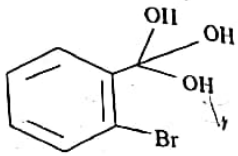
A. 

C. 

B. 

D. 

Answer: A

72. Which of the following is an acceptable IUPAC name for the organic compound shown below?

A. Octanoic acid
B. 7-methyl heptanoic acid
C. 8-methyl heptanoic acid
D. 7-methyl octanoic acid

Explanation:

Each angle represents carbon atom

—— methyl

ethyl

- The parent name is octanoic acid
- The IUPAC name of the compound is 7-methyl octanoic acid.

Answer: D

Which of the following molecules has a dipole moment?

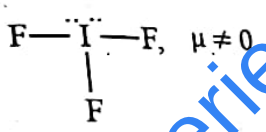
A. XeF_2 C. BF_3

B. IF_3 D. SF_5^+

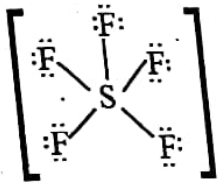
Explanation:

- Polar molecules have dipole moment different different from zero.
- Molecules whose molecular shape is linear, trigonal pyramid, tetrahedral, trigonal bipyramid, octahedral and square planar have zero dipolemoment.

XeF_2, F—Xe—F, $\mu = 0$

IF_3  $\mu \neq 0$

BF_3 F, $\mu = 0$

SF_3^+  , $\mu = 0$

Answer: B

74. Which of the following describes the balanced molecular equation when perchloric acid is mixed with solid iron (III) hydroxide?

A. $HClO_4(aq) + Fe(OH)_3(s) \rightarrow H_2O(l) + FeClO_4(aq)$
B. $3HClO_4(aq) + Fe(OH)_3(s) \rightarrow 3H_2O(l) + Fe(ClO_4)_3(aq)$
C. $3HOClO_2(aq) + Fe(OH)_3(s) \rightarrow 3H_2O(l) + Fe(ClO_3)_3(aq)$
D. $HOClO_2(aq) + Fe(OH)_3(s) \rightarrow H_2O(l) + Fe(ClO_3)(aq)$

Answer: B

75. At 298K the following two gaseous equilibria involving SO_2, SO_3 and O_2 are established.

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g), K_1$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g), K_2$$

Which of the following equilibrium expression is correct?

A. $K_1 = K_2$ C. $K_2 = \frac{1}{K_1^2}$

B. $K_2 = K_1$ D. $K_2 = \frac{1}{K}$

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Solution: The second reaction is obtained by reversing and multiplying by 2.

$\Rightarrow K_1 = \frac{1}{K_2^{\,2}}$ or $K_2 = \frac{1}{K^1}$

Answer: C

76. A compound is formed by the combination of V and X as follows.

V W X Y Z

What is the empirical formula for the compound?

A. VX C. V_3X

B. V_2X D. VX_2

Explanation: Valence number of V = 1

Valence number of X = 8 – 5 = 3

Formula:

$V^1 \times X^3 = V_3X$

Answer: C

77. ΔH for solid to liquid transitions for compound A is 2.73K cal/mol and for compound B is 3.0K cal /mol. The melting point for compound A is 0°C and the melting point for compound B is 30°C. The entropy changes ΔS_A and ΔS_B at the two transition temperatures are related as:

A. $\Delta S_A = \Delta S_B$ C. $\Delta S_B < \Delta S_A$

B. $\Delta S_A < \Delta S_B$ D. $\Delta S_A = \frac{300\Delta S_A}{273}$

Answer: C

78. Which of the following metals forms a volatile compound that is taken as an advantage for its extraction?

A. Co B. Fe C. Ni D. W

Answer: C

79. A 0.1M solution of HCl is dissolved in water. What species of ions are present at equilibrium and what will be their equilibrium concentrations?

A. $[H_3O^+] = 0.1M$ $[OH^-] = 10^{-13}M$ $[Cl^-] = 0.1M$

B. $[H_3O^+] = 0.1M$ $[OH^-] = 0.1M$ $[Cl^-] = 0.1M$

C. $[H_3O^+] = 0.1M$ $[OH^-] = 10^{-13}M$ $[Cl^-] = 0M$

D. $[H_3O^+] = 10^{-13}M$ $[Cl^-] = 0.1M$ $[OH^-] = 0.1M$

Explanation:

For strong monoprotic acid such as HCl

$[H_3O^+] = [Cl^-] = [HCl] = 0.1M$

$[OH^-] = \frac{K_w}{[H_3O^+]}$

$[OH^-] = \frac{1\times10^{-14}}{0.1} = 10^{-13}M$

Answer: A

80. The following bases and their conjugate acids (as the chlorides) are available in the laboratory:

NH_3, $K_b = 1.8 \times 10^{-5}$

C_6H_5N, $K_b = 1.7 \times 10^{-9}$

$CH_3CH_2NH_2$, $K_b = 6.4 \times 10^{-4}$

CH_3NH_2, $K_b = 4.4 \times 10^{-4}$

Which of these acid-base pairs are best to prepare a buffer solution having a PH of about 9?

A. NH_3 + conjugate acid

B. C_6H_5N + conjugate acid

C. $CH_3CH_2NH_2$ + conjugate acid

D. CH_3NH_2 + conjugate acid

Explanation:

Best buffer can be prepared from weakest base and its conjugate acid.

Answer: B

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1. Which of the following is correct when 34495 is rounded to three significant figures?
A. 345 B. 34500 C. 344 D. 3840

Explanation:
When 34495 is rounded to three significance, the result will be 3.45×10^4 or 34500

Answer: B

2. What is the first step of the scientific method?
A. Making observations
B. Forming a hypothesis
C. Performing an experiment
D. Predicting the result of an experiment.

Answer: A

3. Which of the following are **NOT** electromagnetic waves?
A. Infrared waves C. Radio waves
B. Gamma waves D. Sound waves

Explanation:
Gamma rays, x-ray, ultraviolet ray, visible, Infrared, microwaves, Radio waves and Television waves are electromagnetic waves.

Answer: D

4. What is the distance that a radio wave will travel in 0.250s?
A. 1.2×10^7m C. 7.5×10^7m
B. 12×10^7m D. 5.6×10^7m

Solution: $C = \lambda v = \frac{\lambda}{T}$ because $v = \frac{1}{T}$

$\lambda = CT = 3 \times 10^8 \text{m/s} \times 0.25\text{s} = 7.5 \times 10^7\text{m}$

T – period

Answer: C

5. Which one of the following is **NOT** a form of chemical bonding?
A. Covalent bonding C. Ionic bonding
B. Metallic bonding D. Hydrogen bonding

Explanation:
- Ionic, covalent and metallic bonds are the common types of bonds. Hydrogen bond, dipole-dipole force, London force and ion-dipole force are intermolecular forces.

Answer: D

6. Which of the following statements is **NOT** true about covalent bonding?
A. Covalent bonds are least likely to be formed between atoms of the same element.
B. Covalent bonds are least likely to be formed between atoms of different elements on the right side of the periodic table.
C. Covalent bonds are least likely to be formed between an element in group I and an element in Group VII.
D. Covalent bonds are least likely to be formed by head of the group elements with height ionization energies.

Explanation:
- Covalent bond is formed between non-metals, and between less reactive metals and non-metals.
- It is not formed between alkali metals and halogens.

Answer: A

7. If we increase the concentration of a reactant, what happens to the collisions between particles?
A. There are more collisions.
B. There are fewer collisions.
C. There are the same number of collisions, but they have less energy.
D. There are the same number of collisions, but they have more energy.

Explanation: Concentration of reactants and number of collisions are directly related.

Answer: A

8. A drug decomposes by zero-order kinetics with a rate constant of $2\text{mg}.\text{mL}^{-1}$ month^{-1}. If the initial concentration is 100mg mL^{-1}, how long will it take for the drug to decompose by 10%?
A. 2 months C. 5 months
B. 3 months D. 4 months

Solution: $[A]_t - [A]_0 = -kt$

$[A]_0 - [A]_t = kt$ but $\frac{[A]_t}{[A]_0} = 90\% = \frac{90}{100} = \frac{9}{10}$

$[A]_t = \frac{9}{10}[A]_0^9$

$[A]_0 - \frac{9}{10}[A]_0 = kt$

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$$\frac{1}{10}[A]_o = kt$$

$$t = \frac{[A]_o}{10k} = \frac{100 mg/mL}{10 \times 2 mg\ mL^{-1}.month^{-1}}$$

$t = 5$ months

Answer: C

9. The value of K_{eq} for the following reaction is 0.25.

$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$

What is the value of K_{eq} at the same temperature for the reaction below: $2SO_2(g) + 2NO_2(g) \rightleftharpoons 2SO_3(g) + 2NO(g)$?

A. 0.25 B. 0.062 C. 0.50 D. 16

Solution: If the given reversible reaction is multiplied by 2, the neal equilibrium constant is the original the power of 2.

$\Rightarrow K_{eq} = (0.25)^2 = 0.0625$

Answer: B

10. Which of the following would react to form pentylethanoate?

A. 1-propanol and pentanoic acid
B. Ethanol and pentanoic acid
C. 1-pentanol and ethanoic acid
D. Ethanol and ethanoic acid

Solution:

$$CH_3CH_2CH_2CH_2CH_2OH + CH_3COOH \xrightleftharpoons{H^+}$$

1-pentanol Ethanoic acid

$CH_3CH_2CH_2CH_2CH_2OOC\ CH_3^+H_2O$

pentylethanoate

Answer: C

11. The difference between fats and oils is that:

A. oils are liquid at room temperature.
B. oils have more calories.
C. oils are solid at room temperature.
D. fats are liquid at room temperature.

Explanation:

Fats are solid while oils are liquid at room temperature.

Answer: A

12. What kind of solution forms when gasoline evaporates in air?

A. gas in gas solution C. Liquid in liquid solution
B. gas in liquid solution D. Liquid in gas solution

Explanation: When gasoline evaporates it turns into gas. Air is gas. Thus, the solution is gas-gas solution.

Answer: A

13. What is the solvent in 70% alcohol solution?

A. Water B. Alcohol C. Sugar D. Kerosene

Explanation: 70% alcohol solution means 70% alcohol and 30% water. Water is universal polar solvent.

Answer: A

14. How many moles of H_2SO_4 are needed to prepare 5.0 liters of a 2.0M solution of H_2SO_4?

A. 2.5 B. 5.0 C. 20 D. 10

Solution: $n = M \times u = 2\frac{mol}{L} \times 5L = 10 moles$ **Answer: D**

15. What is the balanced **net ionic equation** for the reaction of $CaCl_2$ and $AgNO_3$?

A. $CaCl_2(aq) + 2\ AgNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$
B. $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+\ 2NO_3^-(aq) \longrightarrow Ca^{2+}(aq) + 2NO_3^-(aq) + 2AgCl(s)$
C. $Cl^-(aq) + Ag^+(aq) \longrightarrow 2AgCl(s)$
D. $2Cl^-(aq) + Ag^+(aq) \longrightarrow 2AgCl(s)$

Solution:

→ $CaCl_2(aq) + 2AgNO_3(aq) \longrightarrow 2AgCl(s) + Ca(NO_3)_2(aq)$ molecular equation

→ $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+(aq) + 2NO_3^-(aq) = 2AgCl(s) + Ca^{2+}(aq) + 2NO_3^-(aq)$ over all ionic equation

$= 2AgCl(s) + Ca^{2+}(aq) + 2NO_3^-(aq)$

→ $2Ag^+(aq) + 2Cl^-(aq) \longrightarrow 2AgCl(s)$, net ionic equation

Ca^{2+} and NO_3^- are spectator ions.

Answer: D

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16. The Ka of hypochlorous acid (HClO) is 3.0×10^{-8} at 25°C. What is the % ionization of hypochlorous acid in a 0.015M aquous solution of HClO at 25°C.

A. 2.1×10^{-5} C. 1.4×10^{-2}
B. 0.14 D. 3.3×10^{-1}

Solution:

	$HClO \rightleftharpoons$	H^+	$+ OCl^-$
initial	0.015	0	0
change	-x	+x	+x
equilibrium	0.015 − x	x	x

$$Ka = \frac{[H^+][OCl^-]}{[HClO]}$$

$$3\times10^{-8} = \frac{x \cdot x}{0.015 - x} \quad \text{approximation: } 0.015 - x = 0.015$$

$$x^2 = 0.015 \times 3 \times 10^{-8}$$

$$x^2 = \sqrt{4.5\times10^{-10}} = \sqrt{4.5}\times10^{-5}M = [H^+] \Rightarrow [H^+] = 2.12\times10^{-5}M$$

$$\%HClO = \frac{[H^+]}{[HClO]}\times100 = \frac{2.12\times10^{-5}}{0.015}\times100 = 0.141\%$$ **Answer: B**

17. Which one of the following combinations **CANNOT** produce a buffer solution?

A. $HClO_4$ and $NaClO_4$ C. HNO_2 and $NaNO_2$
B. HCN and NaCN D. NH_3 and $(NH_4)_2SO_4$

Explanation: Buffer solution is a combination of weak acid and its conjugate base or weak base and its conjugate acid.

Answer: A

18. How many moles of NH_4Cl must be added to 1.5L of 0.2M solution of NH_3 to form a buffer whose pH is 9.00 ($Kb = 1.8 \times 10^{-5}$)?

A. 0.36 B. 0.65 C. 0.54 D. 0.8

Solution: $pH = 14 - pOH \Rightarrow pOH = 14 - 9 = 5$

$$pOH = pKb + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$pOH = -10gKb + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$5 = -10g\ 1.8\times10^{-5} + 10g\frac{[NH_4Cl]}{[NH_3]}$$

$$5 - 4.74 = 10g\frac{[NH_4Cl]}{0.2}$$

$$0.26 = 10g\frac{[NH_4Cl]}{0.2}$$

$$[NH_4Cl] = 10^{0.26} \times 0.2M = 0.36M$$

$$n_{NH_4Cl} = M \times V = 0.36M \times 1.5L = 0.54mol$$ **Answer: C**

19. What mass of magnesium is plated out upon electrolysis from molten $MgCl_2$ using a current of 60 A for a period of 4000 seconds?

A. 30g B. 24g C. 60g D. 72g

Solution: $Mg^2 + 2e^- \longrightarrow Mg$

$$M = \frac{MIt}{ZF} = \frac{24g/mol \times 60A \times 4000s}{2\times96485A.s/mol.} = 29.85g$$ **Answer: A**

20. For a voltaic (or galvanic) cell using Ag, Ag+(1.0M) and Zn, Zn^{2+} (1.0M) half-cells, which of the following staatements is **INCORRECT?**

A. The zinc electrode is the anode.
B. Electrons will flow through the external circuit from the zinc electrode to the silver electrode.
C. The mass of the zinc electrode will decrease as the cell operates.
D. Reduction occurs at the zinc electrode as the cell operates.

Explanation:

- More reactive metal is anode (Zn)
- Less reactive metal becomes cathode (Ag)
- Electrons flow from anode to cathode in the external circuit.
- reduction occurs at cathode and oxidations occurs at anode.
- mass of anode decreases and that of cathode increases.

Answer: D

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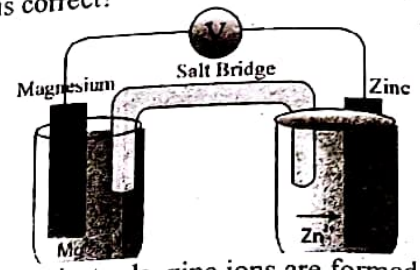
21. For the galvanic cell shown below, which one of the following statements is correct?



A. At the zinc electrode, zinc ions are formed.
B. The electrode potential is measured by the voltmeter.
C. The following reaction takes place at the magnesium electrode: $Mg^2 + 2e^- \longrightarrow Mg$
D. Electrons flow from the zinc electrode to the magnesium electrode.

Explanation: Mg – anode, Zn – cathode

Anode reaction: $Mg \longrightarrow Mg^{2+} + 2e^-$

Cathode reaction: $Zn^{2+} + 2e^- \longrightarrow Zn$

Answer: B

22. Which is the most common ore used for the extraction of copper?
A. CuO B. $CuSO_4$ C. $CuCO_3$ D. $CuFeS_2$

Solution: The principal ore of copper is chalcopyrite, $CuFeS_2$

Answer: D

23. Which of the following material has maximum ductility?
A. Nickel B. Aluminum C. Mild steel D. Copper

Answer: D

24. Which of the following statements about polyvinyl chloride is **NOT** correct?
A. PVC can be used in making water pipes.
B. PVC is stiff.
C. PVC is softened on heating.
D. The monomer of PVC is CHCl = CHCl.

Explanation: The monor of PVC is vinyl chloride, $CHCl = CH_2$

Answer: D

25. Which of the following is a natural polymer?
A. Kertain C. Cellulose
B. Polythene D. Polymethyl methacrylate

Answer: A, C

26. Which of the following does **NOT** affect the solubility of a gas dissolved in a liquid?
A. Nature of solute and solvent
B. pressure
C. temperature
D. rate at which the gas dissolves

Answer: D

27. When a small amount of crystal solute is added to the supersaturated solution, the solute crystals will:
A. grow bigger. C. remain unchanged.
B. slightly dissolve. D. dissolve completely.

Explanation: Super saturated solution develops a crystal easily.

Answer: A

28. What is the molality of a solution that contains 51.2g of naphthalene, $C_{10}H_8$, in 500mL of carbon tetrachloride? The density of CCl_4 is 1.60g/mL.
A. 0.750m B. 0.500m C. 0.840m D. 1.69m

Explanation:

$m_{solution} = d \times v = 1.60g/mol \times 500mL = 800g$

$m_{solvent} = m_{solution} - m_{solvent} = 800 - 51.2 = 748.8g = 0.7488kg$

$$n = \frac{m}{M} = \frac{51.2g}{128g/mol} = 0.4mol$$

$$m = \frac{n}{Kg\ solvent} = \frac{0.4mol}{0.7488kg} = 0.534m$$

Answer: B

29. What is the ionization constant for a weak acid, HA, that is 1.60% ionized in 0.0950M solution?
A. 2.69×10^{-3} C. 2.47×10^{-5}
B. 3.77×10^{-2} D. 9.91×10^{-6}

Explanation: $\%HA = \frac{[H^+]}{[HA]_o} \times 100 \Rightarrow [H^+] = \frac{1.60 \times 0.095M}{100}$

$[H^+] = 1.52 \times 10^{-3}M$

At equilibrium, $[H^+] = [A^-] = 1.52 \times 10^{-3}M$

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$$Ka = \frac{[H^+][A^-]}{[HA]-[H^+]} = \frac{(1.5\times10^{-3})(1.52\times10^{-3})}{0.095-0.00152}$$

$$Ka = \frac{2.3104\times10^{-6}}{0.09348} = 2.47\times10^{-5}$$

Answer: C

30. Which of the following is correct?
A. $1Pa = 10Nm^{-2}$
B. $1N = 10kg\ m\ s^{-2}$
C. $0.00072 = 7.2 \times 10^{-3}$
D. $1L = 1dm^3$

Answer: D

31. Which of the following represents a tentative explanation of a certain scientific law?
A. Hypothesis
B. Observation
C. Experimentation
D. Theory

Answer: D

32. In order to advance to the level of a theory, a hypothesis should
A. be obviously accepted by most people.
B. be repeatedly confirmed by experimentation.
C. be a fully functional experiment
D. report the past experience.

Answer: B

33. Which of the following types of rays combine to form atoms of helium?
A. gamma (γ) rays
B. beta (β) rays
C. alpha (α) rays
D. x-rays

Explanation: Alpha particle is known to be a doublyionized helium atom.

Answer: C

34. What is the relationship between frequency (ν), wavelength (λ) and the speed of light (c)?

A. $\frac{\lambda}{\nu} = c$ B. $\nu c = h\lambda$ C. $\frac{hc}{\lambda} = \nu$ D. $c = \nu\lambda$

Solution: $c = \nu\lambda$

Answer: D

35. What is the magnitude of quantum energy and the frequency for an object whose wavelength is 0.6×10^{-6}m?
A. 3.31×10^{-19}J, $5 \times 10^{14}s^{-1}$
B. 3.98×10^{-40}J, $2 \times 10^{15}s^{-1}$
C. 1.99×10^{-25}J, $3.98 \times 10^{-40}s^{-1}$
D. 9.94×10^{-12}J, $1.99 \times 10^{-25}s^{-1}$

Solution: $E = \frac{hc}{\lambda}$ because $\nu = \frac{c}{\lambda}$

$$\nu = \frac{3\times10^8 m/s}{6\times10^{-7}m} = 5.0\times10^{14}s^{-1}$$

$E = 6.626\times10^{-34}J.s \times 5.0\times10^{14}s^{-1} = 3.313\times10^{-19}J$ **Answer: A**

36. What new concept did Bohr adapt and use to formulate his modelof the atom?
A. Electromagnetic theory developed by Maxwell.
B. The quantum concept developed by Planck.
C. Photoelectric theory developed by Thompson.
D. Neutron theory developed by Chadwick.

Explanation: Bohr made the revolutionary assumption that a certain properties of the electron in a hydrogen atom based on the work of plank and Einstein.

Answer: B

37. What values of l are permitted for an electron with n = 4?
A. 1, 2, 3
B. 1, 2, 3, 4
C. 0, 1, 2, 3, 4
D. 0, 1, 2, 3

Solution: The values of l are from 0 to n – 1
⇒ When n = 4, ℓ = 0, 1, 2, 3

Answer: D

38. Which of the following electrons, identified only by their n and ℓ quantum numbers have the highest energy?
n = 3, ℓ = 0
n = 4, ℓ = 1
n = 3, ℓ = 2
n = 4, ℓ = 2
A. n = 3, l = 2
B. n = 4, l = 1
C. n = 4, l = 2
D. n = 3; l = 0

Explanation: n = 3, ℓ = 0 ⇒ 3s
n = 4, ℓ = 1 ⇒ 4p
n = 3, ℓ = 2 ⇒ 3d
n = 4, ℓ = 2 ⇒ 4d
The energy of these orbitals in decreasing order is:
4d > 4p > 3d > 3s.

Answer: C

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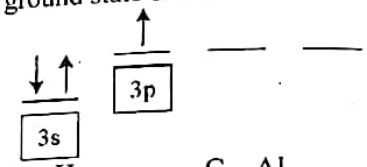
39. What is the maximum number of unpaired electrons in a d subshell?
A. 2 B. 5 C. 3 D. 4
Explanation: d – sub shell has five orbitals. Thus, it has a maximum of five unpaired electrons.

1	1	1	1	1

Answer: B

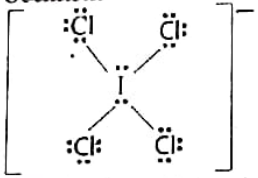
40. The following energy level diagram represents the outermost shell of what ground state element?



A. B B. He C. Al D. Be
Solution: $_5B : 1s^2 2s^2 2p^1$, $_4Be : 1s^2 2s^2$
$_2He : 1s^2$, $_{13}Al : 1s^2 2s^2 2p^6 3s^2 3p^1$

Answer: C

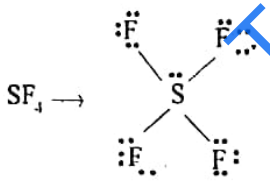
41. How many bonding pairs and lone pairs,, repsectively does the ion ICl_4^- have?
A. 3, 2 B. 4, 2 C. 5, 1 D. 4, 1
Solution:



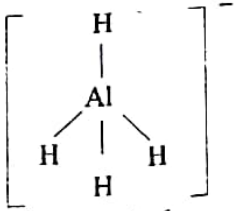
bonding pairs = 4
lone pairs on the central atom = 2
total lone pairs = 14

Answer: B

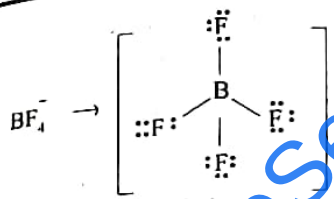
42. Which of the following molecules does **NOT** have a tetrahedral central atom?
A. SF_4 B. AlH_4^- C. BF_4^- D. $SiCl_4$
Solution:



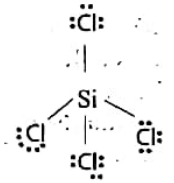
see-saw



tetrahedral



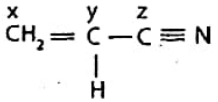
Tetrahedral



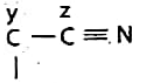
Tetraphedral

Answer: A

43. Acrylonitirile has the follwoig Lewis structure with designation of x, y and z for each carbon atom:



What will be the value of the bond angle and geometry of

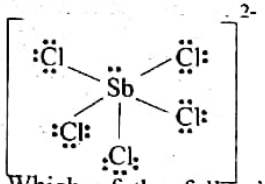


?

A. 109°, tetrahedral C. 180°, linear
B. 120°, trigonal pyramidal D. 90°, T-shaped

Answer: C

44. Antimony (Sb) is a group V element. What will be the molecular gometry and number of lone pair electrons, repsectively that exist in the ion $[SbCl_5]^{2-}$?
A. Seesaw, 1 C. Seesaw, 2
B. Square planar, 2 D. Linear, 3
Explanation:



Molecular geometry = squar pyramid
Lone pair = 1 on the central atom.

Answer:

45. Which of the following molecules does **NOT** have a trigonal bipyramidal electron-pair geometry?
A. SF_4 B. ClF_3 C. XeF_2 D. BrF_5
Explanation:
⇒ SF_4, ClF_3 and XeF_2 have five sets of electrons on their central atom but BrF_5 has six sets of electrons on the central atom. BrF_5 has octahedral electron arrangement. **Answer: D**

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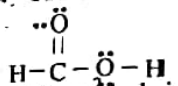
46. How many atomic orbitals are required for an sp^3 hybridization?
A. 2 B. 6 C. 4 D. 8

Solution: One s and three p-atomic orbitals are required for an sp^3 hybridization.

Answer: C

47. Formic acid, which is released by ants, has a molecular formula of HCOOH. What are the possible hybridizations that exist in the molecule?
A. sp^2 and sp^3 C. sp, sp^2 and sp^3
B. sp and sp^3 D. sp and sp^2

Solution: HCOOH has two central atoms, C and O

$\Rightarrow$ 

- C is sp^2 hybridized and 0 – is sp^3 hybridized.

Answer: A

48. What would happen to the O_2 molecule upon ionization to O_2^+?
A. The bond length will increase and the bond energy will increase.
B. The bond length will increase and the bond energy will decrease.
C. The bond length will decrease and the bond energy will increase.
D. The bond length will decrease and the bond energy will decrease.

Explanation:

The bond order of O_2 is 2 but O_2^+ is 2.5. This implies O_2 is less stable than O_2^+. When O_2 is change to O_2^+ bond energy increases but bond length decreases.

Answer: C

49. For a first-order reaction, a plot of _____ versus _____ is linear
A. $\frac{1}{[A]_t}, t$ C. $[A]_t, t$
B. $\text{Ln}\frac{1}{[A]_t}, t$ D. $\text{Ln}[A]_t, t$

Answer: D

50. The rate law of the overall reaction
$A + B \longrightarrow C$
is : rate = $k[A]^2$
Which of the following will NOT increase the rate of the reaction?
A. Increasing the concentration of reactant A.
B. Increasing the concentration of reactant B.
C. Increasing the temperature of the reaction.
D. Adding a catalyst for the reaction.

Explanation:

$A + B \longrightarrow C$, rate = $k[A]^2$ means the reaction is zero order with respect to reactant B. The rate of zero-order reaction is independent on concentration.

Answer: B

51. Which of the following statement (s) is (are) applicable to a balanced chemical equation of an elementary reaction?
i. Order is same as molecularity.
ii. Order is less than the molecularity.
iii. Order is greater than the molecularity.
iv. Molecularity can never be zero.
A. i B. i, ii C. i, iv D. i, iii

Explanation:

- For an elementary reaction order is the same as coefficient of a octant.
- Molecularity cannot to zero.

Answer: C

52. At high pressure, the following reaction is zero order.
$2NH_3(g) \xrightarrow{1130k.pt} N_2(g) + 3H_2(g)$
Which of the following options is (are) correct for this reaction?
i. Rate of reaction = rate constant.
ii. Rate of reaction depends on the concentration of ammonia.
iii. Rate of decomposition of ammonia remains constant until ammonia decomposes completely.
iv. further increase in pressure will change the rate of reaction.
A. i B. i, iii, iv C. i, ii D. i, ii, iv

Answer: B

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53. Which of the following expressions is correct for the rate of reaction given below?

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br(aq) + 3H_2O(l)$$

A. $\frac{\Delta[Br]}{\Delta t} = 5\frac{\Delta[H]}{\Delta t}$ C. $\frac{\Delta[Br]}{\Delta t} = 6/5\frac{\Delta[H]}{\Delta t}$

B. $\frac{\Delta[Br]}{\Delta t} = 5/6\frac{\Delta[H]}{\Delta t}$ D. $\frac{\Delta[Br]}{\Delta t} = 6\frac{\Delta[H]}{\Delta t}$

Explanation:

$$-\frac{\Delta[Br^-]}{5\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{\Delta[H^+]}{6\Delta t} = \frac{\Delta[Br_2]}{3\Delta t} = \frac{\Delta[H_2O]}{3\Delta t}$$

$$\Rightarrow \frac{\Delta[Br^-]}{\Delta t} = -\frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$$

Answer: B

54. Rate law for the reaction $A + 2B \longrightarrow C$ is found to be

$$Rate = k[A][B]$$

If the concentration of reactant 'B' is doubled, keeping the concentration of 'A' constant, what will be the value of the rate constant?

A. the same C. quadrupled

B. halved D. doubled

Explanation:

Rate constant is affected by temperature only. It is not affected by concentration.

Answer: A

55. The following equilibrium constants were determined at 300°C.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), \quad K_c = 4.0\times10^{18}$$

$$N_2O + O_2(g) \rightleftharpoons 2NO(g), \quad K_c = 4.0\times10^{-31}$$

What will be the equilibrium constant at 300°C for the gaseous reaction of

$$2N_2O(g) + O_2(g) \rightleftharpoons 4NO(g)?$$

A. 3.2×10^{-12} C. 5.0×10^{50}

B. 2×10^{-13} D. 1.6×10^{-49}

Solution: Multiply the second by 2 and add the two reactions to get the required.

$$\Rightarrow 2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g) \qquad K_1 = 4.0\times10^{18}$$

$$2N_2(g) + 2O_2(g) \rightleftharpoons 4NO(g) \qquad K_2 = \left(4.0\times10^{-31}\right)^2$$

$$2N_2O(g) + O_2(g) \rightleftharpoons 4NO(g) \qquad K_C = K_1 \times K_2$$

$$K_C = 4.0\times10^{18}\left(4.0\times10^{-31}\right)^2$$

$$K_C = 6.4\times10^{-43}$$

56. When 0.50 mol of N_2O_4 is placed 4,0 liter reaction vessel and heated to 400K, 80% of the N_2O_4 decomposes to NO_2 gas as follows: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

What will be the value of K_p, in units of pressure, at 400K for this reaction?

A. 2.62 B. 13.12 C. 50.48 D. 16.20

Solution: $0.5mol \times \frac{80}{100} = 0.4mol$ of N_2O_4 decomposed

nN_2O_4 left $= 0.5mol - 0.4mol = 0.1mol$

$nNO_2 = 2\times0.4mol = 0.8mol$

$$[N_2O_4] = \frac{0.1mol}{4l} = 0.025M$$

$$[NO_2] = \frac{0.8mol}{4l} = 0.2M$$

$$K_C = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.2)^2}{0.025} = 1.6mol/L$$

$$K_p = K_C(R_T)^{\Delta n}, \Delta n = 2-1 = 1$$

$$= 1.6\frac{mol}{L}\times\left(0.82\frac{atm.L}{mol.K}\times400k\right)^1 = 52.48atm$$

Answer: C

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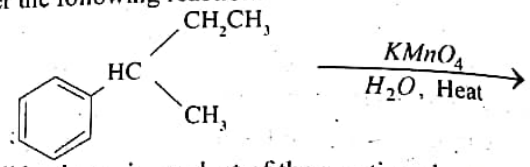
57. Which of the following is **NOT** true about carbonyl compounds?
A. Carbonyl compounds contain 3σ-bond and 1π-bond.
B. The carbon oxygen bond is both longer and weaker.
C. The bond angle in carbonyl is about 120°.
D. Carbonyl compounds may be hydrolyzed.

Explanation: Aldehyde and ketone are carbonyl compounds.

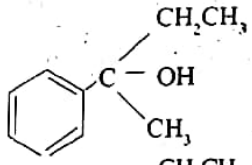
The formula of carbonyl carbon is $-\overset{\overset{O}{\|}}{C}-$. The carbonyl carbon is joined with 3σ and 1π. The carbon oxygen bond is short and strong.

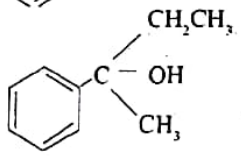
Answer: B

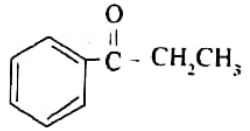
58. Consider the following reaction:

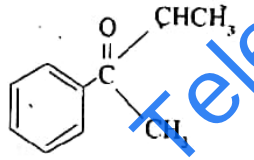


What will be the main product of the reaction above?

A.


B.


C.


D.


Answer: C

59. Which of the following statements concerning the carbonyl group in aldehydes and ketones is **NOT** true?
A. The bond is polar, with a slight negative charge on the oxygen atom.
B. The bond angles about the central carbon atom are 120°.
C. In condensed form, the carbonyl group can be written as CHO.
D. The bond is polar. Therefore, carbonyl groups readily form hydrogen bonds with each other.

Answer: D

60. Which of the following statements concerning fats and oils is **INCORRECT**?
A. They are also called triacylglycerols.
B. They are also called triglycerides.
C. They are fatty acid salts.
D. They are glycerol trimesters.

Explanation:
- Fats and oils are triglycerides, triacylglycerols or trimesters of glycerol.

Answer: C

61. Which of the following types of compounds are expected products from the saponifcation of a fat?
A. Glycerol and fatty acid salts.
B. Glycerol and fatty acids.
C. Fatty acid salts and fatty acids.
D. Glycerol, fatty acid salts and fatty acids.

Explanation: Saponfification of a fat results soap (fatty acid salt) and glycerol.

Answer: A

62. Which one of the following statements best describes the standard enthalpy of formation of any element?
A. The value of $\Delta H°_f$ (element) depends on temperature.
B. The value of $\Delta H°_f$ (element) is zero for any element in the standard state.
C. The value of $\Delta H°_f$ (element) is zero only for elements in the solid state.
D. The value of $\Delta H°_f$ (element) is zero only at absolute zero temperature.

Answer: B

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63. Equal masses of He and Ne are placed in a sealed container. What is the partial pressure of Ne, if the total pressure is 6 atm?
A. 2 B. 3 C. 1 D. 5

Solution: Let the mass of He and Ne be m.

$$n_{He} = \frac{m}{4}$$

$$n_{Ne} = \frac{m}{20}$$

$$x_{He} = \frac{n_{He}}{n_{He} + n_{Ne}} = \frac{\frac{m}{4}}{\frac{m}{4} + \frac{m}{20}} = \frac{\frac{m}{4}}{\frac{6m}{20}} = \frac{20}{6m} \times \frac{m}{4} = \frac{5}{6}$$

$$x_{Ne} = 1 - \frac{5}{6} = \frac{1}{6}$$

$$p_{Ne} = x_{Ne} \cdot p_T = \frac{1}{6} \times 6\text{atm} = 1\text{atm}$$

Answer: C

64. What is the molarity of a solution made by dissolving 10g of glucose ($C_6H_{12}O_6$) in sufficient water to form 200mL solution?
A. 0.18 B. 0.251 C. 0.362 D. 0.278

Solution: $n = \frac{m}{M} = \frac{10g}{180g/mol} = \frac{1}{18}\text{mol}$

$$M = \frac{n}{v} = \frac{\frac{1}{18}\text{mol}}{0.20L} = 0.278M$$

Answer: D

65. In which direction will the following equilibrium shift, if a solution of CH_3CO_2Na is added?

$$CH_3COOH(aq) \rightleftharpoons CH_3CO_2^-(aq) + H^+(aq)$$

A. The equilibrium shifts to the right (more products)
B. No change
C. The equilibrium shifts to the left (more reactant)
D. Cannot be predicted

Explanation: CH_3COONa is a source of $CH_3CO_2^-$. As a result, concentration of $CH_3CO_2^-$ increase and the equilibrium will shift to the left.

Answer: C

66. A 1.0M solution of a weak acid is found to dissociate by only 1.3%. Which of the following acid is it likely to be?
A. HF $K_a = 7.2 \times 10^{-4}$
B. HFO_2 $K_a = 6.9 \times 10^{-4}$
C. CH_3COOH $K_a = 1.8 \times 10^{-5}$
D. HCOOH $K_a = 1.9 \times 10^{-4}$

Solution: Let the weak acid he HA

$$\%HA = \frac{[H^+]}{[HA]_o} \times 100\%$$

$$[H^+] = \frac{\%HF \cdot [HA]_o}{100} = \frac{1.37 \times 1M}{100} = 0.0137M$$

$$K_a = \frac{[H^+][A^-]}{[HA]_o - [H^+]}, \quad [H^+] = [A^-]$$

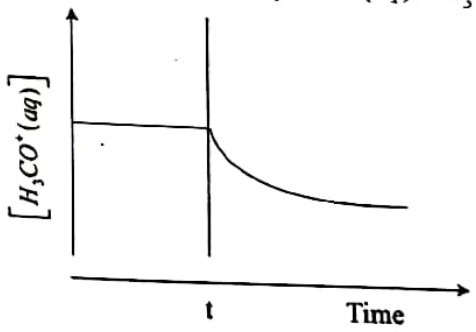
$$K_a = \frac{(0.0137)^2}{1 - 0.0137} = 1.9 \times 10^{-4}$$

Thus, it is the Ka value of HCOOH.

Answer: D

67. Consider the following graph, which relates to the equilibrium system:

$$CH_3COOH(aq) + H_2O \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq) \Delta H < 0$$



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Which of the following actions caused the change in the concentration of $[H_3O^+(aq)]$ at time t?

A. Addition of $CH_3COO^-(aq)$
B. Addition of HCl
C. Decreasing of temperature
D. Increasing the volume of the container

Answer: A

68. Sodium acetate spontaneously crystallizes out of a supersaturated solution o n standing. Which of the following is true for the thermodynamic quantities of this system for such a process?

A. $\Delta S < 0, \Delta H < 0$
B. $\Delta G < 0, \Delta H > 0$
C. $\Delta S > 0, \Delta G < 0$
D. $\Delta S > 0, \Delta H > 0$

Explanation: When sodium acetate crystallizes out of a supersaturated solution, entropy decreases.

Answer: A

69. In which of the following systems will the position of equilibrium shift to the left upon an increase in pressure, but to the right upon an increase in temperature?

A. $CO_2(g)+H_2(g)\rightleftharpoons CO(g)+H_2O(g) \quad \Delta H>0$
B. $C_2H_2(g)+H_2O(g)\rightleftharpoons C_2H_5OH(g) \quad \Delta H<0$
C. $C_2H_6(g)\rightleftharpoons C_2H_4(g)+H_2(g) \quad \Delta H>0$
D. $2SO_2(g)+O_2(g)\rightleftharpoons 2SO_3(g) \quad \Delta H<0$

Answer: C

70. The hydrogen used in the Haber process is made by the following reaction:

$$CH_4(g)+H_2O(g)\rightleftharpoons CO(g)+3H_2(g) \quad \Delta H^\circ=+206kJ$$

Which of the following sets of conditions will favor the formation of H_2?

A. Low pressure and high temperature.
B. Low pressure and low temperature.
C. High pressure and low temperature.
D. High pressure and high temperature.

Explanation:
The equilibrium shifts to the right by lowering pressure and increasing temperature.

Answer: A

71. What is the molar solubility of $Fe(OH)_3$ in a solution that is buffered at pH = 3.50 at 25°C? ($K_{sp}(Fe(OH)_3 = 4\times10^{-38}$).

A. 1×10^{-5} B. 1.1×10^{-6} C. 2.0×10^{-6} D. 1.26×10^{-6}

Solution:

$POH = 14 - pH = 14 - 3.5 = 10.5, [OH^-] = 10^{-POH} = 10^{-10.5}$

$\Rightarrow [OH^-] = 3.16\times10^{-11}M$

$Fe(OH)_3 \longrightarrow Fe^{3+} + 3OH^-$

$K_{sp} = [Fe^{3+}][OH^-]^3 \Rightarrow [Fe^{3+}] = \dfrac{K_{SP}}{[OH^-]^3}$

$$[Fe^{3+}] = \frac{4\times10^{-38}}{(3.16\times10^{-11})^3} = 1.26\times10^{-6}\ mol/L$$

Answer: D

72. What is the purpose of a salt bridge in an electrochemical cell?

A. To provide a source of ions to react at the anode and cathode.
B. To maintain electrical neutrality in the half-cell through migration of ions.
C. To provide means of electrons to travel from the cathode to the anode.
D. to provide means of electrons to travel from the anode to the cathode.

Explanation:

- Salt bridge maintains electrical neutrality between the two solutions.
- It prevent mixing of the true half-cells solution.

Answer: B

73. What is galvanized iron?

A. Iron that is coated with tin.
B. Iron that is coated with zinc.
C. Iron that is coated with chromium.
D. Iron that is coated with aluminum.

Explanation:
Iron that is coated with zinc is called galvanized iron.

Answer: B

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74. The standard cell potential (E°) for the reaction below is 1.10V. What is the cell potential for this reaction hen $[Cu^{2+}] = 1\times10^{-5}M$ and $[Zn^{2+}] = 1M$?

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

A. 1.10 B. 0.95 C. 1.20 D. 1.35

Solution:

$$E_{cell} = E°_{cell} - \frac{0.0592V}{n}\log Q$$

$$E_{cell} = 1.10v - \frac{0.0592V}{2}\log\frac{[zn^{2+}]}{[Cu^{2+}]}$$

$$E_{cell} = 1.10v - 0.0296\log\frac{1M}{10^{-5}M}$$

$$E_{cell} = 0.952V$$

Answer: B

75. Which of the following statements is true?

A. The more positive the value of $E_{red}°$, the greater the driving force for reduction.

B. The more exothermic the value of $E_{red}°$, the greater the driving force for reduction.

C. The more endothermic the value of $E_{red}°$, the greater the driving force for reduction.

D. The more negative the value of $E_{red}°$, the greater the driving force for reduction.

Explanation: The more positive the standard reduction potential, the easier to be reduced.

Answer: A

76. Which of the following types of elements are good oxidizing agents?

A. Alkali metals C. Lanthanides

B. Halogens D. Transition elements

Solution: Halogens (group (VIIA) non-metals are powerful oxidizing agents. **Answer: B**

77. Which of the following statements concerning petroleum is **INCORRECT?**

A. It is a renewable energy source.

B. It is a fossil fuel.

C. It is a mixture consisting mainly of hydrocarbons.

D. It was formed from marine organisms, which died millions of years ago. **Answer: A**

78. A 1.0×10^{-4}M solution has pH of 10.00. The solute is a

A. weak acid C. strong base

B. weak base D. strong acid

Solution:

$pH = 14 + \log[OH^-]$

$pH = 14 + \log 1.0\times10^{-4}$

$pH = 14 - 4 = 10$. The solute must be strong base because $[OH^-]$ = [base] for strong base. **Answer: C**

79. The four most abundant metals in the earth's crust in decreasing order of abundance are:

A. Oxygen, Silicon, Aluminum, and Iron.

B. Aluminum, Iron, Calcium, and Magnesium.

C. Iron, Aluminum, Silicon, and Oxygen.

D. Silicon, Aluminum, Magnesium, and Sodium.

Explanation: The four abundant metals in decreasing order of abundance are aluminum, iron, calcium and magnesium. However, from elements oxygen is the first. **Answer: B**

80. A polysaccharide is a polymer made up of which kind of monomers?

A. Amino acids

B. Nucleotides

C. Simple sugars

D. Alternating sugar and phosphate groups

Explanation: Monosaccharides (simple sugars) such as glucose, galactose and fructose are monomers of polysaccharides.

Answer: C

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1. Which one of the following measurements has the greatest number of significant figures?
A. 0.00046 B. 0.00460 C. 460 D. 4.6×10^6

Solution:
- 0.0046 and 4.6×10^6 have two significant figures.
- 0.00460 has three significant figures.
- 460 is very difficult to determine its significant figure because no one is sure about the last zero.

Answer: B

2. How many significant figures has the sum of 2.453, 4.6 and 8.14?
A. 2 B. 3 C. 4 D. 5

Solution: 2.453 + 4.6 + 8.14 = 15.193 but the answer is 15.2 because during addition and subtraction of significant figures, the answer should be reported with the fewer decimal place.

Answer: B

3. Which of the following is the correct relationship between picometer (pm) and nanometer (nm)?
A. 1 pm = 10 nm C. 1 nm = 10 pm
B. 1 pm = 100 nm D. 1 nm = 10000 pm

Solution:

$$\frac{1pm}{1nm}=\frac{10^{-12}m}{10^{-9}m}$$

$$\frac{1pm}{1nm}=10^{-3}$$

$1pm = 10^{-3}nm \Rightarrow 10^{3}pm$

Answer: D

4. Which of the following are the correct steps of the Scientific method?
A. Observation – Hypothesis – Experimentation – Formulation of Laws – Theories
B. Observation – Formulation of Laws – Experimentation – Hypothesis – Theories
C. Observation – Experimentation – Hypothesis – Formulation of Laws – Theories
D. Observation – Experimentation – Formulation of Laws – Hypothesis – Theories

Solution:
- The correct steps of the scientific method are

  - Making observation
  - Hypothesis
  - Theories
  - Scientific laws

5. Which of the following is **NOT** a laboratory safety rule?
A. You should tie back loose hair.
B. You should never add acid to water.
C. Do not suck solution in the pipette by mouth.
D. When lighting a Bunsen Burner, you should light the match stick before turning on the gas.

Solution:
To dilute concentrated acid, we have to add acid to water but not water to acid.

Answer: B

6. Which of the following particle was discovered by the cathode rays experiment?
A. Electron C. Nucleus
B. Neutron D. Proton

Answer: A

7. What is the frequency of a radiation (in Hz) from an atom if its wavelength is 1.0×10^{-9}nm?
A. 3.3×10^{-27} C. 3.0×10^{8}
B. 3.0×10^{17} D. 3.0×10^{26}

Solution:

$$u=\frac{c}{\lambda}=\frac{3\times10^{8}m/s}{1\times10^{-18}m}=3\times10^{26}Hz$$

Answer: D

8. What is the frequency associated with one photon of a microwave radiation having a frequency of $2.45 \times 10^{9}s^{-1}$?
A. $1.62 \times 10^{24}J$ C. $2.45 \times 10^{-9}J$
B. $2.45 \times 10^{9}J$ D. $1.62 \times 10^{-24}J$

Solution:
$E = h\nu = 6.626 \times 10^{-34}\ J.s \times 2.45 \times 10^{9}s^{-1} = 1.62 \times 10^{-24}\ J$

Answer: D

9. Which of the following transitions will have minimum wavelength?
A. $n_4 \rightarrow n_1$ C. $n_4 \rightarrow n_2$
B. $n_2 \rightarrow n_1$ D. $n_3 \rightarrow n_1$

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Solution:
The smaller the transition gap, the higher is the energy. As energy increases, wave length decreases.
$n_2 \rightarrow n_1$ is the smallest gap and nearest to the nucleus. This transition corresponds to minimum wavelength.

Answer: B

10. The Heisenberg uncertainty principle states that
    A. Two atoms of the same element must have the same number of protons.
    B. No two electrons in the same atom can have the same set of four quantum numbers.
    C. It is impossible to determine both position and momentum of an electron simultaneously.
    D. Electrons of atoms in ground state enter energetically equivalent set of orbitals singly before they pair up in any orbitals of the set.

Answer: C

11. How many different types of orbitals are there in the third energy level?
    A. 2 B. 3 C. 6 D. 7
    Solution:
    Number of orbitals = n^2
    Types of orbitals = n = 3 (s, p and f)

Answer: B

12. The normal human eye responds to visible light of wavelength ranging from about 390 to 710 nm. What is the frequency range of the human eye?
    A. $7.7 \times 10^{14} s^{-1}$ to $4.2 \times 10^{14} s^{-1}$
    B. $4.2 \times 10^{14} s^{-1}$ to $7.7 \times 10^{14} s^{-1}$
    C. $1.3 \times 10^{-15} s^{-1}$ to $2.4 \times 10^{-15} s^{-1}$
    D. $2.4 \times 10^{-15} s^{-1}$ to $1.3 \times 10^{-15} s^{-1}$
    Solution:

$$v_{390nm} = \frac{c}{\lambda} = \frac{3\times10^8 m/s}{3.90\times10^{-7} m} = 7.7\times10^{14} Hz$$

$$v_{710nm} = \frac{c}{\lambda} = \frac{3\times10^8 m/s}{7.10\times10^{-7} m} = 4.2\times10^{14} Hz$$

Answer: A

13. Which of the following is the correct order for electrons filling in orbitals?
    A. 3s, 3p, 4s, 3d
    B. 2p, 3s, 3p, 3d
    C. 3p, 4s, 3d, 5s
    D. 4s, 3d, 4p, 4d

Answer: A

14. What is the ground state electronic configuration of Cu^{2+}?
    A. $[Ar]4s^1 3d^8$
    B. $[Ar]4s^2 3d^7$
    C. $[Ar]4s^1 3d^{10}$
    D. $[Ar]3d^9$
    Solution:
    Electron = 29 – 2 = 27, the two electrons are lost from 4s and 3d.
    ⇒ Electron configuration: $[Ar]3d^9$

Answer: D

15. Which of the following is the most reasonable valence – bond structure of the nitrocyl chloride (NOCl)?
    A. :Ö–N̈–C̈l:
    B. Ö=N–C̈l:
    C. Ö=N–C̈l:
    D. Ö–N=C̈l
    Solution:
    Total number of valence electrons = 5 + 6 + 7 = 18 pairs = 9, central atom = N.
    ⇒ The structure of NOCl

Ö=N–C̈l:

Answer: C

16. What is the chemical formula of the compound formed by aluminum (III) and sulfate ion, SO_4?
    A. $AlSO_4$
    B. Al_2SO_4
    C. $Al_2(SO_4)_3$
    D. $Al(SO_4)_3$
    Solution:

$$Al^{3} \times SO_4^{2} \Rightarrow Al_2(SO_4)_3$$

Answer: C

17. Which of the following is the correct Lewis structure for oxygen atom?
    A. Three lone – pairs of valence electrons.
    B. One lone – pair of valence electrons.
    C. Two lone – pairs of valence electrons and one bonding site.
    D. Two lone – pairs of valence electrons and two bonding sites.

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Solution:

$\ddot{O}=\ddot{O}$

On one oxygen atom, 2 bp and 2Lp are present. **Answer: D**

18. How many electron are shared between two atoms in a double bond?
A. 2 B. 3 C. 4 D. 6

Solution:
In a double bond formation two pairs of electrons or four electrons are shared. **Answer: C**

19. The forces that hold different atoms or ions together are called?
A. Electric forces C. Chemical bonds
B. Physical bonds D. Atomic forces

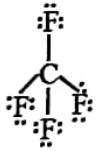
Intra molecular forces (chemical bonds) are forces that hold together atoms or ions. **Answer: C**

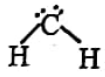
20. Which of the following covalent bonds is the most polar?
A. H – F B. H – C C. H – H D. H – N

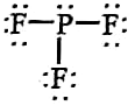
Solution:
The higher the electronegativity difference between the bonded atoms, the more polar is the bond. **Answer: A**

21. Which of the following molecule is **INCORRECTLY** matched with the electronic geometry of the central atoms?
A. CF_4: tetrahedral C. PF_3: pyramidal
B. H_2O: tetrahedral D. $BeBr_2$: linear

Solution:

CF_4:  , tetrahedral, sp^3 hybridized

H_2O:  , bent or angular, sp^3 hybrdized

 , trigonal pyramidal, sp^3hydridized

$BeBr_2$: Br ___ Be ___ Br, linear **Answer: B**

22. What kind of hybrid orbitals are utilized by the carbon atom in CF_4 molecules?
A. Sp B. sp^2 C. sp^3 D. sp^3d

Answer: C

23. What is the bond order of a superoxide anion, with the chemical formula O_2^-?
A. 3.0 B. 2.5 C. 2.0 D. 1.5

Solution:
Electron $= 2 \times 8 + 1 = 17$
$(\sigma 1s)^2(\sigma 1s^*)^2(\sigma 2s)^2(\sigma 2s^*)^2(\sigma 2px)^2(\pi 2py)^2(\pi 2pz)^2(\pi 2py^*)^2(\pi 2pz^*)$

$$B_o = \frac{1}{2}(nb - na) = \frac{1}{2}(10 - 7) = 1.5$$ **Answer: D**

24. What is (are) the F-S-F bond angle (s) SF_6?
A. 109.5° C. 90° and 120°
B. 120° D. 90° and 180°

Solution:
- In SF_6 all bonds are 90° except the one made by the bonds between the central atom and the two atoms diametrically opposite to each other. That angle is 180°.

Answer: D

25. Which of the following refer to a reaction that is **NOT** dependent on the concentration of a reactant?
A. Second order reaction C. Zero order reaction
B. First order reaction D. Third order reaction

Solution:
For zero – order reaction, with rate law, rate = k, the rate is independent on the concentration of reactants. **Answer: C**

26. Consider the following reactions:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

At a certain temperature, the rate of decomposition of $N_2O_5(g)$ is 2.5×10^6 mol/s. What is the rate of formation of NO_2?
A. 1.0×10^{-5} mol/s C. 1.3×10^{-6} mol/s
B. 2.5×10^{-6} mol/s D. 5.0×10^{-6} mol/s

Solution:

$$\text{Rate of } NO_2 = \frac{\text{coefficient of } NO_2}{\text{coefficient of } N_2O_5} \times \text{ rate of } N_2O_5$$

$$= \frac{4}{2} \times 2.5 \times 10^6 \text{ mol/s} = 5.0 \times 10^6 \text{ mol/s}$$

Answer: ___

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27. The rate of a reaction increases with increasing the concentration of reactant molecules, because
A. The average kinetic energy of the molecules increases.
B. The frequency of molecular collisions increases.
C. The order of the reaction increases.
D. The activation energy increases.

Solution:
As concentration of reactants increases, the number of collision between reacting species also increases. This increases the rate of a reaction. **Answer: B**

28. Given the following data for the reaction:

$$NH_4^+(aq) + NO_2^-(aq) \longrightarrow N_2(g) + 2H_2O(l)$$

Experiment	$[NH_4^+]$	$[NO_2^-]$	Rate
1	0.010M	0.020M	0.020 M/s
2	0.015M	0.020M	0.030M/s
3	0.010M	0.010M	0.005M/s

What will be the rate law for the given reaction?

A. Rate = $k[NH_4^+][NO_2^-]$

B. Rate = $k[NH_4^+]^2[NO_2^-]$

C. Rate = $k[NH_4^+]^2[NO_2^-]^2$

D. Rate = $k[NH_4^+][NO_2^-]^2$

Solution:

Let the rate law be: Rate = $k[NH_4^+]^x[NO_2^-]^y$

- Find the values of x using experiment 1 and 2.

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \frac{k[NH_4^+]_1^x[NO_2^-]_1^y}{k[NH_4^+]_2^x[NO_2^-]_2^y}$$

$$\frac{0.020\,m/s}{0.030\,m/s} = \frac{(0.010)^x(0.020)^y}{(0.015)^x(0.020)^y}$$

$$\frac{2}{3} = \left(\frac{2}{3}\right)^x \Rightarrow x = 1$$

- Use experiments 1 and 3 to find the value of y.

$$\frac{\text{Rate 1}}{\text{Rate 3}} = \frac{[NO_2^-]_1^y}{[NO_2^-]_3^y}$$

$$\frac{0.020}{0.005} = \frac{(0.020)^y}{(0.010)^y} \Rightarrow 4 = 2^y$$

$\therefore$ Rate = $k[NH_4^+][NO_2^-]^2 \Rightarrow y = 2$ **Answer: D**

29. The initial rate of a reaction A + B → products was measured for several different starting concentrations of A and B, with the following results.

Exp	[A]	[B]	Initial Rate of Formation of C M/s
1	0.04	0.04	9.6×10^{-6}
2	0.08	0.04	1.92×10^{-5}
3	0.08	0.02	9.6×10^{-6}

What is the initial rate of reaction when $[A]_o$ = 0.12M and $[B]_o$ = 0.015M?

A. 0.006 Ms^{-1}
B. 0.006$M^{-2}s^{-1}$
C. 1.08×10^{-5} Ms^{-1}
D. 1.08×10^{-4} Ms^{-1}

Solution:

Determine the rate law first. Rate = $[A]^x[B]^y$

- Using experiment 1 and 2, when the concentration of A doubles, the initial rate also doubles. This works for first order reaction only. Thus, the value of x is one.
- Using experiment 2 and 3, when concentration of B halves, the rate also halves. As a result, the order in B is also one.

⇒Rate = k[A][B], then find the value of k

$$k = \frac{\text{Rate}_1}{[A]_1[B]_1} = \frac{9.6\times10^{-6}\,M/s}{(0.04\,M)(0.04)\,M}$$

$k = 6 \times 10^{-3}M^{-1}.S^{-1}$

Rate = k[A][B]

$= 6 \times 10^{-3}M^{-1}.S^{-1} \times 0.12M \times 0.015M = 1.08 \times 10^{-5}MS^{-1}$

Answer: C

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30. A possible mechanism for the reaction $A + 2B \rightarrow AB_2$ is

Step 1: $A + B \rightarrow AB$ slow

Step 2: $AB + B \rightarrow AB_2$ fast

Overall: $A + 2B \rightarrow AB_2$

According to the mechanism, what will be the rate law?

A. Rate = $k[A][B]$
B. Rate = $k[B]^2$
C. Rate = $k[A][B]^2$
D. Rate = $k[A]$

Solution:

- The slowest step is the rate determining step and the order of the reaction is the same as the coefficient of the reactants because it is an elementary step.

$\Rightarrow$ Rate = $k[A][B]$ **Answer: A**

31. Which one of the following is correct when a reaction reaches equilibrium?

A. The reaction rate for the forward direction is decreased to zero.
B. The rate of the forward and reverse reaction becomes equal.
C. The rate of the reverse reaction is reached a maximum.
D. The amounts of reactants and products becomes equal.

Solution:

At equilibrium, rate of forward reaction is equal to rate of backward reaction. **Answer: B**

32. What is the equilibrium expression in (M)?

$2C(s) + O_2(g) \leftrightarrows 2CO(g)$

A. $K = \dfrac{2[CI]}{2[C]+[O_2]}$
C. $K = \dfrac{2[CO]}{[O_2]}$
B. $K = \dfrac{[CO]^2}{[C]^2[O_2]}$
D. $K = \dfrac{[CO]^2}{[O_2]}$

Solution:

Solids and liquids (solvents) are not included in the equilibrium constant expression.

$\Rightarrow K_C = \dfrac{[CO]^2}{[O_2]}$ **Answer: D**

33. For the reaction: $2C(s) + O_2(g) \rightleftharpoons 2CO(g)$ $\Delta H < 0$ if the reaction is at equilibrium with excess C(s) remaining, what change will increase the quantity of CO(g) for the reaction at equilibrium?

1) Adding C(s)
2) Decreasing the temperature
3) Increasing the pressure

A. Only 1
B. Only 2
C. 1 and 2
D. 2 and 3

Solution:

$2C(s) + O_2(g) \rightleftharpoons 2CO(g)$, $\Delta H < 0$, formation of CO(g) is favored

- By lowering temperature and pressure.
- By increasing concentration of O_2.
- By removing CO and increasing volume.

Answer: B

34. For one component system, when would the degree of freedom becomes one?

A. When only one phase is resent.
B. When two phases are present equilibrium.
C. When three phases are present at equilibrium.
D. At all times.

Solution:

$F = C - P + 2$

$P = C - F + 2$, $C = 1$, $F = 1$... given

$P = 1 - 1 + 2$

$P = 2$, thus, for one component system, degree of freedom becomes on when two phases are present at equilibrium.

Answer: B

35. Consider the following reaction at equilibrium.

$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$ $\Delta H° = -514$ kJ

What does Le Chatelier's principle predict when the temperature for the reaction is increased?

A. The value of the equilibrium constant will increase.
B. The value of the equilibrium constant will decrease.
C. The partial pressure of $O_2(g)$ at equilibrium will increase.
D. The partial pressure of $CO_2(g)$ at equilibrium wll increases.

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Solution:

$$2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g),\ \Delta H = -514\ kJ$$

Increasing temperature shifts the equilibrium to the left. That is, formation of $CO_2(g)$ is favored.

Answer: D

36. Consider the reaction

$$AB_3(g) \rightleftharpoons A(g) + 3B(g)$$

What is the equilibrium constant expression if the initial concentration of AB_3 is 0.1 M and the equilibrium concentration of A is expressed by x? (Assume the initial concentrations of A and B are both zero.)

A. $\frac{x-3x}{0.1-x}$ C. $\frac{x-x^3}{(0.1-x)^3}$

B. $\frac{x.3}{(0.1-3x)^3}$ D. $\frac{x.(3x)^3}{0.1-x}$

Solution:

$$AB_3(g) \rightleftharpoons A(g) + 3B(g)$$

Initial conc.	0.1	0	0
Change conc.	-x	+x	+3x
Equilibrium conc.	0.1 – x	x	3x

$$K_C = \frac{[A][B]^3}{[AB_3]}$$

$$K_C = \frac{(x)(3x)^3}{0.1-x}$$

Answer: D

37. What products will be formed when a carboxylic acid reacts with an alcohol?

A. Ketone and Water C. Acid chloride and water

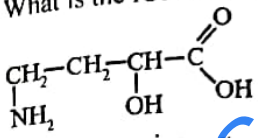
B. Amide and water D. Ester and water

Solution: The reaction between carboxylic acid and alcohol yields ester and water.

$$\underset{\text{Carboxylic acid}}{RCOOH} + \underset{\text{alcohol}}{R'OH} \overset{H^+}{\rightleftharpoons} \underset{\text{Ester}}{RCOOR'} + \underset{\text{Water}}{H_2O}$$

Answer: D

38. What is the IUPAC name for the following molecules?



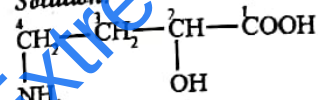
A. γ – amino – α – hydroxybutyric acid

B. 4 – amino – 2 – hydroxybutanoic acid

C. 1 – amino – 3 – hydroxybutanoic acid

D. α – amino – γ – hydroxybutyric acid

Solution



4 – amino – 2 – hydroxyl butanoic acid – IUPAC name

γ – amino – α – hydorxybutyric acid – common name

Answer: B

39. The solubility of compounds containing the carboxylic acid group can be increased by reaction with

A. Sulfuric acid C. Water

B. Sodium hydroxide D. Benzoic acid

Answer: A

40. Which of the following methods is used to convert oils into fats?

A. Esterification reaction using strong bases like NaOH or KOH

B. Saponification reaction using strong bases like NaOH or KOH

C. Hydrolysis reaction in the presence of water and a catalyst.

D. Hydrogenation reaction at high temperature and in the presence of a catalyst.

Solution:

Oils are converted into fats by hydrogenation reaction at high temperature and in the presence of a catalyst.

Answer: D

41. Which of the following general class of organic molecules do triglycerides belong to?

A. Ketones C. Esters

B. Carbonxylic acid D. Ethers

Solution: Triglycerids are fats and oils and they belong to esters.

Answer: C

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42. The chemical makeup of oil is esters of glycerol with three
A. Identical saturated fatty acids.
B. Predominantly saturated fatty acids.
C. Predominantly unsaturated fatty acids.
D. Identical unsaturated fatty acids.

Solution:
- Fats are esters of glycerol and mostly saturated fatty acids.
- Oils are liquid esters primarily derived from unsaturated fatty acids and glycerol.

Answer: C

43. Which of the following types of solution are possible?
1) Solid dissolved in a liquid
2) Gas dissolved in a liquid
3) Gas dissolved in a gas
4) Solid dissolved in a solid

A. 1 and 2 C. 1, 2 and 4
B. 1, 2, 3 and 4 D. 1

There are six main types of solutions based on the states of solute and solvents.
- Gas – gas
- Gas – liquid
- Liquid – liquid
- Liquid – solid
- Solid – liquid
- Solid – solid

Answer: B

44. Which of the following pairs of substances are miscible?
A. Water and gasoline C. Water and salt
B. Water and oxygen D. Water and ethanol

Solution:
Polar dissolves polar and non – polar dissolves non – polar. For solid solute we say soluble but for gas and liquid solutes we use the term miscible, water and ethanol are miscible.

Answer: D

45. Which of the following will increase the rate at which a solid dissolved in a liquid?
A. Supersaturating the solution.
B. Grinding the solid into smaller pieces.
C. Lowering the temperature of the solvent.
D. Placing the solution in opaque container.

Solution:
The rate of dissolution of solid solute can be by increased its surface area. This can be done by grinding or powdering the solid.

Answer: B

46. The solubility of NaCl in water at 20°C is 36g in 100g of water. A solution that contains 18 g of NaCl in 50g of water at 20°C is
A. Saturated C. Supersaturated
B. Unsaturated D. Dilute

Solution:
The solubility of NaCl in water at 20°C is 36g in 100g of water. This is saturated solution. But at the same temperature, 36g/100g is the same as 18g/50g.

Answer: A

47. What will be the normality of a 2.3M sulphuric acid solution?
A. 0.23N B. 0.46N C. 2.3N D. 4.6N

Solution:

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$\therefore Z = 2$ eq/mol

$N = M.Z = 2.3$ mol/L $\times$ 2 eq/mol $= 4.6N$

Answer: D

48. A certain 500g toothpaste sample has 0.2g of fluoride. What is the concentration of the fluoride in ppm?
A. 200 B. 250 C. 400 D. 1,000

Solution:

$$\text{ppm } F^- = \frac{\text{mass of } F^-}{\text{mass of tooth paste}} \times 10^6$$

$$= \frac{0.2g}{500g} \times 10^6 = 400\text{ppm}$$

Answer: C

49. How many grams of methanol would have to be added to water to prepare 150mL of solution which is 2.0M methanol?
A. 2.4 B. 9.6 C. 9.6×10^3 D. 4.3×10^2

Solution:

$$M = \frac{m}{M_{wt}.V}$$

$M = M.M_{wt}.V = 2$ mol/L $\times$ 32 g/mol $\times$ 0.150L

$M = 9.6$g of methanol (CH_3OH)

Answer: B

50. If 0.454 kg of nitroglycerin explodes according to the following reaction:

$$C_2H_5(NO_2)_3(l) \rightarrow CO_2(g) + N_2(g) + H_2O(g) + O_2(g)$$

What is the total volume of gases in dm^3 produced at STP?
A. 44.8 B. 163 C. 325 D. 650

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42. The chemical makeup of oil is esters of glycerol with three
A. Identical saturated fatty acids.
B. Predominantly saturated fatty acids.
C. Predominantly unsaturated fatty acids.
D. Identical unsaturated fatty acids.

Solution:

- Fats are esters of glycerol and mostly saturated fatty acids.
- Oils are liquid esters primarily derived from unsaturated fatty acids and glycerol.

Answer: C

43. Which of the following types of solution are possible?
1) Solid dissolved in a liquid
2) Gas dissolved in a liquid
3) Gas dissolved in a gas
4) Solid dissolved in a solid

A. 1 and 2 C. 1, 2 and 4
B. 1, 2, 3 and 4 D. 1

There are six main types of solutions based on the states of solute and solvents.

- Gas – gas
- Gas – liquid
- Liquid – liquid
- Liquid – solid
- Solid – liquid
- Solid – solid

Answer: B

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B. Water and oxygen D. Water and ethanol

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Answer: D

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Answer: B

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Solution:

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$\Rightarrow Z = 2$ eq/mol

$N = M.Z = 2.3$ mol/L $\times$ 2 eq/mol $= 4.6N$

Answer: D

48. A certain 500g toothpaste sample has 0.2g of fluoride. What is the concentration of the fluoride in ppm?
A. 200 B. 250 C. 400 D. 1,000

Solution:

$$\text{ppm } F^- = \frac{\text{mass of } F^-}{\text{mass of tooth paste}} \times 10^6$$

$$= \frac{0.2\text{g}}{500\text{g}} \times 10^6 = 400\text{ppm}$$

Answer: C

49. How many grams of methanol would have to be added to water to prepare 150mL of solution which is 2.0M methanol?
A. 2.4 B. 9.6 C. 9.6×10^3 D. 4.3×10^2

Solution:

$$M = \frac{m}{M_{wt}.V}$$

$M = M.M_{wt}.V = 2$ mol/L $\times$ 32 g/mol $\times$ 0.150L

$M = 9.6$g of methanol (CH_3OH)

Answer: B

50. If 0.454 kg of nitroglycerin explodes according to the following reaction:

$$C_3H_5(NO_2)_3(l) \rightarrow CO_2(g) + N_2(g) + H_2O(g) + O_2(g)$$

What is the total volume of gases in dm^3 produced at STP?
A. 44.8 B. 163 C. 325 D. 650

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Solution:
By mass – volume relationship find the answer. At STP, the volume of one mole of any gas is 22.4L (22.4dm^3). However the reaction is incorrect. The reaction is $2C_3H_5(NO_2)_3(l) + 5O_2(g) \rightarrow 6CO_2(g) + 3N_2(g) + 5H_2O(g)$

51. When the molecular equation:
$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaCl(aq)$
is written in terms of ionic equation, which one of the following pairs will be the spectator ions?
A. Na^+ and Cl^- C. CO_3^{2-} and Cl^-
B. Ca^{2+} and Cl^- D. Ca^{2+} and CO_3^{2-}

Solution:
$Ca^{2+}(g) + 2Cl^-(aq) + 2Na^+(aq) + CO_3(aq) \rightarrow CaCO_3(s) + 2Na^+(aq) + 2Cl^-(aq)$
Thus, Na^+ and Cl^- are spectator ions. **Answer: A**

52. What will be the ratio of the boiling point elevation of a 0.1M solution of sucrose, sodium sulphate and potassium bromide?
A. 1:3:1 B. 1:1:2 C. 1:3:2 D. 1:2:2

Solution:
- For no n- electrolytes, the value of Vans Hoff factor i is one. But for electrolytes, it is equal to the number of ions produced.
  - ⇒ For sucrose solution, i = 1
  - ⇒ For sodium sulphate (Na_2SO_4) solution, i ≈ 3
    $Na_2SO_4 \rightarrow 2Na^+ + SO_4^+$
- For potassium bromide solution i ≈ 2
  $KBr \rightarrow K^+ + Br^-$
  $\Delta T_b = imk_b$, therefore the ratio becomes 1: 3: 2

Answer: C

53. Which of the following is the conjugate acid of the HS^- ion?
A. H_2S^+ B. S^{2-} C. S^- D. H_2S

Solution:
Conjugate acid = Base + H^+ = HS^- + H^+ = H_2S

Answer: D

54. Which one of the following is a Lewis acid?
A. NH_3 B. CCl_4 C. $AlCl_3$ D. LiCl

Answer: C

55. Among the following acids, which has the lowest pK_a value?
A. CH_3CH_2COOH C. $(CH_3)_2CHCOOH$
B. HCOOH D. CH_3COOH

Solution:
As carbon number increases, strength of carboxylic acids decreases. The stronger the acid, the lower is the Pka value.

Answer: B

56. Nitrous acid, HNO_2 is 0.37% ionized in 3.0M solution. What is the K_a for this acid?
A. $\frac{3.0(0.0037)^2}{99.63}$ C. $\frac{3.0(0.0037)^2}{0.9963}$
B. $\frac{(0.0037)^2}{0.9963}$ D. $\frac{0.9963}{(0.0037)^2}$

Solution:
$[H^+] = 0.37\% \cdot [HNO_2]_0$

$[H^+] = \frac{0.37}{100} \times 3.0M = 0.0111M$

For monoprotic weak acid: $Ka = \frac{[H^-]^2}{[Acid]_o - [H^-]}$

$Ka = \frac{(0.0111M)^2}{3M - 0.011M} = 4.12 \times 10^{-5}$ **Answer: C**

57. The suppression of ionization of a weak acid or a weak base by adding one of its own ions is known as:
A. Buffer effect C. Common ion effect
B. Ionization effect D. Association effect

Answer: C

58. To a 0.0mL of a buffer solution that is 0.25 M in HF an d0.10M in NaF, if 5.0mL of 0.050 M NaOH is added, what si the final pH of this buffer solution? Given the pKa HF = 3.177
A. 2.77 B. 2.80 C. 2.74 D. 3.17

Solution: $HF + NaOH \rightarrow NaF + H_2O$
$n_{HF} = M_{HF} V_{buffer} = 0.25$ mol/L × 5mL = 0.25mmol
$n_{NaOH} = M_{NaOH} \cdot V_{NaOH} = 0.05$ mol/L × 5mL = 0.25mmol
$n_{NaF} = M_{NaF} \cdot V_{buffer} = 0.10$ mol/L × 50mL = 5mmol
after the reaction

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n_{HF} left = 12.5 – 0.25 = 12.25 mmol
n_{NaF} = 0.25mmol + 5mmol = 5.25mmol
$n_{NaOH} = 0$

$$[HF] = \frac{n_{HF}}{V_{NaOH} + V_{buffer}} = \frac{5.25mmol}{55mL} = 0.095M$$

$$pH = Pka + \log \frac{[NaF]}{[HF]}$$

$$pH = 3.17 + \log \frac{0.095}{0.223} = 2.80$$

Answer: B

59. Which statement about indicators is always correct?
A. The mid – point of the pH range of an indicator is 7.0.
B. The pH range is greater with higher pK_a values.
C. The color red indicates an acidic solution.
D. The pK_a values of the indicator is within its pH range.

Answer: B

60. A 0.217 g sample o HgO reacts with excess iodide ions according to the reaction: $HgO + 4I^- + H_2O \rightarrow HgI_4^{2-} + 2OH^-$

How many mL of 0.1M HCl is required to reach equilibrium point during the titration
A. 1.0 B. 10 C. 20 D. 50

Solution:
By mass – mole relationship find the number of mole of OH^-

$$\underset{1\times217g}{\overset{0.217g}{HgO}} + 4I^- + H_2O \longrightarrow HgI_4^{2-} + \underset{2mol}{\overset{x}{2OH^-}}$$

$$\frac{0.217g}{217g} = \frac{x}{2mol}$$

$$x = n_{OH^-} = 0.002mol$$

$$2HCl + 2OH^- \longrightarrow 2H_2O + 2Cl^-$$

For 1 mole HCl, 1 mole OH^- is required. Thus, for 0.002 mol OH^-, 0.002 mole HCl is required $\Rightarrow n_{HCl} = 0.002mol$

$$V_{HCl} = \frac{n_{HCl}}{M_{HCl}} = \frac{0.002mol}{0.1mol/L} = 0.02L = 20mL$$

Answer: C

61. What is a thermodynamic system that exchanges both energy and matter with the surroundings?
A. Adiabatic C. Isolated
B. Closed D. Open

Answer: D

62. At what condition (s) is the equation, $\Delta H = \Delta U + p\Delta V$, applicable?
A. For constant pressure and temperature processes.
B. Only for constant pressure processes.
C. Only for constant temperature processes.
D. Only for constant volume processes.

Solution:
$H = U + PV$
$\Delta H = \Delta U + \Delta(PV)$
$\Delta H = \Delta U + \Delta PV + P\Delta V$, at constant pressure
$\Delta H = \Delta u + P\Delta V$

Answer: B

63. Based on the following information:
$H_2(g) + ½\ O_2(g) \rightarrow H_2O(l)$, $\Delta H° = -286kJ$
$2Na(s) + ½\ O_2(g) \rightarrow Na_2O(s)$, $\Delta H° = -414kJ$
$Na(s) + ½\ O_2(g) + ½\ H_2(g)$, $\Delta H° = -425\ kJ$
What will be the standard enthalpy change for the reaction n
$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(s)$?
A. -978kJ B. -722kJ C. -150kJ D -275kJ

Solution:
Reverse the first and second reactions and multiply the third by 2.

$$H_2O(l) \longrightarrow H_2(g) + \frac{1}{2}O_2(g), \Delta H_1^\circ = 286kJ$$

$$Na_2O(g) \longrightarrow 2Na(s) + \frac{1}{2}O_2(g), \Delta H_2^\circ = 414kJ$$

$$2Na(s) + O_2(g) + H_2(g) \longrightarrow 2NaOH(s), \Delta H_3^\circ$$
$$= 2 \times (-425) = -850kJ$$

$$Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(s),\ \Delta H° = \Delta H_1^\circ + \Delta H_2^\circ + \Delta H_3^\circ$$

$$\Delta H° = 286 + 414 - 850 = -150kJ$$

Answer: C

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64. The K_p for the reaction: $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ is 0.26 at 1,000°C and 40.8 at 1,300°C. Which of the following combinations of ΔH and ΔS are most plausible for the reaction at these temperatures?

	ΔH	ΔS
A.	>0	>0
B.	>0	>0
C.	<0	<0
D.	<0	

Solution:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

The reaction is endothermic because it is decomposition reaction ($\Delta H > 0$). As temperature increases entropy also increases or there are more gaseous products than reactant ($\Delta s > 0$).

Answer: A

65. In the following reaction:

$3Cl^-(aq) + 3CrO_4^{2-}(aq) + 23H^+(aq) \rightarrow 3HClO_2(aq) + 4Cr^{3+}(aq) + 10H_2O(l)$

What is the behavior of Cl^- (aq)?

A. An acid
B. A base
C. An oxidizing agent
D. A reducing agent

Solution:

$$3Cl^-(aq) + 4CrO_4^{2-}(aq) + 23H^+(aq) \longrightarrow 3HClO_2(aq) + 4Cr^{3+}(aq) + 10H_2O(l)$$

The oxidation state of Cl is increased from $-1(Cl^-)$ to $+3(HClO_2)$. Thus, Cl^- is reducing agent.

Answer: D

66. Devices in which oxidation – reduction reaction is induced by an electric current are called

A. Galvanic cells
B. Electrolytic cells
C. Voltaic cells
D. Electromotive cells

Solution:

In electrolytic cell, electric current causes non – spontaneous redox – reaction to take place but in Galvanic (voltaic cell) spontaneous redox reaction generates electricity.

Answer: B

67. What mass of magnesium (Mg) is produced in the electrolysis of molten $MgCl_2$, if a current of 7.50×10^2 A is passed through the molten salt for 1.00 hour?

A. 3.4×10^2 g
B. 9.92×10^2 g
C. 4.7×10^2 g
D. 4.96×10^2 g

Solution:

At cathode: $Mg^{2+} + 2e^- \rightarrow Mg$

$Z = 2$

$I = 7.50 \times 10^2$A, $t = 1$ hour $= 3600$s

$$m = \frac{MIt}{zF} = \frac{24.31 g/mol \times 7.50 \times 10^2 A \times 3600 s}{2 \times 96500 A.s/mol}$$

$m = 340$g

Answer: A

68. In the reaction:

$Pb(s) + PbO_2(s) + H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$

Which substance behaves as the oxidizing agent?

A. Pb B. $PbSO_4$ C. PbO_2 D. H_2SO_4

Solution:

$Pb^0(s) + P^{+4}bO_2(l) + H_2SO_4(aq) \rightarrow 2Pb^{+2}SO_4(l) + 2H_2O(l)$

Pb is oxidized and it is reducing agent.

PbO_2 is reduced and it is oxidizing agnet.

Answer: C

69. What product (s) is/are formed during the electrolysis of a concentrated aqueous solution of sodium chloride?

B. $Cl_2(g)$
C. $Cl_2(g)$ and $H_2(g)$
C. NaOH(aq) and $Cl_2(g)$
D. $Cl_2(g)$, $H_2(g)$ and NaOH(aq)

Solution:

In electrolysis of concentrated NaCl solution

Anode reaction: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode reaction: $2H^+ + 2e^- \longrightarrow H_2$

Cell reaction: $2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{electrolysis}} 2NaOH(g) + H_2(g) + Cl_2(g)$

Therefore, the products formed are NaOH, H_2 and Cl_2.

Answer: D

70. For the electrochemical cell represented by:

$Zn/AnSO_4(aq)//CuSO_4(aq)/CU(s)$

the standard reduction potentials for Cu^{2+}/Cu and Zn^{2+}/Zn are +0.34 and -0.75V, respectively. What will be the cell reaction that occurs spontaneously and its cell voltage?

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		E°(V)
A.	$Cu + Zn^{2+} \rightleftharpoons Cu^{2+} + Zn$	1.10
B.	$Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$	0.42
C.	$Cu^{2+} + Zn^{2+} \rightleftharpoons Zn + Cu$	0.42
D.	$Cu^{2+} + Zn \rightleftharpoons Cu + Zn^{2+}$	1.10

Solution:

$E°\ cell = E°ox + E°\ red$

Anode is $Zn/ZnSO_4$ and cathode is $Cu/CuSO_4$

Cell reaction: $Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$

$E°cell = 0.76V + 0.34V = 1.10V$

Answer: D

71. Consider the reaction:

$$Cu^{2+}(aq) + Fe(s) \rightleftharpoons Cu(s) + Fe^{2+}(aq),\ E° = 0.78V$$

What is the value of E when $[Cu^{2+}]$ is equal to 0.040 M and $[Fe^{2+}]$ is equal to 0.40M?

A. 0.72V B. 0.75V C. 0.81V D. 0.84V

$$E_{cell} = E°cell - \frac{0.0592V}{n}\log Q$$

$$= 0.78V - \frac{0.0592V}{2}\log\frac{0.40}{0.040}$$

$$= 0.75V$$

Answer: B

72. Which one of the following is needed for photosynthesis?

A. Carbon dioxide C. Oxygen
B. Nitrogen D. Phosphorus

Solution:

Carbon dioxide and water are raw materials for photosynthesis in the presence of sunlight. **Answer: A**

73. In what way is the phosphorous cycle differs from the carbon and nitrogen cycle?

A. Phosphorous is more abundant than carbon and nitrogen.
B. Phosphorous is less important to biological system than carbon or nitrogen.
C. Phosphorous, once used by an organism does not cycle back to the environment.
D. The reservoirs for phosphorus exists in mineral form in rocks than in the atmosphere.

Answer: D

74. Which one of the following metal is used for galvanization of iron sheets?

A. Nickel C. Chromium
B. Zinc D. Aluminum

Solution:

Galvanization is coating metal surface with zinc. **Answer: A**

75. Which of the following reactions in used in the production of industrially important chemical?

A. $H_2S_2O_7(l) + H_2O(l) \rightarrow 2H_2SO_4(aq)$
B. $3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO_2(g)$
C. $(NH_4)_2HPO_4(s) + NH_3(g) \rightarrow NH_4H_2PO_4(s)$
D. $2Cr(s) + Al_2O_3(s) \rightarrow Cr_2O_3(s) + 2Al(s)$

Solution:

The most important chemical is sulphuric acid. **Answer: A**

76. The second step in the production of sulphuric acid is given by the reaction:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

Which condition would increase the yield of this reaction?

A. High temperature and low pressure.
B. Low temperature and high pressure.
C. High temperature and high pressure.
D. Low temperature and low pressure.

Solution:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g), \Delta H = -196kJ$$

Formation of $SO_3(g)$ is favored at low temperature and high pressure. **Answer: B**

77. What is the monomer of polyvinyl chloride (PVC)?

A. Chloromethane C. Chloroform
B. Ethylene dichloride D. Ethyl chloride

Solution:

The monomer of polyvinyl chloride is vinyl chloride or chloreethene or ethylene chloride.

$$nCH_2=CHCl \xrightarrow{\text{covalent}} \left[CH_2-\underset{Cl}{CH}\right]_n$$

Vinyl Chloride — PVC

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78. What is the other name for polymethyl methacrylate (PMMA)?
A. Perspex
B. Teflon
C. Bakelite
D. Nylon – 6

Solution:
- Polymethyl methacrylate (PMMA) is also known Perspex, Lucite or Plexi glass.

Answer: A

79. Which of the following is the most abundant polysaccharide?
A. Amylose
B. Cellulose
C. Starch
D. Sucrose

Answer: B

80. What type of polymerization occurs when amino acids react to form peptides and proteins?
A. Addition polymerization
B. Condensation polymerization
C. Substitution polymerization
D. Hydration polymerization

Solution:
Amino acids are monomers of proteins with two functional groups. Such type of monomer form polymers by condensation polymerization.

Answer: B

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1. The reaction between alcohols and acyl chloride produces _____.
A. Ethers
B. Carboxylic acids
C. Aromatic salts
D. Esters

Solution:

$$R-\overset{O}{\overset{\|}{C}}-Cl + ROH \rightarrow R-\overset{O}{\overset{\|}{C}}-OR' + HCl$$

Aceylchloride alcohol Ester Hydrochloric acid

Answer: D

2. Which of the following aqueous solutions will have the LOWEST freezing point?
A. Pure H_2O
B. aq. 0.50 m KF
C. aq. 0.24 m FeI_3
D. aq. 0.60 m glucose

Solution: $\Delta T_f = imk_f = T_f^\circ - T_f$

T_f° – freezing point of sovlent

T_f – freezing point of solution

As the value of i and m increase, T_f decreases, i

i. For $K_F = 2 \Rightarrow i.m = 2 \times 0.5 = 1$

ii. For $FeI_3 = 4 \Rightarrow i.m = 4 \times 0.24 = 0.96$

iii. For glucose $= 1 \Rightarrow i.m = 1 \times 0.6 = 0.60$

Thus, aqueous solution of 0.50m kF has the lowest freezing point.

Answer: B

3. Fats and oils are:
A. esters B. Alcohols C. Acids D. Alkenes

Answer: A

4. Who used the Cathode Ray Tube to discover the electron and determine its charge to mass ratio?
A. Robert. A. Millikan
B. Ernes Rutherford
C. James Chadwick
D. J. J. Thomson

Answer: D

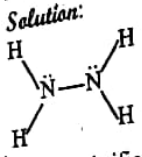
5. Compounds that contain the carboxyl (carbonyl and hydroxyl) group are said to be:
A. Esters
B. Ketones
C. Organic acids
D. Aldehydes

Solution: Compounds containing carboxyl (carbonyl and hydroxyl) group are called carboxylic (organic) acids.

Answer: C

6. When a student draws a plausible Lewis structure for hydrazine molecule (N_2H_4), how many lone pair of electrons are available?
A. 2 B. 1 C. 3 D. 4

Solution:



bp = 5
Lp = 2

Answer: A

7. During nitrification, bacteria convert _____ into _____.
A. N_2, N_2O into nitrous oxide
B. NO_3^-, NH_4^+ or nitrates into ammonium.
C. N_2, nitrous oxide into nitrogen
D. N_2 into nitrates

Answer: D

8. Which of the following metals exists in a free state?
A. Pt B. Mg C. Zn D. Na

Solution: Pt, Au, Pd, Os, Ag and Cu exist in free states.

Answer: A

9. The main ore of lead is called:
A. Cinnabar B. Zinc blend C. Galena D. Chromite

Answer: C

10. Which of the following is **NOT** both a Bronsted – Lowery acid and a Bronsted – Lowery base?
A. HS^- B. HSO_4^- C. OH^- D. HCO_3^-

Solution: Hydroxide ion (OH^-) is a Bronsted – Lowery base but not acid.

Answer C

11. Chlorine has an oxidation number of +5 in:
A. NaClO B. $NaClO_3$ C. $NaClO_2$ D. $NaClO_4$

Solution: The oxidation state of chlorine in NaClO, $NaClO_3$, $NaClO_2$ and $NaClO_4$ is +1, +5, +3 and +7 respectively.

Answer: B

12. A _____ ΔH corresponds to an _____ process.
A. Positive, endothermic C. Negative, endothermic
B. Positive, exothermic D. Zero, exothermic

Solution:
$\Delta H > 0$, endothermic process
$\Delta H < 0$, exothermic process

Answer: A

13. The monomer of neoprene is:
A. Butadiene C. Isoprene
B. 2 – methyl – 1, 3 – butadiene D. Chloroprene

Solution:
- Neoprene is the first synthetic rubber.
- Its monomer is chloroprene (2 – chloro – 1.3 - butadiene),

Answer: D

14. What is the sum of 3.71×10^8 and 4.62×10^7 to the correct significant figure?
A. 4.17×10^8 B. 4.99×10^7 C. 4.172×10^8 D. 4.991×10^7

Solution: $3.71 \times 10^8 + 4.62 \times 10^7 = 37.1 \times 10^7 + 4.62 \times 10^7$
$= (37.1 + 4.62) \times 10^7 = 41.72 \times 10^7 = 4.17 \times 10^8$

Answer: A

15. What is a measure of the closeness of a measurement to its true value?
A. Precision C. Accuracy
B. Reproducibility D. Usefulness

Answer: C

16. When a sodium chromate (Na_2CrO_4) solution is acidified, it is converted to:
A. Cr(s) B. $Cr_2O_7^{2-}$ C. CrO_3 D. Cr_2O_3

Solution: When a sodium chromate (Na_2CrO_4) solution is acidified it is converted to sodium di chromate ($Na_2Cr_2O_7$).
$2Na_2CrO_4 + 2HCl \rightarrow Na_2Cr_2O_7 + 2NaCl + H_2O$

Answer: B

17. Nylons are:
A. Polyamides B. PVC C. Amides D. Polyesters

Solution: Nylons are polyamides. They are condensation polymers.

Answer: A

18. Which of the following is a natural polymer?
A. Perspex B. PVC C. Rubber D. Teflon

Answer: C

19. The maximum kinetic energy of a photoelectron emitted from a metal is 1.03×10^{-19} J when light that has a 656 nm wavelength shines on the surface. What is the threshold frequency for this metal?
A. $4.57 \times 10^{-14}\ s^{-1}$ C. $3.02 \times 10^{-14}\ s^{-1}$
B. $4.57 \times 10^{14}\ s^{-1}$ D. $3.02 \times 10^{14}\ s^{-1}$

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Solution: $\phi = E_{photon} - k_E = \frac{\lambda C}{\lambda} - kE$

$$= \frac{6.626\times10^{-34}\times3\times10^{8}}{6.56\times10^{-7}} - 1.03\times10^{-19}$$

$$= 3.03\times10^{-19} - 1.03\times10^{-19} = 2\times10^{-19}\,J$$

$\phi = hv_o$

$v_o = \frac{\phi}{\lambda}$

$v_o = \frac{2\times10^{-19}\,J}{6.626\times10^{-34}\,J.s}$

$v_o = 3.02\times10^{14}\,s^{-1}$

Answer: D

20. What is the maximum number of electrons in an atom that can have the principal quantum number n = 4?
A. 32 B. 8 C. 18 D. 34
Solution: Number of electrons $= 2n^2 = 2\times4^2 = 32$

Answer:

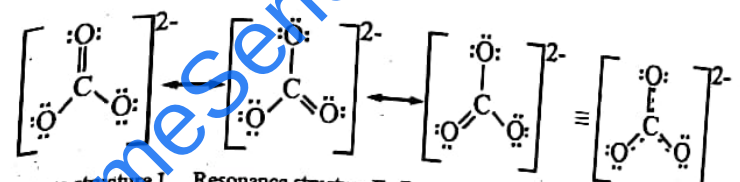
21. Which quantum number is used to determine subshells?
A. Principal quantum number
B. Magnetic quantum number
C. Azimuthal quantum number
D. Spin quantum number
Solution: Angular – momentum or azimuthal quantum number determines sub shells.
$l = 0$, s, $l = 1$, p, $l = 2$, d, $l = 3$, f, $l = 4$, g etc **Answer: C**

22. What skill is a scientist using when he/she listens to the sounds that animals make?
A. Drawing conclusions C. Making a hypothesis
B. Making observations D. Interpreting data
Answer: B

23. Which of the following is fundamentally different from the others?
A. Radio waves C. Light waves
B. Sound waves D. Microwaves
Answer: B

24. Which of the following equations expresses de Broglie's hypothesis?
A. $v = c/\lambda$ B. $\Delta E = hc/\lambda$ C. $\Delta E = c/\lambda$ D. $\lambda = h/mv$
Answer: D

25. The number of resonance structures for CO_3^{2-} are:
A. 3 B. 2 C. 6 D. 9
Solution:



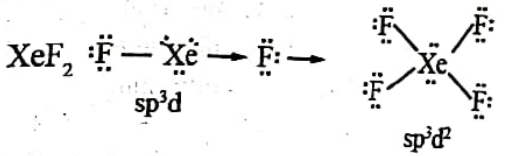
Resonance structure I Resonance structure II Resonance structure III Resonance hybrid

Answer: A

26. In the following equation, what type of hybridization change, if any, occurs at the Xe atom?

$$XeF_2(s) + F_2(g) \rightarrow XeF_4(s)$$

A. sp^3d to sp^3 C. sp^3d to sp^3d^2
B. dsp^2 to sp^3 D. sp^3 to sp^3d
Solution:



Answer: C

27. What is (are) the bond angle (s) in SF_6?
A. 180° B. 109.5° C. 90° and 109.5° D. 90°
Answer: D

28. Which of the following statements about oxygen and fluorine is **NOT** correct?
A. O and F have the same number of core electrons.
B. O has a smaller atomic radius than F.
C. O has a smaller electron affinity than F.
D. O^{2-} has a larger ionic radius than F^-.
Solution:
- $_8O$: $1s^2 2s^2 2p^4$, number of core (inner) electrons = 2
- $_9F$: $1s^2 2s^2 2p^5$, number of core electrons = 2
- Oxygen has larger atomic radius than fluorine.
- O^{2-} has larger ionic radius than F^-

Answer: B

29. What will be the charges on the ions formed when silicon reacts with nitrogen?
A. Si^{2+}, N^{2-} B. Si^{4+}, N^{-3} C. Si^{3+}, N^{3+} D. Si^{4+}, N^{2}

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Solution: $3Si + 2N_2 \rightarrow Si_3N_4$, thus silicon forms Si^{4+} while nitrogen forms N^{3-}.

Answer: B

30. Which of the following compounds does **NOT** contain an ionic bond?

A. K_2S B. NaOH C. HCl D. LiH

Solution: An ionic bond is formed between metals and non – metals or between ammonium ion and non – metals. It is not formed between two non – metals.

Answer: C

31. The rate equation for the decomposition of nitramide, $H_2HNO_2 \rightarrow N_2O + H_2O$, Rate = $k[H_2NNO_2][H^+]^{-1}$. Which of the following mechanisms is consistent with this rate equation?

A. $H_2NNO_2 \rightarrow N_2O + H_2O$ slow

B. $H_2NNO_2 + H^+ \rightleftharpoons H_3NNO_2^+$ fast equilibrium
$H_3NNO_2^+ \rightarrow N_2O + H_3O^+$ slow
$H_3O^+ \rightleftharpoons H^+ + H_2O$ fast equilibrium

C. $H_2NNO_2 \rightarrow OH^- + NH_4^+$ slow
$NH_4^+ \rightleftharpoons NH_3 + H^+$ fast equilibrium
$H_2O \rightleftharpoons H^+ + OH^-$ fast equilibrium

D. $H_2NNO_2 \rightleftharpoons H^+ + HNNO_2^-$ fast equilibrium
$HNNO_2^- \rightarrow N_2O + OH^-$ slow
$H^+ + OH^- \rightarrow H_2O$ fast

Solution: For a mechanism to be consistent for the reaction:

- The rate law must agree with the experimentally determined rate law.
- The overall reaction must be the same as the given reaction.

A. Cannot be consistent because the rate law does not agree.

B. Similar reason as A.

C. The rate law and the overall reaction are not similar with the given reaction.

D. Rate = $k[HNNO_2^-]$, using the slow step, but $HNNO_2^-$ is an intermediate. Replace it using the fast equilibrium

$r_f = k_f[H_2NNO_2], r_b = k_b[H^+][HNNO_2^-]$ at equilibrium $r_f = r_b$

$k_f[H_2NNO_2] = k_b[H^+][HNNO_2^-]$

$$\Rightarrow [HNNO_2^-] = \frac{k_f[H_2NNO_2]}{k_b[H^+]}$$

$$\Rightarrow \text{Rate} = k\frac{k_f}{k_b}\frac{[H_2NNO_2]}{[H^+]} = k[H_2NNO_2][H^+]^{-1}$$

Answer: D

32. Given the equilibrium constant values:

$N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons N_2O(g)$, $K_1 = 3.4\times10^{-18}$

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $K_2 = 4.6\times10^{-3}$

$\frac{1}{2}N_2(g) + O_2(g) \rightleftharpoons NO_2(g)$ $K_3 = 4.1\times10^{-9}$

What is the value of K_C for the following reaction?

$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g)$

A. 2.4×10^{-6} B. 1.2×10^{6} C. 1.2×10^{-6} D. 4.8×10^{6}

Solution: Reverse the first and multiply by 2.
Reverse the second and multiply by 2.
Multiply the third by 4.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), k_1' = \left(\frac{1}{3.4\times10^{-18}}\right)^2 = 8.65\times10^{34}$$

$$4NO_2(g) \rightleftharpoons 2N_2O_4(g), k_2' = \left(\frac{1}{4.6\times10^{-3}}\right)^2 = 4.73\times10^{4}$$

$$2N_2(g) + 4O_2(g) \rightleftharpoons 4NO_2(g), k_3' = (4.1\times10^{-9})^4 = 2.83\times10^{-34}$$

$$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g),\ k_C = k_1'.k_2'.k_3'$$

$$K_C = 1.16\times10^{6}$$

Answer: B

33. The equilibrium constant for the ionization of HCN is 4.9×10^{-10}.

$HCN \rightleftharpoons H^+ + CN^-$, $K = 4.9\times10^{-10}$

Which of the following statements is true regarding this equilibrium?

I. The reaction is product favored.
II. The reaction is reactant favored.
III. Equilibrium lies far to the right.
IV. Equilibrium lies far to the left.

A. II and III C. II and IV
B. I and III D. I and IV

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Solution:

- K is less than one. This implies, formation of reactant is favored and the equilibrium lies far to the left.

Answer: C

34. Which of the following molecular orbital diagrams is correct for the carbide ion (C_2^{2-})?

A. $\sigma_{1s}^2, \sigma_{1s}^2\sigma^{*2}_{1s}, \sigma_{2s}^2, \sigma^{*2}_{2s}\pi^4_{2p}$

B. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2\ \pi^2_{2p}$

C. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2\ \pi^{*4}_{2p}$

D. $\sigma_{1s}^2\ \sigma^{*2}_{1s}\ \sigma_{2s}^2\ \sigma^{*2}_{2s}\ \pi^4_{2p}\ \sigma_{2p}^2$

Solution:

C_2^2, electrons $= 2 \times 6 + 2 = 14$

$(\sigma_{12})^2(\sigma^*_{1s})^2(\sigma_{2s})^2(\sigma^*_{2s})^2(\pi_2 py)^2 = (\pi_2 p_z)^2(\sigma_2 p_x)^2$

$\Rightarrow \sigma_{1s}^2\sigma_{1s}^2\sigma_{2s}^2\sigma^{*2}_{2s}\pi^4_{2p}\sigma^2_{2p}$

Answer: D

35. If a reaction is zero order in A, tripling the concentration of A will cause the reaction rate to:

A. Increase by a factor of 27. C. Remain constant

B. Increase by a factor of 3. D. Increase by a factor of 9.

Solution: The rate of zero – order reaction is independent on concentration.

Answer: C

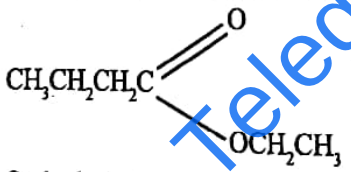
36. Which one of the following factors does **NOT** affect the rate of a chemical reaction?

A. Humidity C. Temperature

B. Concentration D. Nature of reactants

Answer: A

37. An ester has the structural formula



On hydrolysis, the ester would produce:

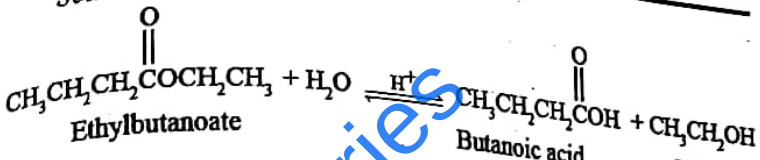
A. Propanoci acid and propan – 1 – ol

B. Ethanoic acid and butan – 1 – ol

C. Butanoic acid and ethanol

D. Propanoci acid and ethanol

Solution:



Ethylbutanoate — Butanoic acid — Ethanol

Answer: C

38. You are given a bottle of solid X and three aqueous solutions of Y, the first saturated, the second unsaturated and the third supersaturated. Which of the following is correct, if you add a small amount of the solid solute to each solution?

A. The solution in which the added solid solute dissolves is the saturated solution.

B. The supersaturated solution is unstable and addition of additional solute causes the excess solute to crystallize.

C. The solution in which the added solid solute remains un-dissolved is the unsaturated solution.

D. In all the three solution; saturated, unsaturated and supersaturated the added solid solute will dissolve.

Solution:

- Unsaturated solution dissolves the solid solute added.
- In saturated solution, the added solute remains un dissolved.
- In supersaturated solution, the size of the added solute will grow bigger.

Answer: B

39. The phrase "like dissolves like" refers to the fact that:

A. Polar solvents dissolve polar solutes and nonpolar solvents dissolve non polar solutes:

B. Polar solvents dissolve nonpolar solutes and vice versa.

C. Solvents can only dissolve solutes of similar molar mass.

D. Gases can only dissolve other gases.

Answer: A

40. For a certain gas phase reaction.

$2A(g) \rightleftharpoons B(g) + C(g)\ \Delta H = +45\ KJ/mol, K = 4.5 \times 10^{-2}$

Which of the following would be true if the temperature was increased from 25°C to 200°C?

I. The value of K would be smaller.

II. The concentration of A(g) would be increased.

III. The concentration of B(g) would increase.

A. III B. II C. I D. I and III

Solution:

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- The forward reaction is endothermic thus increasing temperature favors formation of B and C.
- The value of equilibrium constant increases with increasing temperature.

Answer: A

41. Which of the following compounds would be the most soluble in H_2O?
A. Ethane C. Octanoic acid
B. Pentane D. Ethanoic acid

Solution: Carboxylic acid with smaller mass are miscible with water due to hydrogen bonding but not hydrocarbons.

Answer: D

42. Consider the following equilibrium:
$2CO_{(g)} + O_{2(g)} \leftrightarrows 2CO_{2(g)}$ $K_{eq} = 4.0 \times 10^{-10}$
What is the value of K_{eq} for $2CO_{2(g)} \leftrightarrows 2CO_{(g)} + O_{2(g)}$?
A. 4.0×10^{-10} B. 2.5×10^{9} C. 5.0×10^{4} D. 2.0×10^{-5}

Solution: If the reaction is reversed, the new K_{eq} will be the reciprocal of the original.

$$\Rightarrow k_{eq} = \frac{1}{k_{eq}} = \frac{1}{4\times10^{-10}} = 2.5\times10^{9}$$

Answer: B

43. What species of ions are present in a 0.1M solution of HCl and what will be their equilibrium concentrations?
A. $[H_3O]^+ = 0.1M$; $[OH]^- = 0.1$ M, $[Cl]^- = 0.1M$
B. $[H_3O]^+ = 0.1M$; $[OH]^- = 0.01M$, $[Cl]^- = 0.1M$
C. $[H_3O]^+ = 0.1M$; $[OH]^- = 10^{-13}M$, $[Cl]^- = 0.1M$
D. $[H_3O]^- = 10^{-13}$; $[OH]^- = 0.1M$, $[Cl]^- = 0.1M$

Solution: For strong monoprotic acid, HA, $[HA] = [H^+] = [A^{-1}]$.
$\Rightarrow$ In 0.1MHCl solution, $[H_3O^+] = [Cl^-] = 0.1M$

$$[OH^-] = \frac{K_w}{[H^-]} = \frac{1\times10^{-14}M^2}{0.1M} = 1\times10^{-13}M$$

Answer: C

44. A solution in which propionic acid is 0.94% ionized has a pH of 2.85. What is the value of the acid ionization constant (Ka) for propionic acid?
A. 4.2×10^{-4} B. 6.8×10^{-4} C. 1.3×10^{-5} D. 8.7×10^{-5}

Solution: $[H^+] = 10^{-pH} = 10^{-2.85} = 1.41 \times 10^{-3}M$

$$Ka = \frac{(1.41\times10^{-3})^2}{0.15-0.00141} = 1.3\times10^{-5}$$

Answer: C

45. A 0.2M solution of a weak acid HA is 1% ionized at 25°C. K_a for the acid is equal to:
A. $\frac{0.19}{0.01\times0.01}$ C. $\frac{0.01\times0.01}{0.19}$
B. $\frac{0.02\times0.02}{0.18}$ D. $\frac{0.002\times0.002}{0.198}$

Solution: $[H^+] = 1\% \times [HA]_o = \frac{1\times0.2M}{100} = \frac{0.2M}{100}$

$$K_a = \frac{[H^+]^2}{[HA]_o - [H^+]} = \frac{\left(\frac{0.2}{100}\right)^2}{0.2-\frac{0.2}{100}} = \frac{\frac{0.2\times0.2}{100\times100}}{\frac{20-0.2}{100}} = \frac{0.02\times0.02}{19.8}$$

$$Ka = \frac{0.002\times0.002}{0.198} = 2.02\times10^{-5}$$

Answer: D

46. 2.3 g of ethanol (CH_3CH_2OH) is added to 500g of water. What is the molality of the resulting solution?
A. 0.01m B. 1.0m C. 0.1m D. 10.0m

Solution: $m = \frac{m_{solute}}{M_{solute}.kg\ of\ solvent}$

$$m = \frac{2.3g}{46g/mol\times0.5kg} = 0.1m$$

Answer: C

47. A 0.5L and 0.1M HNO_3 solution is to be prepared by dilution process from a 13M nitric acid. How many mL conc. HNO_3 and how many mL of water are required to prepare the dilute solution?
A. 3.85 mL HNO_3 and 496.15 mL water
B. 2 mL HNO_3 and 498 mL water
C. 10mL HNO_3 and 490 mL water
D. 20 mL HNO_3 and 490 mL water

Solution: $V_1 = ?$ $V_2 = 0.5L$
$M_1 = 13M$ $M_2 = 0.1M$

$$V_1 = \frac{M_2V_2}{M_1} = \frac{0.1M\times0.5L}{13M} = 3.85\times10^{-3}L = 3.85mL$$

$$V_{H_2O\ added} = V_2 - V_1 = 500mL - 3.85mL = 496.15mL$$

Answer: A

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48. Commercial concentrated sulfuric acid (density = 1.831 g cm^{-3}) is 94.0% H_2SO_4, by mass. What is the normality of H_2SO_4 solution? (Molar mass of H_2SO_4 = 98 g mol^{-1})
A. 16.8M B. 28.2 C. 40.4M D. 35.0N

Solution: $N = Z.M = \frac{Z\% \times d \times 10}{M_{H_2SO_4}} = \frac{2 \times 94 \times 1.831 \times 10}{98} = 35.1N$

$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}, Z = 2$ **Answer: D**

49. During the electrolysis of a concentrated aqueous solution of NaCl, what substance is formed at the cathode?
A. Hydrogen C. Chlorine
B. Oxygen D. Sodium
Answer: A

50. In a voltaic cell, electrons flow from the _____ to the _____.
A. Salt bridge, anode C. Anode, slat bridge
B. Salt bridge, cathode D. Anode, cathode
Solution: In both galvanic (voltaic) and electrolytic cell, electrons flow from anode to cathode in the external circuit.
Answer: D

51. During the electrolysis of $CuSO_4$(aq) using carbon electrodes, which of the following is the correct half reaction for the anode electrode?
A. $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
B. $4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$
C. $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$
D. $SO_4^{2-}(aq) + O_2(g) + 4H^+(aq) \rightarrow SO_4^-(aq) + 2H_2O(l)$ **Answer: B**

52. Which of the following statements is true?
A. Addition of $(CH_3NH_3)Cl$ to a solution of CH_3NH_2 will decrease the pH.
B. Addition of $NaNO_2$ to a solution of HNO_2 will decrease the pH.
C. Addition to HCl to a solution of $NaC_2H_3O_2$ will increase the pH.
D. Addition of KBr to a solution of HBr will increase the pH.
Solution:
- Addition of acidic salt or acid to a certain solution will decrease the pH but increase the pOH.
- Addition of basic salt or base to a given solution will increase its pH but decrease its pOH.

- Neutral salts will not affect the pH or pOH of a solution.

Answer: A

53. What is the final temperature when 150.0 mL of water at 90.0 °C is added to 100.0mL of water at 30.0°C?
A. 33.0°C B. 45.0°C C. 66.0°C D. 60.0°C
Solution: Heat lost by hot water = heat gained by cold water
⇒ The initial temperature of hot water is the same as, the final temperature of cold water. Let it be T.
$m_1 = d \times v_1$ and $m_2 = d. v_2$
⇒ $m_1 C \Delta T_1 = m_2 C \Delta T_2$, $\Delta T_1 = 90 - T$, $\Delta T_2 = T - 30$
$dV_1 C\Delta T_1 = dv_2 C\Delta T_2$, $V_1(90 - T) = V_2(T - 30)$
$150 (90 - T) = 100 (T - 30)$
Solving for T, T = 66°C **Answer: C**

54. What is the pH of a mixture of 15.0 mL of 0.126M NaOH and 21.0M H_2SO_4?
A. 1.70 B. 13.60 C. 11.81 D. 2.15
No answer because the volume of NaOH is not given.

55. The relationship between picometer (pm) and nanometer (nm) is:
A. 1 pm = 10 nm C. 1 pm = 100nm
B. 1 nm = 1000 pm D. 1 nm = 10 pm

Solution: $\frac{1pm}{1nm} = \frac{10^{-12}m}{10^{-9}nm}$

$\frac{1pm}{1nm} = 10^{-3} \Rightarrow 1pm = 10^{-3}nm$ or $1nm = 10^3 pm$ **Answer: B**

56. What would be the wavelength of a radio wave having a frequency of 3 MHz?
A. 300nm B. 300m C. 100nm D. 100m

Solution: $\lambda = \frac{C}{V} = \frac{3 \times 10^8 m/s}{3 \times 10^6 s^{-1}} = 100m$ **Answer: D**

57. Which of the following correctly lists electromagnetic waves in order from shortest to longest wavelength?
A. Microwaves, ultraviolet, visible light, gamma rays
B. Radio waves, infrared, gamma rays, ultraviolet
C. Gamma rays, ultraviolet, infrared, microwaves
D. Gamma rays, infrared, ultraviolet, microwaves
Solution:
- Based on wavelength: Gamma rays < x – rays < ultraviolet < visible < infrared < microwaves < radio wave **Answer: C**

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58. Which of the following is **NOT** the decomposition product of HNO_3?
A. N_2O_4 B. NO_2 C. O_2 D. H_2O

Solution: $4HNO_3 \xrightarrow{\Delta} 4NO_2 + 2H_2O + O_2$ **Answer: A**

59. To determine the volume of an irregularly shaped glass vessel, the vessel is weighed empty (121.3 g) and when filled with CCl_4 (283.2 g). What is the volume capacity of the vessel, given that the density of CCl_4 is 1.59 g/cm^3?
A. 76.29cm^3 B. 257.42cm^3 C. 178.11 cm^3 D. 101.82 cm^3

Solution: $m_{CCl_4} = m_{total} - m_{glass} = 283.2g - 121.3g = 161.9g$

$$V = \frac{m}{d} = \frac{161.9g}{1.59g/cm^3} = 101.82cm^3$$ **Answer: D**

60. The process of vulcanization of rubber makes it _____.
A. Less elastic C. Hard
B. Soft D. More soluble in sovlents

Answer: C

61. For the reaction $C_6H_{14}(g) \rightarrow C_6H_6(g) + H_2(g)$, $\Delta P(H_2)/\Delta t$ was found to be 2.5×10^{-2} atm/s, where $\Delta P(H_2)$ is change in the pressure of hydrogen. Determine $\Delta P(C_6H_{14})/\Delta t$ (in units of atm/s) for this reaction at the same time.
A. -6.2×10^{-3} B. 1.6×10^{-3} C. 2.5×10^{-2} D. 6.2×10^{-3}

Solution: $\Delta P(C_6H_{14})/\Delta t = \frac{\text{coefficient of } C_6H_{14}}{\text{coefficient of } H_2} \times \Delta P(H_2)/\Delta t$

$$= \frac{1}{4} \times 2.5 \times 10^{-2} \text{ atm/s} = 6.25 \times 10^{-3} \text{ atm/s}$$ **Answer: D**

62. The decomposition of a compound at 400°C is first order with a half-life of 1570 seconds. What fraction of an initial amount of the compound remains after 4710 seconds?
A. 1/12 B. 1/6 C. 1/8 D. 1/3

Solution: $t_{1/2} = \frac{0.693}{k}$ $\quad k = \frac{0.693}{1570s}$

$$k = \frac{0.693}{t_{1/2}}$$

$$\log\frac{[A]_0}{[A]_t} = \frac{kt}{2.303}$$

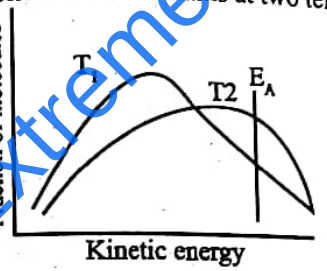
$$\frac{[A]_0}{[A]_t} = 10^{\frac{kt}{2.303}} = 10^{\frac{\frac{0.693}{1570} \times 4710}{2.303}} = 8$$

$$[A]_0 = 8[A]_t$$

$$[A]_t = \frac{1}{8}[A]_0$$

Answer: C

63. The diagram below shows the range of energies of collision of a collection of reactants at two temperatures. T_1 and T_2.



Which of the following is true regarding T1 and T2?
A. T1 = T2, fraction of molecules at both temperatures are equal.
B. T1 < T2, fraction of molecules at T1 is smaller.
C. T2 < T1, fraction of molecules at T2 is smaller.
D. T1 < T2, fraction of molecules at T1 is larger.

Answer: A

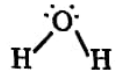
64. From CO_2, H_2O, $BeCl_2$ and N_2O which have the same molecular geometry?
A. CO_2, $BeCl_2$, and N_2O C. CO_2, H_2O, and N_2O
B. CO_2 and $BeCl_2$ only D. H_2O and N_2O only

Solution:

CO_2: $\ddot{O}=C=\ddot{O}$, linear

$BeCl_2$: $:\ddot{Cl}-Be-\ddot{Cl}:$, linear

N_2O: $:\ddot{N}=N=\ddot{O}$, linear

H_2O:  **Answer: A**

65. How many electrons are present in the σ_{2p} molecular orbital of N_2^+?
A. 1 B. 4 C. 3 D. 2

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Solution: N_2^+ : electrons $= 2\times7-1=13$

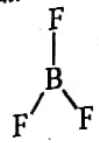
$\sigma_{1s}^2\ \sigma_{1s}^{*2}\ \sigma_{2s}^2\ \sigma_{2s}^{*2}\ \pi_{2py}^2\ \pi_{2pz}^2\ \sigma_{2px}^1$

Answer: A

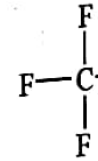
66. Given the following AF_n species, BF_3, BeF_2, CF_4, NF_3, OF what is the correct of F – A – F bond angles?

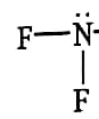
A. $OF_2 < BeF_2 < NF_3 < BF_3 < CF_4$
B. $CF_4 < BF_3 < NF_3 < BeF_2 < OF_2$
C. $OF_2 < NF_3 < CF_4 < BF_3 < BeF_2$
D. $BeF_2 < OF_2 < NF_3 < BF_3 < CF_4$

Solution:

BF_3:  , Bond angle = 120°

BeF_2: F—Be—F , bond angle = 180°

CF_4:  , bond angle = 109.5°

NF_3:  , bond angle = 107°

OF_3: , bond angle = 104.5°

Answer: C

67. Which of the following is true regarding the solution formation process?

A. Intermolecular forces between the solute particles must weaken in which the enthalpy change is exothermic ($\Delta H<0$).
B. Intermolecular forces between the solvent molecules must weaken in which the enthalpy change in exothermic ($\Delta H<0$).
C. Covalent bonds within the solute and solvent molecules must be broken.

D. New Columbic attractions between the solute and solvent form in which the enthalpy change is exothermic ($\Delta H<0$).

Answer: D

68. What volume of 1.40M H_2SO_4 solution is needed to react exactly with 10.0g of aluminum according to the following reaction?

$2Al(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$

A. 2.643mL B. 26.43mL C. 2643mL D. 264.3mL

Solution: The balanced chemical equation given is wrong correct it. By mass – mass relationship, find the mass of H_2SO_4.

$$\underset{2\times27}{\overset{10g}{2Al(s)}} + \underset{3\times98}{\overset{x}{3H_2SO_4}} \longrightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$$

$$\frac{10g}{2\times27} = \frac{x}{3\times98}$$

$$x = m_{H_2SO_4} = 54.4g$$

$$n_{H_2SO_4} = \frac{m}{M} = \frac{54.4g}{98g/mol} = 0.56mol$$

$$\therefore V_{H_2SO_4} = \frac{n_{H_2SO_4}}{M_{H_2SO_4}} = \frac{0.56\ mol}{1.40\ mol/L} = 0.397L = 397mL$$

Answer: _____

69. A solution of NH_4Cl made by dissolving 3.16g NH_4Cl in 30.14g H_2O has a density of 1.0272g cm^{-3}. What is the mole fraction of NH_4Cl?

A. 0.0341 B. 0.9659 C. 0.6500 D. 0.2100

Solution: $n_{NH_4Cl} = \frac{m}{M} = \frac{3.16g}{53.5g/mol} = 0.059mol$

$$n_{H_2O} = \frac{30.14}{18} = 1.67$$

$$X_{NH_4Cl} = \frac{n_{NH_4Cl}}{n_{NH_4Cl} + n_{H_2O}} = \frac{0.059}{0.059 + 1.67} = 0.0341$$

Answer: A

70. Consider the following equilibrium:

$Cl_2(g) + 2NO(g) \leftrightarrows 2NOCl(g)$ $K_{eq} = 5.0$, at equilibrium $[Cl_2] = 1M$ and $[NO] = 2M$ what is $[NOCl]$?

A. 4.5M B. 0.89 C. 0.80M D. 10M

Solution: $k_{eq} = \frac{[NOCl]^2}{[Cl_2][NO]^2}$

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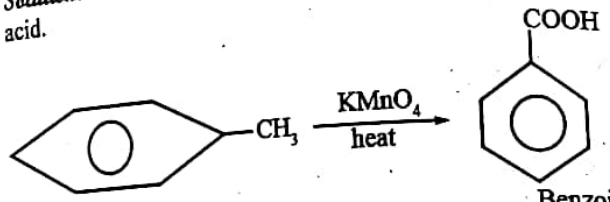
$[NOCl] = \sqrt{k_{eq}[Cl_2][NO]^2} = \sqrt{5 \times 1 \times (2)^2} = 4.47M$ **Answer: A**

71. Which acid is produced when toluene is subjected to $KMnO_4$ oxidation?
A. Toluic acid C. Phenyl acetic acid
B. Benzoic acid D. Phthalic acid

Solution: Alkyl benzene reacts with $KMnO_4$ to yield benzoic acid.



Toluene (Methyl benzene) Benzoic acid

Answer: B

72. Which of the following statements is **NOT** true in relation to the triple point on a single component phase diagram?
A. The point at which the solid, liquid and gaseous phases for a substance co-exist.
B. The system must be enclosed so that no vapour can escape.
C. The triple point exists for a substance occurs at a specific temperature and pressure.
D. The triple point exists at a single temperature and is independent of pressure.

Solution: At the triple point:
- Solid, liquid and gaseous phases coexist at equilibrium.
- The system must be enclosed so that no vapor can escape.
- A substance occurs at a specific temperature and pressure.
- The degree of freedom is zero (invariant system).

Answer: D

73. Calculate ΔH for the following reaction using the bond energies given below:
$H-H(g) + I-I(g) \rightarrow 2H-I(g)$
Bond energies: H – H = 436 kJ/mol, I – I = 151 kJ/mol, H – I = 297 kJ/mol
A. +290kJ B. -7kJ C. +7kJ D. -290kJ

Solution: $\Delta H = \sum$ bond broken $- \sum$ bond formed

$= (1 \times H-H + 1 \times I-I) - (2 \times H-I)$
$= (436 + 151) - (2 \times 297) = -7kJ$

74. Two electrolytic cells were placed in series, one of $AgNO_3$ and the other of $CuSO_4$. If 1.273g of Ag is deposited, how much Cu was deposited at the same time? **Answer: B**
A. 0.0118 B. 0.954 C. 0.748 D. 0.374

Solution: $\frac{m_{Cu}}{E_{Cu}} = \frac{m_{Ag}}{E_{Ag}}$

$m_{Cu} = \frac{m_{Ag}}{E_{Ag}} E_{Cu} = \frac{1.273g}{108} \times 32 = 0.377g$ **Answer: D**

75. What is the standard cell potential for the oxidation of ammonia, given below? $4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$
A. 3.51V B. 1.17V C. 7.02V D. 14.04V

Solution: $4N^{3-} \rightarrow 2N_2 + 12e^-$
⇒The number of electrons lost is 12(n = 12)
$\Delta G = -nE^\circ_{cell}F$

$E^\circ_{cell} = \frac{\Delta G}{-nF} = \frac{-356 \times 10^3 J}{-12 \times 96500 J/V} = 1.17V$ **Answer: B**

76. What is the freezing point of a 0.1 m solution of NaCl (in °C)? (K_f = 1.86°C/m and the freezing point of pure water is 0°C).
A. -1.86 B. -0.186 C. -0.372 D. -0.093

Solution: $T_f = T_f^\circ - imk_f$, i for NaCl = 2
$= 0°C - 2 \times 0.1m \times 1.86°C/m = -0.372°C$

Answer: C

77. During a titration, what volume of 0.500M KOH is necessary to completely neutralize 10.0mL of 2.00M CH_3COOH?
A. 10.0mL B. 20mL C. 40.0mL D. 25.0mL

Solution: $n_{CH_3COOH} = M_{CH_3COOH} \cdot V_{CH_3COOH} = \frac{2mol}{2} \times 10mL = 20mmol$

By mole – mole relationship, find the number of mole of kOH.

$\underset{1}{\overset{20mmol}{CH_3COOH}} + \underset{1}{\overset{x}{KOH}} \longrightarrow CH_3COOK + H_2O \Rightarrow n_{KOH} = 20mmol$

$V_{KOH} = \frac{n_{KOH}}{M_{KOH}} = \frac{20mmol}{0.5mol/L} = 40mL$ **Answer: C**

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78. Which of the following is an organic acid?
A. CH_3CO_2H B. CH_3CH_2OH C. $CH_2=CH_2$ D. CH_3CH_3
Solution: An organic acid contains carboxyl group (–COOH or $-CO_2H$) as a functional groups. **Answer: A**

79. Which of the following statements is true?
A. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will decrease by the same amount.
B. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will also increase by the same amount.
C. If the entropy of the system increases during a reversible process, the entropy change of the surroundings will remain the same.
D. If the entropy of the system decreases during a reversible process, the entropy change of the surroundings will also decrease by the same amount.

Solution: For a reversible process, $\Delta S_{univ} = \Delta S_{syst} + \Delta S_{surr} = 0$
$\Rightarrow \Delta S_{system} = -\Delta S_{surrounding}$ **Answer: A**

80. How long has a current of 3 ampere to be applied through a solution of silver nitrate to a coat a metal surface of 80 cm^2 with 0.05 cm thick layer? Density of silver = 10.5 g cm^3
A. 476s B. 1028s C. 1252s D. 683s
Solution: $m_{Ag} = d_{Ag}.\ V_{Ag} = 10.5\ g/cm^3 \times (80cm^2 \times 0.005cm)$
$m_{Ag} = 4.2g$, $Ag^+ + e^- \rightarrow Ag$
$z = 1$

$$m = \frac{MIt}{ZF}$$

$$t = \frac{mZF}{MI} = \frac{4.2g \times 1 \times 96500A\,s/mol}{108g/mol \times 3A}$$

$t = 1251$ sec **Answer: C**

SECTION 3 – PHYSICS

1. Two non – zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = c\vec{B}$, where c is a scalar. If the two vectors have opposite directions, then one of the following is true about c?
A. c is a positive number C. $c = 1$
B. c is a negative number D. $c = 0$
Solution:
When a vector is multiplied by a positive scalar only its magnitude is changed. When a vector is multiplied by a negative scalar both the magnitude and direction changed. When $\vec{A}$ and $c\vec{B}$ are in opposite direction then the scalar c is a negative number.
Answer: B

2. On which of the following does the induced emf in a coil NOT depend when a magnet is inserted into it?
A. Resistance of the coil.
B. Number of turns in the coil.
C. Speed of approach of the magnet.
D. Magnetic moment of the magnet.
Solution:
The induced electromotive force (emf) in a coil during electromagnetic induction depends on:
- The number turns of the coil.
- The speed of the approaching magnet.
- The magnetic field strength.
- The orientation of the coil to the field.

Answer: D

3. A wave has a speed of 50 m/s and a wavelength of 25,000 m. What is the frequency of the wave?
A. 2.0×10^{-3} Hz C. 5.0×10^{2} Hz
B. 6.0×10^{-1} Hz D. 1.8×10^{6} Hz
Solution: $V = 50$ m/s, $\lambda = 25000$ m
$f = ?$
Using the wave equation:

$$V = \lambda f \Rightarrow f = \frac{V}{\lambda} = \frac{50\ m/s}{25000\ m}$$

$f = 2 \times 10^{-3}$ Hz **Answer: A**

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4. An oscillating system is used to create a succession of waves on a water surface. The rate of wave generation is one every 0.33 seconds. What is the period of these waves?
   A. 0.3 seconds
   B. 1.0 seconds
   C. 2.0 seconds
   D. 3.0 seconds

Solution:
If it generates one wave in every 0.33 second this implies in one second three wave can be produced. So f = 3 Hz

$$T = \frac{1}{f} = \frac{1}{3Hz} = 0.3\,sec$$

Answer: A

5. A vector that represents the position of an object in relation to another object is called
   A. Unit vectors
   B. Position vector
   C. Coplanar vectors
   D. Collinear vectors

Solution:
The vector that represents the position of an object in relation to another object is called position vector. **Answer: B**

6. Consider a series RC circuit for which R = 1.0MΩ, C = 5.0μF and ε = 30.0 V. What is the time constant of the circuit and the maximum charge on the capacitor after the switch is closed respectively?
   A. 5 seconds, 150μC
   B. 7 seconds, 155μC
   C. 10 seconds, 160μC
   D. 9 seconds, 135μC

Solution:
$R = 1Mn = 1 \times 10^6\Omega$
$C = 5\mu F = 5 \times 10^{-6}\ F$
$\varepsilon = 30V$
The time constant 'τ' is given by $\tau = RC\ \tau = 10^6\Omega \times 5 \times 10^{-6} = 5$ second and the maximum charge is $Q = \varepsilon \times C = 30 \times 5 \times 10^{-6}$ $F = 150 \times 10^{-6}C = 150\mu C$
Answer: A

7. What is the direction to which a fish must push the water with its fins in order to propel eastward?
   A. Eastward
   B. Upward
   C. Westward
   D. Downward

Solution:

When the fish needs to propel eastward. It must push the water west ward. This is the consequence of "For every action force there exist an equal and opposite reaction." **Answer: C**

8. Which of the following is correct about the meaning of Photon?
   A. Quantum of matter.
   B. Positively charged particle.
   C. Negatively charged particle.
   D. Quantum of light.

Solution:
A photon is defined as a quantum of energy photons are natural.
Answer: C

9. Suppose two molecules of different mass are kept at the same temperature. Which one of the following statements is NOT true?
   A. Both remain at rest.
   B. Both move with the same speed.
   C. The lighter molecule moves faster.
   D. The heavier molecule moves faster.

Solution:
The molecules have different mass. Let m_1 is the mass of the first molecule and m_2 is the mass of the second molecule

$$m_1 \neq m_2$$

They are at the same temperature at the same temperature, they have the same kinetic energy.

$$E_{k_1} = E_{k_2}$$

Since they have different masses. The higher molecule must move faster than the heavier molecules. Probably both maybe at rest so B and D are necessary false. A = can be true or false and C is true.
Answer: B and D

10. One person drops a ball from the top of a building while another person at the bottom observes its motion. The two people **DISAGREE** on which of the following?
   A. The value of total energy of the ball.
   B. The change in potential energy of the ball.
   C. The value of the gravitational energy of the ball Earth system.
   D. The kinetic energy of the ball at some point in its motion if one of them is walking. S

Solution:

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To observe the motion of the ball by each of them both must be at the same reference. i.e, both person should be at rest. If one of the person is moving, the speed of the ball, therefore the kinetic energy are different as observe by both person.

Answer: D

11. An air – filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance become 2 pF. What is the dielectric constant of the wax?

A. 8.0 B. 4.0 C. 2.0 D. 0.5

Solution:

Let the capacitance initially was $C_1 = 1pF$ the plate separation is d_1 after doubling the separation $d_2 = 2d$, the capacitance c_2 becomes $c_2 = 2pF$, $k = ?$

Initially the capacitance is $c_1 = \frac{\varepsilon_o A}{d_1}$.

Finally, $c_2 = \frac{\varepsilon_o A}{d_2}, d_2 = 2d$

$$c_2 = \frac{\varepsilon_o kA}{2d_1}$$

$$2PF = \frac{\varepsilon_o kA}{2d_1}$$

$$2PF = \frac{1}{2}\left(\frac{A}{d_1}\right) \text{ but } \frac{EA}{d_1} = c_1 = 1\mu F$$

$$2PF = \frac{k}{2}\left(\frac{\varepsilon_o A}{d_1}\right)$$

$$2PF = \frac{k}{2}c_1$$

$$2PF = \frac{k}{2}(1PF)$$

$$k = \frac{2PF \times 2}{1PF} = 4$$

$k = 4$ **Answer: B**

12. A step – up transformer is used to

A. Increase the power C. Increase the voltage

B. Decrease the power D. Decrease the voltage

Solution:

A step up transformer used to increase the voltage supplied. For step up transformer, the voltage in the secondary coil is larger than the voltage in the secondary coil.

$V_s > V_P$ and also $N_s > N_P$

For step down transformer $V_s < V_P$ and $N_s < N_P$ **Answer: C**

13. A 200 – V power line is protected by a 15 – A fuse. What is the maximum number of "220V, 500W" light bulbs that can be operated at full brightness from the line?

A. 0.2 B. 0.4 C. 0.6 D. 0.8

Solution:

The current tolerated by the fuse is 15A. The current supplied by the line is $P = VI$.

$$I = \frac{P}{V} = \frac{500W}{220V}, I = 2.27\ A$$

So $n \times 15A = 2.27\ A$

$$n = \frac{2.27A}{15A} = 0.153 \approx 0.2$$ **Answer: A**

14. In a reversible thermodynamic process, the system

A. Is always close to equilibrium states.

B. Might never be close to any equilibrium state.

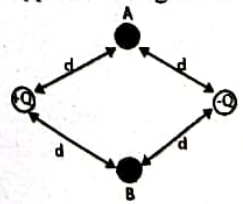
C. Is close to equilibrium states only at the beginning and end

D. Is close to equilibrium states throughout, except at the beginning and end

Solution:

In a reversible thermodynamic process, the system is always close the equilibrium state. **Answer: A**

15. Points A and B are each a distance d from each of two equal and opposite charges as shown below.



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What will be the work required to move a negative charge q from A to B?

A. Zero

B. $\frac{qQ}{4\pi\varepsilon_o d^2}$

C. $\frac{qQ}{4\pi\varepsilon_o d}$

D. $\frac{qQ}{4\pi\varepsilon_o (4d^2)}$

Solution:

The potential at point A is

$$V_A = \frac{k(+Q)}{d} + \frac{(k)(-Q)}{d} = \frac{kQ}{d} - \frac{kQ}{d}$$

$V_A = 0$

$$V_B = \frac{k(+Q)}{d} + \frac{k(-Q)}{d} = \frac{kQ}{d} - \frac{kQ}{d}$$

$V_B = 0$

The potential difference between points A and B is $V_{AB} = V_A - V_B = 0$. Then the work done is $W = Q\Delta V = Q(0) = 0$,

W = zero

Answer: A

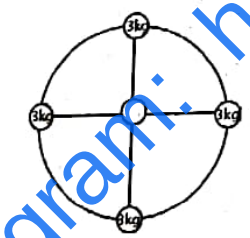
16. The radius of the wheel shown below is 0.75m. What is its moment of inertia?

A. 2.25 kgm^2

B. 3.38 kgm^2

C. 6.75 kgm^2

D. 9.00 kgm^2



Solution:

r = 0.75m, I =?

The total moment of inertia is the sum of the moment of inertia due to each mass.

$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2$ but $m_1 = m_2 = m_3 = m_4 = 3$ kg

$r_1 = r_2 = r_3 = r_4 = 0.75$m

$I = mr^2 + mr^2 + mr^2 + mr^2 = 4mr^2 = 4\ (3\ kg)\ (0.75m)^2 = 6.75\ kg$

Answer: C

17. An ordinary refrigerator is turned on and its door is left open in a thermally insulated kitchen. What is the temperature of the room?

A. Increase according to the first law of thermodynamics.

B. Decreases according to the first law of thermodynamics.

C. Remains constant according to the first law of thermodynamics.

D. Remain constant according to the second law of thermodynamics.

Solution:

As the room thermally insulted, there is NO transfer of heat energy between the room and the refrigerator. This is due to the first law of thermodynamics.

Answer: C

18. If $\vec{A} = 8\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{B} = 5\hat{i} - 3\hat{j} + \hat{k}$, then, what is the value of $\vec{B} \times \vec{A}$?

A. $3\hat{i} + 4\hat{j} - 3\hat{k}$

B. $-3\hat{i} - 4\hat{j} + 3\hat{k}$

C. $\vec{A} = -5\hat{i} - 18\hat{j} - 29\hat{k}$

D. $\vec{A} = 5\hat{i} + 18\hat{j} + 29\hat{k}$

Solution:

$\vec{A} = 8\hat{i} + \hat{j} - 2k$

$\vec{B} = 5\hat{i} - 3\hat{j} + k$

B × A =?

$B \times A = (5\hat{i} - 3\hat{j} + k) \times (8i + j - 2k) = 5\ (i \times j) + (-10)(i \times k) + -24\ (j \times i) + 6(j \times k) + 8\ (k \times i) + 1\ (k \times j) = 5k - 10(-j) - 24\ (-k) + 6\ (2) + 8j + 1\ (-i)$

$\vec{B} \times \vec{A} = 6\hat{i} - \hat{i} + 10\hat{j} + 8\hat{j} + 5k + 24k$

$\vec{B} \times \vec{A} = 5\hat{i} + 18\hat{i} + 29k$

Answer: D

19. The coordinate of a particle in meters is given by $x(t) = 25t - 3t^3$, where the time t is in second. At what value t will the particle become momentarily at rest?

A. 2.78s B. 1.67s C. 0.6s D. 0.36s

Solution:

$x(t) = 25t - 3t^3$

The velocity of the particle at any time 't' is:

$$V = \frac{dx}{dt} = \frac{d(25t - 3t^3)}{dt}$$

$V = 25 - 9t^2$

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$\Rightarrow V = -9t^2 + 25$

To find the time at which the particle comes to rest i.e., the velocity is zero.

$V = -9t^2 + 25$

$O = -9t^2 + 25 \Rightarrow 9t^2 = 25$

$$t^2 = \frac{25}{9}$$

$$t = \sqrt{\frac{25}{9}} = \frac{5}{3}\text{sec}$$

$t = 1.67$ seconds

Answer: B

20. A woman driving a 2000 kg car along a level road at 30 m/s takes her foot off the gas to see how far her car will roll before it slows to a stop. She discovers that it takes 150m. What is the average force of friction acting on the car?

A. 9000N B. 6000N C. 3000N D. 400N

Solution:

$m = 2000$ kg ; $V_i = 30$ m/s; $V_f = 0$; $S = 150$m

$F_{av} = ?$

The acceleration of the car is

$$a = \frac{V_f^2 - V_i^2}{2s} \quad a = \frac{V_i^2}{2s} = \frac{(30\text{m/s})}{2\times 150\text{m}} = \frac{900\text{m}^2/\text{s}^2}{300\text{m}}$$

$a = 3$ m/s^2

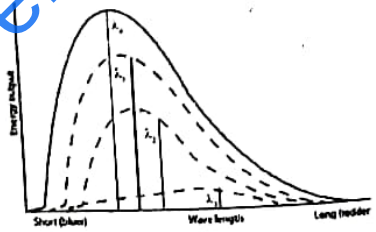
Then From Newton's second law, the force is

$F = ma = 2000$ kg $\times$ 3 m/s^2

$F = 6000$ N

Answer: B

21. The energy spectrum emitted by a black body at different temperature is as shown in the figure below. Which of the following statements is NOT correct about the black body energy spectrum?



A. $T_1 < T_2 < T_3 < T_4$

B. $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$

C. Black body radiate energy over the entire electromagnetic spectrum.

D. The wavelength corresponding to the maximum intensity with temperature.

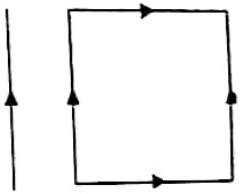
Solution:

As shown in the figure the energy of the spectrum varies directly with the temperature. The energy depends on the intensity of light and frequency and the maximum energy corresponds to the shorter wave length.

Answer: D

22. A rectangular loop carrying a current is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown in figure, the loop will

A. Remain stationary
B. Move toward the wire
C. Move away from the wire
D. Rotate about an axis parallel to the wire



Solution:

The figure show, the direction of the current in the wire and the side of the loop and the wire are parallel, there is attraction between them the loop attracts the wire and so does the wire i.e., the loop moves towards the wire.

Answer: B

23. Which device uses the principle of electromagnetic induction for its operation?

A. AC generator C. Cathode ray tube
B. Moving coil meter D. Light emitting diode

Solution:

AC generator works with the principle of electromagnetic induction.

Answer: A

24. An electron enters a region of uniform electric field of 5×10^3 N/C. What is the force on the electron?

A. 8×10^{-22} N C. 8×10^{-16} N
B. 5.3×10^{-22} N D. 1.88×10^{22} N

Solution:

$E = 5 \times 10^3$ N/C

$Q = e = 1.6 \times 10^{-19}$ C

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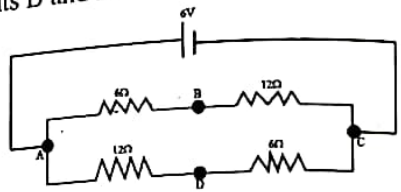
$F = EQ = 5 \times 10^3 N/C \times 1.6 \times 10^{-19} C = 8 \times 10^{-16} N$ is the force on the electron.

Answer: C

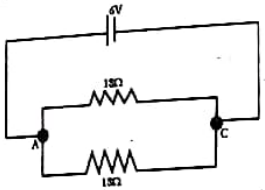
25. In the circuit shown below, what is the potential difference between points B and D?



A. 1V B. 2V C. 3V D. 4V

Solution:

The circuit can be reduced as:



The total resistance is $\frac{1}{R_T} = \frac{1}{18\Omega} + \frac{1}{18\Omega}$

$\Rightarrow R_T = \frac{2}{18\Omega} \Rightarrow R_T = 9\Omega$ are the total current is

$I = \frac{V}{R} = \frac{6V}{9\Omega} = \frac{2}{3}A$ and the current through each of the 18Ω is

$\frac{2}{3}A$ so the current through each resistor in the current is $\frac{1}{3}A$.

The voltage across point B is $V_B = IR = \frac{1}{3}A \times 6\Omega = 2V$ and

voltage across point D is $V_D = IR = \frac{1}{3}A \times 12\Omega = 4V$

The potential difference between B and D is:

$\Delta V = V_D - V_B = 4V - 2V = 2V$

Answer: B

26. A 1.5 kg particle moves in the xy-plane with velocity $v = (4.2\hat{i} - 3.6\hat{j})$ m/s. What is the angular momentum when its position vector is $r = (1.5\hat{i} + 2.2\hat{j})$ m?

A. $-22\hat{k}\ kgm^2/s$ C. $-14.6\hat{k}\ kgm^2/s$

B. $14.6\hat{k}\ kgm^2/s$ D. $22\hat{k}\ kgm^2/s$

Solution:

m = 1.5 kg; $\vec{v} = (4.2\hat{i} - 3.6\hat{j})$ m/s; $\vec{r} = (1.5\hat{i} + 2.2\hat{j})$m

$\vec{L} = ?$

The angular momentum of a particle of mass m is given by:

$\vec{L} = m(\vec{r} \times \vec{v}) = 1.5\ kg\ (1.5i + 2.2j) \times (4.2i - 3.6j)$

$= 1.5\ kg\ (-5.4k + (-9.24k))m^2/s$

$= 1.5\ kg\ (-14.64\hat{k})\ m^2/s = -21.96\ kkgm^2/s \approx -22\ \hat{k}\ kg\ m^2/s$

Answer: A

27. A very large open tank is full of oil whose density is 800 kg/m^3. If the absolute pressure at the bottom of the tank is 1.81×10^5 Pa, then what is the depth of the oil? (Note that atmospheric pressure is 1.01×10^5 Pa).

A. 100.0m B. 75.0m C. 10.0m D. 9.5m

Solution:

$\rho = 800\ kg/m^3$, $P_{tot} = 1.81 \times 10^5$ Pa

$P_{atm} = 1.01 \times 10^5$Pa

h = ?

The total pressure at the bottom of the tank is:

$P_{total} = P_{atm} + \rho gh \Rightarrow \rho gh = P_{tot} - P_{atm}$

$$h = \frac{P_{tot} - P_{atm}}{\rho g} = \frac{1.8\times10^5 Pa - 1.0\times10^5 Pa}{800\ kg/m^3 \times 10\ m/s^2}$$

$$= \frac{8\times10^4 Pa}{8\times10^3 kg/m^2s^2}$$

h = 10m

Answer: C

28. The length of an aluminum rod at a temperature of 20°C is 3m. What will be the ratio of change in length of the rod to its original length if its temperature is raised to 120°C? The coefficient of linear expansion of Aluminum is $2.31 \times 10^{-5} K^{-1}$.

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A. 23% B. 3.2% C. 3% D. 0.23%

Solution:

Initial (original) length $\ell_o = 3m$

Initial temperature $T_o = 20°C$

Final temperature $T_f = 120°C$

$\alpha = 2.31 \times 10^{-5} K^{-1}$

The change in length of the rod is

$\Delta\ell = \ell_o \times \Delta T = 3m \times 2.31 \times 10^{-5} k^{-1} \times 100°C$

$\Delta\ell = 0.00693$ m

The ratio of $\Delta\ell$ to ℓ_o is: $\frac{\Delta\ell}{\ell_o} = \frac{0.00693}{3} = 0.00231$

In percentage it becomes

$\frac{\Delta\ell}{\ell_o} \times 100\% = 0.00231 \times 100\% = 0.231\%$

Answer: D

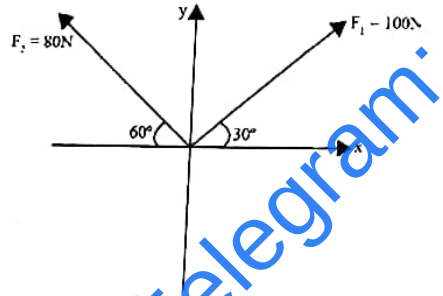
29. A 2 kg object experiences two concurrent forces of 100N at 30° above the positive x-axis 80N at 60° above the negative x-axis. What is the magnitude of acceleration of the object?

A. 68 m/s² B. 64 m/s² C. 48 m/s² D. 46 m/s²

Solution:

m = 2 kg; P_1 = 100N at 30° above positive x

F_2 = 80N at 60° above negative x



The components of the forces are

$F_x = P_{1x} + F_{2x} = F_1 \cos 30 - F_2 \cos 60° = 100\mu\ (0.86) - 80\mu\ (0.5)$

$F_x = 86.6 - 40 = 46.6N$

$F_y = F_{1y} + F_{2y} = F_1 \sin 30 + F_2 \sin 60 = 100\mu\ (0.5) + 80\mu(0.86)$

$F_2 = 50N + 68.8N = 118.8N$

The magnitude of the resultant force is

$|F| = \sqrt{F_x^2 + F_y^2} = \sqrt{(46.6N)^2 + (118.8N)^2}$

$F = 127.5\mu$ then the acceleration is $a = \frac{F}{m} = \frac{127.6\mu}{2m}$

$= 63.8\ m/s^2 \approx 64\ m/s^2$

Answer: B

30. A driver moving at a constant speed of 20 m/s sees an accident up ahead and hits the brakes. If the car decelerates at a constant rate of 05 m/s^2, how far does the car go before it comes to a stop?

A. 120m B. 100m C. 40m D. 20m

Solution:

$V_i = 20$ m/s, $V_f = 0$, $a = -5\ m/s^2$, s =?

The distance moved in the car before coming to rest is given by the equation of uniformly accelerating motion

$s = \frac{V_f^2 - V_i^2}{2a} = \frac{0 - (20m/s)^2}{2x - 5m/s^2} = \frac{-400m^2/s^2}{-10m/s^2}$

s = 40m

Answer: C

31. A series RLC circuit has a resistance of 45.0Ω and an impedance of 75.0Ω. What average power is delivered to this circuit when $\Delta V_{rms} = 210V$?

A. 573W B. 353W C. 255W D. 247W

Solution:

R = 45Ω, Z = 75Ω, $\Delta V_{rms} = 210V$, P_{av} =?

First the root mean square current is

$I_{rms} = \frac{V_{rms}}{Z} = \frac{210V}{75\Omega} = 2.8A$

$I_{rms} = 2.8A$

The average power is given by $P_{av} = (I_{rms})^2 \times R = (2.8A)^2 \times 45\Omega = 352.8W$

$P_{av} \approx 353$ W

Answer: B

32. A container resting on the floor of an elevator contains an incompressible fluid of density ρ. When the elevator has an upward acceleration of magnitude a then what will be the pressure difference between two points in the fluid separated by a vertical distance Δh?

A. ρaΔh B. ρ(g + a)Δh C. ρgΔh D. ρgaΔh

Solution:

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Density of fluid = ρ acceleration of elevator = a distance b/n points = Δh when the elevator accelerates upward the net force is $F_{net} = F' - mg$ when F' is the force on the fluid and mg – is its weight $F_{net} = ma$; $F_{net} = F' - mg \Rightarrow ma = F' = ma + mg$; $F' = m(a + g)$ but $m = \rho v = \rho A\Delta h$, $F' = \rho A\Delta h (a + g)$ then the pressure is $P = \frac{F'}{A} = \frac{\rho A\Delta h(a+g)}{A}$; $P = \rho\Delta h (a + g)$.

Answer: B

33. The planet Jupiter is about 318 times as massive as Earth, yet on its surface you would weigh only about 2.4 times your weight on earth. Why is this happened?

A. Your mass is 132.5 times less on Jupiter.
B. You are 132.5 times more weightless there.
C. Jupiter is significantly farther from the sun.
D. Jupiter's radius is about 11.5 times the Earth's radius.

Solution:

$M_{Jupiter} = 318\ Me$

$W_{earth} = 2.4\ W_{Jupiter}$

$\Rightarrow$ The acceleration due to gravity on Jupiter in 2.4 times that of the earth. i.e., $g_{jupiter} = 24\ g_{earth}$

$$g_{earth} = \frac{GM_e}{R_e^2} \ldots\ldots\ldots (i) \qquad g_{jupiter} = \frac{GM_{jupiter}}{R_{jupiter}}$$

$$g_{jup} = \frac{G(318)(Me)}{R_{jup}^2} \ldots (ii)$$

Taking the ratio of (ii) to (i)

$$\frac{g_i}{g_e} = \frac{\frac{G(318)Me}{R_{jup}^2}}{\frac{GM_e}{R_e^2}}$$

$$2.4 = \frac{318GMe}{R_{jup}^2} \times \frac{Re^2}{GMe}$$

$$2.4 = \frac{318Me}{R_{jup}^2} \times \frac{R_e^2}{GMe} \Rightarrow R_{jup}^2 = \frac{318}{2.4} Re^2$$

$$R_{jup} = \sqrt{\frac{318.}{2.4}}\, Re$$

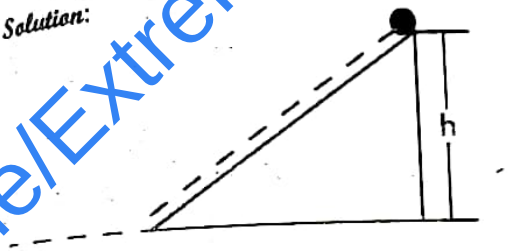
$R_{jup} = 11.5\ Re$

Answer: D

34. A uniform ball of radius r and mass m starts from rest and rolls down a frictionless incline of height h. When the ball reaches the ground, how fast is it going?

A. $\sqrt{\frac{10}{7}gh}$ B. $\sqrt{\frac{9}{10}gh}$ C. $\sqrt{\frac{10}{9}gh}$ D. $\sqrt{\frac{7}{10}gh}$

Solution:



When the sphere rolls down the frictionless inclined plane the total energy is conserved.

$E_{top} = E_{bottom}$

$mgh = E_{krot} + E_{k\ translational}$

$$mgh = \frac{1}{2}Iw^2 + \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{2}{5}mR^2\right)w^2 + \frac{1}{2}mv^2$$

$$= \frac{1}{5}mR^2w^2 + \frac{1}{2}mv^2$$

$$mgh = \frac{7}{10}mv^2$$

$$V^2 = \frac{10gh}{7}$$

$$V = \sqrt{\frac{10}{7}gh}$$

Answer: A

35. A particle of mass m is fired into a magnetic field of strength B at a speed v. The particle travels in a circular path inside the field

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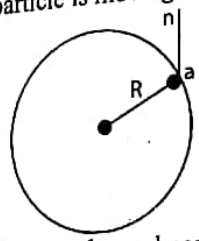
with a radius r. Which of the following expressions gives the magnitude of the charge on the particle?

A. $\frac{mr}{v^2B}$ B. $\frac{mv^2}{Br}$ C. $\frac{vB}{mr}$ D. $\frac{mv}{Br}$

Solution:

The particle is moving in a circular path.



The force on charged particle is given by:

$F_m = qvB$

q = charge; v = speed; B = magnetic field also. The force on the particle along a circle is equal to the centripetal force

$$F_C = \frac{mv^2}{R}$$

Equating the centripetal force and the magnetic force $F_m = F_C$

$$qvB = \frac{mv^2}{R}$$

$$\Rightarrow q = \frac{mv^2}{vBR} = \frac{mv}{BR//}$$

$\therefore$ The charge is $q = \frac{mv}{BR}$ **Answer: D**

36. An alternating current is represented by I (t) = (10.0 A) sin (100π t) what are the amplitude and frequency of the alternating current?

A. 100 A, 75 Hz C. 10.0 A, 50 Hz
B. 10.0 A, 100 Hz D. 5.0 A, 25 Hz

Solution:

I (t) = 10.0A sin (500 Hz) the amplitude of the current is 10A to find the frequency the equation is analogues to I (t) = A sin (w t) = A sin (2π ft) $\Rightarrow 2\pi = 100\pi \Rightarrow f = \frac{100}{2} = 50$ HZ

Answer: C

37. If two particles have equal momenta, then which one of the following statements is TRUE about their kinetic energies?

A. Their kinetic energies are always equal.
B. Their kinetic energies are never equal.
C. Their kinetic energies are equal when their masses are the same.
D. Their kinetic energies are equal when the speed of one of the particle is half the speed of the other particle and their masses are the same.

Solution:

Let the particles are 1 and 2. It is given that they have the same momentum.

$$P_1 = P_2$$

If they have the same mass, it is true that they have the same speed. Now, the particles have the same mass and speed therefore they have the same kinetic energy. **Answer: C**

38. What is the capacitance of a capacitor that should be connected with 2 other capacitors having of 4μF and 8μF all in series to produce an effective capacitance of $\frac{24}{11}$μF?

A. 12μF B. 10μF C. 8μF D. 4μF

Solution:

Let $C_1 = 4\mu F$, $C_2 = 8\mu F$, C_3 is unknown

$$C_{total} = \frac{24}{11}\mu F$$

If all the three capacitors are connected in series.

$$\frac{1}{C_{tot}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{\frac{24}{11}\mu F} = \frac{1}{4\mu F} + \frac{1}{8\mu F} + \frac{1}{C_3}$$

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$$\frac{11}{24}\mu F = \frac{3}{8}\mu F + \frac{1}{C_3}$$

$$\frac{1}{C_3} = \frac{11}{24}\mu F - \frac{3}{8}\mu F = \frac{2}{24}\mu F$$

$$\Rightarrow C_3 = \frac{24}{2}\mu F = 12\mu F$$

Answer: A

39. In Young's double slit experiment both the separation between the slits the distance between the slits and the screen are halved then what is the fringe width?
A. Halved B. unchanged C. Zero D. doubled

Solution:
Let S_1 is the slit separation initially D_1 is the distance between the screen and the slits W_1 is the initial fringe width and S_2, D_2 and W_2 are the quantities after changing. The Yound's double slit experiment provides.

$$W = \frac{\lambda s}{D}, \lambda = \text{wave length}$$

$$\text{So, } W_1 = \frac{\lambda S_1}{D_1}$$

$$W_2 = \frac{\lambda S_2}{D_2} \text{ but } D_2 = \frac{1}{2}D_1$$

$$W_2 = \frac{\lambda S_2}{D_2} = \frac{\lambda \frac{S_2}{1}}{\frac{D_1}{2}} = \frac{\lambda 2 S_1}{2D_1}$$

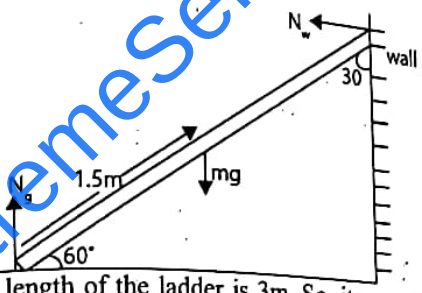
$$W_2 = \frac{2\lambda S_1}{2D_1} = \frac{\lambda S_1}{D_1} = W_1$$

Therefore, the fringes width is not changed. **Answer:** B

40. A ladder of length 3m and mass 20kg leans against a smooth, vertical wall so that the angle between the wall and the ladder is 30°. What are the magnitudes of the normal forces at the contact point on the wall (N_w) and at the contact point on the ground (N_8)?

A. $N_w = 0N, N_8 = 200N$
B. $N_w = 200N, N_8 = 0N$
C. $N_w = 57.7N, N_8 = 200N$
D. $N_w = 200N, N_8 = 57.7N$

Solution:
The ladder can be drawn as below:



The length of the ladder is 3m. So, its center of mass is at 1.5m from the ground. The ladder is in equilibrium, it satisfies the first and second conditions of equilibrium.
First condition:

$$\sum F = 0 \Rightarrow Ng - mg = 0$$

$$N_g = mg = 20\text{ kg} \times 10\text{ m/s}^2$$

$$N_g = 200N$$

Second condition:

$$\sum \tau = 0 \Rightarrow N_w (3m) \sin 60 - mg\,(1.5m)(\cos 60) = 0$$

$$N_w = \frac{mg(1.5)(\cos 60°)}{3m \sin 6} = \frac{200N \times 1.5 \times 0.5}{3 \times 0.86}$$

$$N_w = 57.7N$$

Answer: C

41. Gas at a pressure P is contained in a vessel. If the masses of all the molecules are doubled and their speeds halved, then what will be the resulting pressure?

A. 2P B. $\frac{P}{3}$ C. $\frac{P}{2}$ D. 3P

Solution:
Initial condition pressure = P
Mass = m_1; speed = v_1
Final condition:
Pressure = P_2
Mass = $m_2 = 2m_1$

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Speed $= v_2 = \frac{v_1}{2}$

$P_2 = ?$

Assuming the volume of the gas is constant.

From kinetic theory of gases

$$PV = \frac{1}{3}mv^2$$

$$P_1V = \frac{1}{3}m_1v_1^2$$

$$\Rightarrow P_1 = \frac{1}{3}\frac{m_1v_1^2}{v} \text{ and } P_2V = \frac{1}{3}\frac{m_2v_2^2}{v} = \frac{1}{3}\frac{(2m_1)}{v}\left(\frac{v_1}{2}\right)^2$$

$$= \frac{1}{3}\left(\frac{2m_1 \times v_1^2}{4v}\right) = \frac{1}{3}\left(\frac{1}{2}\right)\left(\frac{m_1v_1^2}{v}\right)$$

$$P_2 = \frac{1}{2}P \Rightarrow P_2 = \frac{P}{2}$$

Answer: C

42. Refraction of a wave does NOT involve change in its

A. Speed
B. Wavelength
C. Frequency
D. Speed and wavelength

Solution:

During refraction:

- The speed of wave changes
- The wave length changes but the frequency remains constant.

Answer: C

43. A rocket moving in free space has a speed of 3×10^3 m/s relative to the earth. It engines are turned on, and fuel is ejected in a direction opposite the rocket's motion at a speed of 5×10^3 m/s relative to the rocket. What is the speed of rocket relative to the earth once the rocket's mass is reduced to half its mass before ignition?

A. 7×10^3 m/s
B. 5×10^3 m/s
C. 4×10^3 m/s
D. 2×10^3 m/s

Solution:

$V_{re} = 3 \times 10^2$ m/s; $V_{fr} = 5 \times 10^3$ m/s

The mass of the rocket reduced to half of its value using the principle of conservation momentum. $MV_1 = \frac{m}{2}V_2 = \frac{m}{2}V$

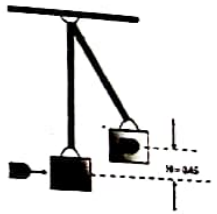
$$m(3 \times 10^3 \text{ m/s}) - \frac{m}{2}(5 \times 10^3 \text{ m/s}) = \frac{m}{2}V$$

$$m(3 \times 10^3 \text{ m/s}) = \left(\frac{m}{2} + \frac{m}{2}\right)(v + 5 \times 10^3 \text{ m/s})$$

$$\Rightarrow v = 2 \times 10^3 \text{ m/s}$$

Answer: D

44. A bullet of mass 10 gm, moving horizontally, strikes and embeddes itself in a box of mass 1 kg suspended from a light string as shown in the figure below. If the composite mass rises to a height, H = 0.45 m, then what is the speed of the bullet before collision?



A. 303 m/s
B. 300 m/s
C. 90.0 m/s
D. 90 m/s

Solution:

m_b = mass of butlet = 10 g = 0.01 kg

m_B = mass of block = 1 kg

H = 0.45 m

U_b = speed of bullet = ?

Using conservation of momentum

$P_{before} = P_{after}$

$$m_bU_b = (m_b + M_b)V \Rightarrow V_b = \frac{(m_b + M_B)V}{m_b} \text{ but}$$

$$V = \sqrt{2gh} = \sqrt{2 \times 10 m/s \times 0.45m}$$

V = 3 m/s

$$\text{Then } V_b = \frac{(m_b + M_B)V}{m_b} = \frac{(0.01kg + 1kg)3m/s}{0.01kg}$$

V_b = 303 m/s

Answer: A

45. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.012 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exists the nozzle?

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A. 63.2 m/s B. 46.8m/s C. 31.6m/s D. 23.4 m/s

Solution:

Diameter$_1$ = d_1 = 6.35 cm = 0.063 m

Flow rate Q = 0.012 m^3/s

d_2 = 2.20 cm

radius r_2 = 1.10 cm = 0.11 m

The flow rate is constant Q = AV

$$V = \frac{Q}{A} = \frac{0.012 m^3/s}{\pi(0.011m)^2}$$

V = 31.6 m/s

Answer: C

46. Suppose that a voltage surge produces 240V for a moment. By what percentage does the power output of 220-V light bulb increase?

A. 38% B. 36% C. 19% D. 18%

Solution:

Let P_1 is the power when V_1 = 240 and P_2 is theh power when V_2 = 220V

$$P_1 = \frac{V_1^2}{R} \text{ and } P_2 = \frac{V_2}{R}$$

$$P_1 = \frac{(240V)^2}{R}, P_2 = \frac{(220V)^2}{R}$$

$$\frac{P_1}{P_2} = \frac{\frac{(240V)^2}{R}}{\frac{(220V)^2}{R}}$$

$$\frac{P_1}{P_2} = \frac{(240V)^2}{(220V)^2} = \left(\frac{24}{22}\right)^2 = 1.19$$

$$\frac{P_1}{P_2} = 1.19$$

$P_1 = 1.19 P_2$

$P_1 = P_2 + 0.19 P_2$

The increased is 0.19 × 100% = 19%

Answer: C

47. If all the resistors in the circuit shown in the figure have equal resistances, and the current flowing into resistor B is 4A, then what is the current flowing into resistor G?

A. 16A B. 8A C. 2A D. 1A

Solution:

I_B = 4A

Since all the resistors have the same resistance. The current through B is equal to the current through D.

So $I_B = I_D$ = 4A

$I_G = I_B + I_D$ = 4A + 4A

I_G = 8A

Answer: B

48. A wire carries a steady current of 2.4A. A straight section of the wire is 0.75 m long and lies along the x axis within a uniform magnetic field, B = 1.6T along +z axis. If the current is in the −x direction, what is the magnetic force on the section of wire?

A. 2.88 $\hat{j}$ N B. 1.62 $\hat{j}$ N C. -1.62 $\hat{j}$ N D. -2.88 $\hat{j}$ N

Solution:

I = 2.4A; ℓ = 0.75 m along x axis; B = 1.6T along z axis

F =?

$F = I(\ell \times B)$

$F = 2.4A\,(0.75\,\hat{j} \times 1.6k) = 2.4A\,(1.2(-\hat{j})Tm = -2.88\,\hat{j}N$

Answer: D

49. Which one of the following changes to parallel plate capacitor would not increase energy stored in the capacitor at a fixed voltage across the plates?

A. Increasing the area of the plates.

B. Increasing the dielectric constant.

C. Decreasing the charges on the plates.

D. [illegible] the plates

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Solution:
Given: constant voltage

The energy stored in a capacitor is $E = \frac{1}{2}CV^2$

E increases as 'C' increase and 'C' increases as area A increase according to $C = \frac{\varepsilon_o A}{d}$.

- When a dielectric constant increases, the capacitance and also the energy increases.
- When the charge decrease the energy also decreases.
- Increasing the distance between the plates decrease the capacitance and also the energy.

Answer: D

50. A thin film of oil (n = 1.25) is located on a smooth we floor. When viewed perpendicular to the floor, the film reflects most strongly red light at 640 nm and reflects no blue light at 512 nm. How thick is the oil film?
A. 640 nm B. 612 nm C. 548 nm D. 512 nm

Solution:
$n = 1.25$, $\lambda_{red} = 640$ nm; $\lambda_{blue} = 512$ nm the thickness of the film

$$t = \frac{\lambda_o}{n} = \frac{640nm}{1.25}$$

$t = 512$ nm

Answer: D

1. The distance between two point charges is doubled. What will happen to the force of interaction between the charges?
A. The force will be doubled.
B. The force will be reduced by half.
C. The force will be reduced by a factor of $1/4$.
D. The force will not be affected.

Solution:
The electrostatic force between charged particle is inversely proportional to the square of the distance between them. $F \sim \frac{1}{r^2}$.

Let r_1 = initial distance and r_2 = final distance. Then $r_2 = 2r_1$.

Thus, $F_1 = \frac{kQ_1Q_2}{r_1^2}$ and $F_2 = \frac{kQ_1Q_2}{r_2^2} = \frac{kQ_1Q_2}{(2r_1)^2}$.

$$F_1 = \frac{kQ_1Q_2}{4r_1^2} = \frac{1}{4}F_1$$

Answer: C

2. What does the slope of a velocity – time graph represent?
A. Distance C. Displacement
B. Acceleration D. Speed

Solution:

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \longleftarrow \text{Slope}$$

Answer: B

3. Moving 2.5×10^{-6} coulomb of charge from point A to point B in an electric field requires 6.3×10^{-4} joule of work. The potential difference between points A and B is approximately.
A. 4.0×10^{-3}V C. 1.6×10^{-19}V
B. 2.5×10^{-2}V D. 1.0×10^{-14}V

Solution:
$Q = 2.5 \times 10^{-6}$C, $W = 6.3 \times 10^{-4}$J, $V_{AB} = ?$

$$V_{AB} = \frac{W}{Q} = \frac{6.3\times10^{-4}\,J}{2.5\times10^{-6}\,C} = 2.5\times10^{2}\,V$$

Answer: B

4. A transverse sinusoidal wave is travelling on a string. Which statement is correct a point on the string?
A. The point moves in the same direction as the wave.

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B. The point moves in simple harmonic motion with a different frequency than that of the wave.
C. The point moves in simple harmonic motion with the same angular frequency as the wave.
D. The point moves I uniform circular motion with a different angular speed than the wave.

Explanation:
The displacement of the particle in a string is perpendicular to the direction of the wave motion. **Answer:**

5. Two non zero vectors **D** and **E** have precisely equal magnitudes. For the magnitude of **D** + **E** to be 3 times larger than the magnitude of D – E, what must be the angle between D and E?
A. 30° B. 37° C. 53° D. 60°

Solution:
We have $|D| = |E|$ and $|D + E| = 3|D - E|$ squaring of magnitude:
$|D + E|^2 = (3|D - E|)^2$
$D^2 + E^2 + 2DE\cos\theta = 9D^2 + 9E^2 - 18DE\cos\theta \ldots (*)$
Since $|D| = |E| \Rightarrow D^2 = E^2$ replacing in (*)
$\Rightarrow D^2 + D^2 + 2D\cos\theta = 9D^2 + 9D^2 + 9D^2 - 18D^2\cos\theta$
$\Rightarrow 2D^2 - 18D^2 = -18D^2\cos\theta - 2D^2\cos\theta$
$-16D^2 = -20D^2\cos\theta \ldots$ simplifying

$$\cos\theta = \frac{16}{20} = \frac{8}{10} = 0.8$$

$\therefore\theta = \cos^{-1}(0.8) = 37°$ **Answer: B**

6. Which element is correct about the pressure under a fluid in a container?
A. It depends on the volume of the fluid.
B. It depends on the density of the fluid.
C. It depends on the shape of the container.
D. It is independent of the depth of the fluid.

Explanation:
The pressure under the fluid in a container depends on the density, depth of the fluid and the acceleration due to gravity. **Answer: B**

7. A capacitor made of parallel plates is charged by connecting it to a battery. The distance between the plates is much less than the surface area of each plate. After fully charged if the plates are disconnected from the battery and then pulled apart a little, what will happen to the total stored energy, the electric field, and the potential difference between the plates?
A. The electric field between the plates increases.
B. The potential difference between the plates decreases.
C. The total energy stored in the capacitor increases.
D. The electric field the potential difference and the stored energy remain the same.

Explanation:
When the capacitor plate is fully charged, the charge on the plates remains constant.

i) Capacitance: $C = \frac{\varepsilon_o A}{d}$

As the distance between the plates increases, the capacitance decrease.

ii) **The voltage (p.d):** $V = \frac{Q}{C}$

As the capacitance decrease the voltage increases. The potential difference increases linearly.

iii) **The electric field:** $E = \frac{v}{d} =$ remains the same. As d – decrease, v – increase.

iv) **Energy stored** in the capacitor: $U_{stored} = \frac{1}{2}CV^2$. The energy stored in the capacitor increase linearly with separation due to the work required to move the plates farther apart. **Answer: C**

8. Which of the following statements is true about the motion of a particle in a circular path?
A. The centripetal acceleration is constant if the particle's speed is constant.
B. The tangential acceleration can be perpendicular to the velocity of the particle.
C. The centripetal acceleration is always in the direction perpendicular to the velocity vector of the particle.
D. The acceleration is always perpendicular to the velocity vector of the particle.

Explanation:

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When a particle moving in a circular path:

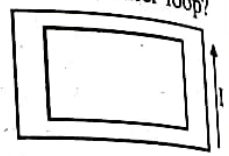
→ Its speed remains constant in magnitude but direction changes continuously.

→ Its centripetal acceleration is perpendicular to the velocity.

→ Its tangential acceleration is parallel to the velocity.

Answer: C

9. Two rectangular loops of wire lie in the plane of this paper as shown in Fig. 1. If the current I in the outer loop is counter clockwise and increases with time, which one of the following statements is correct about the current induced in the inner loop?



A. A current is induced in the inner loop in the clockwise direction.

B. A loop is induced in the inner loop in the counter clockwise direction.

C. The induced current in the inner loop creates a magnetic field inside the area bounded by the inner loop in the same direction of the magnetic field that the outer loop products in the same area.

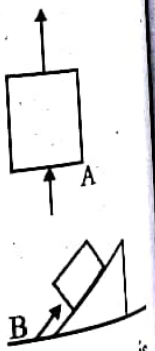
D. It is zero.

Explanation:

As the current I in the outer loop is increasing with time and counter clock wise, the induced current in the inner loop is flows in clock wise direction that opposes the change that caused the current.

Answer: A

10. Two men, A and B, are loading identical boxes onto a truck. A lifts his box straight up from the ground to the bed of the truck, whereas B slides his box up a rough ramp to the truck. Which statement is correct about the change in the potential energy on the box – Earth system?



A. A and B change the potential energy by the same amount.

B. A changes more potential energy than B.

C. B changes more potential energy than A.

D. None of the statements is necessarily true because the force of friction and the angle of the incline is unknown.

Explanation:

As the boxes moves the same height in loading onto the truck the changes in potential energy is the same in each cases. **Answer: A**

11. Two projectiles are fired at an angle of 30 degrees and the other at 60 degrees. The projectile to hit the ground first will be the one fired at (neglect air resistance).

A. 60 degrees

B. 30 degrees

C. Both hit at the same time.

D. Cannot be determined from the given information.

Solution:

The time taken to reach the ground is given by $t = \frac{2v_o \sin\theta}{g}$

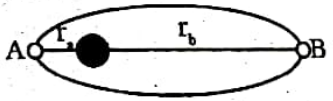
When, $\theta = 30°$, $t_1 = \frac{2v_o \sin 30°}{g} = \frac{v_o}{g}$

When, $\theta = 60°$, $t_2 = \frac{2v_o \sin 60°}{g} = 1.7\frac{v_o}{g}$

$\Rightarrow t_1 < t_2$, so the net that thrown at 30° hits the ground first.

Answer: B

12. A planet of mass m is circling a star in a elliptical orbit as shown below.



If its velocity at point A is 60,000 m/s, and $r_b = 8\ r_a$ what is its velocity at point B?

A. 7500 m/s B. 8,571 m/s C. 60,000 m/s D. 488,000 m/s

Solution:

Using Kepler's law of planetary motion "the principle of conservation of angular momentum"

$L_A = L_B \Rightarrow mr_aV_A = mr_bV_B$

$$V_B = V_A\left(\frac{r_a}{r_b}\right) = (60,000\text{m/s})\left(\frac{r_a}{8r_a}\right) = 7,500$$

Answer: A

13. A small radio receiver uses a batter that delivers a constant current of 25 m/A for 3 hours. Calculate the total charge delivered by the battery.

A. 1080000C C. 540

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B. 2700000C D. 270C

Solution:

$Q = It = (25 \times 10^{-3}A)(3 \times 3600\ sec) = 270\ C$ **Answer: D**

14. A cup of coffee at a temperature of 80°C is placed on a table in a 20°C room. What happens to be the thermal energy and molecular motion of the coffee when it is left on the table?

A. The thermal energy decreases and the molecules move more slowly.

B. The thermal energy increases and the molecules move more slowly.

C. The thermal energy decreases and the molecules move faster.

D. The thermal energy increases and the molecules move faster.

Explanation:

The coffee is at higher temperature. When placed on a table at lower temperature, the thermal energy of the coffee decreases and its molecular speed decreases. **Answer: A**

15. The vector $2\hat{i}+\hat{j}+\hat{k}$ is perpendicular to $\hat{i}-4\hat{j}+\lambda\hat{k}$ if λ is equal to:

A. 0 B. -1 C. 2 D. 3

Solution:

The scalar product of perpendicular vectors is zero:

$\Rightarrow(2i+j+k).(i-4j+\lambda k)=0$

$\Rightarrow 2-4+\lambda=0 \Rightarrow \lambda=2,$ **Answer: C**

16. A garden hose has an inside diameter 0d 16 cm. The hose can fill a liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm.

A. Water exits with a speed twice the speed inside the hose.

B. A 10 liter bucked can be filled in 10 seconds.

C. A 20 liter bucket can be filled in 40 seconds.

D. Water exits with a speed half the speed inside the hose.

Solution:

Inside diameter $d_1 = 16$ cm and $d_2 = 8$ cm. Thus $r_1 = 8$ cm, $r_2 = 4$ cm flow rate $= \dfrac{10\ liter}{20\ sec} = 0.5$ liter/sec.

Equation of continuity: $A_1V_1 = A_2V$

$$V = V\left(\frac{A_1}{A_2}\right) = V_1\left(\frac{\pi r_1^2}{\pi r_2^2}\right) = V_1\left(\frac{64}{16}\right) = 4V_1$$

Since the flow rate is 0.5 liter/sec constant everywhere:

- 10 liter bucket can be filled in 20 second.
- 20 liter bucket can be filled in 40 second.

17. A mass suspended from the end of 16 cm. The hose can fill a 10 litter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm. **Answer: C**

A. Water exists with a speed twice the speed inside the hose.

B. A 10 litter bucked can be filled in 10 seconds.

C. A 20 litter bucket can be filled in 40 seconds.

D. Water exists with a speed half the speed inside the hose.

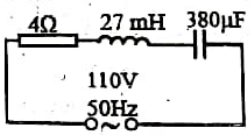
Solution:

$$\text{Frequency} = \frac{\text{number of oscillaiton}}{\text{Time taken}}$$

$$f = \frac{24}{36\ sec} = 0.67Hz$$

$$T = \frac{36\ sec}{24} = \frac{3}{2} = 1.5\ sec$$ **Answer: B**

18. Determine the total impedance for the circuit shown below.



A. 0Ω B. 4Ω C. 27Ω D. 31Ω

Solution:

$R = 4\Omega$, $L = 27mH$, $C = 380\mu F$, $f = 50$ Hz, $V = 110V$

- $X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi fC} = \dfrac{1}{2\pi(50Hz)(380\times10^{-6}F)} = 8.4\Omega$
- $X_L = \sqrt{R^2+(X_L-X_C)^2} = \sqrt{16+(8.5-8.4)^2} = 4\Omega$

Answer: B

19. A hoop of mass M and radius R is at rest at the top of an incline plane of height h. The hoop rolls down the plane without slipping. When the hoop reaches bottom, its angular momentum about its center of mass is:

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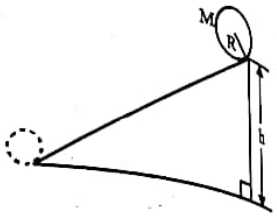
A. Mgh B. $MR\sqrt{2gh}$ C. $MR\sqrt{gh}/2$ D. $MR\sqrt{gh}$

Solution:

$E_{top} = E_{bottom}$



$$Mgh = \frac{1}{2}MV^2 + \frac{1}{2}I\omega^2$$

But $I = MR^2$, and $\omega = \frac{V}{R}$

$$Mgh = \frac{1}{2}M(\omega^2R^2) + \frac{1}{2}MR^2\omega^2$$

$$gh = \omega^2R^2 \Rightarrow\Rightarrow \omega\frac{\sqrt{gh}}{R}$$

$$L = I\omega = MR^2\frac{\sqrt{gh}}{R} = MR\sqrt{gh}$$

Answer: D

20. What is the resistance of a copper cable if it has a cross - sectional area of 1 cm^2 and a length of 2 km? The resistivity of copper is $2 \times 10^{-8}\Omega m$?

A. 0.0004Ω B. 0.04Ω C. 0.04Ω D. 0.4Ω

Solution:

$R = 1\ cm^2 = 1 \times 10^{-4}m^2$, $L = 2\ km = 2 \times 10^3\ m$ and $\sigma = 2 \times 10^{-8}\Omega m$

$$R = \frac{\rho}{A} = \frac{(2\times10^{-8}\Omega m)(2\times10^{3}\ m)}{1\times10^{-4}\ m^2} = 0.4\Omega$$

Answer: D

21. When a load of 500 kg is hanging from a steel wire of length 3m and cross sectional area 0.20 cm^2, the wire stretches beyond its no – load length. If the Young's modulus for the wire made of steel is $Y = 2 \times 10^{11}Pa$, what is the change in length of the wire beyond its no load length?

A. 3.75 mm B. 2.75mm C.1.75mm D. 0.75mm

Solution: Load = 500 kg, $\ell = 3m$, $A = 0.20\ cm^2 = 0.2 \times 10^{-4}m^2$ and $Y = 2 \times 10^{11}Pa$, $\Delta\ell = ?$

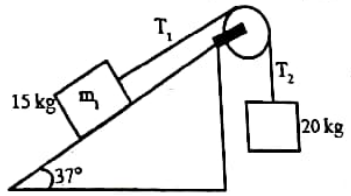
Young's modules, $Y = \frac{\text{stress}}{\text{strain}} = \frac{\frac{F}{A}}{\frac{\Delta f}{\ell}} = \frac{F\ell}{A\Delta\ell}$

$$\Rightarrow \Delta\ell = \frac{F\ell}{Y.A} = \frac{(500kg)(10m/s^2)(3m)}{(2\times10^{11}Pa)(0.2\times10^{-4}m)} = 3.75\times10^{-3}m$$

$\Delta\ell = 3.75 \times 10^{-3}m = 3.75$ mm

Answer: A

22. As shown in figure 5, two blocks are connected by a string of negligible mass passing over a pulley of radius r = 0.2m and moment of inertia I. The block on the frictionless incline is moving with a constant acceleration a = 2.0 m/s^2 up along the incline.



What is the moment of inertia of the pulley?

A. 0.8 kgm^2 B. 1.25 kgm^2 C. 4 kgm^2 D. 40 kgm^2

Solution:

Applying Newton's second law.

For m_2: $T_2 - m_2g = -m_2a$

$\Rightarrow T_2 = m_2g - m_2a = 20\ kg\ (10\ m/s^2 - 2m/s^2) = 160N$

For m_1: $T_1 - m_1\ g\sin 37° = m_1a$

$\Rightarrow T_1 = m_1(g\sin 37° + a) = 15\ kg\ [10(0.6)m/s^2 + 2m/s^2] = 120N$

The net torque on the pulley is:

$\tau_{net} = (T_2 - T_1)r$ and also $\tau = I\alpha$

$$I = \frac{(T_2 - T_2)r}{\alpha} = \frac{(T_2 - T_1)r}{a/r} = \frac{(T_2 - T_1)r^2}{a}$$

$$I = \frac{(160N - 120N)(0.2m)^2}{2m/s^2} = 0.8kgm^2$$

Answer: A

23. A table – tennis ball which is spherical in shape has a diameter of 6 cm and average density to 0.400 g/cm^3. What force is required to hold it completely submerged under water?

A. 0.68N down ward C. 1.13N downward

B. 0.68N upward D. 1.13N upward

Solution:

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$F + W = F_b \Rightarrow F = F_b - W$

$F = \rho_f V_{fg} - \rho_o V_{og} = gV_o(\rho_f - \rho_o)$, since $V_f = V_o$

$\Rightarrow F = (10\ m/s^2)\left[\frac{4}{3}\pi \times (0.03m)^3\right][1000kg/m^3 - 400kg/m^3]$

$F = (10)\left(\frac{4}{3}\right)(3.14)(0.03m)^3(600\ kg/m^3) = 0.68N$ downward

Answer: A

24. A viewing screen is separated from a double slit by 4.80m. The distance between the two slits is 0.30 nm. Monochromatic lights is directed toward the double slit and forms an interference pattern on the screen. What is the wave length of the lights?
A. 1250nm B. 1875nm C. 2500nm D. 5625nm

Solution:

D = 4.8nm, d = 0.03mm = 3×10^{-5}m and W = 4.5cm = 4.5×10^{-2}m

Condition for dark – fringes in double slit interference is d sin

$\theta = \left(m + \frac{1}{2}\right)\lambda, m = 0, +1, +2 \ldots$

For small: $\sin\theta = \tan\theta = \frac{W}{D}$ and m = 1

$\frac{Wd}{D} = \left(1 + \frac{1}{2}\right)\lambda \Rightarrow \lambda = \frac{2Wd}{3D}$

$\lambda = \frac{2(4.5 \times 10^{-2}m)(3\times10^{-5}m)}{(3)(4.8m)} = 1.875\times10^{-7}m = 1875\times10^{-8}m$

25. A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5 cm in diameter. If the pressure of the water in the large pipe is 80 kPa and the pressure in the smaller pipe is 60 kPa, at what rate does the water flow through the smaller pipe?
A. 1.63 m/s B. 6.53 m/sC. 13.6m/s D. 42.67 m/s

Solution:

$d_1 = 10$ cm, $d_2 = 5$ cm, $P_1 = 80$ kPa, $P_2 = 60$ kPa

Equation of continuity

$A_1V_1 = A_2V_2 \Rightarrow V_2 = V_1\left(\frac{A_1}{A_2}\right) = V_2\left(\frac{100}{25}\right) = 4V_1$

Applying Bernoulis principle

$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2 \Rightarrow P_1 - P_2 = \frac{1}{2}\rho(V_2^2 - V_1^2)$

$80{,}000\ Pa - 60{,}000\ Pa = \frac{1}{2}(1000kg/m)(14V_1)^2 - V_1^2$

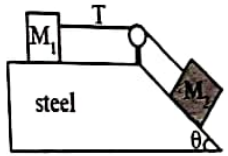
$20{,}000Pa = (7500\ kg/m^3)V_1^2$

$V = \frac{20{,}000}{7500} = 2.7m^2/s^2 \Rightarrow V = \sqrt{2.7m^2/s^2} = 1.63m/s$

But $V_2 = 4V_1 = 4(1.63\ m/s) = 6.53\ m/s$

Answer: B

26. An aluminum block of mass $m_1 = 3$ kg and a copper block of mass $m_2 = 5$ kg are connected by a light string over a frictionless pulley. They sit on a steel surface as shown in Figure 4, where θ = 53°. The coefficient of kinetic friction between the steel surface and each block is 0.3. if the blocks are released from rest. What is the tension T in the string?
A. 29.40N
B. 17.25N
C. 12.25N
D. 8.60N



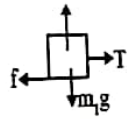
Solution:

Applying Newton's second law for each mass:

i) For M_1, $T - f = m_1a \Rightarrow T = m_1a + f$

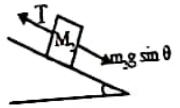
$T = m_1a + \mu m_1g$

$T = 3a + 9 \ldots (1)$



ii) For M_1

$m_2g\sin\theta - T - f = m_2a$, but $f = \mu F_N = \mu mg\cos\theta$



$m_2g\sin53° - T - \mu m_2g\cos53° = m_2a$

$(5\ kg)(10\ m/s^2)(0.8) - T - (0.3)(5\ kg)(10\ m/s^2)(0.6) = 5a$

$40 - T - 9 = 5a \Rightarrow T = 31 - 5a \ldots (2)$

Equating equation (1) and (2)

$\begin{cases} T = 3a + 9N \\ T = -5a + 31N \end{cases} \Rightarrow 3a + 9N = -5a + 31N$

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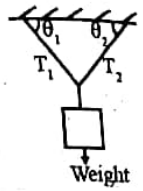
$8a = 22N \Rightarrow a = \frac{22}{8} m/s^2 = 2.75 m/s^2$

To find tension: $T = 9N + 3a = 9N + 3(2.75)N = 17.25N$

Answer: B

27. An object of mass 3 kg hangs from three light inextensible wires as shown in Figure 6. Two of the wires make angles $\theta_1 = 53°$ and $\theta_2 = 37°$ with the horizontal. If the system is equilibrium, what will be the tension T_1 and T_2 in the wires?

A. $T_1 = 18N$ and $T_2 = 24N$
B. $T_1 = T_2 = 18N$
C. $T_1 = T_2 = 24N$
D. $T_1 = 24N$ and $T_2 = 18N$



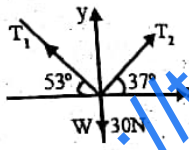
Solution:

$\sum F_x = 0, T_1 \sin 53° + T_2 \sin 37° - 30N = 0$

$\sum F_x = 0, T_1 \cos 53° - T_2 \cos 37° = 0$

From these two equation

$0.8T_1 \div 0.6T_2 = 30$ and



$0.6T_1 = 0.8T_2 \Rightarrow T_2 = \frac{6}{8}T_1 = \frac{3}{4}T_1$

$0.8T_1 + \frac{3}{4}(0.6)T_1 = 30 \Rightarrow T_1 = 24N$ and $T_2 = 18N$

Answer: D

28. An object of mass ms is hung from a string and set into oscillation. The frequency of the oscillation is measured and recorded as f. If the object of mass ms is replaced with an object of mass 2m and set into oscillation, what will be the frequency of the motion?

A. $\frac{f}{\sqrt{2}}$ B. f C. $(\sqrt{2})f$ D. 2f

Solution:

The frequency of spring mass – system:

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

If $m \to 2m$; $f' = \frac{1}{2\pi}\sqrt{\frac{k}{2m}} = \frac{1}{\sqrt{2}}\left(\frac{1}{2\pi}\sqrt{\frac{k}{m}}\right) = \frac{f}{\sqrt{2}}$ **Answer: A**

29. A placekicker must kick a football from a point which is at a distance of 36.0 m from the goal. When kicked, the ball leaves the ground with a speed of 20.0 m/s at an angle of 53.0° to the horizontal. If the ball hits the crossbar of the goal at a height h and bounces back what will be the height of the crossbar?

A. 2.45m B. 2.85 C. 3.00m D. 3.15m

Solution:

The equation of motion of a ball from the origin with an initial speed v_o at an angle θ with respect to the horizontal are:

(i) $x = (v_o \cos\theta)t \Rightarrow t = \frac{x}{v_o \cos\theta}$

$t = \frac{36m}{(20m/s)\cos 53°} = \frac{36m}{(20)(0.6)m/s} = 3sec$

(ii) The height of the cross bar of the goal:

$h = (V_o \sin\theta)t - \frac{1}{2}gt$

$h = [(20\ m/s)\sin 53°](3\ sec) - \frac{1}{2}(10\ m/s^2)(9\ sec)$

$h = 48m - 45m = 3m$ **Answer: C**

30. A body moving with constant acceleration covers the distance between two points 60m apart in 5s. It velocity as it passes the second points is 15 m/s. What is the acceleration?

A. $3\ m/s^2$ B. $2.4 m/s^2$ C. $1.8\ m/s^2$ D. $1.2\ m/s^2$

Solution: $V_{av} = \frac{s}{t} = \frac{60m}{5sec} = 12m/s$

Similarly, V_{av} is expressed as

$V_{av} = \frac{V_A + V_B}{2} \Rightarrow 12m/s = \frac{V_A + 15m/s}{2}$

$V_{av} = 24\ m/s - 15\ m/s = 9\ m/s$

Acceleration: $a = \frac{V - V}{t} = \frac{15m/s - 9m/s}{5sec} = 1.2 m\,s^2$

Answer: D

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31. Which of the following combinations of properties would be most desirable for a heat sink in a machine?
A. High specific heat and conductivity.
B. Low specific heat and high conductivity.
C. High specific heat and high conductivity.
D. Low specific heat and low conductivity. **Answer: A**

32. A 6.00V battery is connected across the primary coil of a transformers having 50 turns. If the secondary coil of the transformer has 100 turns, what voltage appears across the secondary?
A. 24.0V B. 12.0V C. 6.00V D. 3.00V

Solution: Transformer equation: $\frac{N_s}{N_P} = \frac{V_s}{V_P}$

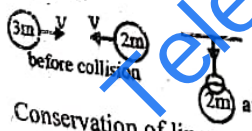
$$\Rightarrow V_s = \frac{N_s V_P}{N_P} = \frac{(100)(6V)}{50} = 12V$$ **Answer: B**

33. The change in entropy is zero for:
A. Reversible isobaric processes.
B. Reversible adiabatic processes.
C. Reversible isothermal processes.
D. Reversible processes during which no work is done.

Solution: In reversible adiabatic process, the change in entropy is zero $\Delta s = \frac{Q}{T} = \frac{0}{T} = 0$ **Answer: B**

34. Two particles with masses 2m and 3m are moving toward each other along the x axis with the same initial speeds v. Particle 2m is travelling to the left, and the particle 3m is traveling to the right. They undergo an electric glancing collision such that particle 2m is moving in the negative y direction after the collision. What are the x component of the final velocity of particle 3m and the kinetic energy of particle 2m, respectively?
A. 0.33V and 0.7 mV^2 C. 0.67V and 0.70mV^2
B. 0.33V and 1.40 mV^2 D. 0.67V and 1.4mV^2

Solution:



Conservation of linear momentum of x – component.

$(P_x)_{before} = (P_x)_{after}$

$$(3m)v - (2m)V = (3m)V_x + \cancel{(2m)V_x} = 0$$

$$mV = (3m)V_x \Rightarrow V_x = \frac{mV}{3m} = \frac{1}{3}V = 0.33V$$ **Answer: B**

35. Two separate coherent sources produce waves whose wavelengths are 0.10m. Which statement is correct about points that are 3.2m away from one source?
A. Constructive interference occurs if the points are 3.0m from the other source.
B. Constructive interference occurs if the points are 3.05m from the other waves.
C. Constructive interference occurs if the points are 2.95m from the other source.
D. Destructive interference occurs if the points are 3.2m from the other source.

Solution:
Since $\lambda = 0.1$ path length difference of construction interference:

$\ell_1 - \ell_2 = m\lambda$, where, $m = 0, 1, 2, 3, \ldots$ $\ell_1 - \ell_2 = \frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda, \ldots$

for $\lambda = 0.1 \Rightarrow \ell_1 - \ell_2 = 0.05, 0.15, 0.25, \ldots$

A. Let $\ell_1 = 3.2m$, $\ell_2 = 3m$, $\ell_1 - \ell_2 = 3.2 - 3 = 0.2$
∴Constructive interference

B. Let $\ell_1 = 3.2m - 2.95m$, $\ell_2 = 3.05$;
$\ell_1 - \ell_2 = 3.2m - 3.05m = 0.15m$
∴Destructive interference

C. $\ell_1 - \ell_2 = 3.2m - 2.95m = 0.25m \leftarrow$ destructive

D. $\ell_1 - \ell_2 = 3.2m - 3.2m = 0 \leftarrow$constructive **Answer: A**

36. A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle of 37° with the direction of the field. When the magnetic field is increased uniformly from 200 mT to 600 mT in 4.00 s an emf of magnitude 160.0mV is induced in the coil. What is the total length of the wire in the coil?
A. 100m B. 200m C. 300m D. 400m

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$$\varepsilon = NA\frac{\Delta B}{\Delta t}\cos\theta = \frac{NA(B_2 - B_1)\cos 37°}{\Delta t}$$

$$160mV = \frac{(50)(A)(600mT - 200mT)(0.8)}{4\,sec}$$

$$A = \frac{16}{400}m^2 \text{ but } A = L^2 = \frac{16}{400}m^2 \Rightarrow L = \frac{4}{20}$$

Total length = perimeter of square = 4NL

$$\therefore \text{Total length} = (450)\left(\frac{4}{20}\right) = 40m$$

37. The half – time of radium – 224 is 3.5 days. What fraction of the sample remains undecayed after fourteen days?

A. ½ B. ¼ C. 1/8 D. 1/16

Solution:

Half – life; $t_{\frac{1}{2}} = 3.5$ days and $t = 14$ days

Decay constant: $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{.0.693}{3.5\ days}$

Sample after, $t = 14$ days; $N = N_e e^{-\lambda t}$

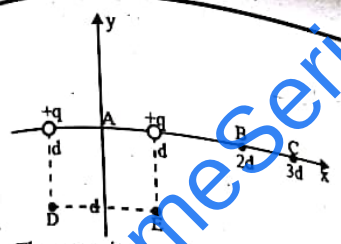
$$\Rightarrow \frac{N}{N_o} = e^{-\lambda t} = e^{\left(\frac{-0.693}{3.5}\right)} = (e^{-0.693})^{\left(\frac{14}{3.5}\right)}$$

$$\frac{N}{N_o} = \left(\frac{1}{2}\right)^4 \text{; recall that: } e^{-\ln 2} = e^{-0.693} = \frac{1}{2}$$

$$\frac{N}{N_o} = \frac{1}{16}$$

Answer: D

38. Two positive charges of magnitude q are such a distance d from the origin A of a coordinate system as shown below.



The magnitude of the electric field is minimum at point.

A. A B. B C. C D. D

Explanation:

An electric field created by a charge always points away from positive charge and toward a negative charge. The magnitude of the electric field that (+q)an d(+q) produce.

i) At point A: $E_A = \frac{k(+q)}{d} - \frac{k(+q)}{d} = 0$

ii) At point B: $E_B = \frac{k(+q)}{d^2} + \frac{k(+q)}{9d^2} = \frac{10kq}{9d^2}$ to right

iii) At point D: $E_D = E_{1D} + E_{2D}$ (non – zero vector sum)

$E_{1D} = \frac{kq}{d^2}$ down ward in the negative y – direction

$E_{2D} = \frac{kq}{\left(\sqrt{2}d\right)^2} = \frac{kq}{2d^2}$ in the x – y direction. **Answer: A**

39. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8W. If the same battery is connected across the same resistors in parallel, what is the total power delivered by the battery?

A. 2W B. 4W C. 15W D. 32W

Solution:

In series case: $R_T = R_1 + R_2 = R + R = 2R$

$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{2R} = 8W \Rightarrow \frac{V^2}{R} = 16W$$

In parallel case: $\frac{1}{R_{Total}} = \frac{1}{R} + \frac{1}{R} = \frac{2}{R} \Rightarrow R_T = \frac{R}{2}$

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$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{R/2} = \frac{2V^2}{R} \text{ but } \frac{V^2}{R} = 16W$$

$$\therefore P = 2\left(\frac{V^2}{R}\right) = 2(16W) = 32W$$ **Answer: D**

40. Wire B has twice the length and twice the radius of wire A. Both wires are made from the same material. If wire A has a resistance R, what is the resistance of wire B?
A. 4R B. 2R C. R/s D. R/4

Solution:

$\ell_B = 2\ell_A$, $r_B = 2r_B$, $\rho_A = \rho_B$ and $R_A = R$

The resistance of

i) Wire A: $R_A = \frac{\rho_A \ell_A}{A_A} = \frac{\rho \ell_A}{\pi r_A^2} = R$

ii) Wire B: $R_B = \frac{\rho \ell_B}{\pi r_B^2} = \frac{\rho(2\ell_A)}{\pi(2r_A)^2} = \frac{2\rho\ell_A}{4\pi_A^2} = \frac{1}{2}\left(\frac{\rho\ell_A}{\pi_A^2}\right)$

Hence $R_B = \frac{1}{2}R$ **Answer: C**

41. The sound level at a distance of 4.00m from a source is 120 dB. At what distance is the sound level 100 dB?
A. 0.40m B. 4.80m C. 40m D. 48m

Solution:

Describe (dB) is a measurement used when comparing two sound intensities. The intensity level $\beta = (dB)\log\left(\frac{I}{I_o}\right)$.

Hence $\beta_1 - \beta_2 = (10\text{ dB})\log\left(\frac{I}{I_o}\right) - (10\text{ dB})\log\left(\frac{I_2}{I_o}\right)$.

$$120\text{ dB} - 100\text{dB} = (10\text{ dB})\log\left(\frac{I_1/I_o}{I_2/I_o}\right) = (10\text{ dB})\log\left(\frac{I_2}{I_o}\right).$$

$$20\text{ dB} = (10\text{ dB})\log\left(\frac{I_1}{I_2}\right) \Rightarrow \log\frac{I_1}{I_2} = 2,$$

$$\Rightarrow \frac{I_1}{I_2} = 10^2 = 100, \text{ since } I = \frac{P}{4\pi r}$$

Since I is inversely related to the square of radius.

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}\frac{r_2^2}{(4m)^2} = 100 \Rightarrow r_2^2 = (100)(16m)^2$$ **Answer: C**

42. An ideal gas is at a temperature of 300 K. If we wish to double the root mean square (rms) speed of the molecules of the gas, to what value must we raise its temperature?
A. 450 B. 600k C. 800K D. 1200K

Solution: $V_{rms} = \sqrt{\frac{3kT}{m}}$

The root mean square speed of molecules is directly proportional to the square root temperature. $V_{rms} = \sqrt{\frac{T_1}{T_2}} = \frac{1}{2}$

$$\frac{T_1}{T_2} = \frac{1}{4} \Rightarrow \frac{300k}{T_2} = \frac{1}{4} \Rightarrow T_2 = 1200k$$ **Answer: D**

43. Suppose that the index of refraction of a soap – bubble film is 1.25, what is. The minimum thickness of the soap – bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600nm$?
A. 120nm B. 150nm C. 300nm D. 450nM

Solution:

In constructive air – film – air interference the film thickness can be calculated by $2nt = \left(m + \frac{1}{2}\right)\lambda, m = 0, 1, 2, \ldots$

For $m = 0$, $t = \frac{\lambda}{4n} = \frac{600nm}{4(1.25)} = 120nm$ **Answer: A**

44. A wire carries a steady current of 2.4 A.A straight section of the wire is 0.75 m long and lies in xy – plane along the x axis within

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$$\text{Power} = \frac{V^2}{R_T} = \frac{V^2}{R/2} = \frac{2V^2}{R} \text{ but } \frac{V^2}{R} = 16W$$

$$\therefore P = 2\left(\frac{V^2}{R}\right) = 2(16W) = 32W$$ **Answer: D**

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A. 4R B. 2R C. R/s D. R/4

Solution:

$\ell_B = 2\ell_A$, $r_B = 2r_B$, $\rho_A = \rho_B$ and $R_A = R$

The resistance of

i) Wire A: $R_A = \frac{\rho_A \ell_A}{A_A} = \frac{\rho \ell_A}{\pi r_A^2} = R$

ii) Wire B: $R_B = \frac{\rho \ell_B}{\pi r_B^2} = \frac{\rho(2\ell_A)}{\pi(2r_A)^2} = \frac{2\rho\ell_A}{4\pi_A^2} = \frac{1}{2}\left(\frac{\rho\ell_A}{\pi_A^2}\right)$

Hence $R_B = \frac{1}{2}R$ **Answer: C**

41. The sound level at a distance of 4.00m from a source is 120 dB. At what distance is the sound level 100 dB?
A. 0.40m B. 4.80m C. 40m D. 48m

Solution:

Describe (dB) is a measurement used when comparing two sound intensities. The intensity level $\beta = (\text{dB}) \log\left(\frac{I}{I_o}\right)$.

Hence $\beta_1 - \beta_2 = (10\text{ dB}) \log\left(\frac{I}{I_o}\right) - (10\text{ dB}) \log\left(\frac{I_2}{I_o}\right)$.

$$120\text{ dB} - 100\text{dB} = (10\text{ dB}) \log\left(\frac{I_1/I_o}{I_2/I_o}\right) = (10\text{ dB}) \log\left(\frac{I_2}{I_o}\right).$$

$$20\text{ dB} = (10\text{ dB}) \log\left(\frac{I_1}{I_2}\right) \Rightarrow \log\frac{I_1}{I_2} = 2,$$

$$\Rightarrow \frac{I_1}{I_2} = 10^2 = 100, \text{ since } I = \frac{P}{4\pi r^2}$$

Since I is inversely related to the square of radius.

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}\frac{r_2^2}{(4m)^2} = 100 \Rightarrow r_2^2 = (100)(16m)^2$$ **Answer: C**

42. An ideal gas is at a temperature of 300 K. If we wish to double the root mean square (rms) speed of the molecules of the gas, to what value must we raise its temperature?
A. 450 B. 600k C. 800K D. 1200K

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$$\frac{T_1}{T_2} = \frac{1}{4} \Rightarrow \frac{300\,k}{T_2} = \frac{1}{4} \Rightarrow T_2 = 1200k$$ **Answer: D**

43. Suppose that the index of refraction of a soap – bubble film is 1.25, what is. The minimum thickness of the soap – bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600nm$?
A. 120nm B. 150nm C. 300nm D. 450nM

Solution:

In constructive air – film – air interference the film thickness can be calculated by $2nt = \left(m + \frac{1}{2}\right)\lambda, m = 0, 1, 2, \ldots$

For $m = 0$, $t = \frac{\lambda}{4n} = \frac{600nm}{4(1.25)} = 120nm$ **Answer: A**

44. A wire carries a steady current of 2.4 A.A straight section of the wire is 0.75 m long and lies in xy – plane along the x axis within

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a uniform magnetic field, B = 1.6k T. If the current is in the positive x direction, what is the magnetic force on the section of wire?

A. 2.88N, in the negative y direction.
B. 2.88N, in the positive y direction.
C. 3.84N, in the negative y direction.
D. 3.84N, in the positive y direction.

Solution:

The magnetic force on a current carrying conductor is given by:

$F = I(\ell \times B)$.

$F = (2.4A)[(0.75mi \times 1.6Tk)] = 2.88(-j)N$

$F = 2.88N$ in the negative y – direction

Answer: A

45. Two different capacitors of capacitances $C_1 = 2\mu F$ and $C_2 = 3\mu F$ are connected in series across a 120V supply line. The charged capacitors are disconnected from the line and from each other and reconnected with terminals of like sign together. What will be the final charged C_1 and C_2, respectively?

A. 57.6μC and 86.4μC
B. 115.2μC and 127.8μC
C. 144μC across each
D. 288μC across each

Solution:

Initial: Capacitor in series; $C_T = \frac{C_1C_2}{C_1 + C_2} = \frac{(2\mu F)(3\mu F)}{2\mu F + 3\mu F} = \frac{6}{5}\mu F$

$$Q_T = C_T V = \left(\frac{6}{5}\mu F\right)(120V) = 144\mu F$$

$$V_1 = \frac{Q_T}{C_1} = \frac{144\mu F}{2\mu F} = 72V \text{ and } V_1 = \frac{Q_T}{C_1} = \frac{144\mu F}{2\mu F} = 72V \text{ and}$$

$$V_2 = \frac{Q_T}{C_2} = \frac{144\mu F}{2\mu F} = 48V \text{ is } V = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{288\mu F}{2\mu F + 3\mu F} = 57.6V$$

$Q_1 = (57.6V)(2\mu F) = 115.2\mu C$ and $Q_2 = (57.6V)(3\mu F) = 172.8\mu C$

46. A glass flask whose volume is exactly 1000 cm^3 at 0°C is totally filled with 1000 cm^3 of mercury at this temperature. When flask and mercury are heated to 100°C, 15.2 cm^3 of mercury is 1.82×10^{-4} per Celsius degree, what should be the coefficient of volume expansion of the glass?

A. 1.52×10^{-4}/°C
C. 3.00×10^{-5}/°C

B. 1.82×10^{-4}/°C
D. 1.00×10^{-5}/°C

Solution:

$V_o = 1000\ cm^3$, $T_o = 0°C$, $V_{o\,mercury} = 1000\ cm^3$, $T_f = 100°C$

$V_{o\ over\ flow} = 15.2\ cm^3$, $\gamma_{mercury} = 1.82 \times 10^{-4}$/°C and

$\Delta T = 100°C - 0°C = 100°C$

The volume of mercury over flow is equal to difference in expansion of mercury and glass.

$V_{o\ over\ flow} = \Delta V_{mercury} - \Delta V_{glass} = V_o\gamma_m\Delta T - V_o\gamma_g\Delta T$

$15.2cm^3 = V_o\Delta T(\gamma_m - \gamma_{glass}) = 1000\ cm^3(100°C)(\gamma_m - \gamma_{glass})$

$$\Rightarrow \gamma_m - \gamma_g = \frac{12.2cm^3}{(1000cm^3)(100°C)} = 1.52 \times 10^{-5}/°C$$

$\gamma_{glass} = \gamma_m - 15.2 \times 10^{-5}/°C = 1.82 \times 10^{-4} - 1.52 \times 10^{-4}$

$\gamma_{glass} = 0.3 \times 10^{-4}/°C = 3 \times 10^{-5}/°C$

Answer: C

47. A heat engine is being designed to have a Carnot efficiency of 65.0% when operating between two energy reservoirs. If the temperature of the cold reservoir is 20.0°C, what must be the temperature of the hot reservoir?

A. 57°C B. 310°C C. 564°C D. 807°C

Solution: Efficiency: $e = 1 - \frac{T_{cold}}{T_{not}}$

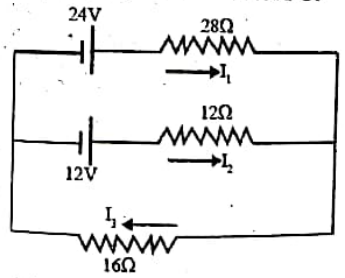
$$0.65 = 1 - \frac{293}{T} \Rightarrow T_{not} = \frac{293}{1 - 0.65} = 873k$$

$T_{hot} = 873 - 273 = 564°C$

Answer: C

48. In Figure 8, the current in the 16Ω resistor is:

A. 0.64A B. 0.15A C. 0.49A D. 1.28A



Solution:

Applying kirchoff's rules: $I_1 + I_2 = I_3$. From kirchoff's loop – rule

The potential drop in upper branch loop

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$24V - (28\Omega)I_1 + (12\Omega)I_2 - 12V = 0$

$\Rightarrow (28\Omega)I_1 - (12\Omega)I_2 = 12V \ \ldots (1)$

The potential drop in lower branch loop;

$12V - (12\Omega)I_2 - (16\Omega)(I_1 + I_2) = 0$

$(28\Omega)I_2 + (16\Omega)I_1 = 12V \ \ldots (2)$

Solving (1) and (2)

$$7\begin{cases}(28\Omega)I_1 - (12\Omega)I_2 = 12V\\ 3\begin{cases}(16\Omega)I_1 + (28\Omega)I_2 = 12V\end{cases}\end{cases} \Rightarrow \begin{cases}196I_1 - 84I_2 = 84V\\ 48I_1 + 84I_2 = 36V\end{cases}$$

$$244I_1 + 0 = 120V$$

$$I_1 = \frac{120V}{244\Omega} = 0.49A \text{ and } I_2 = 0.15A$$

$I_3 + I_1 + I_2 = 0.49A + 0.15A = 0.64A$

Answer: A

49. A 10.0 m uniform ladder weighing 200N rests against a frictionless wall. The ladder makes 53° angle with the horizontal. If the ladder is just on the verge of slipping when a 600.0N fire fighter is 6.00m along the ladder form the bottom, what is the coefficient of static friction between the ladder and ground?

A. 0.3240 B. 0.4313 C. 0.5750 D. 0.7667

Solution: The forces on the ladder can be shown as below.

$\sum F_x = 0 \Rightarrow F_w - f = 0 \qquad f = F_w \ldots (1)$

$\sum F_y = 0 \Rightarrow F_R - 200N - 600N = 0 \qquad F_B = 800N \ldots (2)$

Applying the second condition of equilibrium considering the ground as an axis:

$F_w \times 10m \sin 53° = 200N \times (5m) \cos 53 + 600 \times 6m \times \cos 53°$

$F_w \times 10 \times 0.8 = 1000 \times 0.6 + 3600 \times 0.6;\ 8F_w = 2760N$

$$F_w = \frac{2760}{8}N = 345N \text{ . The coefficient of friction is } f = \mu F_N$$

$$\Rightarrow \mu = \frac{345N}{800N} = 0.4313 \text{ , since } F_N = F_w$$

Answer: B

50. A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 2000 N/m. The spring is compressed 2.0 cm and is then released from rest. At what speed will the block pass through the equilibrium position of a constant friction force of 10.0N retards its motion from the moment it is released?

A. 70 m/s B. 50 m/s C. 70 cm/s D. 50 cm/s

$k = 2000$ N/m, $f = 10$N, $x = 2$ cm, $m = 1.6$ kg

As the block starts from rest its velocity is zero at the release point. At the compressed position the total engine is Elastic potential energy.

$$EPe = \frac{1}{2}kx^2 = \frac{1}{2}(2000)(2\times10^{-2}m)^2 = (1000N/m)(4\times10^{-4}m^2) = 0.4J$$

Work done by friction force is $W_f = fs. -fs = 10N \times 0.02m = 0.2J$

The kinetic energy at the equilibrium position is

$$k.E = \frac{1}{2}mV^2 = \frac{1}{2}(1.6kg)V^2 = 0.8V^2$$

Work done by non – conservative force is equal to the lost in mechanical energy: $\Delta M.E = W_{nC}$

$$0.4J - 0.8V^2 = 0.2J \Rightarrow 0.8V^2 = 0.2J \Rightarrow V = \frac{0.2}{0.8} = \frac{1}{4}$$

$\therefore V = 0.5$ m/s $= 50$ cm/s

Answer: D

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1. What is the angle between vector $\mathbf{A}=\left(a\mathbf{i}+a\sqrt{3}\mathbf{j}\right)$ units $\mathbf{B}=\left(a\sqrt{3}\mathbf{i}+a\mathbf{j}\right)$ units?

A. 90° B. 60° C. 37° D. 30°

Clue:

$$\mathbf{A}\cdot\mathbf{B}=AB\cos\theta$$

$$\Rightarrow\cos\theta=\frac{\mathbf{A}\cdot\mathbf{B}}{AB}=\frac{\left(a\mathbf{i}+a\sqrt{3}\mathbf{j}\right)\cdot\left(a\sqrt{3}\mathbf{i}+a\mathbf{j}\right)}{\left(\sqrt{a^2+\left(a\sqrt{3}\right)^2}\right)\cdot\left(\sqrt{\left(a\sqrt{3}\right)^2+a^2}\right)}$$

$$\cos\theta=\frac{a^2\sqrt{3}+a^2\sqrt{3}}{\left(\sqrt{a^2+3a^2}\right)\left(\sqrt{3a^2+a^2}\right)}$$

$$=\frac{2a^2\sqrt{3}}{4a^2}=\frac{\sqrt{3}}{2}\Rightarrow\theta=\cos^{-1}\frac{\sqrt{3}}{2}=30°$$

Answer: D

2. The magnitudes of two vectors **A** and **B** are 12 units and 5 units, respectively. Which one of the following is **NOT** the possible value for the magnitude of the resultant vector **A + B**?

A. 60 units C. 13 units

B. 17 units D. 7 units

Clue: For any two vectors A and B, the resultant is always, in the range $|A-B|\le R\le|A+B|$, now A = 12 and B = 5

$\Rightarrow|12-5|\le R\le|12+5|\Rightarrow 7\le R\le 17$. Thus, 60 is out of the range

Answer: A

3. What is the vector product $\vec{A}\times\vec{B}$ of two vectors $\vec{A}=7\hat{i}+4\hat{j}-8\hat{k}$ and $\vec{B}=3\hat{i}-2\hat{j}+5\hat{k}$?

A. $36\hat{i}+11\hat{j}-2\hat{k}$ C. $-4\hat{i}+59\hat{j}+26\hat{k}$

B. $4\hat{i}-59\hat{j}-26\hat{k}$ D. $-36\hat{i}-11\hat{j}+2\hat{k}$

Clue:

$$\vec{A}\times\vec{B}=\begin{vmatrix} i & j & k \\ 7 & 4 & -8 \\ 3 & -2 & 5 \end{vmatrix}$$

$$=\left[(4)(5)-(-2)(-8)\right]i+\left[(-8)(3)-(7\times5)\right]j+\left[(7)(-2)-(4)(3)\right]K$$

$$=4i-59j-26k$$

Answer: B

4. Which one of the following experimental errors can be reduced by taking repeated measurements?

A. Parallax errors C. Random errors

B. Systematic errors D. Zeros errors

Answer: C

5. A teacher gave a student a book whose thickness is 4.30 cm, and instructed him to measure the thickness of the book with vernier calipers. The student took four measurements. The average and standard deviation of the measurements were 4.33 cm and 0.04 cm. respectively. Which one of the following is the uncertainty in the accuracy of the student's measurement?

A. 0.03 cm C. 0.04cm

B. 0.035cm D. 0.30cm

Answer: A

6. If a long distance jumper athlete leaves he ground at an angle of 37.0° above the horizontal surface at speed of 10.0m/s how far does he jump in the horizontal direction?

A. 4.8m B. 6m C. 9.6m D. 12m

Clue:

$$R=\frac{2(u\cos\theta)(u\sin\theta)}{g}=\frac{(2)(10)\cos37°-(10)\sin37°}{10}$$

$$=9.6m$$

Answer: C

7. A woman is rotating a bucket of water in a vertical circle of radius 0.9m. the mass of bucket and water is 5kg. What is the bucket's minimum speed at the top of the circle if no water is to spill out?

A. 0 B. 1m/s C. 3m/s D. 9m/s

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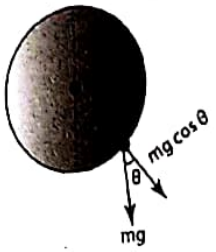
Clue:



At any time, $T - mg\cos\theta = \frac{mv^2}{r}$, and at the top, $\theta = 180°$ and $T = 0$

$$\Rightarrow mg = \frac{mv^2}{r},$$

$$v = \sqrt{gr} = \sqrt{(0.9)(10)} = 3m/s$$

Answer: C

8. A force **F** of magnitude 20N is applied to a block of mass 2 kg that lies on a rough, horizontal surface as shown in Figure 1. The coefficient of kinetic friction between the block and surface is 0.4. what is the magnitude of the acceleration of the block?

A. $10m/s^2$
B. $5.2\ m/s^2$
C. $4\ m/s^2$
D. $2.8\ m/s^2$

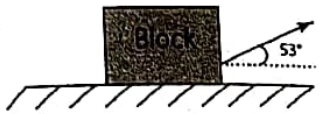


Fig.1

Clue:

$F\cos\theta - f = ma$

$F\cos\theta - \mu F_N = ma$ but $F_N = mg - F\sin\theta$

$F\cos\theta - \mu(mg - F\sin\theta) = ma$

$$\Rightarrow a = \frac{F\cos\theta - \mu(mg - F\sin\theta)}{m}$$

$$a = \frac{(20)\cos 53 - 0.4[(2)(10) - 20\sin 53°)}{2} = 5.2m/s^2$$

Answer: B

9. Which one of the following statements is correct?

A. An object moving toward the east cannot have an acceleration toward the west.

B. If the average velocity of an object is zero in some time interval, the average speed of the object for the time interval is also zero.

C. The velocity-time graph of an object moving with constant acceleration is parallel to the time axis.

D. An object having zero velocity can have acceleration different from zero.

Answer: D

10. A car travels along a straight line at a constant speed of 20m/s for a distance d and then another distance d in the same direction at a constant speed of 36 km/hr. what is the average speed for the entire trip?

A. 23km/h
B. 48km/h
C. 54 km/h
D. 60km/h

Clue: $V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{S_1 + S_2}{\frac{S_1}{V_1} + \frac{S_2}{V_2}}$, but $S_1 = S_2 = d$

$V_1 = 20m/s = (20)(3.6) = 72km/hr$

$$V_{av} = \frac{d + d}{\frac{d}{V_1} + \frac{d}{V_2}} = \frac{2d(V_1V_2)}{d(V_1 + V_2)} = \frac{2V_1V_2}{V_1 + V_2}$$

$$= \frac{(2)(36)(72)}{(36+72)} = 48km/h$$

Answer: B

11. A highway curve is banked (inclined) in such a way that a car travelling at a speed of 13.5m/s can round the curve without skidding, in the absence of friction. If the banking angle is 30° what is the radius of the curve?

A. 10.5m
B. 21.0m
C. 31.7 m
D. 36.7 m

Clue:

$$f = F_c \Rightarrow mg\tan\theta = \frac{mv^2}{r} \Rightarrow r = \frac{v^2}{g\tan\theta} = \frac{(13.5)^2}{(10)\tan 30}$$

$$= 31.6m$$

Answer: C

12. A car mass 1500 kg is travelling at a uniform velocity of 30m/s due east. The driver applies the brakes to slow down the car to a velocity of 10m/s due east. What is the work done in slowing down the car?

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A. 15kJ
B. 300kJ
C. 600kJ
D. 750 kJ

Clue:

$$W = \Delta K_E = \frac{1}{2} \Rightarrow \left(v^2 - u^2\right) = \frac{1}{2}(1500)\left[10^2 - 30^2\right] = 600k.$$

Answer: C

13. Which one of the following statements is true about a spring-mass system oscillating on a horizontal frictionless surface?
A. The kinetic and potential energies are equal at all times.
B. The kinetic and potential energies are both constant.
C. The maximum potential energy is achieved when the mass passes though its equilibrium position.
D. The maximum kinetic energy and maximum potential energy are equal.

Answer: D

Clue: from conservation of energy,

14. A hard ball moving on a horizontal frictionless surface with a speed of 5m/s in the positive x direction strikes a stationary hard ball of the same mass. After the collision, the first ball moves with a speed of 4m/s along a direction that makes an angle of 37.0° with its initial direction of motion. What is the struck ball's speed and the smallest angle between its direction of motion and the positive x axis after the collision?
A. 3m/s and 53.0°
B. 3m/s and 37.0°
C. 4m/s and 53.0°
D. 4m/s and 37.0°

Clue: from conservation of linear momentum,

$$m_1 = \begin{bmatrix} u_1x \\ u_1y \end{bmatrix} + m_2 = \begin{bmatrix} u_2x \\ u_2y \end{bmatrix} = m_1 \begin{bmatrix} v_1x \\ v_1y \end{bmatrix} + m_2 \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}, m_1 = m_2$$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} + 0 = \begin{bmatrix} v_1 \cos 37 \\ v_1 \sin 37 \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$$

$$\begin{bmatrix} 5 \\ 0 \end{bmatrix} + 0 = \begin{bmatrix} (4)(0.8) \\ (4)(0.6) \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix} = \begin{bmatrix} 3.2 \\ 2.4 \end{bmatrix} = \begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$$

$$5 = 3.2 + v_2x \Rightarrow v_2x = 1.8\text{m/s and}$$

$$0 = 2.4 + v_2y \Rightarrow v_2y = -2.4\text{ m/s} \Rightarrow v_2 = \sqrt{v_{2x}^2 + v_{2y}^2}$$

$$v_2 = \sqrt{(1.8)^2 + (-2.4)^2} = 3\text{m/s and } \theta\ 53°$$

Answer: A

15. A man standing on a frictionless surface continuously fired 120 bullets horizontally in 1 minute using a machine gun. If each bullet has a mass of 10 g and leaves the muzzle of the gun with a speed of 800m/s, what is the average force exerted by the gun on the man?
A. 8N
B. 16N
C. 960N
D. 16000N

Clue:

$$F\Delta t = nm\Delta v \Rightarrow F = \frac{nm\Delta v}{\Delta t} = \frac{(120)(0.01)(800)}{60} = 16\text{N}$$

Answer: B

16. Two blocks are connected by a string of negligible mass passing over a pulley of radius $r = 0.2$m and moment of inertia I as shown in Figure 3. The b lock on the horizontal frictionless surface is moving with a constant acceleration $a = 20.0\text{m/s}^2$. If the pulley axis is frictionless, the string is inextensible and does not slip on the pulley, what should be the moment of inertia of the pulley?
A. 0.40 k gm^2
B. 0.80 k gm^2
C. 4.00 k gm^2
D. 8.00 k gm^2

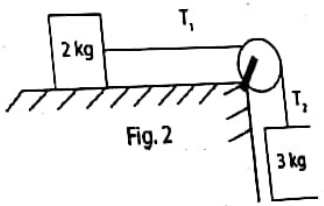

Fig. 2

Clue:

$$T_1 = 4N \text{ and } T_2 = 24N, \Rightarrow \Delta T = F = 20N, \text{ and}$$

$$\tau = Fr = I\alpha, \alpha = \frac{a}{r}$$

$$Fr = I\left(\frac{a}{r}\right) \Rightarrow I = \frac{Fr^2}{a} = \frac{(20)(0.2)^2}{2} = 0.4kgm^2$$

Answer: A

17. An object with mass of $m = 3.0$ kg is attached to the free end of a light string wrapped around a pulley of radius $R = 0.50$ m and moment of inertia $I = 0.75$kg

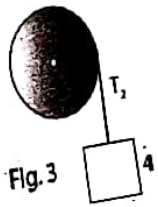

Fig. 3

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m^2. The pulley is free to rotate in a vertical plane about a fixed frictionless axis passing through its center as shown in Figure 4. The suspended object is released from rest at 3.60 m above the floor. What is the angular momentum of the pulley at the time when the object strikes the floor?

A. 4.5 kg m^2 /s
B. 9.0 kg m^2 /s
C. 12.3 kg m^2 /s
D. 18 kg m^2 /s

Clue: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}Iw^2, w = \frac{v}{R}$

$$\Rightarrow mgh = \frac{1}{2}mv^2 + \frac{1}{2}I = \left(\frac{v}{R}\right)^2 = (3)(36) = 1.5v^2 + 1.5v^2$$

v= 6m/s and $w = v/R$=12 rad/s $\Rightarrow L = Iw = (0.75)(12) = 9$

Answer: B

18. Which one of the following statements is **NOT** correct about a particle that is in equilibrium?

A. It can have a velocity different from zero.
B. Its velocity can be equal to zero.
C. It can be under the action of more than one force.
D. It can have an acceleration different from zero.

Answer: D

19. A block of mass 10 kg is pulled up a rough inclined plane with a force of 100N parallel to the inclined plane. The inclined plane makes an angle of 37° with the horizontal. At the instant when the force is applied the block has a speed of 1.5m/s. If the coefficient of kinetic friction is 0.40, and the block is pulled up 5m, what is the change in the kinetic energy of the block?

A. 20J C. 40J
B. 30J D. 45.5J

Hint:

$$F - mg\sin\theta - \mu mg\cos\theta = ma \Rightarrow a = \frac{F - mg\sin\theta - \mu mg\cos\theta}{m}$$

$$a = \frac{100 - (10)(10)(0.6) - (0.4)(10)(10)(0.8)}{10} = 0.8\text{m/s}^2$$ a

nd

$$v^2 = u^2 + 2as = (1.5)^2 + (2)(0.8)(5) = 10.25$$

$$\Delta K_E = \frac{1}{2}m\left(v^2 - u^2\right) = \frac{1}{2}(10)(10.25 - 2.25) = 40\text{J}$$

Answer: C

20. The particles shown in Figure 2 are connected by rectangular rigid rods of negligible mass. The origin of the xy plane about the z axis with an angular speed of 6.00 rad/s what is the rotational kinetic energy of the system?

A. 2574J
B. 2340J
C. 1287 J
D. 1170J

Hint:

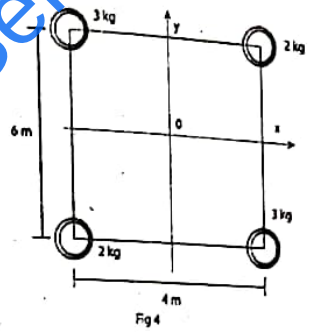

Fig 4

$$K_{Er} = \frac{1}{2}Iw^2 = \frac{1}{2}mr^2w^2 = \frac{1}{2}w^2\left(m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2\right)$$

$$K_{Er} = \frac{1}{2}(6)^2\left((3)(13) + 2(13) + 2(13) + (3)(13)\right) = 2340\text{J}$$

Answer: B

21. Two solid spheres made of gold and silver having equal masses are suspended by strings and fully submerged in a tank of water. Given that the density of gold is greater than that of silver, which one of the following statements is true?

A. The buoyant force on each sphere is the same.
B. The volume of water displaced by the gold sphere is larger than that of the silver sphere.
C. The tension in the string supporting each sphere is the same.
D. The buoyant force on the silver sphere is greater than that on the gold sphere.

Hint: Since $\rho_g > \rho_s$ and $m_g = m_s \Rightarrow v_s > v_g$ The buoyant force equals the weight of the displaced fluid

$\Rightarrow F_b = m_d g = \rho_w v_d g$

Answer: D

22. The cross-sectional areas of the small and large pistons of a hydraulic lift

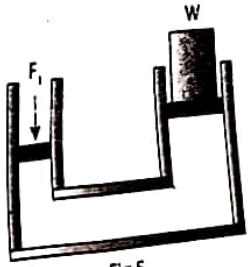

Fig 5

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shown in Figure 6 are 4.0cm^2 and 0.02 m^2, respectively. What is the magnitude of the force that must be applied to the small piston to lift a weight w – 15,000N place on the large piston.

A. 3.0×10^2 N
B. 1.5×10^4 N
C. 7.5×10^5 N
D. 3×10^6 N

Hint: From pascal's principle,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{F_1}{4\times10^{-2}} = \frac{15000}{2\times10^{-2}} \Rightarrow F_1 = 3\times10^2\,N$$

Answer: B

23. An incompressible fluid flows through a horizontal pipe whose diameter varies along its length. The sum of the kinetic energy per unit volume and pressure of the fluid at different sections of the pipe would _____

A. depend on the length of the pipe.
B. increase as the pipe diameter increases.
C. increase as the pipe diameter decreases.
D. remain the same as the pipe diameter changes.

Hint: From Bernoulli's equation $\frac{p}{\rho} + gh + \frac{v_2}{2} =$ constant

$$\Rightarrow \frac{p_1}{\rho} + gh_1 + \frac{v_1^2}{2} = \frac{p_2}{\rho} + gh_2 + \frac{v_2^2}{2}$$

Answer: D

24. a uniform bar of length 2.0 m and mass 30.0 kg is supported by two mass less ropes which are attached to a fixed horizontal bar, as shown in Figure 5. What are the tensions in each rope when a 700N weight is suspended at $d = 0.50$ m from the left end of the bar?

A. $T_1 = 675$ N and $T_2 = 325$ N
B. $T_1 = 325$ N and $T_2 = 675$ N
C. $T_1 = 500$ N and $T_2 = 500$ N
D. $T_1 = 525$ N and $T_2 = 175$ N

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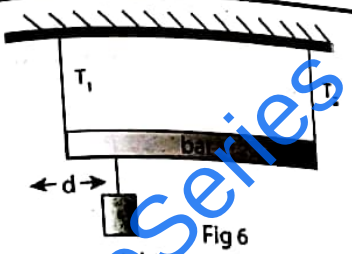

Fig 6

Hint: Taking moment at the left end we have, $\sum\tau = 0$
$\Rightarrow (0.5)(700) + (1)(300) = (T_2)2 = T2 = 325\text{N}\ T_1 = 635\text{N}$

Answer: A

25. Which one of the following laws best describes the behavior of elastic materials?

A. Gay Lussac's law
B. Hook's Law
C. Newton's second law
D. Newton's law of gravitation

Answer: B

26. The first law of Thermodynamics take the form dU = dW. This equation is valid if the process is _____.

A. isobaric
B. isochoric
C. isothermal
D. adiabatic

Hint: $\Delta u = \Delta Q + \Delta w$. When $\Delta u = \Delta w$, adiabatic

Answer: D

27. An ideal gas undergoes an isothermal process starting with a pressure of 2×10^5Pa and a volume of 6cm^3. Which one o the following might be the pressure and the volume of the final state?

A. 4×10^5Pa and 4 cm^3
B. 4×10^5Pa and 3 cm^3
C. 5×10^5Pa and 3 cm^3
D. 10^5Pa and 4 cm^3

Hint: From ideal gas equation $P_1V_1 = P_2V_2$

Answer: B

28. Which one of the following statement is **NOT** correct?

A. The entropy of the universe increases in all natural processes.
B. No heat engine operating in a cycle can absorb energy from reservoir and use it entirely to do work.
C. When a system undergoes a change in state, the change in its internal energy is the difference between the energy transferred to it by heat and the work done on it.

D. The entropy of an isolated system reaches its maximum value when the system is in equilibrium state.

Answer: D

29. The amount of heat required to raise the temperature of an object of mass m = 1kg by 1 Kelvin is known as _____.
A. Specific heat capacity
B. Heat capacity
C. Expansion coefficient
D. Internal energy

Answer: A

30. A given weight of gas has a temperature of 27°C and volume V. keeping the pressure constant, the volume is increased to 3V by varying the temperature. What is the final temperature of the gas?
A. 81°C
B. 546 °C
C. 627 °C
D. 900 °C

Clue: $\frac{P_1V_1}{T_1}=\frac{P_2V_2}{T_2}=\frac{V}{300}=\frac{3V}{T_2}=T_2=900\,k=627°C$

Answer: C

31. A given minimum point (through) of a sinusoidal wave is traveling 480 cm along a rope in 12 s. if the oscillator that generates the wave completes 48 vibrations in 30 s. what is the wave length of the wave?
A. 25 m
B. 15 m
C. 0.25m
D. 0.15m

Hint:

$V=\frac{(4.8)}{(12)}=0.4m/s$ and $f=\frac{48}{30}=1.6\,H_z, \lambda=\frac{V}{f}=0.25\,m$

Answer: C

32. A wave traveling from medium of index of refraction n_1 toward a medium of index of refraction n_2 undergoes partly transmission and partly reflection. If n_2 is greater than n_1 which one of the following statements is not correct?
A. The speed of the wave in the medium with refractive index n_1 is smaller than with n_2.
B. The incident and the reflected waves are in-phase.
C. The angle of reflection is the same as the angle of incidence.
D. The refraction angle is less than the incidence angle.

Answer: A

Hint: from snell's law,

33. In Young's double-slit experiment interference pattern which one of the following activities move the successive bright fringes farther apart.

A. Decreasing the wavelength of the light.
B. Decreasing the distance between the viewing screen and the double slit.
C. Increasing the slit spacing.
D. Decreasing the slit spacing.

Clue: from the relation, $\frac{\lambda}{s}=\frac{w}{D}$, w increases as s decreases.

Answer: D

34. The position of a particle moving with simple harmonic motion is given by an expression $x = A \cos(\omega t + \varnothing)$, ω where A is the amplitude of the oscillation, t is time, ω is angular frequency $\varnothing$ is the phase constant. If $\varnothing = \pi/2$, what will be the sign of the particle's position and velocity at $t = 7T/8$, where T is the period of oscillation?
A. The position is positive, and the velocity is negative.
B. The position and velocity are both positive.
C. The position and velocity are both negative
D. The position is negative, and the velocity is positive.

Clue: $x=A\cos(wt+\phi)=A\left(\frac{2\pi}{T}t+\frac{\pi}{2}\right), t=\frac{7T}{8}$

$x=A\cos\left(\frac{2\pi}{T}\left(\frac{7T}{8}\right)+\frac{\pi}{2}\right)=A\cos\left(\frac{9}{4}\pi\right)$ and $V=A\sin\left(\frac{9\pi}{4}\right)$

Answer: B

35. If a long rope of uniform linear density is hung from a ceiling and waves are sent up the rope from its lower end, which one of the following is true about the speed of the ascending waves?
A. The speed decrease as the wave ascend.
B. The speed increase as the wave ascend.
C. The waves ascend with a uniform speed.
D. The speed of the waves is larger around the middle of the rope than around the beginning and end.

Answer: A

36. A parallel plate capacitor has a capacitance of 10μF. if the distance between the plates is doubled, keeping all other quantities constant, what is the new capacitance of the capacitor?
A. 0.2 μF
B. 5 μF
C. 10 μF
D. 20 μF

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Clue:

$$C_1 = \epsilon_r \epsilon_0 \frac{A}{d_1}, \text{ if } d_2 = 2d_1, C_2 = \epsilon_r \epsilon_0 \frac{A}{2d_1} = \frac{10}{2} = 5\mu F$$

37. If several resistors are connected in parallel, the equivalent resistance is

Answer: B

A. greater than any of the resistance in the group.
B. is equal to the average resistance of the group
C. less than any of the resistances in the group.
D. equal to the sum of the resistances in the group.

Answer: C

38. If aluminum and copper wires with circular cross sections have equal lengths and equal resistance, then the ratio of the radius of aluminum to that of copper should be the same as that of the _____.

A. Ratio of the resistivity of aluminum to copper
B. Ratio of the resistivity of copper to aluminum
C. Square root of the ratio of the resistivity of aluminum to copper
D. Square root of the ratio of the resistivity of copper to aluminum

Clue: $R = \frac{\rho \ell}{A} \Rightarrow \frac{\rho_A \ell_A}{\pi r_A^2} = \frac{\rho_C \ell_C}{\pi r_C^2} = \frac{r_A}{r_C} = \sqrt{\frac{\rho_A}{\rho_C}}$

Answer: C

39. Which of the following statements is correct?

A. The electric field strength due to two point charges of equal magnitude and the same sign is zero only at one point on the line joining the two charges.
B. If a point charge of mass m is released from rest in a uniform electric field, it will move along a line of force with uniform speed.
C. The variation of the strength of the electric field due to an electric dipole is inversely proportional with the square of the distance from its center of mass.
D. An electric dipole moment $\boldsymbol{P}$ in a uniform external electric field $\boldsymbol{E}$ is in stable equilibrium if $\boldsymbol{P}$ and $\boldsymbol{E}$ are in opposite directions.

Answer: B

40. In a certain region of space, a uniform electric field is directed in the positive x direction. If a particle with negative charge is carried from $x = 20$ cm to $x = 60$ cm, what can we conclude about the electric potential and the electric potential energy of the charge-field system?

A. The electric potential at $x = 20$cm is larger than that at $x = 60$ cm.
B. The electric potential at $x = 20$cm is the same as that at $x = 60$ cm.
C. The electric potential energy decrease.
D. The electric potential energy increase.

Answer: D, $u = EQd$

41. An electron moving at a speed of 1.5×10^7 m/s enters a region of uniform magnetic field of magnitude 2×10^{-3} T, in a direction perpendicular to the field. The path followed by the electron is circular. What is the radius of the circular path?

A. 0.43 cm
B. 4.3 cm
C. 43 cm
D. 430 cm

Hint:

$$qB = \frac{mv}{r} \Rightarrow r\frac{mv}{qB} = \frac{(9.11\times10^{-31})(1.5\times10^{7})}{(1.6\times10^{-19})(2\times10^{-3})} = 4.3\times10^{-2}$$

Answer: B

42. If a wire carrying current in the plane of this paper toward the top of the page experiences a magnetic force toward the right edge of the page what will be the direction of the magnetic field causing this force?

A. In the plane of the page and toward the left edge.
B. In the plane of the page and toward the bottom edge.
C. Perpendicular and directed out of the plane of the page.
D. Perpendicular and directed into the plane of the page.

Hint: using right hand rule.

Answer: B

43. A square flat loop of wire is placed in a uniform magnetic field that is in the y direction. The magnetic flux through the loop is a maximum if the plane of the loop is in the

A. xy plane.
B. xz plane.
C. yz plane
D. *inclined 45° with respect to the xy* plane

Answer: B

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44. In the circuit shown in Figure 7, what is the potential difference between point b and a?

A. 19.2 V
B. 24.0 V
C. 13.0 V
D. 12. 0 V

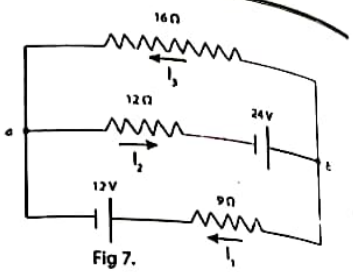

Fig 7.

Hint: from point rule. $I_1 + I_3 = I_2$ and form loop rule $24 = 16I_3 + 12I_2$ and $-24 + 12 = 12I_2 - 9I_1$
$\Rightarrow I_2 = 1A$ and $V_a = I_2R_2 = (1)(12) = 12V$, $V_b = 24_V$,
$\Delta V = V_b - V_a = 12V$

Answer: D

45. Which of the following statements is correct about the effect of a magnetic field?

A. A magnetic field exertes a nonzero force on a stationary charged object.
B. A magnetic field exerts a nonzero force on a conducting wire placed in the field
C. A uniform magnetic field can be created at the point around long current carrying wire
D. A magnetic field can be created in the space surrounding any charged object that is in motion.

Answer: D

46. A series *RLCI* circuit has a total impedance of 75.0 Ω. If the resistance is R = 45Ω. And the average power delivered to the circuit is P = 352.8W, what is the root-mean square value of ac voltage source?

A. 83 V
B. 116 V
C. 210 V
D. 297 V

Clue:

$P = V_{rms} I_{rms} \cos\theta$, $I_{rms} = \frac{V_{rms}}{Z}$ and $\theta = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$ but

$Z = \sqrt{R^2 + X_L - X_C{}^2} \Rightarrow X_L - X_C = \sqrt{Z^2 - R^2} = \sqrt{75^2 - 45^2}$

$X_L - X_C = 60$. Then,

$$P = (V_{rms}) = \left(\frac{V_{rms}}{Z}\right)\cos\theta = \frac{V_{rms}^2}{Z}\cos\left(\tan^{-1}\left(\frac{60}{45}\right)\right)$$

$$\Rightarrow V_{rms}^2 = \frac{PZ}{\cos 53^\circ} = \frac{(352.8)(75)}{0.6} \Rightarrow V_{rms} = 210V.$$

Answer: C

47. Which of the following phenomena most clearly demonstrates the wave nature of electrons?

A. The photoelectric effect
B. Blackbody radiation
C. The Compton effect
D. Diffraction of electrons by crystals

Answer: D

48. When the radioactive element $^{144}Nd_{60}$ decays to $^{140}Ce_{58}$, which one of the following particles is emitted?

A. A beta particle
B. An alpha particle
C. A proton
D. A gamma ray

Answer: B

49. Which one of the following actions increases the magnetic energy density in the interior of a very long current-carrying solenoid?

A. Increasing the number of turns per unit length of the solenoid
B. Increasing the cross sectional area of the solenoid.
C. Increasing the length of the solenoid keeping the number of turns per unit length fixed.
D. Decreasing the cross sectional area of the solenoid.

Hint:

Since $U_B = \frac{B^2}{2\mu o}$, $B = \mu \times In$, $n = \mu/\ell$ $\in = BLV$ and $I = \frac{\in}{R}$

Answer: A

50. Figure 8 (below) shows a rectangular loop of wire of width L and resistance R. One end of the loop is in a uniform magnetic field of strength B in directed perpendicular into the plane of the

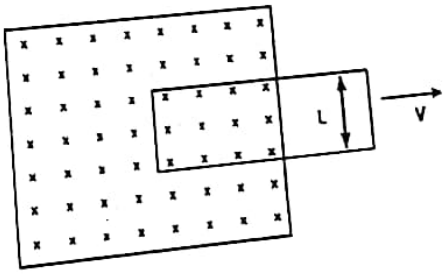

Fig 8.

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page. The loop is pulled to the right at a constant speed v. what are the magnitude and direction of the induced current in the loop?

A. $I = BLvR$ and clockwise
B. $I = BLv/R$ and clockwise
C. $I = BLvR$ and counterclockwise
D. $I = BLv/R$ and counterclockwise

Explanation:

The flux is decreasing as the loop leaves the region of the magnetic field. To oppose, a decreasing the induced current flux generate a magnetic field in the same direction as an applied magnetic field into the page. Then the induced current must be clock wise direction.

$\varepsilon = BLV$ and $I = \frac{\varepsilon}{R}$

$\therefore I = \frac{BLV}{R}$ clockwise

Answer: B

Physics

DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question and carefully **blacken** the letter of your best choice on the separate answer sheet provided.

You may refer to the information given below when you work on some of the questions.

Constant	Symbol	Value
Acceleration due to gravity	G	$10m/s^2$
Permittivity of vaccum	ε_o	$8.85\times10^{-12}F/m$
Permeability of free space	μ_o	$4\pi\times10^{-7}Tm/A$
Faraday's constant	F	$96500C\ mol^{-1}$
Charge of an electron	ė	1.6×10^{-19}Coulomb
Universal gravitational constant	G	$6.67\times10^{-19}\ \frac{Nm^2}{Kg^2}$
Planck constant	h	$6.63\times10^{-34}Js$

$\sin 30° = \cos 60° = 0.5$

$\sin 37° = \cos 53° = 0.6$

$\sin 53° = \cos 37° = 0.8$

$\sin 60° = \cos 30° = 0.866$

$\frac{1}{\sqrt{2}} = 0.707$

1. The graph below illustrates the position and time for a cat that runs to catch a rat and then returns with it. The cat caught the rat after 2 seconds. What was the cat's average speed as it returned with the rat?

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A. Its average return speed was 2 m/s.
B. Its average return speed was 4 m/s.
C. Its average return speed was 0.7 m/s.
D. Its average return speed was 0.9m/s.

Explanation: The return average speed is

$$v_{av} = \frac{S_T}{t_T} = \frac{7}{8} = 0.875 \approx 0.9 m/s$$

Answer: D

2. A man is standing on a spring scale in an elevator. when the elevator is at rest, the scale reads 560N. When the elevator starts to move, the scale reads 600N. The elevator has:

A. an upward acceleration
B. a constant downward speed
C. a constant upward speed
D. downward acceleration

Explanation: $T - mg = ma \Rightarrow T = mg + ma$
The scale reading is bigger when the elevator accelerates upward.

Answer: A

3. What is the necessary and sufficient condition for conservation of linear momentum?

A. The presence of more linear than nonlinear forces in the system.
B. Net force on the system is zero.
C. the presence of only non-linear forces
D. Internal forces exceed external forces.

Explanation: The necessary and sufficient condition for the conservation of linear momentum is that, the net force on the system, should be zero.

Answer: B

4. Which statements is **WRONG** about units and dimensions?

A. Two quantities can have the same units but different dimensions.
B. Two quantities can have the same dimensions but different units.
C. It is always possible to add two numbers that have the same units.
D. There are quantities that have no unit and dimension.

Explanation: It is impossible to add speed and velocity even if they have the same unit (m/s) since one is scalar and the other is vector.

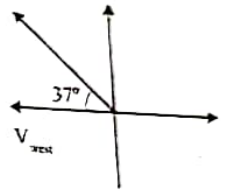
Answer: C

5. An airplane flies at speed of 720.0km/hr at 37° North of west direction. How far does the plane travel to the west in 1 hour?

A. 432 km B. 720 km C. 576 km D. 504 km

Explanation: The displacement of the plane to the west can be computed by

$s = v \Rightarrow s_{(west)} = v_{(west)} \times t$
$s = v_{(cos37)}\, t \Rightarrow s_{(west)} = v_{(west)} \times t$
$s = v_{(cos37)}\,(t) = (720)(cos37)(1)$
$s = 576km$



Answer: C

6. Abrehet (A): Bekele (B) and Chaltu (C) are pulling a box horizontally with forces $\vec{F}_A$; $\vec{F}_B$ and $\vec{F}_C$ respectively as shown below. Which statement is correct if the box is in equilibrium?

A. $\vec{F}_A = \vec{F}_B \times \vec{F}_C$
B. $-\vec{F}_A = \vec{F}_B + \vec{F}_C$
C. $\vec{F}_A = \vec{F}_B \cdot \vec{F}_C$
D. $\vec{F}_A = \vec{F}_B - \vec{F}_C$

Explanation: $\sum F = 0 \Rightarrow \vec{F}_A = \vec{F}_B \times \vec{F}_C = 0$
$-\vec{F}_A = \vec{F}_B + \vec{F}_C$

Answer: B

7. Determine the torque due to force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ N acting at a point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about the origin?

A. $6\hat{i} - 6\hat{j} + 12\hat{k}$ C. $17\hat{i} - 6\hat{j} + 13\hat{k}$

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B. $-6\hat{j}+6\hat{j}+12\hat{k}$ D. $-17\hat{i}+6\hat{j}+13\hat{k}$

Solution: $\tau=\vec{r}\times\vec{F}=\begin{vmatrix} i & j & k \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$

$\tau=17\hat{i}-6\hat{j}+13\hat{k}$

Answer: C

8. An airplane travelled from Addis Ababa to Mekele, 780km, in 45 minutes. What was its average speed?

A. 0.28m/sec C. 17.3 m/min
B. 1040 km/hr D. 585 km/hr

Explanation: $v_{av}=\frac{S_T}{t_T}=\frac{780}{45/60}=1040\text{km/hr}..,\ 45\text{min}=\frac{45}{60}\text{ hr}$

Answer: B

9. An example of a body moving with constant speed but still accelerating is:

A. A body moving on a straight road
B. A body moving on a straight railway track
C. A body moving in a circular path
D. A body falling in a viscous fluid

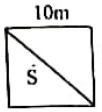
Explanation: A body is said to be accelerating when it changes speed, or direction or both. Thus a body in circular path continuously changes its direction while moving at constant speed.

Answer: C

10. A student moves along the boundary of a square field of side 10m in 40 sec. If he started from one of the corners of the student at the end of 2 minutes and 20 seconds from his initial position?

A. 10m. B. $10\sqrt{2}$m C. 40m D. 30m

Explanation: If they body takes 40 seconds to move the square filed once, he will move the field 3 and half times in 2 minutes and 20 seconds. Thus, his displacement after 140 seconds is



$S=\sqrt{40^2+10^2}=\sqrt{200}$

$S=10\sqrt{2}$m

Answer: B

11. If the impulse of a rotating disc is non-zero, what can you conclude?

A. The change in linear momentum is zero.
B. The torque acting on object is constant.
C. Impulse of the disc equals its moment of inertia.
D. The change in angular momentum is non zero.

Explanation: Impulse, $I=F\Delta t=\Delta P$. or

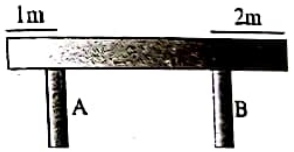
$I=\tau\Delta t=\Delta L$

it is non-zero when $\Delta L\neq 0$.

Answer: D

12. A uniform, 6m long, and 600-N beam, rests on two support, as shown. The force exerted on the beam by the right support B is closest to:

A. 75N
B. 240N
C. 400N
D. 320N

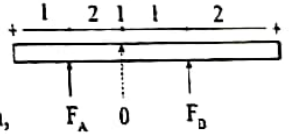


Solution:

from first condition of equilibrium.

$\sum F=0$

$\Rightarrow F_A+F_B=W=600....(*)$



From second condition of equilibrium, taking moment at the center of the rod,

$\Rightarrow 2F_A=F_B$ ...(**)

$\Rightarrow$There fore, from equation (*) and (**)

$$\frac{F_B}{2}+F_B=600\Rightarrow F_B=400\text{N}$$

Answer: C

13. A rod PQ is attached at P to a vertical wall, as shown in the figure below. The length of the rod is 1.80m. The weight W of the rod acts at 0.60m from P. The rod is kept horizontal and in equilibrium by a wire attached to Q and to the wall at R. The wire

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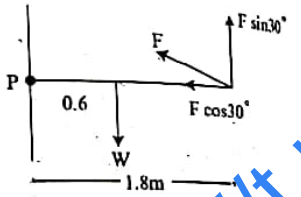
provides a force F on the rod of 50N at 30° to the horizontal. By taking moments about P, determine the weight W of the rod.

A. 130.0N B. 150.0N C. 75.0 D. 50.0N

Solution: Now, using the second condition of equilibrium and taking the pivot point at P, $0.6w = 1.8(50°)(\sin 30°)$.

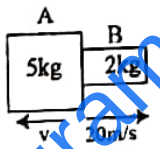


$$w = \frac{(0.8)(50)(0.5)}{0.6}$$

$$w = 75N$$

Answer: C

14. A large mass (A) and a small mass (B) are pushing one another apart, as shown in the figure below. The small mass moves away at a velocity of 20m/s to the right. What is the recoil velocity of the large mass?



A. -20m/s B. 8m/s C. 20m/s D. -8m/s

Explanation: from the conservation of momentum, $\Delta p = 0$

$p_f - p_i = 0$, $p_i = 0$

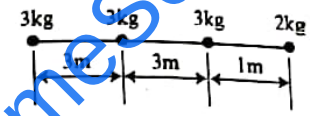
$\Rightarrow p_f = 0$

$m_A v_A + m_B v_B = 0$

$$v_A = \frac{-m_B v_B}{m_A} = \frac{-(2)(20)}{5} = -8m/s$$

Answer: D

15. Find the distance of the centre of mass of the system shown in the diagram from a point A. the rod is massless.



A. 3.73m B. 2.09m C. 2.45m D. 5.86m

Solution: $c \cdot m = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$

measuring the distances of each mass from point A,

$$c \cdot m = \frac{3(0)+3(3)+3(6)+2(7)}{3+3+3+2} = \frac{41}{11} = 3.73m$$

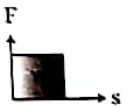
Answer: A

16. The area under a force versus distance graph represents:

A. acceleration
B. the change in energy
C. the change in momentum
D. work done

Explanation: The area under a force versus distance graph represents work done.

since, $w = Fs$



Answer: D

17. The net external non zero force acts on a particle that is moving along a straight line. Which one of the following statement is correct?

A. The velocity, but not the kinetic energy, of the particle is changing.
B. Both the velocity and the kinetic energy of the particle are changing.
C. The kinetic energy, but not the velocity, of the particle is changing.
D. The momentum is conserved.

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Explanation: A net external non zero force causes a particle to accelerate and thus to change its velocity and hence its kinetic energy.

Answer: B

18. Which statement is correct about conservative and dissipative forces?
   A. The work done by conservative forces is path dependent.
   B. The work done, along a closed path, by non-conservative forces is nonzero.
   C. Gravitational force is an example of non-conservative force.
   D. Friction is an example of conservative force.

Explanation: The work done, along a closed path, by non-conservative forces such as friction is non zero.

Answer: B

19. The rotational analogue of mass is:
   A. moment of inertia.
   B. torque.
   C. angular momentum.
   D. angular acceleration.

Answer: A

20. A fly wheel is rotating with an angular velocity of 2.0 rad/s and acted by an acceleration of $1/\pi$ rad/s^2. How long will it take to complete three revolutions?
   A. 0.5π s. B. π s. C. 2π s. D. 4π s^2.

Explanation: $\theta = \text{wot} + \frac{1}{2}\alpha t^2$

in 3 revolutions, $\theta = 3(2\pi) = 6\pi$rad

$\Rightarrow 6\pi = 2t + \frac{1}{2}\left(\frac{1}{\pi}\right)(t^2)$, solving this

$t = 2\pi$ sec

Answer: C

21. A device for transforming heat energy to mechanical energy is referred to as:
   A. transformer
   B. refrigerant
   C. generator
   D. engine.

Answer: D

22. the number of joules of heat energy required to raise the temperature of a body by 1 K is:
   A. Heat capacity.
   B. specific heat capacity
   C. expansion coefficient
   D. calorific value

Answer: A

23. A simple pendulum is made using a long, thin metal wire. What happens to the period of the pendulum when the temperature drops?
   A. Increase
   B. Remain the same
   C. Decrease
   D. Period of oscillation has random dependence on temperature.

Explanation: $T = 2\pi\sqrt{\frac{l}{g}}$, as the temperature decreases, the metal wire contracts and decreases in length which inturn causes the period to drop.

Answer: C

24. Pressure at the bottom of a liquid column is independent of:
   A. shape of the column.
   B. the density of the fluid.
   C. the depth of the point.
   D. the acceleration due to gravity.

Explanation: pressure of fluid is given by, $p = \rho gh$

Answer: A

25. The definition of Young's Modulus is given by:
   A. force per unit area.
   B. ratio of strain to stress.
   C. ratio of stress to strain.
   D. force per unit elongation.

Explanation: Youngs modulus $= \frac{\text{stress}}{\text{strain}}$

Answer: C

26. A water pipe has diameter of 5.0cm in the inlet side and 1.6cm in outlet side. Water flows at 0.50m^3 per minute at the inlet point. What is the peed of the water at the outlet point?
   A. 26.67m/s B. 13.33m/s C. 74.1m/s D. 148.1m/s

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Solution: from continuity equation,
$A_1V_1 = A_2V_2$ = discharge

$$\Rightarrow \frac{\pi\left(1.5\times10^{-2}\right)^2}{4} V_2 = \frac{0.5m^3}{min} = \frac{0.5m^3}{60sec}$$

$V_2 = 47.15m/s$

No answer

27. A metal ball has a diameter that is lightly greater than the diameter of a hole that has been cut into a metal plate. The coefficient of linear expansion for the metal from which the ball is made is greater than that for the metal of the plate. Which of the following procedures can be used to make the ball pass through the hole?

A. Raise the temperatures of the ball and the plate by the same amount.
B. Heat the ball and cool the plate.
C. Lower the temperatures of the ball and the plate by the same amount.
D. There is no way of passing the ball through the hole.

Answer: A

28. A billiard ball of radius 4cm is falling through a liquid of viscosity 0.985 at a velocity of 2.5m/s. The drag force on the ball is

A. 0.591π N
B. 0.118πN
C. 1.182π N
D. 0.059π N

Solution: $F = \pi 6\ y\ vr = \pi 6\ (0.985)(2.5)(0.04) = 0.591\pi$ N

Answer: A

29. Which one of the following statements is correct?

A. Elastic materials retain their shape after deformation.
B. Plastic materials can recover the original size or shape after being distorted.
C. Plastic materials change their shape by pressure.
D. Elastic materials cannot recover the original size or shape after being distorted.

Answer: A

30. Two bodies that are separately in thermal equilibrium with theird body must be in thermal equilibrium with each other. This statement is statement of:

A. the first law of thermodynamics
B. the zeroth law of thermodynamics
C. the second law of thermodynamics
D. the third law of thermodynamics.

Answer: B

31. Which one of the following statements is NOT correct about galvanometer?

A. It is used to measure small amount of current.
B. it has infinite resistance.
C. It can be used to measure voltage if shunted with large load in series.
D. Its working principle is the same as that of the electric motor.

Answer: B

32. Three resistors connected in series have individual voltages labeled ΔV_1, ΔV_2 and ΔV_3 respectively. Which of the following expresses the value of the total voltage Δv_T, taken over the three resistors together?

A. $\Delta V_T = \Delta V_1 + \Delta V_2 + \Delta V_3$
B. $\Delta V_T = \Delta V_1 = \Delta V_2 = \Delta V_3$
C. $V_T = \left(1/\Delta V_1 + 1/\Delta V_2\ 1/\Delta V_3\right)$
D. $V_T = \left(1/\Delta V_1 + 1/\Delta V_2\ 1/\Delta V_3\right)^{-1}$

Explanation: For resistors in series, the total potential drop is equal to the sum of the potential drops across individual resistors.

Answer: A

33. Which instrument should, ideally, have zero resistance?

A. Wheatstone bridge
B. Oscilloscope
C. Voltmeter
D. Ammeter

Answer: D

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34. What fundamental frequency can a string vibrate at if it is 30cm long and the velocity of travelling waves along the string is 120m/s?

A. 400 Hz B. 600Hz C. 800Hz D. 200Hz

Solution: $f = \frac{v}{2L} = \frac{120}{2(0.3)} = 200\text{Hz}$

Answer: D

35. The hearing curve, shown below, indicates the variation of the threshold of hearing with frequency. The ear is most sensitive at the point indicated by _____.

A. A B. C C. B D. D

Explanation: The ear is most sensitive to 2000 – 5000Hz.

Answer: C

36. Away length is:

A. a lien that represents the direction of propagation of light.
B. the minimum distance between two points that are in phase.
C. the locus of all points of a travelling wave that are in phase.
D. the number of waves crossing a given plane in a unit time.

Answer: B

37. When viewed in white light, soap bubbles show color because of:

A. interference C. scattering
B. dispersion D. diffraction

Explanation: Colorful patterns in soap bubbles is due to a thin film interference

Answer: A

38. In the sketches below, which best describes the location of the negative charges (electrons) when a negatively charged rod is brought near but not touching is conducting sphere?

A. C B. D C. A D. B

Explanation: Induction causes polarization of charges in a neutral body.

Answer: D

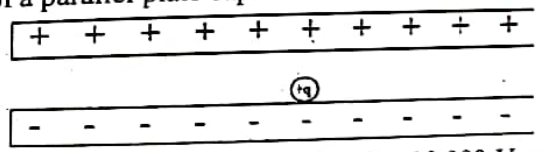
39. What is the potential (V) at a distance of 10m from a charge of +5.0C?

A. -4.45×10^9V C. $+4.45 \times 10^9$V
B. -0.5V D. +0.5V

Solution: $V = \frac{KQ}{r} = \frac{8.9\times10^9(5)}{10} = 4.45\times10^9\,\text{V}$

Answer: C

40. A positive charge of 1.0×10^{-4}C is on the negatively charged plate of a parallel plate capacitor as shown below.



If the potential (V) on the top plate is +10,000 V and that of the bottom plate is -10,000V, how much work (W) is required to move the charge from the bottom to the top plate?

A. 2.0J B. 1.0J C. Zero D. 2.0×10^4J

Solution: $W = Q(\Delta V) = 10^{-4}\left(10,000 - \left(-10,000\right)\right) = 2\text{J}$

Answer: A

41. In photoelectric effect, the energy of photoelectrons is proportional to the:

A. frequency of light. C. speed of light.
B. intensity of light. D. absorbance of the metal

Explanation: $E = hf$

Answer: A

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42. The activity of a radioactive source falls to one sixteenth of its original value in 32 minutes. What is the half-life, in minutes, of this decay process?

A. 2 min
B. 8 min
C. 4 min
D. 16 min

Explanation: $N = N_o e^{-\lambda t}$, $\frac{1}{16}N_o = N_o e^{-\lambda(32)} \Rightarrow \lambda = \frac{\ln 16}{32}$

for half life, $\frac{1}{2}N_o = N_o e^{-\lambda(t)} \Rightarrow t = \frac{\ln 2}{\lambda} = (32)\left(\frac{\ln 2}{\ln 16}\right) = 8\text{min}$

Answer: B

43. Which statement is **NOT** correct about emissions form radioactive isotopes?

A. Alpha particles are positively charged and less penetrating than beta and gamma.
B. Beta particles are negatively charged and more penetrating than alpha particles.
C. Alpha emissions are least influenced by magnetic field.
D. The most penetrating emissions are gamma particles.

Answer: C

44. The torque experienced by a current carrying loop in an external, magnetic field is;

A. independent of the number of turns of the loop.
B. independent of the current in the loop.
C. dependent on the orientation of the loop.
D. dependent on magnetic field generated according to Lenz's law.

Explanation: $\tau = NIAB\sin\theta$

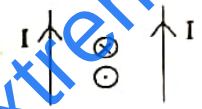
Answer: C

45. Two long wires of the same length and cross-sectional area carry the same amount of current. Which one of the following statements is correct about the magnetic field and the magnetic forces on the wires?

A. The magnetic field is zero at the midpoint between the wires when the currents are in the same direction.

B. The force between the wires is attractive when they carry current in opposite direction.
C. The magnetic fields between the wires reinforce each other when the currents are in the same direction.
D. The magnetic field due to one is proportional to the current through the other wire.

Explanation: The magnetic field produced by wires carrying current in the same directions do have opposite directions. Thus the resultant magnetic field is zero at mid point between the wires is zero.



Since $B_1 = \frac{N_o I}{2\pi r}$ in to and

$B_2 = \frac{N_o I}{2\pi r}$ out of

$\vec{B} = \vec{B}_1 + \vec{B}_2 = 0$

Answer: A

46. Self inductance is quantity that describes:

A. the number of turns in a circuit.
B. the induction of an electric field by an AC current in a lone coil.
C. the induction of a magnetic field by an AC current in a lone coil.
D. the steady state current in a coil

Answer: C

47. Which conversion process forms the basis for the function of an electric generator?

A. alternating current to direct.
B. mechanical energy to electrical.
C. electrical energy to mechanical.
D. low voltage to high or vice versa.

Answer: B

48. What is the change in magnetic flux in a coil of area $5m^2$ as its orientation relative to the perpendicular of a uniform 3.0T magnetic field changes form 45° to 90°?

A. 0 Wb
B. -11 Wb
C. 7.5 Wb
D. 11 Wb

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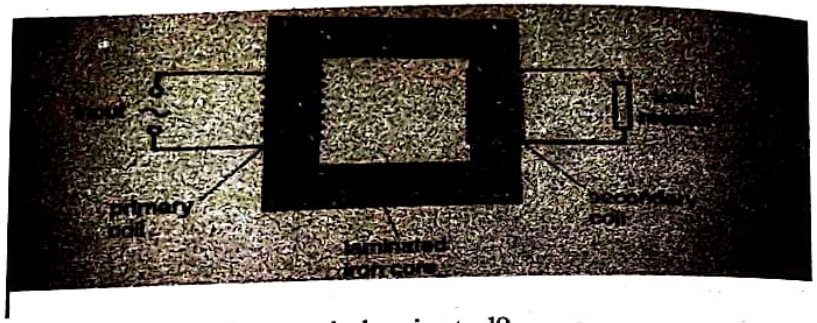
Solution: $\phi = \Delta BA\cos\theta = (5)(3)\cos 0° - (5)(3)\cos 45°$

$\phi = 7.5\,Wb$

Answer: C

49. A simple transformer is illustrated in the figure below.



Why is it that the core is laminated?

A. To increase the induced current in the primary coil.
B. To increase the induced current in the secondary coil.
C. To avoid heat exchange between the primary and secondary coils.
D. To reduce the loss in power in the core due to eddy currents.

Answer: D

50. A common unit of energy in atomic physics is:

A. plank's constant
B. electron volt
C. de Broglie wave length
D. uncertainty principle

Answer: B

1. A block of mass 5kg hanging at the end of spring scale is totally submerged in a beaker of water. The beaker is large enough to avoid overflow of water. If the density of the block is $2.5 \times 10^3 Kg/m^3$, then what is the reading of the scale?

A. 3kg B. 2kg C. 1kg D. 0 kg

Solution:

$W_{air} - W_{water} = F_{bounce}$

$W_{water} = W_{air} - F_b = \delta_b V_b g - \delta_\omega V_\omega g$, here $V_\omega = V_B V_b$

$V_b = \frac{m_b}{s_b} = \frac{5}{2500}$

$\Rightarrow W_{water} = 5(10) - 1000\left(\frac{5}{2500}\right)(10) = 50 - 20$

$W_{water} = 30N$ or $W_{water} = 3kg$ **Answer: A**

2. On an unknown scale X, the ice – point temperature is 50°X and the steam – point temperature is 200°X. If the temperature is 60°C, then what is the reading in °X?

A. 90°X B. 140°X C. 15°X D. 333°X

Solution:

$$\frac{T_x - T_x ice}{T_{x,boiling} - T_{x,ice}} = \frac{T_c - T_{c,ice}}{T_{c,boiling} - T_{c,ice}}$$

$$\frac{T_x - 50}{200 - 50} = \frac{T_c - 0}{100 - 0}$$

$\Rightarrow \frac{T_x - 50}{150} = \frac{T_c}{100}$ when $T_c = 60°C$, then

$\frac{T_x - 50}{15} = \frac{60}{60} \Rightarrow T_x = 140°$ **Answer: B**

3. Metal A has a coefficient of linear expansion that is three times the coefficient of linear expansion of metal B. How does their coefficient of areal expansion, γ, compare?

A. $\gamma_A = 2\gamma_B$ C. $\gamma_A = 4\gamma_B$
B. $\gamma_A = 3\gamma_B$ D. $\gamma_A = 6\gamma_B$

Solution:

$\alpha_A = 3\alpha_B$, and we know $\beta = 2\alpha$

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$$\therefore \frac{\beta_A}{\beta_B} = \frac{2\alpha_A}{2\alpha_B} = \frac{6}{2} \Rightarrow \beta_A = 3\beta_B$$

Answer: B

4. Three metals are in contact and are in thermal equilibrium. The metals have different specific heat capacity. What can be concluded about the temperature of the metals?
   A. Since they have different specific heat capacity, they must have different temperature at thermal equilibrium.
   B. They must have the same temperature at thermal equilibrium.
   C. They must have different internal energy.
   D. They must have different kinetic energy.

Answer: B

5. A certain heat engine draw 500 cal/s from a water bath at 27°C and rejects 400 calories to a reservoir at a lower temperature. What is the efficiency of this engine?
   A. 80% B. 75% C. 25% D. 20%

Solution:

$$\eta = \left(1 - \frac{Q_C}{Q_H}\right)100\% = \left(1 - \frac{400}{500}\right)100\%$$

$$\eta = (0.2)100\% = 20\%$$

Answer: D

6. Which of the following statements is correct?
   A. Irreversible process is a process that can be reserved through finite change in system.
   B. Complete conversion of energy from a hot source into work is possible.
   C. There are processes in which there is an overall decrease in entropy.
   D. Entropy is a measure of the amount of disorder.

Answer: C

7. Which of the following is **NOT** correct about a simple harmonic oscillator?
   A. Total mechanical energy is conserved.
   B. The acceleration is directly proportional to and opposite in direction to displacement.

   C. The period of small oscillation of a pendulum with string length l, a bod mass m attached to its end is $T = 2\pi\sqrt{\frac{l}{g}}$.
   D. The frequency of an oscillating mass attached to a massless spring is independent of the spring constant.

Answer: D

8. Which one of the following indicates with wavelength of a travelling wave?
   A. The distance between an adjacent trough and a peak.
   B. The amplitude of the wave.
   C. The distance between a node and antinodes.
   D. The distance between two adjacent points which are in phase.

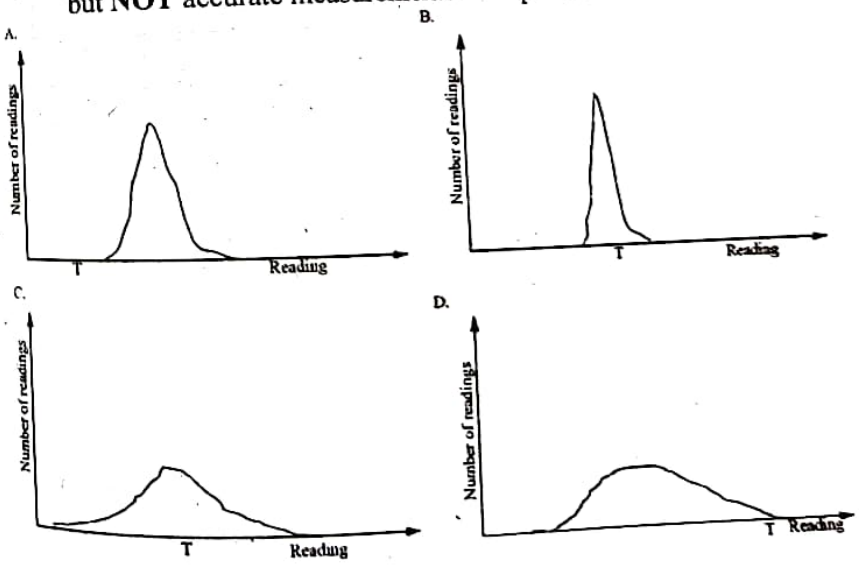
Answer: D

9. If a precision of surveyor measured a distance of 500 meters/between two points in a village/using a meter that has centimeter division on it, then what is the measurement result?
   A. 500m B. 500.0m 500.00m D. 500,000m

Hint: Since centimeter is the smallest division it is 0.01m.

Answer: C

10. From the figures given below, which one of them shows precise but **NOT** accurate measurements? T represents the true value.



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Hint: Precision is the measure of the closeness of values to each other.

Answer: A

11. If vector $\vec{p}$ is 6m in the x – direction and vector $\vec{q}$ is 8m in the y – direction. Then what is the value of the vector $\vec{p}+\vec{q}$?

A. 10m at 53° from x – direction.
B. $\sqrt{136}$ m at 45° from x – direction.
C. 10m at 37° from x – direction.
D. $\sqrt{136}$ m at 30° from x – direction.

Solution:

$$\vec{R}=\vec{P}+\vec{q}$$

$$|R|=\sqrt{P^2+q^2}=\sqrt{(8)^2+(6)^2}=10m$$

and, $\tan\theta=\frac{8}{6}\Rightarrow\theta=\tan^{-1}\left(\frac{4}{3}\right)=53°$

Answer: A

12. Given vectors $\vec{A}=\sqrt{3}\hat{i}+\hat{j}$ and $\vec{B}=\sqrt{3}\hat{i}-\hat{j}$. What is the smaller angle between the vectors?

A. 60° B. 53° C. 45° D. 30°

Solution: $A.B = AB\cos\theta$

$$\cos\theta=\frac{A.B}{AB}=\frac{(\sqrt{3})(\sqrt{3})+(1)(-1)}{\left(\sqrt{(\sqrt{3})^2+(1)^2}\right)\left(\sqrt{(\sqrt{3})^2+(-1)^2}\right)}=\frac{2}{4}=\frac{1}{2}$$

$$\cos\theta=\frac{1}{2}\Rightarrow\theta=\cos^{-1}\left(\frac{1}{2}\right)=60°$$

Answer: A

13. If $\vec{A}=\hat{x}$ m/s + 3m/s $\hat{z}$, and $\vec{B}$ = 2m/s $\hat{y}$ + 3 m/s $\hat{y}$ + 3m/s $\hat{z}$, then what is the value of $|\vec{A}\times\vec{B}|$?

A. $\sqrt{10}$m/s B. $\sqrt{13}$m/s C. 7m/s D. $\sqrt{130}$m/s

Solution: $A\times B=\begin{vmatrix} i & j & k \\ 1 & 0 & 3 \\ 0 & 2 & 3 \end{vmatrix}$ (i, j, k +ve)

$= (0-6)i + (0-3)j + (2-0)k = -6i - 3j + 2k$

$$|A\times B|=\sqrt{(-6)^2+(-3)^2+(2)^2}=7m/s$$

Answer: C

14. A cricketer throws a ball vertically upwards with initial speed of 20m/s. How long is it in the air before it returns to the cricketer's hands?

A. 2s B. 10s C. 1.5s D. 4s

Solution:

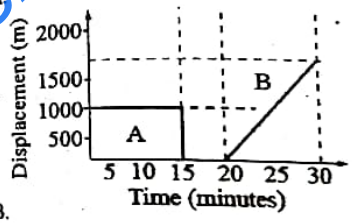
From, $y=ut-\frac{1}{2}gt^2$, $y=0$

$$\Rightarrow t=\frac{2u}{g}=\frac{2(20)}{10}=4\sec$$

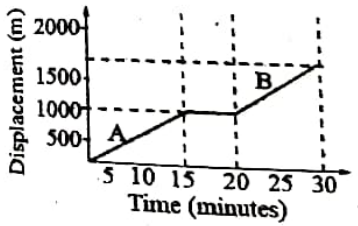
Answer: B

15. Abebe walks to school. She walks 1km in 15 minutes. She meets her friend Makeda – they talk for 5 minutes and then carry on walking to school. They walk 800m in 10 minutes. Which of the following displacement – time graph shows Abeba's journey to school?

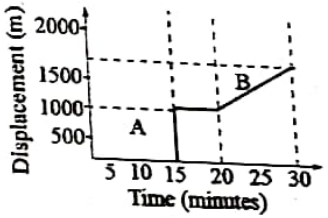
A.



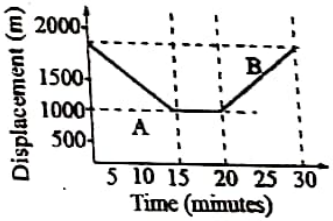
B.



C.



D.



Answer: B

16. A fun making boy swings a blinking ball tied to the end of a string in a vertical circle. If the radius of the circle is 0.5m, then what should be the minimum speed of the ball at the bottom of the circle to make a complete circular motion at the top?

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A. 5m/s B. $\sqrt{5}$m/s C. $\sqrt{15}$m/s D. $2\sqrt{5}$m/s

Solution: $mg = \frac{mv^2}{r} \Rightarrow v = \sqrt{gr} = \sqrt{(10)(0.5)} = \sqrt{5}$m/s

Answer: B

17. The range of cannon – ball fired horizontally from a laboratory table is equal to 8/3 times the height of the table. What is the direction of velocity vector of the projectile as it strikes the ground?

A. 45° from the horizontal C. 60° from the horizontal
B. 53° from the horizontal D. 37° from the horizontal

Solution: $x = \frac{8}{3}y$

$v_x = \frac{x}{t}$, but $t = \sqrt{\frac{2y}{g}}, x = \frac{8}{3}y$

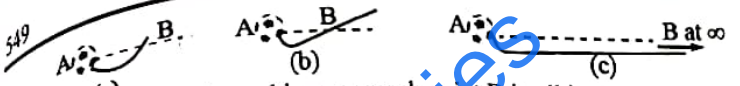
$v_x = \frac{\frac{8}{3}y}{\sqrt{\frac{2y}{g}}} = \frac{8y}{3\sqrt{\frac{2y}{g}}}$, and

$v_y = gt = g\sqrt{\frac{2y}{g}}$

$\tan\theta = \frac{v_y}{v_x} = \frac{g\sqrt{2y/g}}{\frac{8y}{3\sqrt{\frac{2y}{g}}}} = \frac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right) = 37°$

Answer: D

18. Galileo carried out a 'thought' experiment on motion. The line between A and B represents a surface and the center of the ball is placed at A. What would happen in each of the three situations shown in the figure below, without friction when the ball is released?

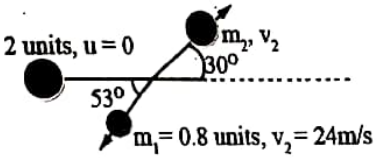


A. The ball would never reach point B in all instances.
B. The ball would reach B in all instances even if point B keeps on receding to infinity in (c).
C. The ball would reach point B in (a) and (b) and not in (c).
D. The ball would reach point B only in (a).

Hint: In the absence of external force, the ball will move forever.

Answer: B

19. A Uranium atom initially at rest has a mass of 2 units and spontaneously disintegrated into two fragments. One of the fragments has a mass of 0.8 units and moves at a speed of 24 m/s as shown in figure below. What is the speed of the second fragments?



A. 25.6m/s
B. 21.3m/s
C. 16m/s
D. 14.8m/s

Solution: From conservation of momentum

$Pi = Pf \Rightarrow m\begin{bmatrix} u_x \\ u_y \end{bmatrix} = m_1\begin{bmatrix} v_1x \\ v_1y \end{bmatrix} + m_2\begin{bmatrix} v_2x \\ v_2y \end{bmatrix}$

$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0.8\begin{bmatrix} -14.4 \\ -19.2 \end{bmatrix} + 1.2\begin{bmatrix} 0.87v_2 \\ 0.5v_2 \end{bmatrix}$

$11.52 = 1.04v_{2x} \Rightarrow v_{2x} = 11.08$m/s

$15.36 = 0.6v_{2y} \Rightarrow v_{2y} = 25.6$m/s

$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2}$

20. A 1 kg object explodes and breaks up into three pieces. One piece has a mass of 400g and has a velocity of $(15\vec{i}+10\vec{j})$m/s. A second piece has a mass of 250 g and has a velocity of $(25\vec{i}-10\vec{j})$m/s. What is the velocity the third piece?

A. $(-4.3i+35\vec{j})$m/s C. $(35\vec{i}-4.3\vec{j})$m/s

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B. $(-35\vec{i}-4.3\vec{j})$m/s D. $(-35\hat{i}-4.3\hat{j})$m/s

Solution: $p_i = p_f$

$$\begin{bmatrix}0\\0\end{bmatrix}=0.4\begin{bmatrix}15\\10\end{bmatrix}+0.25\begin{bmatrix}25\\-10\end{bmatrix}+0.35\begin{bmatrix}v_x\\v_y\end{bmatrix}$$

21. Which of the following statements is **NOT** correct?
   A. Dynamic equilibrium is an equilibrium in which a body is moving at steady velocity without net force or torque acting on it.
   B. Static equilibrium is an equilibrium in which a body is at rest without net force or torque acting on it.
   C. Couples are set of forces with a resultant torque but no net force.
   D. The necessary and sufficient condition for equilibrium of a rigid body is to have the sum of all torques acting on it is zero.

Answer: D

22. A spring with a spring constant of 300N/m is stretched 20cm in 2 seconds. What is the power applied to stretch the spring?
   A. 3W B. 6W C. 30kW D. 60kW

Solution:

$$P=\frac{\omega}{t}=\frac{\frac{1}{2}kx^2}{t}=\frac{(300)(0.2)^2}{2(2)}=3W$$

Answer: A

23. A 50 kW machine is used to lift heavy objects to the top of a building under construction. If this machine is replaced with a new machine of 100kW power, then which of the following statements is **NOT** true about the new machine?
   A. The new machine can raise objects with the same mass at twice the speed of the old machine.
   B. The new machine can raise objects with twice as much mass at the same speed as the old machine.
   C. The new machine takes twice as much time to do the same work as the old machine.

   D. The new machine can do twice as much work in the same time as the old machine.

Hint: $P=\frac{W}{t}$ **Answer: C**

24. Which of the following is correct about a work done for dissipative forces?
   A. Path dependent.
   B. The same as the loss in kinetic energy.
   C. The same as the gain in potential energy.
   D. The same as the potential energy stored in spring.

Answer: A

25. A large (very large in height and cross – section) water tank open at the top has a small hole through its side near the bottom. To what height should the water be filled, to double the speed of water that flow out of the hole when it was filled to a height h?
   A. h B. 2h C. 3h D. 4h

Hint: $P_E = k_E$

$$mgh=\frac{1}{2}mv^2, v_2=2v_1$$

$$v_1=\sqrt{2gh}$$

$$v_2=\sqrt{2gh}$$

$$\frac{v_2}{v_1}=\frac{\sqrt{2gh'}}{\sqrt{2gh}}\Rightarrow\frac{v_2}{v_1}=\sqrt{\frac{h'}{h}} \text{ or } 2=\sqrt{\frac{h'}{h}}\Rightarrow h'=4h$$

Answer: A

26. Which of the following is the application of Pascal's law?
   A. Hydraulic brake in vehicles.
   B. Siphon to remove fluids.
   C. Working of submarines.
   D. Sucking of liquid into a syringe.

Answer: A

27. Which of the following refers to the definition of strain as a ratio?
   A. Extension to the original length.
   B. Force applied to area of the object normal to the force.
   C. Stress to Young modulus.
   D. Tensile to applied force.

Answer: A

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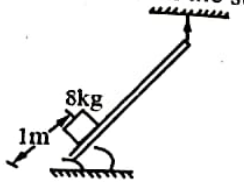
28. A 4m long uniform board of mass 10 kg hinged on the ground at one end is hanging by a string at the other end. A block of mass 8 kg is placed at 1m from the hinge on the ground as shown in the diagram below. What is the tension in the string?



A. 10N B. 50N C. 70N D. 110N

Solution: From 2nd condition of equilibrium, taking moment at the base.

$80(1\cos 60°)v + 100(2\cos 60°) = T(4\cos 60°)$

$80(0.5) + 100(2 \times 0.5) = 2T$

$40 + 100 = 2T \Rightarrow T = 70°$

Answer: C

29. Which of the following statements is correct about the first condition of equilibrium?

A. The net force on an object must be zero.

B. The sum of all torques must be zero.

C. Clockwise moments should balance counter-clockwise moments.

D. The torque produced by a force should be perpendicular to the force.

Answer: A

30. A block of mass $m_1 = 10kg$ placed on a frictionless incline plane connected to mass $m_2 = 15kg$ using a string of negligible mass passing over a pulley of moment of inertia $I = 0.25\ kgm^2$ and radius $R = 10cm$ as shown in the diagram. What is the torque on the pulley?

A. 2.5Nm B. 4Nm C. 5Nm D. 12Nm

Solution:

For m_2, $m_2g - T = m_2a$

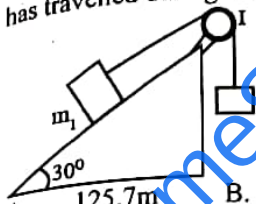
For m_1, $T - m_1g\sin\theta = m_1a$

$$\Rightarrow a = \frac{m_2g - m_1g\sin\theta}{m_1 + m_2} = \frac{150 - 50}{25} = \frac{1000}{25} = 4m/s^2$$

$$\text{And, } T = I\alpha = I\left(\frac{a}{r}\right) = (0.25)\left(\frac{4}{0.1}\right) = 10Nm$$

No answer

31. A pulley is rotating at a frequency of 20 rev/min. A motor speeds up the wheel so that 30s later the frequency of the motion is 60 rev/min. How far will a point on the pool 0.30m from its center has travelled during that period?



A. 125.7m B. 0.13m C. 37.68m D. 4.2m

Solution:

$$\theta = \left(\frac{\omega + \omega_o}{2}\right)t = \left(\frac{20+60}{2}\right)\left(\frac{1}{2}\right) = 20rev$$

$$s = r\theta = 20(2\pi)(0.3) = 20(2 \times 3.14)(0.3) = 37.68m$$

Answer: C

32. A disk of radius R with uniform distribution of mass M, rotates about an axis perpendicular to its plane at the rim with angular speed ω. The moment of inertia of the disc about the axis through the center is ½ MR^2. What is the kinetic energy of the disk?

A. $\frac{1}{4}MR^2\omega^2$ C. $\frac{1}{2}MR^2\omega^2$

B. $\frac{1}{3}MR^2\omega^2$ D. $\frac{3}{4}MR^2\omega^2$

Hint: $k_E = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2 = \frac{1}{4}MR^2\omega^2$

Answer: A

33. The activity of radioactive substance reduces from 2000Bq to 1000Bq in 10^6 seconds. What is the decay constant?

A. 6.93×10^5 decay/s C. 2.00×10^6 decay/s

B. 6.93×10^{-7} decay/s D. 5.00×10^{-7} decay/s

Answer: B

34. Which of the following phenomena clearly demonstrates the particle nature of light?

A. Refraction of light by prism.

B. The photoelectric effect

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C. Diffraction of light by double slit.
D. Dispersion of white light.

35. The capacitance of a parallel plate capacitor is 10μF. If the distance between the plates is halved, keeping all other quantities constant, then what is the new capacitance of the capacitor? **Answer: B**

A. 0.2μF B. 5μF C. 10μF D. 20μF

Hint: $C = \varepsilon_r \varepsilon_o \frac{A}{d} \Rightarrow C \sim \frac{1}{d}$

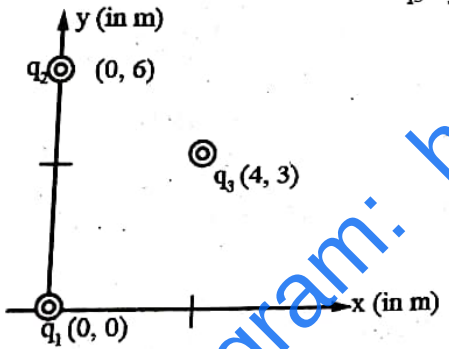
36. Two positive charges of magnitude 2.4×10^{-9}C and 1.6×10^{-9}C are separated in air by a distance of 2.0cm. How much electrostatic potential is produced by the two changes at midway between them? **Answer: D**

A. 3.6KV B. 0.72KV C. 2.16KV D. 1.44KV

Solution: $v = v_1 + v_2 = \frac{kQ_1}{r_1} + \frac{kQ_2}{r_2} = 2160 + 1440$

$v = 3600 = 3.6kv$

37. A charge of $q_1 = 10.0 \times 10^{-9}$C, $q_2 = +10.0 \times 10^{-9}$C and $q_3 = -2.5 \times 10^{-8}$C are placed on x – y plane as shown in the figure below. What is the electric force on q_3 by q_1 and q_2? **Answer: A**



A. 1.08×10^{-7}N in +ve y – direction.
B. 9.00×10^{-7}N in –ve y – direction.
C. 1.42×10^{-7}N in +ve x – direction.
D. 1.42×10^{-7}N in –ve x – direction

Solution:

$$F_{13} = \frac{kQ_1Q_3}{r^2} = \frac{(9\times10^9)(10\times10^{-9})(2.5\times10^{-8})}{(5)^2} = 9\times10^{-8}N$$

$$F_{23} = \frac{kQ_2Q_3}{r^2} = 9\times10^{-8}N$$

$F_x = F_{1x} + F_{2x}$ & $F_y = F_{1y} + F_{2y} = 0$ &

$F_y = 9 \times 10^{-8}\left(\frac{3}{5}\right) + 9 \times 10^{-8}\left(\frac{4}{5}\right)$

$F_R = F_y = 1.08 \times 10^{-7}$N in +ve y direction.

38. In a Young's experiment, there are two slits, rather than one. What do you think the reason? **Answer: A**

A. To increase the intensity.
B. One slit is for frequency, the other for wavelength.
C. To create path length difference.
D. One slit is for electric fields, and the other is for magnetic fields.

39. A light of wavelength 590nm is incident normally on a grating with spacing 1.67×10^{-6}m. How many orders of diffraction maxima can be obtained (including the zero order)? **Answer: C**

A. 1 B. 2 C. 3 D. 4

Solution: $m \le \frac{d}{\lambda}, m \le \frac{1.6\times10^{-6}}{590\times10^{-9}} = 2.8 = 2$ including the zero order, m = 3 **Answer: C**

40. Which of the following statements is correct?

A. Interference is not superposition of waves.
B. Constructive interference results in a decrease in the intensity.
C. Destructive interference results in an increase in the intensity.
D. Path difference which is integer multiple of wavelength results in constructive interference.

Answer: D

41. An electric generator has a coil of 400 turns and cross – sectional area $5m^2$ rotating at angular frequency 0.52rad/s in a uniform magnetic field 0f 0.2T. What is the induced e.m.f produced from the electric generator in each second?

A. 209.6V B. 104.8V C. 200V D. 182.4V

Solution: $\varepsilon = NAB\omega = (400)(5)(0.2)(0.524) = 209.6V$

Answer: A

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42. The primary coil of a transformer is connected to a battery, a resistor, and a switch. The secondary coil is connected to an ammeter. What does the ammeter show when the switch is thrown closed?

PICTURE

A. Zero current as transformer is no working on DC.
B. A nonzero current for a short instant.
C. A steady current.
D. Emf will be induced but not current.

Answer: A

43. Which of the following statement is correct?

A. The direction of induced e.m.f. is in such as direction as to re-enforce the change produces it.
B. Self-inductance of a coil is e.m.f. divided by magnitude or rate change of current with time.
C. The induced e.m.f. in a coil is inversely proportional to its number of turns.
D. Self-inductance of a coil is inversely proportional to the cross – sectional area of the coil.

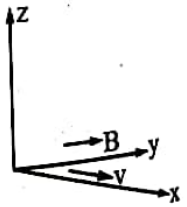
Answer: B

44. The induced electromotive force in a current carrying coil placed in a magnetic field does **NOT** depend on which of the following?

A. The speed at which the coil moves if the relative motion does not result in changing the magnetic flux.
B. The change of magnetic flux density.
C. The length of the coil that is moving in a direction perpendicular to that of the magnetic field.
D. The area of the coil outside the field.

Answer: D

45. A positivity charged particle of charge $q = 3.2 \times 10^{-19}C$ is moving parallel to x – axis with speed 1000m/s in a magnetic field of 0.5T parallel to y – axis as shown in the diagram below. What is the magnetic force on the charge?



A. 8.0×10^{-16} N in – ve z
B. 3.2×10^{-16} N in – ve x
C. 8.0×10^{-17} N in +ve x
D. 1.6×10^{-16} N in +ve z

Solution: $\vec{F} = q(\vec{v} \times \vec{B}) = 1.6 \times 10^{-16} N\hat{k}$

Answer: D

46. A charged particle moves perpendicular to a uniform magnetic field. Which statement is correct about the motion of the particle?

A. Both kinetic energy and linear momentum of the particle change.
B. The kinetic energy of the particle changes but not its linear momentum.
C. The linear momentum of the particle changes but not its kinetic energy.
D. Both the energy and momentum do not change.

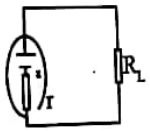
Answer: C

47. Which of the following is **NOT** correct about magnetic field?

A. Magnetic field lines are closed.
B. Magnetic monopoles exist.
C. The closer the field lines are, the stronger the magnetic field.
D. Magnetic field is produced by moving charge.

Answer: B

48. Which of the following is **NOT** correct about the circuit shown below?



A. Voltage across R_L is $\frac{R_L}{r+R_L}\varepsilon$
B. Current flowing through R_L is $\frac{\varepsilon}{r+R_L}$
C. Current flowing through r is $\frac{r}{r+R_L}\varepsilon$
D. Power drop in R_L is $\frac{R_L}{(r+R_L)^2}\varepsilon^2$

Answer: C

Hint: $I = \frac{\varepsilon}{r+R}$, both in R and r.

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49. If a cell supplies a current of 0.9A when connected to a 2Ω resistor and a current of 0.3A, then what will be the internal resistance of the cell when connected to a 7Ω resistor?
A. 2Ω B. 1.2Ω C. 1Ω D. 0.5Ω

Solution:

$$I = \frac{\varepsilon}{r+R} \Rightarrow 0.9 = \frac{\varepsilon}{2+r} \text{ and, } 0.3 = \frac{\varepsilon}{7+r} \Rightarrow r = 0.5\Omega$$

Answer: D

50. A galvanometer of full – scale deflection 10mA is to be converted into a 0 – 10.0A ammeter. If its coil has a resistance of 100Ω, then what value of shunt resistance must be fitted?
A. 0.1Ω B. 10Ω C. 9.99Ω D. 0.999Ω

Solution:

$$R_S = \frac{R_C I_C}{I - I_C} = \frac{(100)(10\times10^{-3})}{10 - 10\times10^{-3}} = 0.1\Omega$$

Answer: A

1. Four vectors are listed below.

$$\vec{A} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}, \vec{B} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}, \vec{C} \begin{bmatrix} -6 \\ 15 \end{bmatrix}, \vec{D} = \begin{bmatrix} 15 \\ 6 \end{bmatrix}$$

Which pair of the vector is orthogonal?
A. $\vec{A}$ and $\vec{D}$ C. $\vec{C}$ and $\vec{D}$
B. $\vec{B}$ and $\vec{D}$ D. $\vec{A}$ and $\vec{B}$

Solution: Two non-zero vectors are orthogonal if their scalar product is zero.

$$\vec{A} \cdot \vec{D} = (4)(5) + (10)(-6) = 0$$

Answer: A

2. A race car is moving at a constant speed of 35 m/s. A security car was moving at a speed of 5 m/s as the race car passes by it and was accelerating at constant rate of 5 m/s^2. What was the speed of the security car when it took over the race car?
A. 35 m/s B. 60 m/s C. 5 m/s D. 65 m/s

Solution: $S_{race} = S_{security}$

$$Vt = Ut + \frac{1}{2}at^2$$

$$V = u + \frac{1}{2}at$$

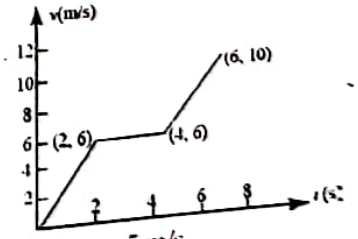
$$35 = 5 + \frac{1}{2}(5)t$$

$$t = 12 \text{ sec.}$$

At $t = 12$ sec, $V_{security} = u + at = 5\ (5)\ (112) = 65$m/s

Answer: D

3. A car start moving from rest and its motion on a straight line is shown in the figure below. What is the average velocity of the car within the first 6 seconds?



A. 5.67 m/s
B. 1.67 m/s
C. 5 m/s
D. 6 m/s

Solution: $\overline{V_{av}} = V_{av} = \frac{S_T}{t_T} = \frac{\text{Area}}{\text{time}}$

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$$= \frac{\frac{1}{2}(2)(6)+(2)(6)+\frac{1}{2}(2)(4)+(2)(6)}{6}$$

$= 5.67$ m/s

Answer: A

4. Which one of the following statements is correct regarding the motion in a plane?
   A. When a body moves in a horizontal circle, its velocity is constant.
   B. When a body moves in a vertical circle, its speed is constant.
   C. The centripetal force for a body moving in either vertical or horizontal circle is toward the center.
   D. In projectile motion the horizontal component of the motion is uniformly accelerated motion.

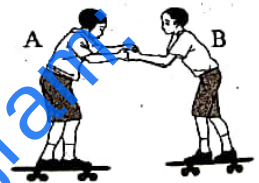
Answer: C

5. A bullet is fired at an angle of 30° above the horizontal. If air resistance is neglected, what should be the bullet's horizontal acceleration $(\vec{a}_x)$ and vertical acceleration $(\vec{a}_y)$ when it reaches its maximum height?
   A. Both $\vec{a}_x$ and $\vec{a}_y$ are zero.
   B. $\vec{a}_x$ is 10m/s$_2$ downward, and $\vec{a}_y$ is zero
   C. Both $\vec{a}_x$ and $\vec{a}_y$ are 10m/s^2, downward
   D. $\vec{a}_x$ is zero and $\vec{a}_y$ is 10m/s^2 downward.

Hint: For a projectile motion, the horizontal velocity is constant, and hence $\vec{a}_x = 0$.

Answer: D

6. Two students A and B, with masses m_A and m_B respectively, stand on a wheeled trolleys as shown in the figure below. If student A has larger mass and pushes student B, then what can be said about their accelerations?



   A. The acceleration of student B is less than that of student A.
   B. Both boys will be accelerated by the same magnitude but in opposite directions.
   C. The acceleration of student A is less than that of student B.
   D. There would be no acceleration because the forces are equal and opposite.

Explanation: Even if the force on each is the same, acceleration is inversely proportional to mass.

Answer: C

7. Which of the following statement is correct about errors in measurement?
   A. Experimental error is the same as a mistake.
   B. Errors make measured values same as accepted value.
   C. Errors occur in every scientific investigation.
   D. Errors can be avoided by using modern measuring instruments.

Hint: Errors can not be avoided in a measurement but can be minimized.

Answer: C

8. What can be concluded when the number of significant figures increases?
   A. True reading increases
   B. Precision increases
   C. Uncertainty increases
   D. Accuracy increases

Answer: B

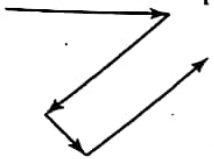
9. Two forces $\vec{F}_1 = (8\vec{i}+3\vec{j})$N and $\vec{F}_2 = (4\vec{i}+6\vec{j})$N are acting on an object. What is the magnitude and direction of the resultant force?
   A. 225 N, 143°
   B. 5 N, −37°
   C. 5 N, 143°
   D. 15 N, 37°

Solution:

$$\vec{F}_R = \vec{F}_x + \vec{F}_y$$
$$= (8+4)\mathrm{i} + (3+6)\mathrm{j}$$
$$= (12\mathrm{i} + 9\mathrm{j})\ \mathrm{N}$$
$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{12^2 + 9^2} = 15\mathrm{N}$$
$$\sin\theta = \frac{F_y}{F} = \frac{9}{15} = 0.6$$
$$\theta = 37°$$

Answer: D

10. The diagram below shows a path taken by a flying bird.



Which one of the following vector may represent a resultant displacement?

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A. ↗ B. → C. ↙ D. ↘

Hint: displacement is the shortest path between the start and end of a journey

Answer: D

11. A body that is released from the top of a tall building reaches the ground after six seconds. What is the ratio of the kinetic energy acquired by it at the end of the three seconds to the increment in the kinetic energy during the next three seconds?

A. 3/2 B. 1 C. 1/3 D. 2

Solution: $\frac{K_{E1}}{K_{E2}} = \frac{\frac{1}{2}mV_1^2}{\frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2}$

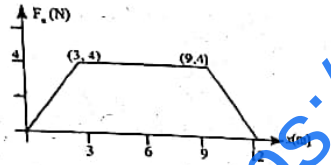
Since $V = gt$

$V_1 = 3g$ and $V_2 = 6g$

$$\frac{K_{E1}}{K_{E2}} = \frac{\frac{1}{2}m(3g)^2}{\frac{1}{2}m(6g)^2 - \frac{1}{2}m(3g)^2} = \frac{1}{3}$$

Answer: C

12. A boy pushes a block over 12 m distance within 6 seconds. The force applied by the boy varies as shown in the diagram below. What is the average power loss by the boy?



A. 4 w B. 8w C. 7w D. 6w

Solution: Power $= \frac{\text{Work}}{\text{time}} = \frac{\text{Area}}{\text{time}}$

$$= \frac{\frac{1}{2}(3)(4) + (6)(4) + \frac{1}{2}(3)(4)}{6}$$

$= 6w$

Answer: D

13. A thin rod when rotated about an axis at the end of the rod has a moment of inertia $\frac{1}{3}ML^2$. The rod has a mass of 1.5 kg and length of 1.0 m. If the rod initially at rest gains a rotational kinetic energy of 1000 J, at what speed does it rotate?

A. $10\sqrt{5}$ rad / s C. $20\sqrt{10}$ rad / s

B. 10 rad /s D. 18.8 rad/ s

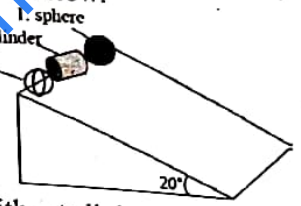
Solution: $K_{Er} = \frac{1}{2}Iw^2$, $I = \frac{1}{3}mL^2 = \frac{1}{3}(1.5)(1)^2 = 0.5$

$$1000 = \frac{1}{2}(0.5)\omega^2$$

$\omega = 20\sqrt{10}$ rad / sec

Answer: C

14. A ring, a solid cylinder and a solid sphere, all having the same mass and the same radius, are held at the top of an inclined plane as shown in the figure below.



If the objects roll without sliping and the coefficient of static friction between each object and the plane is 0.3, which one of the following shows the order in which they will get to the bottom of the inclined plane?

A. All will reach the bottom at the same time
B. Ring, solid cylinder, sphere
C. Solid cylinder, sphere, ring
D. Sphere, solid cylinder, ring

Hint: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$, let $I = Cmr^2$

$$gh = \frac{1}{2}V^2 + \frac{1}{2}Cmr^2\frac{V^2}{r^2}$$

$V = \sqrt{\frac{2gh}{(1+c)}}$, small $c \Rightarrow$ large V

Answer: D

15. If in a given rotational system of a body the angular momentum is increased with time, which one of the following statement is **NOT** correct?

A. The radius of rotation may increase.
B. There is no torque exerted on the system.
C. The moment of inertia may increase.
D. There is a torque exerted on the system.

Answer: B

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16. Which of the following is a correct statement about the first condition of equilibrium?
A. The sum of all torques must be zero.
B. Clockwise moments should balance counter clockwise moments.
C. The net force on an object must be zero.
D. The torque produced by a force should be perpendicular to the force.

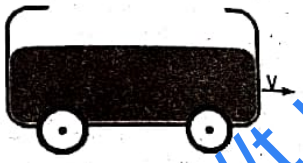
Answer: C

17. For a constant force F, if the impulse imparted to the system is large, then what can you conclude about the collision?
A. Linear momentum s conserved.
B. The contact or impact time is large.
C. Kinetic energy is conserved.
D. The charge in velocity is small.

Hint: Impulse = F (Δt)

Answer: C

18. An open trolley is rolling on a level surface without frictional loss through a vertical downpour of rain, as shown in the figure below



As the trolley rolls, an appreciable amount of rainwater accumulates in the cart. What will be the speed of the cart?
A. It increases because of conservation of mechanical energy.
B. It decreases because of conservation of mechanical energy.
C. It remains the same because the raindrops are falling perpendicular to the direction of the cart's motion.
D. It decreases because of conservation of momentum.

Explanation: momentum is conserved since there is no friction force. Hence,

$$P_i = P_f \Rightarrow m_1 u = m_2 v \Rightarrow v = \frac{m_1 u}{m_2}$$

Since $m_2 > m_1$, speed decreases

Answer: D

19. A satellite moves at a constant speed in a circular orbit about the center of the Earth at an altitude half the radius of the Earth above its surface. If g is the gravitational acceleration at the surface of the Earth and R its radius, then what is the speed of the satellite?
A. $\sqrt{2gR}$ B. $\sqrt{gR}$ C. $\sqrt{3gR/2}$ D. $\sqrt{2gR/3}$

Solution: $\frac{mv^2}{r} = \frac{GmM}{r^2}$, $r = R + \frac{R}{2} = \frac{3R}{2}$

$$V^2 = \frac{2GM}{3R}$$

And, $mg = \frac{GmM}{r^2}$

$$g = \frac{4GM}{9R^2} = \frac{2}{3R}\left(\frac{2GM}{3R}\right) = \frac{2}{3R}V^2$$

$$\Rightarrow V = \sqrt{\frac{3gR}{2}}$$

Answer: C

20. An object of mass 1 kg moving with a speed of 2 m/s was acted on by a force that produces an acceleration of 2 m/s^2. What is the change in kinetic energy of the object if the force is exerted on the object for 2 seconds in its direction of motion?
A. 1 J B. 16J C. 8J D. 4J

Solution:

$$V = u + at = 2 + 2(2) = 6$$

$$\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}(1)\left[6^2 - 2^2\right] = 16J$$

Answer: B

21. In the figure below, what is the speed of water as it leaves A_2?
A. 0.9 m/s
B. 0.3 m/s
C. 3.3 m/s
D. 3.6 m/s

Solution: $A_1 V_1 = A_2 V_2$

$(4)(0.9) = 12(V_2)$

$V_2 = 0.3$ m/s

Answer: B

22. An electric immersion water heater is rated at 400 W. How long will it take to heat one kilogram of water from 10°C? (The specific heat of water is 4.2 J/g K.)
A. 45 min B. 1min C. 15 min D. 3.5 min

Solution: $\Delta Q = mc\,\Delta T$, $c = 4.2\frac{J}{gk} = 4200\frac{J}{kgk}$

$$P = \frac{\Delta Q}{\Delta t} = \frac{(1)(4200)(20)}{\Delta t}$$

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$$400 = \frac{84000}{\Delta t}$$

$\Delta t = 210 \text{ sec} = 3.5 \text{ min}$ **Answer: D**

23. If the internal energy of the system decreases, then what can be concluded about the heat and the work done?
   A. Heat is removed from the system and work is done by the system.
   B. Heat is added to the system and work is done by the system.
   C. Heat is removed from the system and work is done on the system.
   D. Heat is added to the system and work is done on the system.

Hint: $\Delta u = \Delta Q + \Delta W$ **Answer: A**

24. If a thermodynamic system undergoes a process in which its internal energy increased by 400 J while doing 200 J of work on its surrounding, then what is the energy absorbed in or extracted from it in the form of heat?
   A. −600 J B. −200 J C. 200 J D. 600 J

Solution: $\Delta u = \Delta Q + \Delta W$

$400 = \Delta Q - 200$

$\Delta Q = 600 J$ **Answer: D**

25. Which of the following statement is **NOT** correct?
   A. A system can absorb heat from a hot reservoir and can convert it entirely into work without additional changes in the system and its surrounding.
   B. The spontaneous transfer of energy from a cooler body to hotter body is quite possible.
   C. No process is possible in which there is an overall decrease in the entropy of the universe.
   D. The completer conversion of energy from hot source into work is not possible.

Hint: second law of thermodynamics **Answer: A and B**

26. A heat engine operating between 100 °C and 700 °C has efficiency equal to 40% of the maximum theoretical efficiency. How much energy does this engine extract from the hot reservoir in order to do 5000 J of mechanical work?
   A. 20259.32 J C. 8104. J
   B. 81.0 J D. 20. 3J

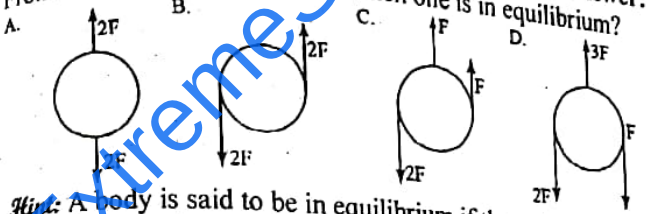
Solution: $\eta = 40\% \eta_{max}$

$$\frac{W}{Q_H} = \frac{40}{100}\left(1 - \frac{T_C}{T_H}\right), \quad T_C = 373k, \quad T_H = 973k$$

$$\frac{5000}{QN} = \frac{40}{100}\left(1 - \frac{373}{973}\right)$$

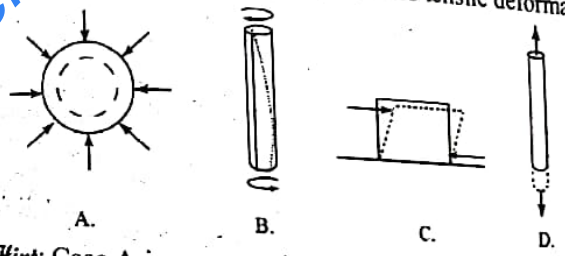
$QH = 20259.32$ J

27. From the figures given below which one is in equilibrium? **Answer: A**



Hint: A body is said to be in equilibrium if there is no net force and net torque acting on it. **Answer: A**

28. Which of the following figure indicates tensile deformation?



Hint: Case A is compression
B is Torsion
C is shear **Answer: D**

29. A beaker with water resting on a scale weighs 40 N. A block suspended on a hanging spring weighs 20 N. The spring scale reads 15 N when a block is fully submerged in the water. What is the reading of a scale on which the beaker with water rests, while the block is submerged in the water after detached from the hanging spring?
   A. 25 N B. 60 N C. 55 N D. 45 N

Hint: Since $S_{block} > S_{water}$ **Answer: B**

30. An object with a mass of 150 kg and a volume of 0.75 m^3 is floating in a liquid of density 0.8 g/cm^3. What percentage of the object's volume will be submerged below the surface of the fluid?

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A. 100% B. 50% C. 75% D. 25%

Solution: Weight = bouyant force

o, object $\rho_o v_o g = \rho_f v_f g$

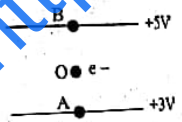
f, fluid $V_f = \frac{\rho_o v_o}{\rho_f} = \frac{150}{800} = 0.1875 m^3$

$$\frac{V_f}{V_o} \times 100\% = \left(\frac{0.1875}{0.75}\right)100\% = 25\%$$ **Answer: D**

31. Why is an interference pattern **NOT** observed between the lights produced from a pair of car headlights?
   A. The light sources are not collimated.
   B. Interference is observed only in laboratory conditions.
   C. The light sources are not coherent.
   D. The spread of light as it travels out of the source. **Answer: C**

32. Suppose that three point charges are placed along a straight line and the electrostatic force exerted on the middle charge is zero. What can we conclude about the charges?
   A. The signs of the three charges should be the same.
   B. The charges at the ends must have the same sign.
   C. The middle charge should be placed at the center between the two charges
   D. The charge at the middle and at the left end should have the same **Answer: B**

33. Two parallel plates are kept at 3 V and 5V as shown in the diagram below. If an electron is released at O, then which way will the electron move?



   A. towards B C. remains at O
   B. towards A D. parallel to the plates

Hint: Attracted by the higher potential **Answer: D**

34. A capacitor of 12μF capacitance can be fully charged after it is connected between a potential difference of 9 V. How much electrical energy is used in charging the capacitor?
   A. 1.0×10^{-3} J C. 9.72×10^{-4} J
   B. 54×10^{-4} J D. 4.86×10^{-4} J

Solution: $u = \frac{1}{2}CV^2 = \frac{1}{2}(12\times10^{-6})(9)^2$

$= 4.86 \times 10^{-4}$ J **Answer: D**

35. A copper wire of cross – sectional area 2 mm^2 carrying a current of 1.5 A. If the drift velocity of the electron is 0.00028 m/s, what is the number of charge carriers (free electrons) per unit volume in the copper wire?
   A. $1.5 \times 10^{25}/m^3$ C. $1.5 \times 10^{32}/m^3$
   B. $1.67 \times 10^{28}/m^3$ D. $1.67 \times 10^{32}/m^3$

Solution: $V_d = \frac{I}{nAq} \Rightarrow n = \frac{I}{V_d Aq}$

$$n = \frac{1.5}{(0.00028)(2\times10^{-6})(1.6\times10^{-19})}$$

$= 1.67 \times 10^{28}/m^3$ **Answer: B**

36. Which of the following statement is correct?
   A. Galvanometer can be converted to voltmeter of different scale by connecting low values resistance in parallel.
   B. Low valued internal resistance increases accuracy of voltmeter.
   C. High valued internal resistance increases the accuracy of ammeter.
   D. Galvanometer can be converted to ammeter of different scale by connecting variety of low valued shunt resistance in parallel. **Answer: D**

37. What makes the oscillatory motion a simple harmonic motion?
   A. The acceleration of the motion is directly proportional to the velocity.
   B. The acceleration of the motion is directly proportional in magnitude but opposite in direction to the displacement.
   C. The velocity of the motion is directly proportional to the displacement.
   D. The velocity of the motion is inversely proportional to the displacement. **Answer: B**

38. The two end fixed string of length 0.5 m has a mass per unit length 2 g/m. If the tension in the string is 80 N, what is the second harmonic frequency?
   A. 400 Hz B. 200 Hz C. $100\sqrt{2}$ Hz D. 600 Hz

Solution: $V = \sqrt{\frac{T}{\mu}}$, $\mu = \frac{m}{\ell} = \frac{29}{m} = \frac{2}{1000}\frac{kg}{m}$

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$= \sqrt{\frac{80}{0.002}} = 200$

$f_n = \frac{nV}{2L} \Rightarrow f_2 = \frac{V}{L} = \frac{200}{0.5} = 400H_z$

Answer: A

39. Which one of the following is **NOT** the use of Doppler effect?
A. Astronomy
B. Energy transmission
C. Radar
D. Blood flow measurement

Answer: B

40. A 632.8 nm laser light produces an interference pattern on a screen of 4.0 m from a pair of slits. The slit space is equal to 0.2 mm. Calculate the fringe width.
A. 15.3 cm
B. 25.3 mm
C. 6.33 mm
D. 12.7 mm

Solution: $\frac{\lambda}{S} = \frac{W}{D} \Rightarrow W = \frac{\lambda D}{S} = \frac{(632.8\times10^{-9})(4)}{0.2\times10^{-3}}$

$= 12.7 \times 10^{-3}$m

Answer: D

41. A magnetic field of strength 5×10^{-5}T passes through an area of 20 cm^2 that is at an angle of 60° to the magnetic field. What is the magnetic flux?
A. 0 Wb
B. 5×10^{-8} Wb
C. 2.5×10^{-8} Wb
D. 8.7×10^{-8} Wb

Solution: $\emptyset = BA\cos\theta, \quad \theta = 30°$

$= (5\times10^{-5})(20\times10^{-4})(0.87)$

$= 8.7 \times 10^{-8}$ Wb

Answer: D

42. An inductor has a self inductance of 12 *mH*. The current in the circuit decreases at a rat of 2.0 *A/s*. What is the magnitude of the induced electromotive force?
A. 2.4π.V
B. 1.2π.V
C. 1.2×10^{-2} V
D. 2.4×10^{-2} V

Solution: $|\epsilon| = \left|L\frac{\Delta I}{\Delta t}\right|$

$= (12\times10^{-3})(2)$

$\varepsilon = 2.4 \times 10^{-2}$ v

Answer: D

43. Which of the following is correct about the characteristic of current and voltage in a.c. circuits?
A. If the current leads the voltage, its capacitive reactance is greater than inductive reactance.

B. If the current is in phase with the voltage, the circuit is inductive.
C. If the current leads the voltage, it is inductive circuit.
D. If the current lags the voltage, the circuit is capacitive.

Answer: A

44. An *RLC* is designed to have a resistance of 400Ω, capacitive reactance of 350 Ω and inductive reactance of 50Ω. The circuit is connected to an alternative voltage source of effective (rms) value 50 V, and frequency 50 Hz. What is the output average power?
A. 62.5 W
B. 4.0 W
C. 400 W
D. 40 W

Solution: $P_{av} = I^2_{rms}R, \quad I_{rms} = \frac{V_{rms}}{Z}$

$Z = \sqrt{R^2 + (x_L - x_c)^2} = \sqrt{400^2 + 300^2}$

$= 500\ \Omega$

$I_{rms} = \frac{50}{500} = 0.1A$

$P_{av} = (0.1)^2(400) = 4W$

Answer: B

45. Which of the following has an electron volt unit?
A. charge
B. energy
C. potential difference
D. electromagnetic intensity

Answer: B

46. In Rutherford experiment, a beam of positively charge alpha particles was projected into a thin piece of gold foil. One of the observation of this experiment is that some charged particles deflected through a large angle (10° to 90°) are those concentrated in one place. What conclusion can be draw from this observation?
A. Most of the mass and all positive charge is in tiny central nucleus
B. Most of the atom is empty space
C. All the atom is positive charge
D. All the atom's negative charge

Answer: A

47. In the circuit shown below, each resistor has a value of 3Ω. What is the value of the current through R_4?
A. 0.75 A
B. 1.0 A
C. 0.625 A
D. 0.125 A

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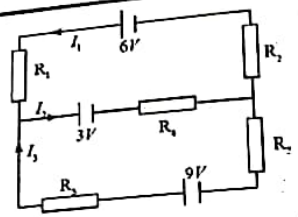

Solution: Applying Kirchhoff's rules

Point rule

$I_1 + I_3 = I_2$ ..... (1)

Loop – 1 (counter – clock wise traverse)

$6 - 3 = 3I_1 + 3I_2 + 3I_1$

$2I_1 + I_2 = 1$ .....(2)

Loop – 2 (clock – wise traverse)

$9 - 3 = 3I_2 + 3I_3 + 3I_3$

$2I_3 + I_2 = 2$ .....(3)

Solving the above equations, we get

$I_2 = 0.75$ A **Answer: A**

48. Suppose you obtain a piece of iron from your laboratory shelf. How can you experimentally determine, for sure, that it is magnetized?

A. If it attracts another magnet
B. If it diverges a changed electroscope
C. If it repels another magnet
D. There is no single and sure way to determine its magnetization. **Answer: C**

49. If a charged particle enters into a uniform magnetic filed perpendicular to its direction, then what will happen?

A. It moves in an elliptical path
B. The radius of its path is proportional to the magnetic field strength
C. The charged particle will not be affected by magnetic field
D. The period of the motion is inversely related to the magnetic field strength

Solution: $\frac{mV^2}{r} = qvB$

$\frac{mV}{r} = qB \quad , V = \frac{qBr}{m}$

$T = \frac{2\pi r}{V} = \frac{2\pi m}{qB}$ **Answer: D**

50. Two parallel wires each carrying 5 A current in the same direction are separated by 2.5 cm. What is the magnitude of the force per unit length acting on the wires? Is the force attractive or repulsive?

A. 2×10^{-4} N/m, attractive
B. 4×10^{-6} N/m, repulsive
C. 8×10^{-5} N/m, repulsive
D. 5×10^{-6} N/m, attractive

Solution: $F = \frac{\mu_0 I_1 I_2 \ell}{2\pi r}$

$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r} = \frac{4\pi \times 10^{-7}(5)(5)}{2\pi(2.5 \times 10^{-2})}$

$= 2 \times 10^{-4}$ N/m **Answer: A**

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SECTION 4 – BIOLOGY

The questions and answers grouped in to two parts. Part one question from Grade 11 are arranged according to the units of the lesson = 42 questions part two questions from Grade 12 topics = 58 questions.

1. Which of the following steps in the scientific method comes following observation?
   A. Prediction
   B. Hypothesis
   C. Experiment
   D. Theory

Explanation:
Scientific method is a process in which scientists can approach to their work.

- This scientific method consists of several steps.
- These steps are:
  1) Asking questions about the problem.
  2) Observation/ doing the background research.
  3) Construct the hypothesis and can made a prediction.
  4) Experiment to test the hypothesis.
  5) Drawing the conclusion.
  6) Making the theory.

Answer: B Hypothesis comes after observation.

2. Which of the following field equipment is used to determine an exact location of a place?
   A. Theodolite
   B. Global positioning system
   C. Filed microscope
   D. Field pH kit

Explanation:
Biologists can grouped the basic tools (equipment) into two. These are, the laboratory equipments and field equipments.
From field equipments:

- Theodolite is used to measure the height of trees and slopes of the area.
- Field microscope is used to investigate the structure of specimen in the field.
- Field pH kit is used to measure the pH of soil or water.

Answer: B – Global positioning system (GPS) helps to makes it possible to record positions or location of a place.

3. What is a theory in biology?
   A. The outcome of the hypothesis is true.
   B. A proof that shows the hypothesis is true.

   C. A hypothesis supported by experiments and/or observations.
   D. An opinion or educated guess resulting from observations.

Explanation:
Theory in biology which can be followed those the scientific method steps.

- The most successful hypothesis become a ruling hypothesis and it may become theory.
- As a knowledge increases and techniques of investigation improve and even well-established theories.

Answer: C – A hypothesis supported by experiment and/or observation.

4. For which of the following is a theodolite used in biology education?
   A. Measuring the height of trees.
   B. Recording position where a species is found.
   C. Measuring the rate of flow of water in a cell.
   D. Measuring the pH of water or soil.

Explanation:
From biological field equipment:
1) Global positioning system (GPS) used to record the position where a species is found.
2) A flow meter equipment used to measure the rate of flow of water.
3) A pH kit is used to measure the pH of water of soil .

Answer: A – theodolite is used to measuring the height of trees or stopes of area.

5. Which of the following is a recently active area of research in biology today?
   A. Taxonic study
   B. Study about the cell theory
   C. Ecological research
   D. Stem cell research

Explanation:
Biology is the science of life and living organisms. Biologists engage in all kinds of research to try to find evidence of life on earth and other planets. To find out this evidence biology has many branches. From three branches taxonomy, cell theory and

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ecology and their researches were earlier studies in the field of biology.

Answer: D – The stem cell research is the recently active in biology to day.

6. How many carbon atoms are there in one disaccharide molecule?

A. 6 B. 12 C. 18 D. 24

Explanation:

Biological molecules can be classified into two main groups. These are in organic and organic molecules.

- In organic molecules many contain carbon or hydrogen but not both elements.
- Organic molecules contain different combinations of carbon and hydrogen.
- Carbohydrates is one of organic molecules which is made up of carbons, hydrogen and oxygen.
- The ration of these elements are 1: 2 = 1 (CH_2O).

The different types of carbons are:

- *Monosaccharide (a single sugar molecule)* which contain from 3 – 6 carbon molecules.
- *Disaccharide molecules* are made up of two monosaccharide molecules joining together.
- *Polysaccharide molecules* are built by many hundreds of monosaccharide molecules join together.

Answer: B – Disaccharide molecules are the combination of two mono saccharide molecules.

$C_6H_{12}O_6 + C_6H_{12}O_6 \longrightarrow C_{12}H_{22}O_{11} + H_2O$

There are 12 carbon atoms in disaccharide molecule.

7. Which energy rich organic compound contains adenine in its molecule.

A. Lipids C. Carbohydrate

B. Glucose D. ATP

Explanation:

The most organic compounds are carbohydrate, lipids, proteins and nucleic acid.

- All carbohydrates contain three elements carbon, hydrogen and oxygen.

- All lipids also contain three elements carbon, hydrogen and oxygen. Their difference is the proportion of oxygen is very less in lipids.
- Proteins are made up of carbon, hydrogen oxygen nitrogen and sometimes contain sulfur.
- Nucleic acids are grouped in to DNA and RNA. Both are made up of nucleotides which have the same three compounds.

These components are:

- Phosphate group
- Pentose sugar (deoxyribose and ribose sugar)
- Nitrogenous base

ATP is energy rich organic compound made up of phosphate groups, ribose sugar and adenine from nitrogenous base.

Answer: D – ATP

8. All proteins contain carbon, hydrogen, oxygen and what other elements?

A. Chlorine B. Florine C. Nitrogen D. Sulphur

Explanation:

Proteins are the most important substances that are needed to form all living cells. Proteins are contain the elements carbon, hydrogen, oxygen like carbohydrate and lipids, but they also contain nitrogen and most contain sulphur. Plants take nitrate from the soil to make their own proteins. Animals can lat plants to make their own protein.

Answer: C – All proteins contain nitrogen rather than sulphur.

9. What makes unsaturated fatty acids different from saturated fatty acids?

A. The presence of long chain of carbon.

B. The presence of one or more double bond.

C. The presence of large number of hydrogen atoms.

D. Their occurrence as solid at room temperature.

Explanation:

- Fatty acid molecules consists of a covalently bonded hydrocarbon chain at the end of carboxyl group.
- This carboxyl group has acidic properties.
- Saturated fatty acids are the carbon – carbon bonds in the hydrocarbon chain are single bonds.

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- Unsaturated fatty acids contain one or more carbon – carbon bonds are double bonds. **Answer: B**

10. What is the name of the sugar found in milk?
A. Glucose B. Lactose C. Maltose D. Sucrose

Explanation:

- Glucose is the most common monosaccharide which is made up during photosynthesis and used in respiration to produce energy.
- Maltose is a disaccharide formed by the combination of two glucose molecule.
- Sucrose is made of one glucose molecule and one fructose molecule – which is a disaccharide.

Glucose + fructose → sucrose (sugar cane)

Answer: B – Lactose is a disaccharide found in milk which is formed by the combinations of one glucose molecule with one galactose molecule glucose + galactose → lactose (milk sugar).

11. What can one conclude about the contents of the food staff if a mixture of Benedict solution and a food staff remain blue after heating?
A. It contains proteins C. It contain fats
B. It lacks starch D. It lacks reducing sugar

Explanation:

Different types of food staffs are tested by different types of chemical substances. Example 1: Carbohydrates are tested by Benedict (Fehling's solution) and iodine (potassium iodide).

a) *Reducing sugars – (glucose)*
The reducing sugars include all mono saccharide such as glucose, fructose and some disaccharide such as maltose. These reducing sugars are tested by Benedict solution which contain copper sulphate. Reducing sugars reduce soluble blue copper sulphate containing copper (II) ions (Cu^{2+}) to insoluble red brown copper oxide containing copper (I).

b) *Non reducing - (sucrose)*
The only common non reducing sugar is sucrose which is disaccharide. This sucrose is hydrolyzed to glucose and fructose by boiling with diluted hydrochloric acid and then tested by Benedict solution. The result is similar to reducing agents.

c) *Starch*

Starch is a polymer of a glucose. It is a major fuel of plants and not found in animals. Starch is tested by using of iodine or potassium iodide and observed a blue black colouration.

Example – 2 – lipids: Lipids include oils (olive oil) and fats. To test lipids, we can use emulsion test and grease spot test. Emulsion test is adding of oil or fat and ethanol in test tube and then add cold water, the result is a cloudy white suspension. Grease spot test is rub a drop of the sample (oil) into a place of paper. The result is a permanent transparent spot on the paper.

Example – 3 – proteins – A soluble protein such as egg albumin is tested by:

1) *Million's test*
Add protein solution (egg abbamin) in to a test tube and add million's regent and boil. The result is a white suspensioin (precipitate) forms which coagulates on heating and turns red or salmon pink color.

2) *Biuret test*
Adding protein solution (egg albamin) to a test tube add equal volume of potassium hydroxide solution and mix. Add two drops of copper sulphate and mix without heating. The result is purple color develops slowly. Therefore the answer is – D. It lacks reducing sugar. Because the Benedict's solution which contain copper (II) an not reduced into copper (I).

12. In which of its contents RNA differs from DNA?
A. Deoxyribose and guanine
B. Ribose and uracil
C. Ribose and thymine
D. Phosphate and adenine

Explanation:

DNA and RNA are the nucleic acids of organic compounds. These nucleic acids are made up of nucleotides. A nucleotide has three components:

1) Pentose (five carbon) sugar
In DNA deoxyl – ribose sugar reducing of one oxygen on the second carbon chain.
In RNA – ribose sugar contains oxygen on the second carbon.

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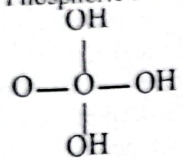
2) Phosphoric acid (phosphate)



3) Nitrogenous bases:
Each nucleic acid (RNA & DNA) contains four different bases. Two of them derived from purine and two of them derived from pyrimidine's. Purines have two rings and pyrimidine have one ring in their structure. The two purine are Adenine (A) and Guanine (G). The two pyrimidine are Thymine (T) and cytosine (c). In RNA instead of thymine it replaced by uracil. Therefore the answer is B – RNA is differs from DNA by its Ribose sugar and uracil its nitrogenous bases.

13. Which of the following groups of substances are all in organic?
A. Water, sugar, calcium, carbonate
B. Sugar, fatty acid, amino acid
C. Water, calcium carbonate, carbon dioxide
D. Carbon dioxide, amino acid, fatty acid

Explanation:
Biological molecules can be classified in to two main types. These are inorganic molecules and organic molecules. In organic molecules are contain carbon or hydrogen. Example: H_2O no carbon (water), CO_2 (no hydrogen), $CaCO_3$. Organic molecules contain both carbon and hydrogen elements which join together. Example – Sugar ($C_{12}H_{22}O_{11}$) fatty acid ($C_{17}H_{35}COOH$), amino acid ($NH_2 - CH_2 - COOH$). Then choice letter A contains sugar which is organic. Choice letter B, all are organic molecules. Choice letter D amino acid and fatty acids are organic molecules. **Therefore, the answer is C** all are inorganic molecules.

14. In which of the following compounds are both members of the pair are polymers of carbohydrates that naturally occur in plants?
A. Starch and chitin
B. Starch and cellulose
C. Glycogen and cellulose
D. Glycogen and chitin

Explanation:
Polymers are a complex organic molecules which are formed from monomers (simple molecules). The polymers of carbohydrates are:

- Starch is a polymer of alpha (α) glucose and stored in plants.
- Cellulose is a polymer of Beta (β) glucose all plant cell walls are made up of cellulose.
- Glycogen is the animal equipment to starch which is made up of alpha (α) glucose.
- Chitin – is closely related to cellulose in structure and function. It is occur in some fungi in some animals particularly in Arthropods. Then choice letter A chitin is not found in plants. Choice letter C glycogen is not found in plants. Choice letter D both glycogen and chitin are not occur in plant cell. Therefore, the answer is letter B, both starch and cellulose are naturally occur in plants.

15. Which of the following is generally expected to give better health benefit when present in human diet?
A. Animal fat
B. Saturated fatty acid
C. Mono unsaturated fatty acid
D. Poly unsaturated fatty acid

Explanation:

- Approximately 95 percent of the lipids in human diet are triglycerides (three fatty acids attached to a glycerol molecule).
- The remaining 5 percent are mainly cholesterol and the phospholipid.
- *Fatty acids* are divided into salurated (single bonds) and unsaturated (double bonds).
- Saturated fatty acids are contain too much of hydrogen than unsaturated fatty acid.
- *Unsaturated fatty acids* have only one double bond is mono saturated and more than one double bond are poly unsaturated fatty acids. Saturated fatty acids are solid at room temperature mostly found in animal body. Unsaturated fatty acids are liquids at room temperature and found in plant cells. They are very important used to

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present cholesterol being laid down in the lining of arteries and helps to prevent heart disease.

- *Animal fat and saturated fatty acids* are contain to much of hydrogen and can not give better health benefits because they are solids. Mono unsaturated fatty acid is found in plans and do not reduce hydrogen. **Therefore the answer is – D** – *poly unsaturated fatty acid* has to give better health benefit, because of having less amount of hydrogen which forms oil in plants.

16. Which of the following classes of enzymes digests carbohydrates?
A. Amylase
B. Lipades
C. Protease
D. Nucleases

Explanation:
- Lipases are enzymes that acts on lipids.
- Protease are enzymes that acts on proteins.
- Nucleases may catalyze nucleic acids.

Therefore, the answer must be A amylase because Amylase involve in the digestion of carbohydrates.

17. One of the following molecules is building units of an enzyme molecule.
A. Amino acids
B. Glucose
C. Nucleic tides
D. Fatty acids

Explanation:
- All enzymes are proteins.
- Glucose in monomer of carbohydrate.
- Nucleotide is nucleic acid and fatty acids are lipids do not contain enzymes.

Therefore, **the answer A**. Because amino acid is a monomers of protein.

18. Which of the following pairs of molecules are known to have catalytic activity?
A. Lipids and proteins
B. Carbohydrates and proteins
C. Proteins and RNAs
D. Proteins and DNAs

Explanation:
- Catalyst is a substance that speeds up a chemical reactions and remains unchanged at the end of the reaction.

- An enzyme is a globular protein with a uniquely shaped active site and acts as biological catalyst for a specific reaction. Lipids carbohydrates and DNA molecules have no catalytic activity in biological reactions. **Therefore answer is C** proteins and RNAs are biological catalysts in biological reactions.

19. To which class of enzymes do the digestive enzymes belong?
A. Esterase's
B. Transferases
C. Hydrolases
D. Isomerizes

Explanation:
Classification of also based on their catalytic reactions. For example:
1) Esterase's enzyme acts on the hydrolysis of lipids to break ester bond.
2) Transfereses – catalyze the transfer of a chemical group from one compound to another compound.
3) Isomerases – catalyze the transformation of one isomer in to another. For instance the conversion of glucose 1, 6 – diphosphate into fructose 1, 6 – diphosphate.
4) Hydorlases – These catalyze hydrolysis (splitting of molecules by use of water) reactions. Most of digestive enzymes are hydrolases.

Therefore, the answer is C hydrolases.

20. Which of the following types of enzyme inhibition can be removed when the end product of the metabolic pathway is depleted?
A. Allosteric inhibition
B. Non reversible inhibition
C. Completive inhibition
D. Reversible competitive inhibition

Explanation:
- Inhibitors are substances which slow down or stop enzyme action that binds to enzymes and prevent them from forming enzyme – substrate complexes. The two main types of inhibitors are:
  - Irreversible inhibitors and
  - Reversible inhibitors

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- Irreversible (nonreversible) inhibitors are bend strongly and the bond that helds them breaks easily and releasing the inhibitors.

1. Competitive inhibitors is a molecule that inhibits enzyme actively by competing with the substrate for the active site.
2. Noncompetitive inhibitor is a molecule that alters the conformation of the active site by binding with the allosteric site of the enzyme.
   - Allosteric inhibitors are designed to change the shape allo means different and steric shape.
   - Allosteric inhibitors are regulated by compounds which act as non – competitive inhibitors.

   These noncompetitive inhibitors produced in a cell as a result of metabolic bath ways in cells. Therefore, the answer is A allosteric inhibitors, because it regulates by noncompetitive inhibitors.

21. The main component of the plant cell wall is:
A. Starch B. Cellulose C. Protein D. Chitin

Explanation:

- Starch is the stored polymers in the internal part of plant cells.
- Proteins are important to form the structure of plasma membrane, immune system, the structure of chromosomes, enzymes etc.
- Chitin has similar function to plant cellulose which is found in fungi.
- Many cellulose molecules lies side by side by hydrogen bond to each other.
- Cellulose has fibrous in nature that give the cell walls their strength.

Therefore the answer is B – Because plant cell wall is made up of cellulose.

22. Which one of the following is the main constituent of biological membranes?
A. Phospholipids C. Glycolipids
B. Glycoproteins D. Cholesterols

Explanation:

Biological membrane or cell membranes separate the content of cells from their external environment. It is used to controlling exchange of materials is and out from the cell.

Cell membranes are contain:

1. Phospholipids bilayer consists of phosphate glycerol and two fatty acid molecules.
   - The phosphate groups are hydrophilic polar heads.
   - The glycerol and fatty acid molecules are hydrophobic non polar tail sides of bilayer.
2. Protein molecules are consists of integral proteins and glycoproteins (the combination of glycogen and protein) molecules.
3. Glycolipids are molecules which contain lipids combined with carbohydrates.
4. Cholesterol is closely related to lipids and is slightly polar at one end.

Therefore, the answer is A, because for all biological membrane the main constituents are phospholipid bilayers.

23. Which unit is the best to use for measuring the smallest cells and organelles?
A. Micro meter C. millimeter
B. Milliliter D. Nanometer

Explanation:

The measuring units commonly used to describe microscope objects are the micrometer (mm) and the nanometer (nm).

- Starting with similar unit, the millimeter (mm).
- One thousands (1/100) of one millimeter is one micrometer (μm).
- One thousand (1/100) micrometer or one millions (1/1000,000) millimeter is one nanometer. The micrometer is used to describe cells and organelles in the microscope.

Therefore the answer is – A – micrometer.

24. If a suspension of a mixture of cellular organelles is spun in a centrifuge which organelle settles to the bottom first:
A. Mitochondria C. Chloroplasts
B. Nuclei D. Ribosomes

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Explanation:
Cell fractionation is a technique used to separate different organelles depending on their size spinning in a centrifuge. The organelles will separate out in a particular order, according to their density and shape.

1) According to their density and shape, the nuclei settle out forming a pellet at the bottom.
2) The suspension containing can spun at higher speed the chloroplasts settle out.
3) The supernatant is spun at a higher speed still the mitochondria settle out.
4) At the last process the ribosomes separate out.

Therefore, the answer is – B – b/se the nuclei settle out first at the bottom of the mixture.

(25) If the size of a cell increases, which one of the following gets smaller?

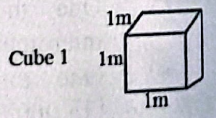
A. The volume of cell.
B. The surface area of the cell.
C. The surface area to volume ratio of the cell.
D. Volume to surface area ration of the cell.

Explanation:
When the size of a cell increases, it increases:

1) The volume of a cell.
2) The surface area of a cell.
3) The volume to surface area ratio of a cell but it decreases the surface area ratio of the cell.

Example – There are six sides to a cube.

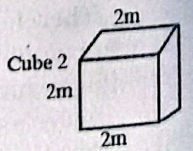
Surface area = $1m \times 1m = 1m \times 6 = 6m^2$
Volume = $1m \times 1m \times 1m = 1m^3$



Surface area to volume ration $= \frac{6}{1} = 6$

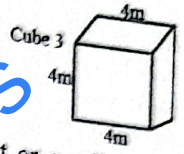
$A = 2m \times 2m \times 6 \text{ side} = 24\ m^2$
$V = 2m \times 2m \times 2m = 8m^3$

S. a. to volume ratio $= \frac{24}{8} = 3$



$A = 4m \times 4m \times 6 = 96m^2$
$V = 4 \times 4 \times 4 = 64m^2$

$S = \frac{96}{64} = 1.5$



Then when the size of an object or a cell increase, the surface area to volume ration gets smaller. Therefore, the answer is C.

(26) Which of the following is an important function of the Golgi apparatus?

A. Proteins synthesis
B. Packaging of protein for export out of the cell.
C. Removing of debris from the cell.
D. Storage of waste materials not needed by the cell.

Explanation:

- *The Golgi apparatus* consists of a number of flattened membrane bound sacs in which proteins are modified.
- For example, proteins may be converted into glycoproteins.
- Golgi apparatus acts as a cellular post office of a cell.
- Ribosomes are organelles used for protein synthesis vesicle is a small spherical organelle, bounded by a single membrane connected with Golgi apparatus and endoplasmic reticulum used to store and transport materials around the cell.
- The Golgi apparatus are involved processing, packaging and storing proteins, lipids and carbohydrates molecules with in a cell. **Therefore the answer is – B.**

(27) In which of the following features are eukaryotic cells distinguished from prokaryotic cells?

A. They have mitochondria.
B. Their nuclei lack membranes.
C. They have no DNA.
D. They have smaller ribosomes.

Explanation:
All cellular organism fall into two natural groups.

1) Prokaryotes and
2) Eukaryotes

These two groups are fundamentally different. The main difference b/n prokaryotes and eukaryote is the location of DNA.

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Both cells have DNA as agnatic material. In prokaryotes the DNA is not enclosed by nuclear membrane and lie freely in the cytoplasm. The cell lack true nuclei pro means before and karyon is nucleus. In eukaryotes contain true nuclei bounded by nuclear membrane. Both cells also have ribosomes. In prokaryotes there are 705 ribosomes smaller in size. In eukaryotes 805 (larger). **Therefore the answer is – A.** Because eukaryotic cell have mitochondria and prokaryotic cells have no mitochondria.

28. What will happen if human red blood cells are kept in a hypotonic solution?
   A. Lose water by osmosis by brust.
   B. Lose water by osmosis and shrink.
   C. Taking in water by osmosis, swell and brust.
   D. Take in water by osmosis, swell and remain turgid.

Explanation:

1) When red blood cells placed in hypertonic solution which means the solution has lower water potential than the cell contents, then water leaves the cell by osmosis and the cell shrinks.
2) In an isotonic solution, water potential of the cell equal that of the external solution and no net movement water occurs.
3) In hypotonic solution, the solution has a higher water potential than the content of the red blood cell, then water enters to the cell by osmosis and the cell brusts due to lack of cell wall like plant cells. **Therefore, answer is C.**

29. In which types of solution is the water potential more negative than in the cells.
   A. Hypotonic
   B. Hypertonic
   C. Isontonic
   D. Equal solute and solvent concentration.

Explanation:

1) Isotonic solution have the same water potential as the cell.
2) Hypertonic solution having a lower (more negative) water potential than the cell.
3) Hypotonic solution having a higher (less negative) water potential than the cell.

Therefore, the answer is – B – *hypertonic solution has more negative water potential to the cell.*

30. Which of the following paired organelles are membrane bounded?
   A. Ribosomes and peroxisomes.
   B. Chloroplasts and ribosomes.
   C. Mitochondria and ribosomes.
   D. Chloroplasts & mitochondria.

Explanation:

The main features of eukaryotic cells have membrane bounded organelles. Most organelles such as nucleus, mitochondria, Golgi apparatus, Endoplasmic reticulum chloroplasts and vacuole in plants have single and double membrane. Ribosomes and peroxisomes lack membrane bounded organelles. Chloroplasts and mitochondria are membrane bounded organelles. **Therefore, then answer is – D.**

31. Most cell membranes are primary composed of which compounds?
   A. Proteins and lipids
   B. DNA and ATP
   C. Chitin and starch
   D. Nucleotides and amino acids

Explanation:

Most cell membranes are composed of phospholipids, proteins, glycolipids and glycoproteins not DNA, ATP, Chitin, starch and nuclei lipids. **The answer is – A.**

32. Choose the one that is not a requirement of all living things.
   A. Ability to think
   B. Organization of parts
   C. Response to the stimuli
   D. Maintenance of internal constancy

Explanation:

All living organisms have common features to distinguishes from nonliving things. Some of them are – Reproduction, respiration, growth, nutrition, photosynthesis (not for all), movement, irritability (response to stimuli, homeostasis, maintenance of internal constancy, etc). Therefore, the answer is – A – b/se not all living things require the ability to think.

33. If the red blood cells shrink when placed in a certain solution, what is the strength of the solution relative to the strength of the protoplasm of the cells?
   A. Hypotonic
   B. Isotonic
   C. Hypertonic
   D. Isoosmosis

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Explanation:
Hypotonic solution has a lower solute concentration than the cell, then water enter to the cell. Hypertonic solution has a greater solute (less dilute) concentration than the cell, then water leaves from the cell and the cell become shrink. Isotonic solution has an equal solute concentration, then there is equal net movements of molecules into and out of the cell. **Therefore, the answer is C.**

34. Which of the following is the correct route that connects a stimulus and a response.

A. Receptor → Coordinator → Effector
B. Coordinator → Receptor → Effector
C. Receptor → Effector → Coordinator
D. Effector → Receptor → Coordinator

Explanation:
Receptors are sense organs which detect any stimulus to their surroundings.

- Coordinators are the central nervous system (the brain and spinal cord) which interpirate the stimulus.
- Effectors are organs (muscles and glands) which give response to the stimuli.

Therefore the correct pathway is receptor → coordinator → effector. **The answer is A.**

35. Which of the following cell types can be rich in lysosomes?

A. Red blood cells
B. Nerve cells
C. Phagocytic cells
D. Muscle cells

Explanation:
Lysosomes are small vesicles 0.2 – 0.5 µm in diameter. They contain a mixture of digestive enzymes called lytic enzymes. One of the functions of lysosomes is to digest material taken into the cell by the process of phagocytosis. White blood cell are known as a phagocyte is ingesting a bacterium. **Therefore, the answer is – C** b/se phagocytic cells are rich in lysosomes.

36. One of the following would be harder to see under the ordinary light microscope that is more likely to the available in school laboratories.

A. Nucleus
B. A bacterium
C. A mitochondria
D. Ribosome

Explanation:

- The nucleus is the largest and most prominent organelle in the cell. It has 10µm in size. It can be easily seen under light microscope than other organelles.

- A bacterium has different size and shape which is greater than ribosome and can be seen under high magnification of light microscope.
- Mitochondria are relatively large from 0.5 – 1.5 mm in diameter and 3 – 10 µm long and can be seen under higher magnification of light microscope.
- Ribosomes are small dense organelles about 20nm in diameter and it would be harder to see under light microscope.
- **Therefore the answer is – D.**

37. During chemiosmosis, what substance diffuses from one side of the other side of the membrane?

A. Water molecules
B. Protons
C. Electrons
D. ATP molecules

Explanation:
Chemiosmosis is the diffusion of hydrogen ions through the ATP synthase.

- Protons accumulate in the inner membrane space creating a higher concentration there than in the matrix (on the other side of the membrane).
- This proton gradient results in protons diffusion through ATP synthase.
- During chemiosmosis it diffuses protons through ATP synthase not water molecules, electrons and ATP molecules. **Therefore, the answer is – B – protons.**

38. Which of the following is not true about photo system II?

A. Its reaction center molecules P680.
B. It passes its excited electrons to photosystem I.
C. The energy lost from its excited electrons reduces NADP.
D. It replenishes its lost electrons from photolysis of water.

Explanation:
The chlorophyll and accessory pigment molecules are located in two types of photosystems called photosystem I (PSI) and photosystem II (PSII). In these photosystems light dependent reaction take place. Each photosystem has sensitive to light of different wave length.
In photosystem II (PSII):

- **The reaction center molecule of wave length is P680.**

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- The splitting of water molecules (hydrolysis can take place).
- It passes its excited electrons to photosystem I.
- Reduced NADP is not produced in the photosystem II.

Therefore, the answer is – C.

39. What is the importance of chemiosmosis is photosynthesis and cellular respiration?
A. Splitting of water molecules.
B. Operating the proton pump.
C. Combining hydrogen of carbon.
D. Synthesizing ATP.

Explanation:
The importance of chemiosmosis is the diffusion of hydrogen ions (protons) from the inner membrane space to the other side of the membrane through ATP synthase.

- Splitting of water molecule take place before chemiosmosis.
- Combining hydrogen and carbon take place after chemiosmosis.
- Synthesizing ATP also take place after chemiosmosis.
- During chemiosmosis operating the proton pump take place.

Therefore, the answer is – B.

40. Where does the light dependent reaction of photosynthesis occur in the chloroplast?
A. In the thylakoid membrane.
B. In the fluid of the stroma.
C. In all part of the chloroplast.
D. In the stomatal opening.

Explanation:
The process of photosynthesis is divided into two main stages. These are:
1) Light dependent reaction stage.
2) Light independent reaction.

Both stages are take place in the chloroplast of the legal. In the chloroplast there are many membrane systems which are the site of light dependent reaction.

- The membrane are covered with chlorophyll and other pigments, enzymes and electron carriers.

- The system consists of many flattened fluid filled sacs are called thylakoids. The stack of thylakoids are called grana where light dependent reaction take place.
- Light independent reaction take place in the stromal of the chloroplast.

Therefore the answer is – A.

41. For which of the following is the sugar produced by photosynthesis not used?
A. To produce biomass.
B. To make new DNA.
C. To produce ATP in espiration.
D. To produce enzymes.

Explanation:
During light independent reaction sugar is produced. The sugar produced by photosynthesis is used to produce ATP which involve in respiration.

- To produce enzymes together with proteins.
- To make new DNA molecule b/se DNA molecule consist of Deoxyribose sugar.
- Sugar is not used to produce biomass.

Therefore, the answer is – A.

42. In which process is ATP generated during short distance high speed running?
A. Aerobic respiration
B. Mitochondrial energy transformation
C. Anaerobic respiration
D. The Krebs cycle

Explanation:

- Long distance runners are slower than short distance runners.
- These long distance runners generate ATP by aerobic respiration, whereas short distance high speed runners generate ATP aerobically and to produce more energy they generate ATP also an aerobically.
- Aerobic respiration is producing energy with the presence of oxygen.
- Mitochondria energy transformation also we oxygen.
- The Krebs cycle also need oxygen.

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- An aerobic respiration is generating ATP with the absence of oxygen.

Therefore, the answer is – C – short distance high speed running generate additional ATP without oxygen or by using an aerobically respiration.

43. Which of the following kingdoms of life is consisting of prokaryotic organisms.
A. Fungi B. Monera C. Protista D. Plantae

Explanation:
Now a days there are five kingdom classification system. These are:
1) Kingdom plantae – all green plants.
2) Kingdom animalia – all animals which include vertebrates and invertaberates animals.
3) Kingdom fungi – including mucor (common nucleus) yeast, mush rooms etc.
4) Kingdom protistate all protozoans and algae. These from no. 1 – no. 4 kindoms are under eukaryotic organisms.
5) Kingdom monera – which include Archa bacteria, blue green algae (cyano bacteria) and lubacteria (true bacteria). This kingdom is under prokaryotic organism. **Therefore, the answer is – B** k. monera.

44. Which of the following human diseases can be prevented by taking proper diet?
A. Degenerative diseases C. Social diseases
B. Genetic diseases D. Deficiency diseases

Explanation:
The term health is define as eh state of complete physical metal and social wellbeing. Disease can be grouped into infectious (contagious) and noninfectious (noncontagious) diseases. Infectious disease are diseases which transmitted from an infected organism to infected organism by different factors. Non infections disease which do not transmitted.
Human disease can be grouped:
1) Human induced disease – caused by a person's life style of working condition.
2) Degenerative disease – the result of ageing.
3) Genetic diseases – transmitted through genes.

4) Social disease – social activities (addition).
5) Deficiency disease – lack of nutrition.

Therefore, the answer is – D – b/se deficiency disease are prevented by proper diet.

45. Which of the following particles does not normally transmit HIV?
A. Sexual intercourse through anus.
B. Sharing infection needles
C. Blood transfusion
D. Shaking hands

Explanation:
HIV/AIDS is an infectious disease which transmitted through.
- Sexual intercourse with many partners.
- Sharing infected any materials.
- Blood to blood contact and transfusion.
- From mother to her child during delivery.

But HIV/AIDS not transmitted during shaking hands, in swimming pools, toilet room etc. Therefore, the answer is – D.

46. To which of the following groups do those bacteriophages that integrate their DNA into the chromosome of their bacterial host belong?
A. Virulent viruses C. Lysogenic viruses
B. Lytic viruses D. Non parasitic viruses

Explanation:
Bacteriophages are viruses which attacks bacteria. These viruses have three reproductive life cycle.
1) Lytic cycle – It attach the host cell (the bacteria) and rupture (lysis) the host DNA and kill the bacteria. In this cycle it kills the host cell is called virulent viruses.
2) Chronic release cycle:
In this cycle the virus (bacleriophaphes) reproduce with in the host cell and release the viruses all at once. It release of new viruses is ongoing through the plasma membrane.
3) Lysofenic cycle
In this cycle the viruses DNA becomes in corporated or integrated with the host DNA. When the host cell's DNA with including the viral DNA.

Therefore, the answer is – C – b/se the bacteriophage or the viral DNA integrated with the bacterial DNA.

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47. Which group of microorganisms causes the disease known as athlete's foot?
A. Bacteria B. Fungi C. Protozoa D. Viruses

Explanation:
Different infectious disease can caused by different microorganism. For example: bacteria can caused diseases like pneumonia, cholera, typhoid, diphethec, tuberculosis etc.
Viruses – caused influenza, measles, AIDs, common cold, mumps, polio mellitus etc.
Protozoa – caused malaria, Bliharizia, sleeping sickness, etc.
Fungi – causes thrush, farmer's lung, Athlete's foot.
Therefore, the answer is – B.

48. Under which of the following groups can the fungi be more conveniently placed?
A. Autotrophs C. prokaryotes
B. Heterotrophs D. Plants

Explanation:
Autotrophs are those organisms which make their own food by the process of photosynthesis. Examples – all green plants and algae.
Heterotrophs are organisms which do not make their own food. They are directly or indirectly depending on all through. Examples – animals some prokaryotes and fungi. **Therefore, the answer is – B** under heterotrophs.

49. What is the advantage of using HAART (Highly Active Anti Retroviral Therapy) for the treatment of HIV?
A. It gives a lasting immunity to HIV.
B. It prevents mutation of HIV.
C. It prevents reinfection by HIV.
D. It helps to break the life cycle of HIV.

Explanation:
AIDS is cannot be treated and cured which caused by the virus called HIV.
- No vaccine to give immunity against infection.
- But there are a number of drugs called antiretroviral drugs. These drugs can be effective in slowing down the progression of AIDS.
- The advantage of these drugs work by blocking or breaking the reproduction of the virus in the CD4 lymphocytes because it can be taken in different way and different stages. These HAART cannot give a lasting immunity to HIV, cannot prevents mutation of HIV and cannot prevents reinfection by HIV. The main advantage of these HAART is it help to break the life cycle of HIV in different stages. Therefore the answer is D.

50. On which of the following principles aer most of the anti – HIV drugs currently in use working?
A. Inhibition of enzyme action
B. Degradation of viral DNA
C. Digesting of viral particles
D. Phagocytosis of the viruses

Explanation:
- Anti – HIV drugs currently is used to inhibition of enzyme action.
- These drugs are not degradation of the viral RNA and not digest the viral particles. Therefore, the answer is – A.

51. If a few anti – HIV drug is to be developed to prevent the virus from entering the host cell, which one of the following processes should be drug target?
A. Reverse transcriptase.
B. Binding of GP 120 and CD4.
C. Integration of viral DNA into host DNA.
D. Assembly of viral parts into a whole virus.

Explanation:
- Anti – HIV drugs are different drugs that act in different ways at different stages of the cycle of reproduction.
- It developed to prevent the virus from entering the host cell.
- It prevents the integration of viral DNA into host DNA.
- It prevents the assembly of viral particles.
- The anti – HIV drugs developed to prevent the entry to the host cell to do not binding of GP 120 and CD4. **Therefore, the answer is B.**

52. Which component of soil fertility is improved when farmers grow legumes in crop rotation?
A. Phosphorus B. Nitrogen C. Sulfur D. Carbon

Explanation:
- Crop rotation is very important to improve soil fertility.

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- It is used to add different nutrients to the soil.
- During crop rotation, when it is using by legumes they contain in bacteria which involved in the recycle of nitrogen. **Therefore, the answer is B.**

53. Which of the following biomes in Africa is supporting large wild mammals such as elephants, giraffes and lions.

A. The congo rain forest
B. The rain forest of western
C. Savanna grass land
D. The Sahara desert

Explanation:

There are different types of biomes in the world.

- **A biome** is a geographical or regional areas in the world.
- From different biomes which support large wild mammals in Savanna grass land. **Therefore, the answer is C.**

54. What is the important role player by microorganisms such as bacteria and fungi in the ecosystem?

A. Antibiotic production
B. Recycling of nutrients
C. Forming organic substances
D. Supplying energy to the ecosystem.

Explanation:

- Microorganisms such as bacteria and fungi are played different activities used for humans.
- Some of these activities are antibiotic production and production energy in the biogas process.
- These bacteria plays a great role in the decomposition of dead organisms in the recycling of nutrients in the ecosystem. Therefore, the answer is B.

55. In which one of the following aspects in the tropical rain forest biome poor?

A. Species diversity
B. Amount of sun light
C. Annual precipitation
D. Soil fertility

Explanation:

- The tropical rain forest biome is rich in species diversity, the amount of sun light and have high annual rain forest.
- This biome is poor in soil fertility b/se many of the minerals from the soil remain locked up in the huge trees.

Therefore, the answer is – D.

56. Which of the following is not usually true as an ecological succession progress to advanced seral stages?

A. More ecological niches are formed.
B. Species become more diverse.
C. The depth of the soil increases.
D. Less population are supported.

Explanation:

As an ecological succession developed into advanced seral stages:

- More ecological niches are formed.
- It increases the species diversity.
- The depth of the soil increases and
- More populations are supported and reach in to climax community.

Therefore, the answer is – D.

57. In which one of the four phases of population growth is the number of population the highest?

A. Lag phase
B. log phase
C. constant phase
D. decline phase

Explanation:

- In lag phase – popn is low and stable b/se they start to adapt a new surroundings.
- In log phase – the adapted popn increase rapidly b/se they have abundant resources.
- In stationary (constant) phase – number of popn are high and stable b/se the deaths are matched to birth.
- In decline phase – the numbers of popn decreases b/se of over use of resources and accumulation of toxic substances in the surrounding.

Therefore the answer is C.

58. What is the type of community called when it has reached the final and most complex stage of a succession?

A. Pioneer community
B. Several community
C. Climax community
D. Secondary community

Explanation:

A pioneer species (community) is species which is colonized a hostile area or base land. After several stages the most complex community. The development of different stage is called seres.

The answer is C.

59. Which of the following is an ecosystem?

A. A tropical rain forest.

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B. The African continent.
C. All the organisms in a given area.
D. The nonliving components of an environment.

Explanation:
Ecosystem is a system of interaction b/n living components to each other and a nonliving compounds. It contains biotic (living) and abiotic (nonliving) components. Therefore, the answer is D.

60. What is the main reason for high species richness of plants and mammals observed in Ethiopia?
A. Lack of predators.
B. Lack of disturbance.
C. Presence of several biomes with in the country.
D. Efficient management of the ecological resources.

Explanation:
Ethiopia is an important regional center of biodiversity. The country has a wide range of altitude and climate with the isolation of high lands. There are many different biomes present in the country.

61. Which one of the following concepts contains all the others?
A. Species B. Genus C. population D. Community

Explanation:
- Species is a group of related organisms that can reproduce to each other and give fertile off springs.
- Genus is a contains a group of species.
- Population is a group of organisms of one species occupying a defined area.
- Community is any group of organisms belonging to a number of different species that coexist in the same area.

Therefore, the answer is D – b/se community contains the concepts of all others.

62. If an age pyramid of a certain country is narrowing at the base, what does this tell about the trend of the popn size of the country?
The popn size is:
A. Declining C. Increasing slowly
B. Growing fast D. establishing

Explanation:
When an age pyramid of a certain country is broader at the base, it tell the rapid growth of the popn size. When an age pyramid is

narrowing at the base it tell about the trend of increasing slowly the popn size of a country. Therefore, the answer is C.

63. Before making crosses which part of the flower did meadel remove to avoid self-pollination?
A. Stigma B. Ovule C. Ovary D. Stamens

Explanation:
To make cross breeding to find out the breeding line and avoid self-pollination Mendel remove the statement of the flower. This process is called emasculation. Therefore, the answer is D.

64. Which of the following is the correct F_2 phenotypic ratio of a monohybrid cross?
A. 1: 2 B. 1: 1 C. 3: 1 D. 2: 2

Explanation:
Phenotype is an external appearance of an organism which determine by a particular genotype. Genotype is the genetic make up of traits/ organisms. The F_2 phenotypic ration of a monohybrid cross is 3: 1.

Example – P_1 (parental generation phenotype) Tall × Short
" " genotype TT × tt
" " gametes T × t
F_1 (First filial generation) phenotype all are tall
" " genotype – Tt
Cross b/n F_1 generation = $F_1 \times F_1$ = Tt × Tt
F_2 generation genotype 1 TT = 2Tt = 1 tt
Phenotype ration 3: 1
Genotype ratio 1: 2: 1
Therefore, the answer is C = 3: 1.

65. What are the most likely causes of variations within species?
A. Mitosis and a sexual reproduction.
B. Over population and over production.
C. Vegetable propagation and cloning.
D. Mutations and sexual reproduction.

Explanation:
Variations of species mostly caused by mutation and sexual reproduction which produce gametes by miosis and forming crossing over. Therefore, the answer is D.

66. One of the following is an important cause of gene mutation.
A. Old age C. smoking
B. Alcoholic drinks D. Radiation

Explanation:

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There different causes of gene mutation.

- The most causes of mutation electromagnetic radiation such as ultraviolet light, x – rays, gama rays.
- Other causes of mutations are called mutagenes, which include mustared gas, formal dehyded caffeine, to bacco constituents colichiline, some drugs and food preservers. Therefore, the answer is D.

67. How many chromosomes do humans inherit for each of their parents?
A. 23 chromosomes
B. 23 pairs of chromosomes
C. 46 chromosomes
D. 46 pairs of chromosomes

Explanation:
Each body cell has 23 pairs or 46 chromosomes. But sex cells half number chromosomes which passes to the off spring. Humans off spring inherited half number of 23 chromosomes from their parents through sex cells which produced during meiosis. Therefore, the answer is A – 23 chromosomes.

68. Which of the following is not true true about the gene called sex?
A. It found on the y – chromosome.
B. It determine maleness.
C. Tastes develop in its presence.
D. Females have two copies of this gene.

Explanation:
- SRY is an abbreviation of sex determining region of y chromosome.
- When the SRY gene is activated the bi potential gene develop in to a testes and the embry is male.
- When the SRY gene is absent (in activated) the bi potential gonads develop into ovary and the embryo in female. Then the SRY gene is found on Y chromosome that determine sex.

Answer: D

69. The sheep "dolly" is an example of which biotechnological manipulation of animals by human?
A. Transgenic animal
B. Genetically engineered animal
C. Cloned animal
D. Hybrid animals

Explanation:
To transfer gene in transgenic animal and genetically engineered animal it needs the vector cell and the host. But in cloned animal

it is in direct transfer of gene b/n animals. Dolly sheep is the first cloned animal by human. Therefore, the answer is C.

70. Choose the one that is different from all the others.
A. Genetically modified organisms.
B. Genetically engineered organisms.
C. Pathogenic organisms.
D. Transgenic organisms.

Explanation:
Pathogenic organisms are different from the others b/se these organisms are disease causing organisms. Therefore, the answer is C.

71. Which of the following crops is considered to be the best choice for a better of essential amino acids in a human diet?
A. Maize B. Quino C. Rice D. Wheat

Explanation:
There are two types of amino acids in our body. These are:
1) Nonessential amino acids which can made by our body by the process of transamination.
2) Essential amino acids which obtained from our diet. The best choice to balance these amino acid is taking non animals resource is guinoa. Therefore, the answer is B.

72. In which of the following are flowers and fruits found?
A. Ferns and relatives C. Mosses and confiers
B. Gymno spersm & ferns D. monocots and dicots

Explanation:
Ferns are reproduce by frons which consists of spores, mosses also bears archegonia (male) and antheriadia (female) organs.
- Gymnosperms plants are cone bearing plants and their seeds are found in cones.
- Angiosperms plants (monocots and dicots) are reproduce by flowers and producing fruits.

Answer: D

73. In a cross b/n heterozygotes what proportion is expected to be homozygotes recessive?
A. 25% B. 50% C. 75% D. 100%

Explanation:
When you cross b/n heterozygous the proportion of homozygous is 25%.
Given = Tt × Tt = both side heterozygous

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25% pure dominant
50% dominant
25% homozygous recessive
tt homozygous recessive

Answer: A

74. Gene silencing is the function of one of the following molecules:
A. Molecules
B. mRNA
C. SiRNA
D. ERNA

Explanation:

- ds RNA is double strand RNA produced in the nucleus with their range of gene.
- SiRNA short interfering RNA can act to express gene action and impotant in gene silencing.
- tRNA transfer RNA which transfer information from the cytoplasm to the nucleus.

Answer: C

75. Which process is held responsible for chronic myelogenous leukemia?
A. Translocation
B. Translation
C. Transcription
D. Duplication

Explanation:

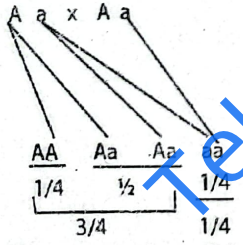
In chromosomal mutation translocation is a process that held responsible to form for chronic myelogenous leukemia.

Answer: A

76. Two parents of genotype Aa are corss bred. The alleles show complete dominance what proportion of the offspring wil phenotypically look like their parents?
A. 0 B. ¼ C. ½ D. ¾

Explanation:

The cross b/n the two heterozygous form complete dominance.



The proportion of their offspring which phenotypically similar to their parents are ¾.

Answer: D

77. Among the following mating, the ABO blood genotypes of the parents are shown, identifying the mating in which all the children will have the same blood type.
A. AO × BO B. AA × OO C. AB × BO D. BB × AO

Explanation:

The children which have the same blood type to their parents having AO × BO blood group.

Answer: A

78. If a new mutant allele a rises in a certain popn which of the following factors determines if the allele is going to be adaptive or non-adaptive?
A. The environment in which the populations lives.
B. The rate at which the gene mutates.
C. The popn in which the gene is found.
D. The use and disuse of the gene by the popn.

Explanation:

The factors that determine the allele is going to be adaptive or non-adaptive is the rate at which the gene mutates.

Answer: B

79. Suppose the amino acid coding region in a mRNA is 1200 nucleotides long, how long is the protein in terms of amino acid number?
A. 1200 amino acids
B. 600 amino acids
C. 400 amino acids
D. 300 amino acids

Explanation:

Genetic code is the sequence of bases in the nucleotides of the DNA that makes a gene that codes for the protein and that each amino acid in the protein is coded for by a tri (sequence of three) of bases.

- A gene is a sequence of base triplates in the DNA molecule that carries the code for a proton.
- A mRNA is 1200 nucleotides and divided by the triplet of protein codes and have 400 amino acids.

Answer: C

80. Which one of the following choices show the end products of a miltotic cell division?
A. Gametes and ordinary cells
B. Four genetically different cells
C. Two genetically different cells
D. Two genetically identical cells

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Explanation:
The three choices (Letter A, B and C) are the end products of meiosis (meiotic cell division). Choice teller D is the end products miotic cell division.
Answer: D

81. What exactly happens during a substitution mutation?
A. One base is replaced by different base.
B. An extra base is added during DNA replication.
C. A base is missed out during DNA replication.
D. Changes occur in the arrangement of structure of a chromosome.

Explanation:
- One base is replaced by different base substitution mutation.
- An extra base in adding during DNA replication is addition mutation.
- When a base is missed out it is deletion mutation.

When changes occur in the arrangement or structure of a chromosome is chromosome mutation.
Answer: A

82. In which homid species did scientists find the smallest brain size (cranial activity)?
A. Homo sapiens
B. Homo habilis
C. Homo erectus
D. Homo neander the lensis

Explanation:
Brain capacity of Homosapiens is 1400 cm^2 in size.
" " of Homo habilis is 700 cm^2 in size.
" " of Homo erectus 840 cm^2 in size.
" " of Homo neander the lensis 1200 cm^2
The smallest brain size observed in homo habilis.
Answer: B

83. What is the specialist in biology called if he/she studies?
A. Geneticist
B. Ecologist
C. Palenotolgist
D. Comparative Biochemist

Explanation:
- Geneticist study about gene and heredity.
- Ecologist study about ecology.
- Paleontologist study about chemical in living things.

Answer: C

84. When two species are compared which of the following sources of evidence is least informative about the degree of relationships b/n the species?
A. Nucleotide sequence of DNAs
B. Amino acid sequence of proteins
C. Glucose sequence of polysaccharides
D. DNA – DNA hybridization

Explanation:
Nucleotide sequence of DNAs, amino acid sequences of protein and DNA – DNA hybridization are the source of evidence of evolution. Glucose sequences of polysaccharides cannot uses as a source of evolution.
Answer: C

85. What is the reproductive isolating mechanism called if two species of frogs do not interbreed become they cannot understand the melting cells of one another?
A. One another
B. Seasonal isolation
C. Temporal isolation
D. Isolation by distance

Explanation:
- Seasonal isolation is members of two species (popn) reproduce at different times of the year.
- Temporal isolation is members of the two popns reproduce at different time of the day.
- Behavioral isolation is members of the two popns have different courtship patterns.
- Isolation by distance is geographical isolation. Therefore the answer is B. Because the two species of frogs have different courtship patterns.

Answer: B

86. Which of the following theories explains evolutionary changes of living things in terms of changes in their allele frequencies?
A. Darwin's natural selection.
B. Spontaneous generation.
C. Inheritance of acquired characteristics.
D. Neo – Darwinism.

Explanation:
Natural selection is related to members of species will be adapted to their environment better to survive and reproduce. The inheritance of acquired characteristic is that traits (alleles) changed (acquired) during the individuals life time.
Answer: C

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87. Among the following which one is the best criteria to show that the populations belong to the same species?
A. Morphological similarity
B. Physiological similarity
C. In habiting the same geographical area
D. Production of fertile offspring

Explanation:
Species is a group of closely related species (organisms) which interbreed to each other and produce fertile off springs.

Answer: D

88. In the process of the evolution of life on earth, which of the following four processes evolved last?
A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo anto trophism

Explanation:
Photosynthesis is a process in which plants produce their own food and oxygen by using of sun light. Plants are then they are photo autotrophs.

Answer: A

89. What is the reproductive isolating mechanism called it two species of frogs do not inter breed because they cannot understand the mating calls?
A. Seasonal isolation
B. Behavioral isolation
C. Temperature isolation
D. Isolation by distance

Explanation:
- Seasonal isolation is members of two species (popn) reproduce to different times of the year.
- Temporal isolation is members of the two popn reproduce at different time of the day.
- Behavioral isolation is members of the two popn have different courtship patters.
- Isolation by distance is geographical isolation.

Answer: B

90. Which of the following theories explains evolutionally changes of living things in terms of changes in their allele frequencies?
A. Darwin's natural selection
B. Spontaneous generation
C. In heritance of acquired characterstics
D. Neo Darwinism

Explanation:
Natural selection is related to members of species will be adopted to their environment better to survive and reproduce. the inheritance of acquired characteristic is that trait (alleles) changed (acquired) during the individuals life time.

Answer: C

91. Among the following which one is the best criterion to show that the populations belong to the same species?
A. Morphological similarity
B. Physiological similarity
C. In habiting the same geographical area
D. Production of fertile offspring.

Explanation:
Species is a group of closely related species (organisms) which interbreed to each other and produce fertile offsprings.

Answer: D

92. In the process of the evolution of life on earth, which of the following four process evolved last?
A. Photosynthesis
B. Aerobic respiration
C. Chemosynthesis
D. Photo auto tropism

Explanation:
Photosynthesis is a process in which plants produce their own food and oxygen by using of sun light plants are then they are photo autotrophs.
Aerobic respiration is evolved last than others b/se heterotrophic organisms can use oxygen and used to release energy than others.

Answer: B

93. Which of the following characteristics can show the evolutionary relationships among organism?
A. Structures having similar functions.
B. Structures having same size.
C. Structures having common origin.
D. Structures having different origins.

Explanation:
The evolutionary relationships among organisms can show those structures having common anscestors or common origin.

Answer: C

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94. Which one of the following is not important for evolutionary change of a population?
A. Over reproduction
B. Insufficient of natural resources
C. Existence of heritable variation
D. Survival of all that are born.

Explanation:
The theory of natural selection that was proposed by Darwin have main observations:
1) Over production (fecundity)
2) Variations among the offspring
3) The struggle for existence b/se resource are insufficient the survival of the filest.

Answer: D

95. In some human liver cells there are 92 chromosomes per cell what is the prodigy level of such cells?
A. Haploid C. Tetraploid
B. Diploid D. Hexaploid

Explanation:
Each gamete cells aer half chromosomes are called haploid (n). Body cells are diploid (2n), and some cells in humans like liver cells are tratrapliod (4n). For example, humans such sex cell is 23 chromosome. Each body cell (nerve cell) has 46 chromosomes. Some body cells (liver cell) has 46 chromosomes.

Answer: C

96. During the course of evolution, which of the following events came before all the others?
A. Origin of dinosaurs and other reptiles.
B. Availability of free oxygen in the atmosphere.
C. Origin of the oldest eukaryotic organisms.
D. Origin of the first multicellular animals and plates.

Explanation:
To be survive on the earth mostly higher and lower animals need oxygen. The availability of free atmospheric oxygen is produced by photosynthesis.

Answer: B

97. What kind of natural solution is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle?
A. Stabilizing selection
B. Directional selection
C. Disruptive selection
D. Selection that removes the extremes.

Explanation:

- In stablizing selection, individuals at both endes of the range of valucs for a features are the least adapted and favor the middles.
- In directional selection, individuals at one extreme could have disadvantage and the other extreme have advantage.
- In disruptive selection, individuals at both extremes have some advantage over those displaying the mean value and the middle of the range have disadvantage.

Answer: C

98. In which of geologic period does the fossil record show more diverse and relatively higher forms of organism?
A. Devonian C. Jurassic
B. Cretaceous D. Permian

Explanation:
In cretaceous geological period there were observed more fossil records and more diverse higher forms of organisms tha others geological periods.

Answer: B

99. What is the most frequent cause tine agents of food poisoning?
A. Bacteria C. Viruses
B. Protozoa D. Worms

Explanation:
Bacteria is the most causative agents of food poisioning than other microorganisms.

Answer: A

100. Which one of the following do bees use to inform other bees about the location and distance of a new nector discover?
A. Pheromones C. Buzzing noise
B. Waggle dance D. Vibration of wings

Explanation:
Waggle dance is in which worker bees used to inform the location and distance of nectar where it is found for other bees.

Answer: B

101. Which of the following is not classified as a learned behavior?
A. Insight C. Latent
B. Innate D. Conditioned

Explanation:
Insight learning, latent behavior and conditioning are grouped in learned behavior but innate behavior is not under learned behavior.

Answer: B

102. What do we call the learned behavior is a mouse that had just escaped from the mouth of a cat Jumped violently at a slight touch by a trivial object?
A. Latent learning C. Conditioning
B. Sensitization D. Imprinting

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Explanation:

- Latent learning's laking about hidden learning orlearing that is not apparent as it takes place.
- Conditioning is a technique used to get an animal to lear a particular behavior.
- Sensitization is an increases in response to a harmless stimulus, when that stimulus occurs after a harmful stimulus.
- Imprinting is another kind of instinctive behavior in which the fixed action pattern in organized. For example a newly born (hatched) organism followed any movement.

Answer: B

103. The group of the Ethiopia wolf which does not contribute to territory marking with their urine containing pheromones is _____.

A. Adult males
B. Adult females
C. Sub – adult males
D. Sub – adult females

Explanation:

The Ethiopia wolf in a social animal. They maintain a group territory by making with urine containing pheromones. In this territory making behaviors, the adult males contribute to marking the territory but sub – adult females do not have contribution.

Answer: D

104. To what kind of animal behavior can the spinning of a web by a spider be classified?

A. Learned behavior
B. Experimental behavior
C. Instinctive behavior
D. Accidental behavior

Explanation:

There are two types of behavior. These are innate behavior include instinctive behavior. It involve a complex sequence of actions. A good example is spinning of a web by spider.

Answer: C

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1. While conducting research, which of the following should a biologist do first?

A) Generate hypothesis
B) Conduct experiments
C) communicate the results
D) Indentify a research question/problem

Explanation: To conduct scientific research biologists follow the scientific methods.

- The stage of scientific method starts from
→ observation
→ questioning and exploration about the problem
→ constructing experiments to test the hypotheses
→ conducting experiments to test the hypotheses
→ the development of theories and laws

Therefore the answer is D identify a research question

2. Suppose in an experiment conducted to test a new drug, 20 people were given the drug and 20 people a placebo, what is the control of the experiment?

A. people who received the placebo
B. people who received the new drug
C. the new drug itself
D. the new drug and the placebo.

Explanation: In a typical controlled experiment there are two groups

a) The first group is in which the variable is manipulated in a particular way and bring change during experiment is an experimental group
b) The another group is in which there is no manipulation or change is called the control group

The people those who are taking the new drug are experimental group and those people who received the placebo are the control group. Therefore the answer is A

3. Which molecule in the cell is the constituent of the gene?

A. Nucleic acid
B. protein
C. lipid
D. carbohydrate

Explanation: The answer is A. Because the genes are the constituent of Nucleic acid composed of DNA and RNA.

4. Which of the following is a distinguishing feature of amino acids that is not found in carbohydrates
A. carbon B. oxygen C. hydrogen D. Nitrogen
Explanation: Amino acid is the monomer of protein, which made up of carbon, hydrogen oxygen, nitrogen and sometimes sulphur, But carbohydrates are only made up of three element such as carbon, hydrogen and oxygen. **Therefore the answer is D**

5. Which one of the following molecules in living things is **not** an organic molecule?
A. sugar B. Nucleotide C. amino acid D. water
Explanation: Different organic molecules contain different combinations of carbon and hydrogen in addition to oxygen. Sugar (carbohydrate), lipids, proteins are organic molecules made up of carbon and hydrogen with other elements. Inorganic molecules may contain one or the other (carbon or hydrogen) but not both. Therefore the answer is D water contains hydrogen not carbon.

6. How many carbon and oxygen atoms are there in a molecule of maltose?
A. 6 carbon and 6 oxygen C. 11 carbon and 12 oxygen
B. 12 carbon and 11 oxygen D. 24 carbon and 12 oxygen
Explanation: Maltose is a double sugar (disaccharide) which is derived from two 2-glucose molecules and a molecule of water (H_2O) is formed by the process of condensation.
$C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$ **The answer is B**

7. Which of the following is true about saturated fatty acids?
A. They have single bonds between carbon atoms
B. They are the same as polyunsaturated fatty acids
C. They are the same as monounsaturated fatty acids
D. They have double bonds between the carbon atoms
Explanation: The natural of the hydrocarbon chains in fatty acids can differ based on the bonds between the carbon atoms.
* When one of the carbon – carbon bonds is a double bond, then it is a mono unsaturated fatty acid.
* When more than one carbon – carbon bond is a double bond then the fatty acid is polyunsaturated fatty acids.
* When all the carbon – carbon bonds in the hydrocarbon chain are single bonds the fatty acid is a **saturated fatty acid.** ⇒ **Therefore the answer is A**

8. Which of the following is **true** about the substance represented by the chemical formula "$C_2H_5NO_2$"?
A. simple sugar C. fatty acids
B. inorganic substance D. amino acids
Explanation: The chemical formula of $C_2H_5NO_2$ is represented an amino acid because it contains nitrogen element.
Therefore the Answer D

9. Choose the one that represents a chemical formula of an organic compound?
A. $C_{12}H_{22}O_{11}$ C. NH_3
B. $Mg(OH)_2$ D. NaCl
Explanation: Organic compounds contain carbon and hydrogen in the addition to oxygen.
Then the answer is A because it contains carbon hydrogen and oxygen elements

10. How are unsaturated fats are distinguished?
A. They are made up of glucose and fructose
B. They are made up of amino acids and glycerol
C. They have double bonds in their carbon chains
D. They have only single bonds at the points of attachment
Explanation: Unsaturated fats are consists one double bond which is mono unsaturated fatty acids and more than one double bond with in the carbon chains which are polyunsaturated fatty acid. ⇒ **The answer is C**

11. Which of the following is produced when a lipid molecule is hydrolysed?
A. Amino acid and water C. Fatty acids and glycerol
B. Amino acid and glucose D. Glucose and glycerol
Explanation: Ehen lipid molecule is hydrolysed, it produce fatty acid glycerol. **Therefore the answer is C**

12. In the cause of enzymatic chemical reactions, what do you call the substance that is acted up on by on enzyme?
A. polypeptide B. Coenzyme C. vitamin D. substrate
Explanation: The substance that is acted up on by an enzyme is the substrate. **Therefore the answer is D**

13. Which of the following is the common characteristic of all enzymes?
A. They act inside cells only C. They require cofactors
B. They are proteins D. They operate at any pH

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Explanation: The answer is B because all enzymes are proteins.

14. Which group of organic compounds includes the enzymes?
A. proteins B. lipids C. starches D. carbohydrates
Explanation: Enzymes are globular proteins that have a tertiary structure that resembles a globule or ball.
Protein is one of the organic molecule that makes enzyme.
Therefore the answer is A

15. Which of the following pH values represents the strongest base?
A. 2 B. 7 C. 13 D. 15
Explanation: The pH scale is starting from zero (0 – 14) up to fourteen bellow 7 is acidic and 7 is neutral and from 7 up to 14 is basic condition. Then the strongest base is around 14
Therefore the answer is C

16. Which region of human digestive tract contains enzymes that perform well at low pH?
A. mouth B. stomach C. small intestine D. large intestine
Explanation: The pH of the intestinal tract of human changes from one region to the next.
⇒ The concentration of acid (proton donor or electron acceptor) is greatest when the pH number is lowest.
⇒ As the pH number increases the concentration of base (proton acceptor or electron donor) increases.
⇒ At a pH of 7.0 the concentration of H^+ and OH^- are equal
Therefore the answer is B because enzymes which are found in stomach works in acidic

17. For a biological detergent to effectively remove oily and greasy dirt, which of the following enzymes should it contain?
A. cellulose C. amylase
B. protease D. Lipase
Explanation: Oils and greasy dirt are lipid containing molecules Lipase is an enzyme used to hydrolys lipids
Therefore, the answer D

18. Which one of the following cellular forms did Robert Hooke observe under his crude microscope?
A. Bacteria B. protozoa C. yeast Dd. Empty cell wall
Explanation: Robert Hooke was the first person who use the term cell. He choose to study thin slices of cork from the bark of a cork oak tree.

He saw only the cell walls that surrounded the living portions of plant cell under his crude micro scope.Therefore the answer is D

19. One of the following organelles of the eukaryotic cells does protein synthesis take place?
A. The nucleus C. The chloroplast
B. The ribosome D. The mitochondria
Explanation: The ribosomes are the organelles of the eukarioplic cells in which protein synthesis can take place. The answer is B

20. Which of the following structures is not present in animal cells
A. cell wall B. nucleus C. protoplasm D. nucleic acid
Explanation: From the given structures cell wall is not present in animal cells. Then the answer is A.

21. What is the general term for the pat of the protoplasm that lies outside the nucleus?
A. cytosol C. central vacuole
B. cytoplasm D. plasma memberane
Explanation: Every Eukaryotic cells have two main parts. these parts are the cytoplasm and the nucleus. The general term of these parts is protoplasm. Then the out side portion of the nucleus is cytoplasm. **The answer is B**

22. Which of the following cellular structures is possessed by all cells?
A. cell membrane C. cell wall
B. nucleus D. Golgri apparatus
Explanation: From the given choice nucleus is not found in prokaryotic cells, the cell wall is **not** present in animal cells and Golgi apparatus also **not** present in prokaryotic cells. The cell membrane is the only cellular structures found in all cells.
Therefore the answer is A

23. Which of the following carbohydrates has structural function?
A. cellulose B. Glycogen C. starch D. sucrose
Explanation: cell wall is the structural function of plants which is made up of polysaccharide called cellulose. Therefore the answer is A

24. Which one of the following eukaryotic cell organelles was a free living cell before eukaryotic cells evolved?
A. nucleus B. Ribosome C. chloroplast D. Nucleolus

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Explanation: From the given choice Ribosomes are the only organelles possessed in both prokaryotic and eukaryotic cells. Then the answer is B

25. Which of the following mechanisms moves digested amino acids and glucose across the plasma membrane of the cells lining the wall of the small intestine?
A. osmosis C. facilitated diffusion
B. simple diffusion D. Active transport

Explanation: The digested amino acides and glucose are move across the plasma membrane of the cells lining the wall of the small intestine by facilitated diffusion. Because the plasma membrane has the carrier proteins or channel proteins with ion pores. Therefore the answer is C

26. Which one of the living characteristics UNNECESSARY for an object to be considered a living things?
A. Ability to respond to stimuli C. Ability to grow
B. Ability to reproduce D. Ability to move

Explanation: The answer is D because not all living things show the ability to move.

27. If a cell fails to clear its cellular debris, which one of its organelles is most likely not functioning?
A. Nucleus C. Endoplasmic relialum
B. mitochondria D. lysosomes

Explanation: Lysosomes are formed in the Golgi apparatus and contain digestive enzymes that break down cellular waste and debris. Then a cell fails to clear its cellular debris, when hysosomes are not function properly. Therefore the answer is D

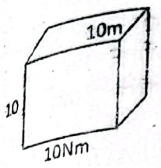
28. Among the following identify the organelle in which nucleic acid is not found?
A. mitochondria C. Ribosome
B. chloroplast D. Golgi apparatus

Explanation: From the given alternatives Golgi apparatus do not consists of nucleic acid (both DNA and RNA). Nucleus, mitochondira and chloroplast have their own DNA, but ribosome conteins RNA of nucleic acid. Then the answer is D

29. Suppose a hypothetical cube-shaped cell has sides of 10 micrometers, what is the surface area to volume ratio of this cell?
A. 6:10 B. 10:10 C. 3:6 D. 4:8

Explanation: Area of the cube is a = L × w

$10 \times 10 = 100 \times 6 = 600 Nm^2$



The volume of the cube (V) = L × w × h
V = 10Nm × 10Nm × 10m = 1000N

Then the surface area to volume ratio is = $\frac{A}{V}$

$= \frac{600}{1000} = 6:10$

Then the answer is A

30. In which of the following solution does an animal cell under go haemolysis?
A. In hypotonic solution
B. In hypertonic solution
C. In isotonic solution
D. In both hypotonic and hypertonic solution

Explanation: When an animal cell put in hypotonic solution, the cells gain water by osmosis and the hypotonic solution, the cells due to the weak plasma membrane. This is condition is called haemohysis. Therefore the answer is A

31. Which of the following will happen if a plant cell is kept in a solution that is stronger than its protoplasm?
A. The cell will become liquid
B. The central vacuole will expand
C. The protoplasm will get plasmolysed
D. The cell will swell and brust

Explanation: when the plant cell kept in a strong solution the cytoplasm of the cells loses its water by osmosis. Then the plant become shrink and the cell has been plasmolysed
Therefore the answer is C

32. How many net ATP molecules are generated through an aerobic respiration when as single glucose molecule is changed to pyruvale in the human body?
A. two B. three C. four D. six

Explanation: The first stage of the cellular respiration process take place in the cytosol of cytoplasm. This first step, known as glycolysis conists of the enzymatic break down of glucose without oxygen. Glycolysis is called an anaerobic process.
In glycolysis a single glucose molecule changed to pyruvate and generate four ATP molecules because to ATPs were utilized to

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start the process and a net two ATP molecule are stored in the process. Therefore the answer is A

33. During the krebs cycle, which of the following molecules temporarily stores most of the energy released from food molecule?
A. ADP B. ATP C. NADH D. FADH

Explanation: During the krebs cycle most of the energy, six $NADH^+$ released. From a glucose molecule stores temporarily. Only two $FADH^+$ and 2ATP molecule in krebscycle. Then the answer is C.

34. At which stage is most of the ATP generated in aerobic respiration?
A. Glycolysis C. krebs cycle
B. link reaction D. electron transport

Explanation: Most of the ATP produced by aerobic respiration comes from the electron transport system (ETS), because 10NADH and 2 FADH which are produced during glycolysis, link reaction and krebs cycle are deliver electrons to the enzymes responsible for the ETS and released out 34 ATPs during this stage. Therefore the answer is D

35. Which of the following substances is not necessary for photosynthesis to take place?
A. chlorophyle C. oxygen
B. carbon dioxide D. water

Explanation: Water and carbondioxide are the two raw materials of plants to photosynthesis take place. The necessary conditions of photosynthesis are chlorophyle and sunlight. Glucose is the main products of photosynthesis and oxygen is the byproduct of hotosynthesis of plants oxygen is the byproduct of photosynthesis of plants. Therefore the answer is C. oxygen is not necessary for photosynthesis to take place.

36. The molecule of which pigment is located at the reaction center of a photosystems?
A. chlorophyll a C. carotenid
B. chlorophyll b D. Accessary pigments

Explanation: The pigments in antenna complex contains chlorophyll a center of photosystem II (PSII) which PGSO and the reaction center of photosyptem I (PSI) P700 molecule is always chlorophyll a. Therefore the answer is A

37. What is the ultimate source of energy for all organisms living on the earth?
A. water B. oxygen C. Nitrogen D. the sun

Explanation: The sun light is the ultimate source of energy for all organisms living on the earth. The answer is D

38. Where exactly in the cell does the krebs cycle take place?
A. mitochondrial matrix C. cytoptosmic fluid
B. inner mitochondrial membrane D. outer mitochondrial membrane

Explanation: There are four stages of cellular respiration, they take place in different region of the cell. For example
1. glycolysis is take place in the cytosol of cytoplasm with the absence of oxygen (anaerobic)
2. The link reaction and the krebs cycle are take place in the fluid matrix of mitochondria with the presence of oxygen (aerobic)
3. The electron transport system (ETS) can take place in the inner membrane of the mitochondria, known as the cristae with the presence of oxygen (aerobic)

Therefore the answer is A. the krebs cycle take place in the mitochondrial matrix.

39. If present in the fermentation syptem, which one of the following would negatively affect alcohol production by yeast?
A. water B. oxygen C. Glucose D. Fermentation enzyme

Explanation: Fermentation system is anaerobic process which occur with out oxygen. If there is oxygen it affects the process.
The answer is B

40. Which term describes the process by which water is lost by evaporating through the stomata of leaves?
A. Transport C. Anchorage
B. Transpiration D. Adhesion

Explanation: The answer is B

41. Which of the following is an accurate representation of the relationship between the terms metabolism, anabolism and catabolism?
A. Anabolism = Catabolism
B. Metabolism = Catabolism
C. Anabolism = Catabolism + metabolism
D. Metabolism = catabolism + anabolism

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Explanation: Metabolism is the sum total of all the chemical changes that occur in the body of organisms. It consists of anabolism and catabolism. Anabolism is the energy requiring process by which small molecules are joined together (building up) to form larger molecules. Example . photosynthesis. Catabolism is the energy releasing process by which large molecules are broken down in the smaller ones. Example – respiration. Therefore the answer is D

42. In which of the following ways is the carbon dioxide of the atmosphere fixed in to the carbon found in organic molecules?
A. In the breathing processes of all animals
B. In the decomposition of organic molecules
C. In the process of photosynthesis by green plants
D. In all the cellular respiration processes of organisms

Explanation: The carbon atom which found in organic molecules such as in carbohydrates, lipids and proteins are processes of photosynthesis by algae and green plants. Therefore the answer is C.

43. Which of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells?
A. C – 3 plants C. CAM plants
B. C – 4 plants D. plants without chlorophyll

Explanation: C – 3 plants are those plants living in temperate environments such as those found in Europe. The first compound formed in the light independent reactions of the calvin cycle is glycerate phosphate (Gp) which contains three carbon atoms.
- Both light dependent and light independent (calvin cycle) take place in the mesophyll cells, specially in the palisade cells.

C – 4 plants are mostly living in tropical environments including in Ethiopia.
The first compound formed in the light independent reactions is a C_4 compound (contains four carbon) is called oxalo acetate.
The light dependent and the light independent reactions take place in different (separate) cells. The light dependent reaction takesplace in the mesophyll cells, where as light independent reactions take place in the bundle sheath cell of vascular tissues. Therefore the answer is B

44. From which of the processes of cellular respiration is the majority of the ATP generated?
A. Anaerobic fermentation
B. Glycolysis and link reaction

C. Electron transport and chemiosmosis
D. krebs cycle and glycolysis

Explanation: The processes of cellular respiration the majority of ATP are generated in the electron transport chain and chemiosmosis. Because the molecule if reduced NAD and reduced FAD passing through ATP synthesis of electron transport chain.
Then the answer is C

45. One which of the following does the rate of algal photosynthesis in a Lake depend?
A. The oxygen content of the water
B. The nitrogen content of the water
C. The elevation where the lake is found
D. The amount of light that penetrates the take water

Explanation: The rate of algae photosynthesis in lakes depend on the amount of light that penetrates the lake water, because photosynthesis mostly take place in the presence of sun light. Therefore the answer is D.

46. During aerobic respiration, what is the route through which protons return from the mitochondria inter membrane space beck to its matrix?
A. proton pump C. ion channel
B. ATP synthase D. membrane lipid

Explanation: During the Electron transport chain of aerobic respiration, the protons accumulate in inter membrane space and creating a higher concentration there than the matrix. This proton gradient results in protons diffusing through the ATP synthase molecule which is called chemiosmosis. Therefore the Answer is B.

47. What does the absence of chloroplasts in a unicellular organism indicate?
A. Reproduction by sexual means
B. Autotrophic nutrition
C. Heterotrophic nutrition
D. complete lack of metabolic processes

Explanation: The absence of chloroplast in all living organisms including unicellular organisms are related with heterotropic nutrition. The presence of chloroplast is used to trap sun light and able to produce their own food by the process of photosynthesis. Therefore the answer is C

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48. Which of the following did the early heterotrophs do that probably assisted the origin of autotrophs?
A. They added oxygen and chlorophyle to the environment
B. They added carbon dioxide to the environment
C. They stored energy in the bonds of inorganic compounds
D. They manufactured food from carbondioxide and oxygen

Explanation: The early hetrotrops provide carbondioxide for the autotropic organisms. Auto trophs are able to make their own food by the process of photosynthesis. Then carbondioxide is one of the raw material for the process of photosynthesis. Therefore the answer is B

49. Which of the groups of bacteria consists of rod shaped cells?
A. Gram-positive
B. bacilli
C. cocci
D. spirilla

Explanation: Bacilli (singular bacillus) are rod shaped bacteria
The answer is B

50. Which of the followings do viruses share with cells?
A. cell wall
B. cytoplasm
C. Nucleic acid
D. protoplasm

Explanation: Viruses do not possess cell wall, cytoplasm and protoplasm. They only shared nucleic acid with other living cells.
Therefore the answer is C

51. Which of the following is true about protozoa?
A. They are a group of bacteria
B. They are one celled plants
C. They represent both plants and animals
D. They are unicellular animals

Explanation: The protozoans form a large group of kingdom protoctista, kingdom protoctista are grouped in to algae (plant like) and protozoa (animal like) eukaryotes. Protozoans are unicellular, animal like cells with heterophic nutrition. Therefore the answer is D

52. Choose the organisms that belongs to the eukaryotes?
A. Bacteria
B. Blue green
C. Amoeba
D. virus

Explanation: Bacteria and blue green algae found in kingdom monera and they are prokaryotic unicellular organisms. Viruses are neither prokaryotic nor eykaryotes. Amoeba are protozoans and they are eukaryotic organisms.
Therefore the answer is C

53. Choose the characteristics that viruses share with living things.
A. They are made up of many specialized cells
B. They contain genetic material
C. They reproduce by mitosis
D. They contain chlorophyll

Explanation: Viruses share genetic material which is nucleic acid such are DNA or RNA but not both. The answer is B

54. Which stage in the life cycle of HIV is inhibited by the antiretroviral drug known as integrase – inhibitor?
A. conversion of viral RNA to DNA
B. Replication of the viral genome
C. Insertion of the viral genome in to host genome
D. Release of the viral progeny from the cell

Explanation: There are several different antiretroviral drugs that act in different ways at different stages of the life cycle of reproduction
For example: 1) Reverse transcriptase inhibitors attach to HIV's enzyme preventing it from converting RNA to DNA.
2) protease inhibitors interfere with the action of HIV's protease enzyme preventing the assembly of new viruses particles.
3) Integrase inhibitor preventing the insertion of the viral genome in to host genome. Therefore the answer is B

55. From which organism is the first antibiotic used in medicine produced?
A. Gram negative bacteria
B. members of the fungi
C. Genetically modified protozoa
D. higher plants

Explanation: The first antibiotic which is called penicillin was discovered in 1928 by Dr. Alexander Fleming
Alexander Fleming was growing the mold penicillium notatum on some of the petridish. Penicillium notatum is a unicellular fungi.
Therefore the answer is B

56. What is the difference between HIV and AIDS?
A. AIDS is the virus and HIV is the disease
B. Both AIDS and refer to the disease
C. AIDS is the disease and HIV is the virus
D. Both AIDS and HIV refer to the virus

Explanation: AIDS (Acquired Immune Defiance Syndrome) in the disease and caused by HIV (Human immune Deficiency Virus)

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which attack our immunity system specially T-helper cells. Therefore the answer is C

57. Biotechnology can be applied in environmental remediation. To which of the major application areas of biotechnology is this most related?
A. Medicine and nutrition C. Biodiversity and industry
B. medicine and industry D. Agriculture and environment
Explanation: Medicine and industry are the major application areas of biotechnology. Biotechnology in medical products include, hormone (insulin), human growth hormone, drugs, antibiotics, monoclonal antibodies and vaccines. Industry products include, food industry, brewing industry in dairy farm industries etc. Therefore the answer is B

58. Which of the following parts of plant roots is harbouring nitrogen fixing bacteria?
A. Root nodule C. Root tip
B. Root hair D. Root cap
Explanation: Nitrogen fixing bacteria belonging to the genus Rhizobium are mostly found in nodules on the roots of legume plants. Then the answer is A

59. Which of the following statement is true about matter and energy in the ecosystem?
A. Both matter and energy are recycled
B. Matter is not recycled; energy is recycled
C. Matter is recycled; energy is not recycled
D. energy is recycled more than matter does
Explanation: Energy is continually lost as heat to the ecosystem and has to be replaced as light, where as nutrients or matter recycled in the ecosystem. Energy is not recycled. Energy flow in the ecosystem unidirectional and matter or nutrients recycled multidirectional. Therefore the answer is C

60. Which of the following forms of nitrogen is readily utilizable by green plants.
A. Atmospheric nitrogen C. ammonion ion
B. nitrite D. Nitrate
Explanation: Animals obtained nitrogen from the food which they consumed to build up their own proteins. Plants obtained nitrogen in the form of nitrate in the soil. Then the answer is D

61. In which one of the following ways the Tundrabiome differs from desert biome?
A. It is a treeless biome C. It receives little precipitation
B. It has little biodiversity D. It has permanently frozen subsoil
Explanation: Desert biome is divided in to hot desert and cold desert, when comparing with tundra regarding to the soil type thee tundra biome contains permafrost or frozen type than poor fertile soil desert biome. Therefore the answer is D

62. In Ethiopia, which one of the following vertebrate groups has the largest number of genera?
A. mammals B. Birds C. Reptiles D. Amphibians
Explanation: Ethiopia is an important regional center of biological diversity. This is because of having the wide range of altitude and climate. Ethiopia has many plant species and animal species. Regarding to the vertebrate biodiversity the country has a large number of genera of birds than other vertebrate.
Then the answer is B

63. Which of the following applies to the tropical rainforest?
A. Low rain fall and high temperature
B. Low rain full and low temperature
C. High rain fall and high temperature
D. High rain fall and low temperature
Explanation: In the tropical rain forest became has the characteristic of high rain fall, always warm temperature poor soil, many plant and animal species. Therefore the answer is C.

64. Which of the following resources is need by an ecosystem from out side in order to sustain itself?
A. produces C. micro nutrients
B. macro nutrients D. sun light
Explanation: To sustain life in an ecosystem the sun light is the main resources for all living things. Because, sun light is the primary condition for photosynthesis for produces to make their own food. The answer is D

65. Which of the following can create strong inter specific competition is shared by two or more species?
A. Habitat B. Ecosystem C. Niche D. Predator
Explanation: Habitat is an area where a population lives and finds the nutrients, water, living space and other essential needs to survive. Ecosystem is the interaction between the biotic (living) and a biotic

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(non living) components in the given area. In the given habitat organisms have their on function. The role of organism in the ecosystem is ecological niches. Which create strong inter specific competition. Therefore the answer is C.

66. Suppose sites A, B, C and D have Simpson's. Index of diversity (d) values of 20, 8, 16 and 3 respectively, which site is dominated by one or just a few species.

A. site A B. site B C. site D. site D

Explanation: Site D is dominated by on species. This is because a low value for the index of diversity suggests that an area dominated on or just a few species. Therefore the answer is D.

67. Among the following, which one has little or no contribution to the loss of biodiversity?

A. Overgrazing by domestic animals
B. Clearing forests to grow crops
C. Growing crops of improved varieties
D. growing crops land races

Explanation: From the given choice, overgrazing, clearing forests and growing crops of improved varieties have a great contribution to the loss of biodiversity. Growing crops landraces has a little contribution to the loss of biodiversity. Therefore the answer is D.

68. Which of the following is the largest ecological unit?

A. A community C. A population
B. A biome D. An ecosystem

Explanation: Ecosystem is the largest ecological of the plant, because it is made up of biotic (community and population) and a biotic (physical) component of the planet. Therefore the answer is D

69. What is the ecological succession in which plant communities establish after an area has been completely demolished by fire?

A. Tree succession C. primary succession
B. secondary succession D. Bare rock succession

Explanation: Succession is the process of changing from one types of community to another community. There are two types of successions. These are primary and secondary succession. Primary succession is in which a community of animals and plants develops on the bare land or rock. Secondary succession is in which community of organisms is disturbed by natural or human related events for example hurricane, volcano, fire, forest harvest etc, and

return to the previous stage in the succession. Therefore the answer is B.

70. Which of the following groups of animals have males with zz and females with zw sex chromosome constitution?

A. Grass hoppers B. Birds C. mammals D. honey bees

Explanation: There are reversal systems of sex determination in different organisms. For example in humans, the sex chromosomes of females are homogametic, which designated by letter xx and mates are hetero gametic which designated by xy. In birds the females are hetero gamatic represented by the letters of zw and the mates are homogametic which are zz. Therefore the answer is B.

71. In some crosses of maize, the progeny produces better yield than the parents. What is this phenomenon known as?

A. Dominance C. Hybrid vigour
B. Heterozygosity D. In breeding depression

Explanation: Hybrid vigour occurs when unrelated breeds or lines of the same species are cross bred. This hybrid vigour cases the offspring from this crosses are more productive than the average of their parent breeds. The answer is D.

72. In across between round grean pea of RRyy genotype and wrinkled yellow pea of rrYY genotype, what is the expected genotype of the F_1?

A. RRyy B. rrYY C. RRyy D. RrYy

Explanation: The inheritance of two genes at the same time is called dihybrid cross. Then when you cross between round green pea with the genotype of RRyy and wrinkled yellow pea of rrYY phenotype. The expected genotype of F_1 is RrYy. The answer is D

73. Which of the following is more true about the mall of the honey bee?

A. It has no father C. It is diploid
B. It is sterile D. It regulates the colony

Explanation: In honey bees, a queen bee mates only once and stores the sperm, she thin can control the release of the sperm.

⇒ If no sperm are released the eggs develop parthenogenetically in to drones, which are males.
⇒ If sperm are allowed to fertilize the eggs the fertile eggs develop in to other queens or workers bees which are females.
⇒ The queens and workers are diploid and drones (mates) are haploid.

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The queen and drones are fertile and the worker (females) are sterile. The male drones are produced by unfertilized eggs are called parthenogenesis. Therefore the answer is A

74. In to which of the following organic compounds can lipase, maltase and sucrose be grouped?
A. Hormones
B. carbohydrates
C. nucleic acid
D. enzymes

Explanation: These organic molecules are grouped in enzymes because all their suffix are ended by ase. Then the answer is D

75. In which of the activities of the honey bee colony are worker bees involved?
A. Laying eggs
B. serving as queens
C. Reproductive functions
D. Taking care of the carve

Explanation: The honey bees are social animals and have division of labor among themselves. For example, a few individuals of queens and drones (males) specialize in reproduction.

⇒ Whereas large numbers of sterile worker honey bees are involved in collecting food, defending the hive, and caring for the larve Therefore the answer is D.

76. Which one of the following is the mechanisms by which two genes located on the same chromosome are separated
A. Independent assortment
B. segregation
C. linkage
D. crossing over

Explanation: Mendel formulated several genetic laws to described how characteristic are passed from one generation to the next.

⇒ These mendel laws are

1. **Mendel's law of dominance**
   - When an organism has two different alleles for a given trait, the allele that is expressed, overshadowing the expression of the other allele is known as dominant.

The gene whose expression is overshadow wed is said to be recessive.

2. **Mendel's law of segregation:** When gametes are formed by a diploid organism, the alleles that control a trait separate from one another in to different gametes, retaining their individuality, because they are monohybrid crosses.
3. Mendel's law of independent assortment
   - members of one gene pair separate from each other independently of the members of other gene pairs. Here the

two genes located on the same chromosomes are separated and which is called dihybrid cross can take place. Therefore, the answer is A

77. During protein synthesis, where in the cell does transcription take place?
A. ribosome
B. Nucleus
C. Endoplasmic retialum
D. Golgi apparatus

Explanation: Transcription is the process in which the genetic information converts from a DNA code in to a mRNA code.

⇒ It take place in the nucleus.

Translation is the process in which the mRNA code is converted in a sequence of amino acids, it take place in the cytoplasm where ribosome are located. Therefore the answer is B.

78. Which of the following is not a term used to describe organisms that have had foreign genes added to them?
A. Genetically modified organisms
B. Transgenic organisms
C. Genetically engineered organisms
D. pathogenic organisms

Explanation:

⇒ From the given choice, genetically modified organisms, transgenic and genetically engineered organisms are the same terms (phases) which described, adding of foreign genes in organisms, such as in plants and animals to improve the desired characteristics.

⇒ Where as pathogenic organisms, those organisms which cause different diseases in plants, animals and including human beings. Therefore the answer is D

79. Which field studies the way in which individual traits of organisms are transmitted from one generation to the next?
A. Genetics
B. Ecology
C. Evolution
D. morphology

Explanation: Ecology is the branch of biology that studies about the relationships between organisms and their environment. Evolution is the changes in the frequency of genetically determined characteristics with in a population over time.
- morphology is the study to of the shape, size and the structure of living organisms.

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Genetics is also the branch of biology that studies of genes, how genes produce characteristics and how the characteristic are transmitted from one generation to the next. Then the answer is B.

80. Which molecules carry the instructions for protein synthesis?
A. carbohydrates and lipids
B. Amino acids
C. DNA-and RNA
D. Enzymes

Explanation: DNA and RNA are molecules that carry the instruction for protein synthesis. Therefore the answer is C

81. In protein synthesis, what is produced during transcription?
A. protein B. mRNA C. DNA D. polypeptide

Explanation: Transcription is the process that converts genetic information from a DNA code in to an mRNA code. In this process

$$DNA \xrightarrow{transcription} mRNA \xrightarrow{translation} proteins$$

⇒ Then the answer is B

82. What is a perfect flower?
A. A flower with petals
B. A flower with stames and sepals.
C. A flower with stamens and pistil
D. A flower with sepal and petals

Explanation: A flower which contains both male (stamens) and the female (pistil) part are called perfect flower. Where as a flower which contains male or a female part is called imperfect flower. So the answer is C

83. Which of the following types of chromosome mutations gives rise to an extra chromosome?

Explanation: Mutation is any change of nucleotide sequence of DNA in the genetic information. There are four types of chromosomal aberrations. These are:

1. Inversion = occurs when a chromosome is broken and a piece becomes reattached to it's the original chromosome.
2. Duplication occur when a portion of a chromosome is replaced and attached to the original section in sequence.
3. Deletion occurs when a broken piece becomes cost or destroyed before it can attached.
4. Translocation occurs when on broken segment of DNA becomes integrated in to a different chromosomes. From the given choice the answer is B.

84. Which of the following pairs of individuals have identical DNA finger prints?

A. Father and son
B. mother and daughter
C. Brother and sister
D. monozygotic twins

Explanation: An individual's finger prints are unique and do not change throughout life. An identical twins have identical DNA finger prints. Therefore the answer is D

85. Which of the following mating produces children all having the same ABO blood phenotypes?
A. AO × AB
B. BB × OO
C. AO × BO
D. AA × BO

Explanation: In mating process those children producing all having the same ABO blood phenotypes are blood group AO × BO. This mating type produce all blood types. By using pennet squr.

	B	O
A	AB	AO
O	BO	OO

Then it is possible for two parents with blood groups A and B to have four children, each with a different blood group. Their children may have the blood type AB, AO(A) and OO or AB, A, B and O blood. Therefore the answer is C

86. If a DNA contains, 10% thy mine what is the percentage of cytosine in the DNA?
A. 10% B. 30% C. 40% D. 90%

In the DNA molecules Adenine (A) is paired with thymine (T), then A = T and Guanine (G) is paired with cytosine (C) G = C
The percentage of thymine is equal to the percentage of adenine = A(10%) + T(10%) = 20%
The percentage of Guanine is equal to the percentage of cytosine
100% - 20% = 80% = Then percentage of cytosine is 40%. Therefore the answer is C

87. What is the branch of biology that studies about the origin and gradual changes of living things?
A. microbiology
B. Evolution
C. mutation
D. Reproduction

Explanation: Microbiology is a branch of biology that studies about microorganisms. Mutation is the change of the DNA sequence of organisms. Reproduction is a process in which the production of a new generation of individuals of the same species. Evolution is also the branch of biology that studies about the origin and gradual changes of living organisms. Therefore the answer is B

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88. Which of the following techniques is used to separate DNA fragments according to their size on a gel?
A. Southern blotting C. Radioactive labeling
B. Electrophoresis D. Digestion by restriction enzyme
Explanation: Restriction enzymes have the ability to cat DNA at places where specific sequences of nucleotides occur. The fragments are separated by gel electrophoresis. This is because the smaller fragments move further than larger molecules. Therefore the answer is B.

89. Which of the following can be taken as case for evidence of evolution from the field of plant and animal breeding?
A. Artificial breeding always creates new species
B. Artificial breeding only produces new organisms with higher yield
C. Artificial breading shows that natural selection can produce variations.
D. Artificial breeding can produce new variations in organisms
Explanation: Artificial breeding is used to produce the variation of domestic animals and plants with desirable characteristics such as improving the quality and quantities of yield, disease and drought resistance varieties and eliminating undesirable characteristics. Therefore the answer is D

90. If a radioactive substance that weighs one kilogram has a half life of 100 years what would be the percentage of the substance left after 300 years?
A. 50 B. 30 C. 25 D. 12.5
Explanation: When the radioactive of substance increase in duration or time it decrease it percentage and reduce its weight.
For example:- half timed ½ (50%) ½ is equal to 100 years
(¼) 25% is also 200 years the (1/8) = 12.5% is 300 years
Therefore the answer is D

91. According to lamark's theory of evolution, what is the mechanism by which evolving organisms acquire new structures?
A. mutation C. use-and-disuse of parts of the body
B. heredity variation D. Recombination of ancestral genes
Explanation: The French biologist Lamark proposed in 1809, a hypothesis to account for the mechanism of evolution based on two conditions. These conditions are
1. The use – and disuse of parts
2. The inheritance of acquired characteristics.

Therefore the answer is C

92. In which kingdom of life, is it most likely to find many examples of organisms that can undergo metamorphosis?
A. monera B. Fungi C. plantae D. Animalia
Explanation: Metamorphosis is derived from two words, meta and morphe meta means change and morphe is form. Then metamorphosis is a process in which a change of form or structure of the animal during the course of its life cycle. For example some insects can show complete metamorphosis. These insects have four stages of development, including egg, Larvae, maggot and imago (adult) eg. house fly, others can show incomplete metamorphosis, this include egg, nymph and adult eg. grass hopper and locust. Then the answer is D

93. What do you call structures that have the same evolutionary origin but now are different in structure or have different functions?
A. Endemic B. Analogous C. Homologous D. Indigenous
⇒ Organisms with similar anatomical features are closely related evolutionary because, they shared a common ancestor.
⇒ Some organisms have anatomical structures that are very similar inform and evolutionary origin, but have very different function. Such structures are called homologous structures.
⇒ Other organisms having the same function but different anatomy and different evolutionary origin. These structures are called analogous structures. Therefore the answer is C.

94. Which of the following is prevented from taking place if populations are separated by a geographic barrier?
A. mutation B. Evolution C. Gene flow D. Natural selection
Explanation:
- mutation is the change of the sequence of DNA
- Evolution is the gradual change of organisms through the over period of time.
- Natural selection is the theory that explain of species in terms of survival of those best adapted to a specific environment.
- Gene flow is the movement of genes from generation to the next or from one region to another.
- Then the answer is C

95. The half life of carbon -14 is about 5730 years. If a fossil is 17200 years old, about what percent of its original carbon – 14 is still present in the fossil?

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A. 75% B. 50% C. 25% D. 12%

Explanation: Half life ½ or 50% = 5730 years.

¼ or 25 , 11460 years or 5730 × 2

1/8 or 12.5% = 5730 × 3 = 17190 it is around 17200years, then the percentage is 12.5%. The answer is D

96. Which of the following pairs are analogous structures?

A. wing of a bird and wing of butter fly
B. front leg of a horse and a human arm
C. Awing of bird and a wing of bat
D. front leg of a frog and a wing of but

Explanation: based on the anatomical structure organisms are classified in to homologous and analogous structures.

⇒ Homologous structure are those structures having similar origin and the same form but have different functions.

For examples:- the human arm used for handling lools

- the front leg of a horse used to riding
- the front leg of a frog used for jumping
- the wings of birds and bal used for flying
- the flipper of wholes for swimming

These evidences can tell as these organisms can arose on the same ancestor.

⇒ Analogous structures are those structures have different origin but have the same function:

For example:- the wing of birds and the wing of butterfly. Therefore the answer is A.

97. Among the following fossil hominid species, which one is the oldest of all?

A. Homo erectus C. Australopithecus afarensis
B. Ardipithecus ramidus D. Australopthecus africanes

Explanation: There is a great deal of fossil evidence that several species of hominids of the genera ardipithecus, Australopithecus and genus homo. The earliest fossil record stats from ardipithecus ramidus (selam) found before 5 million years ago, australopithecs a farensis (Luci) around 3 million years ago

⇒ After Australopithecus afarensis it develop in to Australopithecus Africans.

Genus Australopithecus can developed in to genus homos such as homo habilis, homo erectus and then to homo sapiens.

Then from hominid species ardipithecus ramidus or selem is the oldest of all gnera of human ancentors. Therefore the answer is B

98. Which of the following conclusions can be drawn from the structural similarities observed between the flippers of wholes and the arms of humans?

A. The human species began life in the ocean
B. The human species and whales share a common ancestry
C. Whales are older than the human species
D. whales have evolved from early humans that went back to the ocean

Explanation: The flippers of whales and the arms of humans have similar anatomical strucutes and closely related evolutionary. This is because, they shared a common ancesters. These anatomical structures are called homologous structure. For example – the flippers of whales used for swimming where as the arms of humans for manipulation (handling) therefore, the answer is B

99. An experimental animal stopped responding to a stimulus that has been repeated to many times. What type of learned behavior is this?

A. Latent learning C. Imprinting
B. Habituation D. Operant conditioning

Explanation:

⇒ Latent learning is a hidden learning and it happens when the brain acquires knowledge at a certain time without reinforcement.

⇒ Imprinting is not learned behavior, it is innate behavior and instinctive behavior which inborn with an organism.

⇒ Operant conditioning is classical conditioning involves modifying an innate response by pairing it with a previously neutral stimulus.

⇒ Habituation is a type of learned behavior results in a decreased (stopped) response to a stimulus after repeated exposure to that stimulus over a period of time. Therefore the answer is B

100. Behaviour that is repeated on a daily basis is referred to as:

A. Lunar B. Circadian C. circannual D. seasonal

Explanation: Monthly cycle is lunar shythums, yearly rhythms is called cireannual rhythm. Daily rhythms are called circadian rhythms.

Answer: B

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DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question carefully and **blacken** the letter of your choice on the separate answer sheet provided.

1. Which of the following step of the scientific method comes before all the rest?
   A. Hypothesis C. Conclusion
   B. Experiment D. Reporting the result

   Explanation: In scientific method the correct sequential order of the steps involved is:
   Hypothesis → Experiment → Reporting the result of the experiment → Conclusion **Answer: A**

2. What do we call a scientist who specializes in studying insects
   A. An ecologist C. An entomologist
   B. A paleontologist D. A microbiologist

   Explanation:

Scientist	Field of specialty
Ecologist	Environment study
Paleontologist	Studies about fossils
Entomologist	Studies about insects
Microbiologist	Studies about microorganisms

   Answer: C

3. To which one of the following organic molecules do enzymes belong
   A. Carbohydrates C. Proteins
   B. Amino acids D. Lipids

   Explanation: Enzymes are biological catalysts which are composed of globular proteins. **Answer: C**

4. Which of the following is a coenzyme?
   A. NAD C. Water
   B. Systematic errors D. Proteins

   Explanation: coenzymes are the non-proteins parts a cofactor which used to bind to an enzyme to give it its catalytic activity. The most commonly known examples of coenzyme are **NAD** (Nicotine amide adenine dinucleotide) and **FAD** (Flavin adenine dinucleatide)
   Answer: A

5. Which of the following units of measurement is more convenient to express the size of cellular organelles?
   A. Meter C. Millimeter
   B. Centimeter D. Micrometer

   Explanation: cells vary by their size some cells like egg cells of birds are large and can be easily seen by our naked eyes. However,

most cells are small and can only be seen by microscope. Such cells are measured by a unit called **micrometer (μm).**

6. Which tool of the biologist is more frequently used to study cell structure in the laboratory? **Answer: D**
   A. Dissecting kit C. Petri-dish
   B. Centrifuge D. Microscope

   Explanation: All the equipments listed are important in biological studies. However, microscope is the basic tool of a biologist which is most frequency used in biological laboratories. **Answer: D**

7. What colour do gram-positive bacteria stain with Gram's stain?
   A. Red C. Purple
   B. Pink D. White

   Explanation: the major difference between Grham positive and Grham negative bacteria is that, Grham positive bacteria have ability to easily interact and retain their primary stains. Hanu Christian Gram stained gram-positive bacterial purple.
   Answer: C

8. Which group of organisms in the ecosystem release nutrients locked up in dead bodies of organisms?
   A. Parasites C. Autotrophs
   B. Decomposers D. Carnivores

   Explanation: Decomposes (bacterial and Fungi) play a role in recycling activities by breaking down dead bodies of organisms, inorder to return nutrients locked in organisms body back to the nature. **Answer: B**

9. Which one of the following are the two major constituents of eukaryotic chromosomes?
   A. DNA and RNA
   B. DNA and protein
   C. DNA and carbohydrate
   D. RNA and lipid

   Explanation: the chromosome of eukaryotic cells contain DNA that is coined around proteins called histones. **Answer: B**

10. Which one of the following behavioral biologists is known for his study about imprinting behavior in animas?
    A. W.Kohler C. Ivan Pavlov
    B. B.F. Skinner D. Konrad Lorenz

    Explanation: Konrad Lorenz imprinted the goslings on himself during their sensitive phase. They formed along lasting bon to him and followed him readily.

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When an animal or person learns the characterstics of some stimulus, it is said to be "imprinted" on to the subject.

Answer: D

11. Which group of animals has the highest numbers of total and endemic species in Ethiopia?
A. Amphibians
B. Birds
C. Mammals
D. Reptiles

Explanation:

Animals endemic to Ethiopia	Number of species
Birds	27
Mammals	22
Amphibians	17
Fish	4
Reptiles	3

Answer: B

12. What is the molecule that supplies the quickest and suitable source of energy to cells?
A. Lactose
B. Sucrose
C. ATP
D. Lipid

Explanation: **ATP**(Adenosme triphosphate) is the immediate or the cash form of energy that can be directly used by our cells.

Answer: C

13. A carbohydrate compound is known to have 12 carbon atoms in the whole molecule. What could this compound be?
A. A polypeptide
B. A disaccharide
C. A monosaccharide
D. A polysaccharide

Explanation: Disaccharides are groups of carbohydrate that have two sugar units. One sugar unit contains 3 – 8 carbon atoms. The most commonly known examples of disaccharides are:

lactose
sucrose } $C_{12}H_{22}O_{11}$
maltose

Answer: B

14. Among the following organisms, which one belongs to the prokaryotes?
A. Paramecium
B. *Streptococcus*
C. Spirogyra
D. Tapeworm

Explanation: Prokaryotes are groups of organisms that have nuclear material lacking nuclear membrane, thus they freely suspend in the

cytoplasm. In addition, prokaryotes lack some organelles which are found in eukaryotic cell and they also lack double membrane bounded organelles. Prokaryotes belong to all organisms kingdom protista (bacteria and blue green).

Answer: B

15. Which one of the following crops has its centre of origin and diversity in Ethiopia has become a leading international commodity of commerce?
A. Teff
B. Enset
C. Coffee
D. Anchote

Explanation: All the plants listed are originated form Ethiopia. Above all, coffee (coffee Arabica) which is originated form Ethiopia has nowadays become the leading international commodity of commerce.

Answer: C

16. Baby ostriches tend to follow the first moving object that they see as they hatch out of the eggs. What kind of animal behavior does this demonstrate?
A. Positive taxis
B. Innate behaviour
C. Positive kinesis
D. Learned behaviour

Explanation: Innate behaviour is the behaviour that is genetically determined and the behaviour is fully functional at first attempt. Thus, the ability of baby ostriches to follow the first moving object that they see as they hatch out of eggs is an example of innate behavior.

Answer: A

17. Which of the following is true about mutations that occur in normal body cells?
A. They never lead to cancerous cells.
B. They never pass to the next generation.
C. They never damage the affected cells.
D. There is no way that they kill the affected cells.

Explanation: Mutation is a sudden change in the amount, arrangement or structure of the DNA of an organism. It may occur on mormal body cells (somatic cells) or sex cells (gametes). If mutations occur on somatic cells they do not pass to the next generation, but, mutation on sex cells are inheritable.

Answer: B

18. What is the process called when two bacteria directly contact cell to cell and exchange their genetic information?
A. Conjugation
B. Transformation
C. Co-transformation
D. Transduction

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Explanation: Bacterial conjugation is the transfer of genetic material between bacterial cells by direct cell to cell contact or by a bridge-like connection between two cells. **Answer: A**

19. Why is it that mutations are considered as one of the raw materials of evolution?
   A. They contribute to new variations in organisms.
   B. They are usually related to the environment in which they appear
   C. They are mostly beneficial to the environment in which they appear
   D. They usually become the causes for species extinction.

Explanation: If a mutated gene is advantageous to the organism, its frequency increases in successive generations. The number of organisms with the new gene will also increase as natural selection favours them. Thus, mutations are the raw materials for evolution because they contribute to variation. **Answer: A**

20. Which one of the following is consistent with understanding of human evolution?
   A. Bipedalism was never important in human evolution.
   B. Larger brain size had no contribution to the evolution of the human species
   C. Human ancestry had no relation whatsoever with that of the chimpanzees.
   D. Fossils of Lucy and Ardi provided evidence for human origin.

Explanation: Lucy and Ardi are both important explanation of modern man and chimpanzee evolution. Lucy was bipedal and the brain of the same size as chimpanzee. Ardi is bipedal and closer to the common ancestor of human and chimpanzees. **Answer: D**

21. If certain food stuff gave a positive result with Benedict's test only after it was hydrolyzed with hydrochloric acid, which of the following substances could it be?
   A. Non-reducing sugar C. Lipid
   B. Nucleic acid D. Protein

Explanation: Non reducing sugars such as are disaccharides such as sucrose. This, substances can get hydrolyzed when they react with hydrochloric acid. **Answer: A**

22. In scientific method what is the importance of a background research for a given problem
   A. To formulate a hypothesis
   B. To make a prediction
   C. To prepare report of the result
   D. To make a conclusion

Explanation: Inorder to formulate a hypothesis the researcher must have to do a back ground research for a collecting sufficient information about the problem under investigation. **Answer: A**

23. Which of the following is NOT true about mitochondria and chloroplasts?
   A. Both contain chlorophyll
   B. Both contain nucleic acid
   C. Both have double membrane
   D. Both transduct energy

Explanation:

Mitochondria	Chloroplast
✓ Have no chloroplast ✓ Have its own nucleic acid (DNA) ✓ Have double membrane energy transducting organelles.	✓ Contain chloroplast ✓ Have its own nucleic acid (DNA). ✓ Have double membrane ✓ Energy tranducting organelles.

Answer: A

24. Which of the following molecules is NOT a polymer? ?
   A. Protein C. RNA
   B. Lipid D. Starch

Explanation: Polymer is a molecule that is made from many smaller identical molecules called monomers.

Polymer	Monomer units
Protein	Aminoacids
RNA	nucleotides
Starch	Simple sugars (glucose

∴ Lipids are composed of two different types of molecules, fatty acid and glycerol. Thus, they are not ~~polysaccharides~~ polymers **Answer: B**

25. Which of the following is formulated as a possible outcome of an experiment?
   A. Hypothesis C. Data analysis
   B. Prediction D. Recording the data

Explanation: Hypothesis is a tentative explanation about an outcome of an experiments, which will be accepted or rejected as a result of the experiment. Hypothesis is sometimes called an educated guess: **Answer: B**

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26. Which one of the following classes of molecules CANNOT pass easily across the cell membrane by simple diffusion?
A. Small non polar molecules
B. Lipid soluble molecules
C. Non-polar molecules
D. Polar molecules

Explanation: Lipid soluble molecules are too large to pass easily across the plasma membrane by simple diffusion. **Answer: B**

27. Which one of the following substances has a shape which is similar to that of the substrate of an enzyme?
A. The reaction product
B. A competitive inhibitor
C. A cofactor
D. An allosteric inhibitor

Explanation: competitive inhibitors are molecules that have similar shape to that of the substrate. They reduce enzymic reaction by preventing substrate form binding to the active site taking its place.
Answer: B

28. At which level of structural organization do proteins have the alpha-helix shape?
A. Primary structure
B. Secondary structure
C. Tertiary structure
D. Quaternary structure

Explanation: In secondary structure of protein a peptide chain folds in to a α -helix plated sheet. Such as wool, hair, skin, nails etc.
Answer: B

29. Among the following scientists who contributed to the cell theory, identify the one who stated that '*a cell can arise only from another cell like it*'.
A. Robert Hooke
B. Theodor Schwann
C. Matthias Schleiden
D. Rudolf Virchow

Biologist	Contribution in cell biology
Robert Hooke	Discovered cells and coined the name cell
Theodor Schwann Matthias Schleiden	Formulated the cell theory
Rudolf Virchow	Developed on idea which says that' cells arise only from previously existing cells.

Answer: D

30. What does it mean when biologists express the cell membrane as a unit membrane?
A. A cell is covered by a single membrane.
B. A membrane is only one lipid layer thick.
C. All cells have essentially similar membrane.
D. A membrane is covered by a single layer of protein.

Answer: D

31. Which one of the following features makes human T-lymphocyte cells more vulnerable to HIV attack? Presence of;
A. Cell membrane.
B. DNA on the chromosomes.
C. CD4 on the membrane surface.
D. Pores in the cell membrane.

Explanation: HIV infects T-helper cells, specialized of the immune system. The virus attaches to protein receptors know as CO_4 on the host plasma membrane. **Answer: C**

32. Which one of the following terms refers to how fast an enzymes acts on its substrate?
A. Turn over number
B. Substrate number
C. Enzyme number
D. Product number

Explanation: Turn over rate is the number of enzyme that binds to substrate within a given period of time **Answer: A**

33. Which one of the following factors determines the rate at which organelles settle out of cell homogenate if spun in a centrifuge?
A. Mass of the organelle.
B. Function of the organelle in the cell.
C. Location of the organelle in the cell.
D. Thickness of the membrane covering the organelle.

Explanation: Centrifuge is a machine that spins at high speed to separate components of different molecular weights. **Answer: A**

34. Which of the following classes of organic molecules is the least important source of energy for cellular respiration?
A. Nucleic acids
B. Lipids
C. Carbohydrates
D. Proteins

Explanation: Carbohydrates and fats are the primary source of energy for the body. Nucleic acids (RNA and DNA) contain a five carbon sugar (carbohydrate). However, proteins are concerned with building and repairing the body, but are not source of energy.
Answer: D

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35. Which one of the following modes of transport is used by cells to move substances against their concentration gradients?
A. Osmosis
B. Simple diffusion
C. Facilitated diffusion
D. Active transport

Explanation: Osmosis and diffusion are passive transport systems that involve the movement of substances along the concentration gradient. However, active transport system occurs against the concentration gradient. **Answer: D**

36. Which of the following is **NOT** true about instinctive behavior?
A. It can be developed further through learning.
B. It is triggered by a key stimulus.
C. It has a fixed action pattern.
D. It is adaptive for the species.

Explanation: **Instnictive behavious** are generally pre programmed or pre-set, unlearned behaviours. They donot require learning or experience. **Answer: A**

37. In which of the following ways do retroviruses differ from other RNA viruses?
A. Their genetic material is DNA.
B. Their genetic material is RNA.
C. They copy RNA to DNA molecule.
D. They copy RNA from DNA molecule.

Explanation: Retro viruses have RNA as their genetic material in their host. They convert their RNA in to DNA by the process called reverse transcription. **Answer: C**

38. Which one of the following groups of micro-organisms does not contain parasitic members?
A. Bacteria
B. Algae
C. Fungi
D. Protozoa

Explanation: Algaes are not parasite, because all algaes have chloroplast and are photosynthetic. **Answer: B**

39. Which one of the following is NOT the correct characters of tropical rainforests?
A. Low biodiversity
B. High temperature
C. Heavy precipitation
D. Trees of different heights

Explanation: Tropical rain forest is a biome which gets the highest amount of rainfall than any other biomes. Thus, it contains variety of plant and animal species. Accordingly, tropical rain forest it the richest biome by its biodiversity. **Answer: A**

40. Which one of the following processes has a decreasing effect on the concentration of atmospheric carbon dioxide?
A. Cellular respiration
B. Combustion of fossil fuels
C. Decomposition of dead organisms
D. Photosynthesis

Explanation: the process of photosynthesis requires carbondixoide as one of the raw materials to synthesize sugar. Thus, the process of photosynthesis has declining effect on atmospheric CO_2. **Answer: D**

41. For what purpose do molecular biologists use the technology known as polymerase chain reaction or PCR?
A. To insert DNA into plasmids
B. To insert plasmid into bacteria
C. To multiply copies of DNA molecule.
D. To produce DNA form RNA.

Explanation: **Polymerase chain reaction** (PCR) is a procedure by which free nucleotides are assembled into a nucleic acid chain by enabling the activity of bacteria DNA polymerase to bind them together. **Answer:**

42. What is the long term primary effect of the current tree planting activities that Ethiopia is undertaking?
A. It will protect from harm of solar rays
B. It will increase the global temperature.
C. It will reduce the atmospheric CO_2.
D. Absence of natural selection

Explanation: Tree planting is vital in order to reduce the level of carbondixoide in the atmosphere which is causes global warming. **Answer: C**

43. Suppose when you first enter a room you notice an unpleasant smell which you eventually forget about its presence, what is this behavior called?
A. Latent learning
B. Insight learning
C. Habituation
D. Operant conditioning

Explanation: Habitation is a process which results in a decreased response to a stimulus after repeated exposure to that stimulus over a period of time. **Answer: C**

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44. Which one of the following is an evolutionary requirement for two sub-populations of a species to evolve into independent species?
A. Free exchange of genev
B. Geographic isolatoin
C. Free migration between populations
D. Absence of natural selection

Explanation: Geographic isolation is an example of allopatric speciation that results when a population from the existing species becomes geographically isolated and the isolated population develops into the new specie. **Answer: B**

45. Form evolutionary point of view, which of the following animals is expected to have hemoglobin proteins that are least similar to that of human?
A. Ape
B. Cow
C. Chicken
D. Frog

Explanation: Human beings have close evolutionary relationship with apes. Thus, there are various biochemical similarities among organisms from closer evolutionary tree. **Answer: A**

46. Which element is found in nucleic acids?
A. Calcium
B. Iron
C. Magnesium
D. Phosphorus

Explanation: Nucleic acids (DNA and RNA) are composed of chain of nucleotides which contain:
✓ Phosphate unit
✓ Nitrogenous bases
✓ Pentose sugar
Answer: D

47. From where do plants get most of their nutrients?
A. Chlorophyll
B. Soil
C. Light
D. Atmosphere

Explanation: plants absorb water and minerals that are used for synthesis of various components in plants. **Answer: D**

48. What do you call a group of genetically identical plants produced by vegetative reproduction?
A. Family
B. Clone
C. Hybrid
D. Genus

Explanation: Cloning is the production of identical organisms by asexual reproduction. **Answer: B**

49. What is the base found in RNA in place of thymine of DNA?
A. Cytosine
B. Guanine
C. Thymine
D. Uracil

Explanation: RNA and DNA are both nucleic acid, but they differ in various ways.

RNA	DNA
✓ Single stranded	✓ Double standard
✓ Contain base uricil instead of thymine	✓ Has thymine base but not uracil
✓ Cannot replicate	✓ It ruplicates
✓ Has ribose sugar	✓ Has deoxy ribose sugar

Answer: D

50. What does a restriction enzyme do?
A. Restricts transcription
B. Cuts DNA at specific sites
C. Prevents DNA from replicating
D. Hydrolyzes the DNA molecule

Explanation: Restriction enzymes are groups of enzymes that are capable of separating DNA at specific points along the chain. **Answer: B**

51. Which one of the following is the smallest of all?
A. A red blood cell
B. A virus
C. A bacterium
D. An amoeba

Explanation: Viruses are the smallest of all in size which ranges in between 0.01 – 0.1μm in diameter **Answer: B**

52. What is the ultimate source of all scientific knowledge?
A. Observation
B. Guessing
C. Trial and error
D. Observation and experimentation

Explanation: The primary source of scientific knowledge is observation, which will be later tested through experimentation. **Answer: D**

53. The optimum temperature of enzymes found in thermophilic bacteria is:
A. Lower than for enzymes in the human body
B. Higher than enzymes in the human body.
C. The same as enzymes found in human body.
D. Lower than enzymes found in warm blooded animals.

Explanation: Thermophilic bacterias are those bacterias that live at extreme temperatures which might be above 70°C. **Answer: B**

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54. A cow was found to yield much higher milk than any of the breeds of the parental cattle. What could be the most probable reason for this?
A. Dominant genes
B. Hybrid vigour
C. Recessive genes
D. Co-dominance genes

Explanation: Hybrid vigour is making two parents of different pure breeds in order to produce an organism that has an increase in such characterstics a rate, fertility and yield over those of its parents.

Answer: B

55. The most complex structure of proteins is called
A. Primary structure
B. Secondary structure
C. Tertiary structure
D. Quaternary structure

Explanation: Quaternary structure of proteins in the most complex one which is formed when two or more polypeptide chains become associated in the final structure of the protein. **Answer: D**

56. Which are the four most abundant elements in living cells?
A. Carbon, oxygen, sulfur, phosphorus.
B. Carbon, oxygen, hydrogen, nitrogen.
C. Carbon, oxygen, nitrogen, sulfur.
D. Carbon, oxygen, sulfur, magnesium.

Explanation: The four most abundant elements in living cells are **carbon, hydrogen, oxygen** and **nitrogen**. Because they are elements that make the organic substances (carbohydrate, lipid and protein).

Answer: B

57. When the F1 hybrid of a monohybrid cross is back crossed with the homozygous recessive parent, what percentage of the offspring would be homozygous recessive?
A. 0%
B. 25%
C. 50%
D. 75%

Explanation: for instance let us take Mendels hybrid tall pea plant when crossed with homozygous short pea plant as an example.

Hybrid tall pea plant = Tt
Homozygous short pea plant = tt

Tt × tt

	T	T	2Tt: 2tt → genotype
t	Tt	tt	50%hybrid tall : 50% short → phenotype
t	Tt	tt	

Answer: C

58. Which of the follwins is an inorganic molecule?
A. $CaCO_3$
B. CH_4
C. $C_6H_{22}O_{11}$
D. $C_{18}H_{36}O_2$

Explanation: All organic compounds necessarily contain carbon and hydrogen.

Answer: A

59. Which one of the following expression is more related to the phrase "survival of the fittest"?
A. Natural selection
B. Mendelian inheritance
C. Gene mutation
D. Inheritance of acquired characteristics

Explanation: The theory of natural selection which is stated by Charles Darwin has for major ideas:
✓ **Fecundity** – producing more offspring than that can possibly survive.
✓ **Variation** – differences among offspring.
✓ **Struggle for survival**
✓ **Survival of the fittest** – better adapted organism can survive among organisms of the same species.

Answer: A

60. In the formation of a macro molecules, what type of reaction should join two subunits together?
A. Hydrolysis reaction
B. Dehydration reaction
C. Denaturation reaction
D. Hydrophobic reaction

Explanation: Dehydration is the combination of two molecules which releases water as a byproduct. **Answer: B**

61. What did Francesco Redi prove through his scientific experiment?
A. Maggots appear spontaneously on foods placed anywhere.
B. Maggots do not appear in foods kept in jars that are protected with a cover.
C. Maggots do not appear in foods kept in open jars.
D. Flies appear spontaneously on rotting meat kept in closed or open jars alike.

Explanation: **Francesco Redi** was able o disprove the theory that maggots could be spontaneously generated from meat. What he did was, he took eight jars, placed meat in all the jars, but covered four of the jars. Maggots were developed in open jars, but did not develop in covered jars.

Answer: B

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62. Which one of the following shows the feeding method of decomposers?
A. Saprobiotic nutrition
B. Autotrophic nutrition
C. Parasitic nutrition
D. Intracellular digestion

Explanation: Decomposers are organisms such as bacteria and fungi which break down dead bodies of organisms. These groups of organisms are called **saprophytes.** **Answer: A**

63. What happens when the carrying capacity of an ecosystem is reached?
A. Excretory product accumulates and population numbers increase.
B. Population number decline rapidly.
C. Population number remain more or less constant.
D. Resources are plentiful and populations shoot up.

Explanation: **carrying capacity** is the condition where the number of organisms of a species which an environment can support at a particular time. At this stage population number stays constant. **Answer: C**

64. In which industrial products is pyruvate fermentation by yeast practically applied?
A. Brewing beer
B. Swiss cheese making
C. Production of vinegar
D. Yoghurt making

Explanation: At anaerobic situation yeast acts on sugar inorder to produce alcohol (ethanol). So, this process involves brewing of beer. **Answer: A**

65. To which group of organic compounds do the triglycerides and waxes belong?
A. Carbohydrates
B. Proteins
C. Lipids
D. Vitamins

Explanation: Triglycerides are neutral fats which are formed when glycerol chemically combines with three fatty acids.
Wax is a fatty acid which is combined with along chain of alcohol. **Answer: C**

66. What does an ethologist study?
A. Insect diets
B. Soil types
C. Fossil fuel
D. Animal behaviour

Explanation: **Ethologist** is a person who studies the behaviour of animals in natural settings. **Answer: D**

67. Which one of the following is a biotic factor operating within an ecosystem?
A. The amount of helium gas in the air
B. The rate of flow of water in a river
C. The carnivores that consume other animals
D. The type of climate in a given region

Explanation: In an ecosystem the biotic factors are living things that operate within an ecosystem. Thus, carnivores that consume other animals are the biotic factors. **Answer: C**

68. When athletes take part in short distance running, how do the cells generate most of the energy that is quickly needed?
A. Aerobic respiration in muscle cells.
B. Mitochondrial respiration in any cells.
C. Anaerobic respiration in muscle cells.
D. Yeast fermentation in the stomach.

Explanation: an aerobic respiration in humans takes place where people do not in hale sufficient amount of oxygen to break glucose as quickly as needed. **Answer: C**

69. What do you call the reasoning technique in science where general principles are used to analyze specific cases?
A. Induction
B. Deduction
C. Pseudo-deduction
D. Pseudo-induction

Explanation:
✓ **Inductive reasoning:** is a reasoning technique that process form specific to general.
✓ **Deductive reasoning:** a reasoning which proceeds from general to specific. **Answer: B**

70. If a substance that weighs 2,000 grams and has a half-life of 100 years is left with only 250 grams, for how long has the radioactive decaying activity been undergoing?
A. 200 Years
B. 250 Years
C. 300 Years
D. 500 Years

Explanation: **Half-life** is the time needed for half atoms of the radioactive element to decay. Thus, the substance having weight of 200 grams and half-life of 100 years decays into 250 grams within 300 years.

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2000 gram $\xrightarrow{100\text{ years}}$ 1000 gram $\xrightarrow{100\text{ years}}$

500 gram $\xrightarrow{100\text{ years}}$ 250 gram

Answer: C

71. Which one of the following will primarily happen if the enzymes in the isosomes of a cell are defective?
   A. Cellular debris will not be removed
   B. Chromosome replication will cease
   C. ATP production will stop
   D. Diffusion process will stop

Explanation: Lysosomes are cellular organelles that contain enzymes which are used:
- To remove cellular debris
- For intercellular digestion

Answer: A

72. Which of the following is true about all proteins?
   A. They are twenty amino acids long.
   B. They are globular in shape.
   C. They perform the same function.
   D. The monomers are held together by peptide bonds.

Explanation: substances which are produced form monomer units called **aminoacids.** The aminoacids that make protein molecule are joined together by a **peptide bond.**

Answer: D

73. Which of the properties of water allows mosquito larvae to protect themselves from sinking down into the water from the water surface?
   A. Its high specific heat capacity
   B. Its high latent heat of vaporization
   C. Its high surface tension
   D. Its low density in solid state.

Explanation: **High surface Tension** is the tension on the surface of a liquid resulting from un balanced forces acting on the molecules at the surface. This property allows small organisms such as insects to walk on the water without breaking its surface.

Answer: C

74. Which of the following properties of enzymes makes it possible that a single enzyme molecule can act on many substrate molecules?
   A. Enzymes are proteins
   B. Enzymes lower than energy of activation
   C. Enzymes are reused over and over again
   D. Enzymes are substrate specific

Explanation: Enzymes can be reused over and over again. Thus, because of this property of enzyme can react with many substrate molecules.

Answer: C

75. If the rate of the enzyme catalyzed reaction remains constant even when more substrate is added to the reaction, which of the following might be the reason?
   A. Saturation of the enzyme
   B. Inactivation of the enzyme
   C. Inhabitation of enzyme-substrate complex formation
   D. Loss of substrate specificity by the enzyme

Answer: D

76. Which stage in the life cycle of HIV is disrupted, if AIDS patients are treated with a drug that has a protease inhibiting activity?
   A. Entry of the virus into the host.
   B. Conversion of viral RNA to DNA
   C. Integration of viral DNA into host DNA.
   D. Assembly of viral parts into whole virus.

Answer: D

77. Of the following four cells whose surface area to volume ratio is given, which tell can more efficiently transport its needs of materials across the cell surface?
   A. 24 : 8 ratio
   B. 54 : 27 ratio
   C. 96 : 64 ratio
   D. 150 : 125 ratio

Explanation: An organism with large surface are to volume ratio can more efficiently transport its needs of materials across its cell surface.

Answer: A

78. Cells immediately use the energy that electrons lose as they pass along the chain of electron carriers to:
   A. Produce ATP
   B. Pump protons
   C. Spin the rotor of ATP synthase
   D. Reduce NAD

Explanation: In electron transport chain the electrons pass along a series of electron carriers that form a transport chain to lose energy as they pass from one carrier to the next. As electrons are transferred through the proton pumps, they lose energy which pumps the protons into the inner membrane space.

Answer: B

79. Which one of the following is a routine method used by clinics to test people for HIV infection?
   A. Microscopic examination for the virus

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B. Counting the number of white blood cells.
C. Testing for human anti-HIV antibody.
D. Measuring the amount of hemoglobin.

Explanation: the lymphocytes make antibodies (anti-HIV antibody) to fight free HIV in the blood when the virus is presenting the body. Thus, the presence of those antibodies is detected and the person is diagnosed being HIV-positive. **Answer: C**

80. The T-lymphocyte cells of AIDS patients are destroyed by:
A. Multiplication of HIV inside the cell
B. Infection of opportunistic organisms
C. The immune system of the host organism itself
D. The CD4 receptors on the cell surface.

Explanation: HIV infects the T-lymphocyte cells of AIDS patients by binding to the CD_4 receptors on the surface of T-helper cells. **Answer: D**

81. Suppose a fossil initially contains 100,000 atoms of a certain radioactive element whose half life is 10,000 years, after how many years would the number of the atoms be 12500?
A. Ten thousand years
B. Twenty thousand years
C. Thirty thousand years
D. Forty thousand years

Explanation: Initially the Fossil had 100,000 atom and if the half life is 10,000 years, the number of atoms be 12500 after thirty thousand years.

$$100,000 \xrightarrow{10,000\ years} 50,000\ atom \xrightarrow{10,000\ years} 25,000\ atom \xrightarrow{10,000\ years} 12,500\ years$$

Answer: C

82. The following are Simpson's Index of species diversity calculated for four areas containing the same types of species. Which index value is from the area dominated relatively by a fewer number of species?
A. 8.00 C. 3.5
B. 6.00 D. 2.5

Explanation Low value of diversity index shows that an area is dominated by one or just few species or organisms, while an area is dominated by more number of successful species, the diversity index will be higher. **Answer: D**

83. Which one of the following happens in both cyclic and non-cyclic photophosphorylation?
A. ATP is formed
B. Oxygen is generated
C. NADP is reduced
D. Water molecule splits

Explanation: The products of noncyclic phosphorylation are ATP, $NADPH^+$ and oxygen. While, the product of cyclic photophosphorylation is ATP only. **Answer: A**

84. Which of the following is NOT a true characteristic of the populations of most of the developing countries of the world?
A. High fertility rate
B. Increasing population size
C. More number of old people than young people
D. Birth rate greater than mortality rate

Explanation: In developing countries the population number is increasing at an alarming rate. Thus, there are more children than older people, because the fertility rate is greater than death rate. **Answer: C**

85. Which of the following is NOT true about the nature of the first form of organism on earth? They were:
A. prokaryotic C. aerobic
B. unicellular D. anaerobic

Explanation: The first form of organisms on the earth are:
- Prokaryotic
- Unicellular
- Anaerobic

Answer: C

86. Choose the step that comes at the last step in the scientific method
A. proposing the research hypothesis
B. formulation of a scientific theory
C. conducting experiments
D. making predictions

Explanation: The correct sequence of steps in scientific method is:
D → A → C → B **Answer: B**

87. Which of the following is NOT true about C4 plants such as tef (*Eragrostic tef*)?
A. CO_2 is harvested during the night time.
B. The bundle sheath cells contain chloroplasts.
C. Light-dependent reaction occurs in mesophyll cells.
D. Chloroplasts of bundle sheath cell lack thylakoids.

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Explanation: In C_4 plants and all other plants CO_2 is fixed in to mesophyll cells during the day time. **Answer: A**

88. What is the reason that plants do not use nitrogen directly from the atmosphere?
A. Nitrogen concentration is low in the atmosphere.
B. The molecular size of nitrogen is too large to pass through the stomata.
C. Nitrogen can enter plants only through the root hairs.
D. Plants lack the necessary process to use elementary nitrogen.

Explanation: Plants cannot use nitrogen directly from the air. Plants can absorb nitrogen only by their roots from the soil in the form of nitrates (NO_3^-). **Answer: C**

89. What does the fermentation of glucose by yeast normally yield?
A. Lactic acid, CO_2, and ATP
B. CO_2, H2O, and 35 ATP
C. Alcohol, CO_2, and 2ATP
D. Alcohol, CO_2, and 36 ATP

Explanation:

$$C_6H_{12}O_6 \xrightarrow{\text{yeast}} C_6H_6OH + 2CO_2 + 2ATP$$ **Answer: C**

90. Which of the following came first in the course of organic evolution?
A. Photosynthetic organisms
B. Free oxygen in the atmosphere
C. Land plants
D. Multicellular organisms

Explanation: The correct sequential order is:

$A \rightarrow B \rightarrow C \rightarrow D$ **Answer: A**

91. Which tow nitrogenous bases belong to the purines?
A. Adenine and thymine
B. Adenine and guanine
C. Guanine and cytosine
D. Thymine and uracil

Explanation: Adenine and guanine are purines. They are double ringed. Uracil, cytosine and thymine are pyramines and are single ringed. **Answer: B**

92. Which one of the following is true?
A. Recessive alleles are only expressed in the homogygote.
B. Dominant alleles are expressed only in the heterozygote.
C. Recessive alleles are expressed in the heterozygote.
D. Genetically modified organisms are never used to manufacture vaccines.

Explanation: Recessive alleles are expressed only when both alles are homozygous for the recessive trait. **Answer: A**

93. Which of the following is a functional group of a fatty acid?
A. A ketone group
B. An aldehyde group
C. An amino group
D. A carboxyl group

Explanation: Fatty acids are made up of a long chain carbon skeleton that has a carboxylic acid functional group. **Answer: D**

94. Which of the following is an example of an orientation innate behaviour?
A. Kinesis in woodlice
B. Blinking of the eyes
C. Sudden withdrawal of limbs from hot object
D. Nest building by weaver birds. **Answer: A**

95. Which hormone promotes human sleepfulness in darkness and controls the sleep-wakecycle?
A. Insulin
B. Adrenaline
C. Melatonin
D. Thyroxine

Explanation: Melatonin is a hormone that is used as a sleep aid in the treatment of some sleep disorders. **Answer: C**

96. In Pavlov's experiment on the classical conditioning of dogs, what does the reaction of the dogs to the sound of the bell represent?
A. The conditioned stimulus
B. The conditioned response
C. The unconditioned stimulus
D. The unconditioned response

Explanation: In Ivan Pavolv's experiment
i) Before conditioning
   - ✓ Food was unconditioned stimulus
   - ✓ Salivation was unconditioned response

ii) After conditioning
   - ✓ Bell was conditioned stimulus.
   - ✓ Salivation was conditioned response.

Answer: B

97. What is the purpose of the infoldings of the inner-membrane of the mitochondrion?
A. Increasing the photosynthetic capacity of the cell
B. Speeding up the loss of CO_2 during fermentation.
C. Spending up the process of glycolysis.
D. Increasing the surface area for ATP production.

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Explanation: The inner folds of mitochondoria (cristae) are used to increase the surface area for cellular respiration, which is used in the production of ATP.

Answer: D

98. Which of the following is a worthwhile biological problem for scientific investigation?
   A. Estimation of the amount of carbon in a forest ecosystem.
   B. How a crying dog can tell who among elders of the village is going to die next.
   C. How witchcrafts inflict bad health by cursing a healthy person
   D. How the traditional medicine men/women cure a possessed person.

Answer: D

99. Which one of the following biomes of the Earth has the greatest diversity of species?
   A. Deciduous forest
   B. Tropical rain forest
   C. Desert
   D. Tundra

Explanation: Tropical rain forest is the biome that gets the highest amount of annual rainfall. It is the richest biome because it contains diversity of plant and animal species.

Answer: B

100. One of the following is true about bacteriophages that have lysogenic life cycle.
   A. They are RNA viruses
   B. They integrate their nucleic acid into that of the host
   C. They multiply in the host immediately after infection
   D. Progeny viruses are released by chronic release method

Explanation: In lysogenic cycle the assembled viruses leave the cells (are released) either by causing cellysis (splitting) or by chronic (on going) exocytosis form the plasma membrane.

Answer: D

Biology (EUEE) 2008

DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question carefully and blacken the letter of your choice on the separate answer sheet provided.

1. What is the molecule in plant cells that first captures the radiant energy from sunlight?
   A. ATP B. DNA C. Chlorophyll D. Carbon dioxide

Explanation: Chlorophyll is one of the requirement for the process of photosynthesis which used for absorbing sunlight.

Answer: C

2. Of the following functions, which one do triglycerides accomplish in cells?
   A. increase density of tissues and cells
   B. increase thermal insulation of cells
   C. yield limited amount of ATP for cells
   D. facilitate entry of excess water into cells.

Answer: B

3. Which of the following laboratory glassware is used for culturing bacteria?
   A. Petri dish
   B. Test tube
   C. Filter paper
   D. Measuring cylinder

Answer: A

4. Of the following, which one is the main source from which plants get the nutrients necessary for their growth and development?
   A. Light B. Chlorophyll C. Atmosphere D. Soil

Explanation: Plants get nutrients necessary for their growth and development from the soil by active transport system.

Answer: D

5. Which of the following is the correct sequence of the steps in scientific research?
   A. Hypothesis formulation → conclusion → experiment → question
   B. Question → hypothesis formulation → experiment → conclusion
   C. Hypothesis formulation → question→ experiment → conclusion
   D. Question → hypothesis formulation → conclusion → experiment

Explanation: The correct sequence of the steps in scientific research is

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